

# **AOS-CX 10.16.xxxx IP Routing Guide**

**4100i, 5420, 6000, 6100, 6200 Switch Series**



**Hewlett Packard  
Enterprise**

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# Chapter 1

## About this document

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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

### Applicable products

This document applies to the following products:

- HPE Aruba Networking 4100i Switch Series (JL817A, JL818A)
- HPE Aruba Networking 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A, R9Y03A, S4R20A, S4R21A, S4R22A, S4R23A, S4R24A, S4R25A, S4R26A, S4R27A, S4R28, S4R29A)
- HPE Aruba Networking 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)
- HPE Aruba Networking 5420 Switch Series (S0U67A, S0U55A, S0U63A, S0U64A, S0U65A, S0U75A, S0U72A, S0U78A, S0U58A, S0U73A, S0U74A, S0U71A, S0U76A, S0U70A, S0U77A, S0U60A, S0U61A, S0U62A, S0U66A, S0U68A)
- HPE Aruba Networking 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A, R8Q67A, R8Q68A, R8Q69A, R8Q70A, R8Q71A, R8V08A, R8V09A, R8V10A, R8V11A, R8V12A, R8Q72A, JL724B, JL725B, JL726B, JL727B, JL728B, S0M81A, S0M82A, S0M83A, S0M84A, S0M85A, S0M86A, S0M87A, S0M88A, S0M89A, S0M90A, S0G13A, S0G14A, S0G15A, S0G16A, S0G17A)

### Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

### Command syntax notation conventions

| Convention                                                                                                                                                                                                        | Usage                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>example-text</code>                                                                                                                                                                                         | Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([ ]).                                                                                             |
| <b>example-text</b>                                                                                                                                                                                               | In code and screen examples, indicates text entered by a user.                                                                                                                                                                                                                                                                                                                       |
| Any of the following: <ul style="list-style-type: none"><li>■ <code>&lt;example-text&gt;</code></li><li>■ <code>&lt;example-text&gt;</code></li><li>■ <i>example-text</i></li><li>■ <i>example-text</i></li></ul> | Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: <ul style="list-style-type: none"><li>■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (&lt; &gt;). Substitute the text—including the enclosing angle brackets—with an actual value.</li></ul> |

| Convention    | Usage                                                                                                                                                                                                                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <ul style="list-style-type: none"> <li>For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.</li> </ul>                                                                                                   |
|               | Vertical bar. A logical <b>OR</b> that separates multiple items from which you can choose only one.<br>Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.                                                                                                                    |
| { }           | Braces. Indicates that at least one of the enclosed items is required.                                                                                                                                                                                                                                                                                           |
| [ ]           | Brackets. Indicates that the enclosed item or items are optional.                                                                                                                                                                                                                                                                                                |
| ... or<br>... | Ellipsis: <ul style="list-style-type: none"> <li>In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information.</li> <li>In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.</li> </ul> |

## About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

### Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the **interface** context.

### Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where **<VLAN-ID>** is a variable representing the VLAN number.



# Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: ***member/slot/port***.

## On the HPE Aruba Networking 4100i Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

## On the HPE Aruba Networking 6000 and 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

## On the HPE Aruba Networking 6200 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

## On the HPE Aruba Networking 6400 and 5420 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
  - Management modules are on the front of the switch in slots 1/1 and 1/2.
  - Line modules are on the front of the switch starting in slot 1/3.
- *port*: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

## Chapter 2

# Virtual Routing and Forwarding (VRF)

VRF is a technology that allows multiple instances of a routing table to co-exist within the same router. Because the routing instances are independent, the same or overlapping IP addresses can be used without conflicting with each other. Network functionality is improved because network paths can be segmented without requiring multiple routers.

## VRF support

The 4100i, 6000, 6100, 5420 and 6200 Switch Series support predefined VRFs:

- **mgmt**: Assigned to the management port and is used to isolate management traffic (5420 and 6200 Switch Series only).
- **default**: All other interfaces are automatically assigned to the VRF **default** (6000, 6100, 5420 and 6200 Switch Series).



User-defined VRFs are not supported on the 4100i, 6000, 6100, and 6200 Switch Series.

## ipv6 route gc interval

```
ipv6 route-gc-interval <INTERVAL>
no ipv6 route-gc-interval
```

### Description

Sets the garbage collection interval timer to remove invalid or old route entries from kernel route cache. The **no** form of this command resets the garbage collection interval timer to default (30 seconds).

| Parameter               | Description                                                        |
|-------------------------|--------------------------------------------------------------------|
| <i>&lt;INTERVAL&gt;</i> | Specifies time interval in seconds. Range: 30 to 600. Default: 30. |

### Examples

Setting garbage collection interval timer to 300:

```
switch(config)# ipv6 route-gc-interval 300
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification       |
|---------|--------------------|
| 10.10   | Command introduced |

## Command Information

| Platforms     | Command context     | Authority                                                                          |
|---------------|---------------------|------------------------------------------------------------------------------------|
| All platforms | <code>config</code> | Administrators or local user group members with execution rights for this command. |



Loopback interfaces are not supported on the HPE Aruba Networking 6000 and 6100 Switch Series.



Loopback interfaces are not supported on the HPE Aruba Networking 4100i Switch Series.

A loopback interface is a virtual interface supporting IPv4/IPv6 address configuration. The loopback interface can be considered stable because once created, it remains up. The loopback interface can then be configured with an address to use as a reference or identifier independent of the physical interfaces.

- **Device identification:** As long as the router is operational, the state of the loopback interface is always up. Even if only one link to the router is active, the loopback interface can be reached. This functionality makes it possible to identify an active device in the network using the IP address configured on the loopback interface.
- **Routing:** Since the loopback interface is always active, a routing session (such as a BGP session) can continue on an alternate path even if the outbound interface fails. In OSPF, a loopback interface address is advertised as an interface route into the network. This functionality increases reliability by allowing traffic to take alternate paths if there is a link failure. In OSPF and BGP, the router ID can be set to the loopback address to avoid reassignment of the router ID when physical interfaces are added or removed.
- **Device management:** Loopback interface is always reachable and can be used for sending and receiving management information such as logs and SNMP traps without interruption.

## Loopback commands

### interface loopback

```
interface loopback <INSTANCE>  
no interface loopback <INSTANCE>
```

#### Description

Creates a loopback interface and enters loopback configuration mode.  
The **no** form of this command deletes a loopback interface.

| Parameter  | Description                                      |
|------------|--------------------------------------------------|
| <INSTANCE> | Selects the loopback interface ID. Range: 0 to 4 |

#### Examples

```
switch(config)# interface loopback  
switch(config-loopback-if)#
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                                                                                                              |
|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config          | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## ip address (interface loopback)

```
ip address <IPv4-ADDR/MASK> [secondary]
no ip address <IPv4-ADDR/MASK> [secondary]
```

## Description

Sets the IPv4 address for a loopback interface.

The **no** form of this command reverses the set of the IPv4 address for a loopback interface.

| Parameter        | Description                                                                                                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <IPv4-ADDR>      | Specifies an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.                    |
| <MASK>           | Specifies the number of bits in the address mask in CIDR format ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 128. |
| <i>secondary</i> | Indicates that the IPv4 address is a secondary address.                                                                         |

## Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 16.93.50.2/24
switch(config-loopback-if)# ip address 20.1.1.1/24 secondary
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                                                                                                              |
|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config          | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## ipv6 address (loopback)

ipv6 address <IPV6-ADDR/MASK>

### Description

Sets the IPv6 address for a loopback interface.

| Parameter   | Description                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <IPV6-ADDR> | Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where <b>x</b> is a hexadecimal number from 0 to F. |
| <MASK>      | Specifies the number of bits in the address mask in CIDR format ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 128.       |

### Examples

switch(config)# interface loopback 1

switch(config-loopback-if)# ipv6 address fd00:5708::f02d:4df6/64



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms     | Command context | Authority                                                                                                                                                              |
|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config          | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show interface loopback

show interface loopback [brief | instance <ID>]

### Description

This command displays the configuration and status of loopback interfaces.

| Parameter     | Description                                                                    |
|---------------|--------------------------------------------------------------------------------|
| brief         | Displays brief information about all configured loopback interfaces.           |
| instance <ID> | Displays the configuration and status of a loopback interface ID. Range: 1-255 |

## Examples

```
switch# show interface loopback
Interface loopback1 is up
IPv4 address 192.168.1.1/24
Interface loopback2 is up
IPv4 address 182.168.1.1/24
```

```
switch# show interface loopback brief
-----
Loopback      IP Address                      Status
Interface
-----
loopback1    10.1.1.1/24                     up
loopback1    1111:2222:3333:4444::6666/128  up
```

```
switch# show interface loopback 1
Interface loopback1 is up
IPv4 address 192.168.1.1/24
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                                                                                                              |
|---------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## Chapter 4

### Static routing

Static routes are manually configured. If the network topology is simple enough, you can use static routes for the network's routing requirements. The proper configuration and usage of static routes can improve network performance and ensure bandwidth for important network applications.

The disadvantage of using static routes is that they cannot adapt to network topology changes. If a fault or topological change occurs in the network, the relevant routes will be unreachable and a network administrator must modify the static routes manually.

These 'long prefix' IPv6 routes are programmed in a switching ASIC's TCAM table instead of the usual route table.

The access switch series are comprised of multiple ASICs of different capabilities with a mixture of line cards with a varying number of capable ASICs. For consistency across platforms, and to support a fair usage scheme amongst TCAM users, there is an imposed maximum long prefix IPv6 route capacity of 510 for all access switch series. Once this hardware limit is exhausted, long prefix IPv6 routes can still be configured/learned but will only be programmed in the software (CPU) path.

Once TCAM/long prefix IPv6 resources are freed up there is no automatic migration of a software-only long prefix IPv6 route to hardware. The route must be removed and recreated, at which point it consumes the freed hardware resource.

Note that since the TCAM table is shared between multiple applications, the 510 long prefix IPv6 route capacity is not guaranteed to be available, depending on current switch configuration/conditions. Use the 'show' resources' command to display current TCAM resource usage & availability.

Each configured/learned long prefix IPv6 route consumes four TCAM entries and these are noted in the 'Long Prefix IPv6 Unicast Route Lookup' row. The maximum 510 TCAM programmed long prefix IPv6 routes would therefore consume 2040 TCAM entries. The remaining Reserved space is necessary for internal TCAM management.

Below is an example of resource usage:

```
show resources
Resource Usage:

Mod  Description
Resource                                Used    Reserved    Free
-----
1/3  Long Prefix IPv6 Unicast Route Lookup
      Ingress TCAM Entries                16        2048
      Total
          Ingress TCAM Entries            16        2048    18432
          Egress TCAM Entries              0           0    8192
          Ingress Lookups                   1           8
          Ingress Flex Lookups              0           1
          Egress Lookups                    0           4
          Ingress Policers                  0        2047
          Egress Policers                   0        2047
1/4  Long Prefix IPv6 Unicast Route Lookup
      Ingress TCAM Entries                16        2048
      Total
          Ingress TCAM Entries            16        2048    18432
```



|       |                                       |    |      |       |
|-------|---------------------------------------|----|------|-------|
|       | Egress TCAM Entries                   | 0  | 0    | 8192  |
|       | Ingress Lookups                       | 1  |      | 8     |
|       | Ingress Flex Lookups                  | 0  |      | 1     |
|       | Egress Lookups                        | 0  |      | 4     |
|       | Ingress Policers                      | 0  |      | 2047  |
|       | Egress Policers                       | 0  |      | 2047  |
| 1/5   | Long Prefix IPv6 Unicast Route Lookup |    |      |       |
|       | Ingress TCAM Entries                  | 16 | 8192 |       |
| Total |                                       |    |      |       |
|       | Ingress TCAM Entries                  | 16 | 8192 | 57344 |
|       | Egress TCAM Entries                   | 0  | 0    | 8192  |
|       | Ingress Lookups                       | 1  |      | 8     |
|       | Ingress Flex Lookups                  | 0  |      | 1     |
|       | Egress Lookups                        | 0  |      | 4     |
|       | Ingress Policers                      | 0  |      | 8191  |
|       | Egress Policers                       | 0  |      | 4095  |

## Default route

Without a default route, a packet that does not match any routing entries is discarded and an ICMP destination-unreachable packet is sent to the source. A default route is used to forward packets that do not match any routing entry.

The network administrator can configure a default route with the destination as 0.0.0.0 and the mask as 0. The router forwards any packet whose destination address fails to match any entry in the routing table to the next hop of the default static route.

## Recursive static routes

A recursive static route is a route for which the next hop is learned from another routing look up (for example, dynamic protocol or from another another static route). For example, the following commands create an unsupported recursive static route:

- `ip route 99.0.0.0/24 30.0.0.2` - Main static route with gateway 30.0.0.2.
- `ip route 30.0.0.0/24 20.0.0.2` - Static route to reach the nexthop of the previously configured route.

## Route manager

Route Manager (or Routing Table Manager) is the central repository for all routing information. It maintains the best routes selected by each routing protocol along with various route attributes in the Routing Information Base (RIB). Route Manager also manages routes from various routing protocols and determines the best route to each destination network which needs to be programmed into Forwarding Information Base (FIB).



If the number of routes is larger than 120,000, displaying all routes will require a significant amount of time to display output. A warning message is displayed to suggest using filters to display routes for prefixes for faster results. Use Ctrl+C to interrupt and return to prompt.

# Configuration concepts

Before configuring a static route, you must understand the following concepts:

- **Destination address and mask:** In the `ip route` command, an IPv4 address is in dotted-decimal format. A mask can be in the form of mask length - the number of consecutive 1s in the mask.
- **Output interface and next hop address:** When configuring a static route, specify the output interface or next hop address. The next hop address cannot be a local interface IP address or the route configuration will not take effect.
- **Other attributes:** You can configure different priorities and administrative distance for different static routes to make route management policies more flexible. For example, specifying the same priority for different routes to the same destination enables load sharing, but specifying different priorities for these routes enables route backup.

## Configuration example procedure

Before configuring a static route, complete the following tasks:

- Configure the physical parameters for related interfaces.
- Configure the link-layer attributes for related interfaces.
- Configure the IP addresses for related interfaces.

---

**6000 and 6100 Switch Series:** The number of IPv4 and IPv6 static routes that can be configured is limited to 512 combined.

---

**5420 and 6200 Switch Series:** The number of IPv4 and IPv6 static routes that can be configured is limited to 1024 combined. 16K routes are not supported.

---

6000, 6100, 5420 and 6200 Series Switches do not support IPv6 addresses with prefixes greater than 64. Prefixes between 65 and 127 will be software-forwarded.

---



## Basic static route configuration example

The IP addresses and masks of the switches and hosts are displayed here. Static routes are required for interconnection between any two hosts.

### Procedure

1. Configure IP addresses for interfaces (details not shown).
2. Configure static routes.

- a. Configure a default route on Switch A.

```
6300# config
6300(config)#
<SwitchA> config
[SwitchA] ip route 0.0.0.0/0 1.1.4.2
```

- b. Configure two static routes on Switch B.

```
<SwitchB> config
[SwitchB] ip route 1.1.2.0/24 1.1.4.1
[SwitchB] ip route 1.1.3.0/24 1.1.5.6
```

- c. Configure a default route on Switch C.

```
6300# config
6300(config)#
[SwitchC] ip route 0.0.0.0/0 1.1.5.5
```

3. Configure the hosts. The default gateways for hosts A, B, and C are 1.1.2.3, 1.1.6.1, and 1.1.3.1. The configuration procedure is not shown.
4. Display the configuration.
  - a. Display the IP routing table of Switch A.
  - b. Display the IP routing table of Switch B.
  - c. Use the `ping` command on Host B to check the reachability of Host A, assuming Windows XP runs on the two hosts.

```
C:\Documents and Settings\Administrator>ping 1.1.2.2
Pinging 1.1.2.2 with 32 bytes of data:
Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
15
Ping statistics for 1.1.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

- d. Use the `tracert` command on Host B to check the reachability of Host A.

```
8320# tracert
tracert Trace the IPv4 route to a device on the network
tracert6 Trace the IPv6 route to a device on the network
8320# tracert
A.B.C.D Enter IPv4 address of the device to tracert
WORD Enter host name of the device to tracert
8320# tracert
```

## Static routing commands

### ip route

```
ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |
nullroute}
no ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject
|
nullroute}
```

### Description

Adds an IPv4 static route on the default VRF.

The **no** form of this command deletes a IPv4 static route.




---

You can configure a maximum of 32 next hops per route.

---

| Parameter                                           | Description                                                                                                                                      |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;DEST-IPV4-ADDR&gt;/&lt;NETMASK&gt;</code> | Specifies the IPv4 route destination.                                                                                                            |
| <code>&lt;NEXTHOP-ADDR&gt;</code>                   | Specifies the next hop address for reaching the destination in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255. |
| <code>&lt;NEXTHOP-PORT-LAG-VLAN&gt;</code>          | Specifies the next hop as an outgoing interface.                                                                                                 |
| <code>nullroute</code>                              | Specifies that packets matching the destination route are silently                                                                               |

| Parameter | Description                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------|
|           | discarded and no ICMP error notification is sent to the sender.                                                           |
| reject    | Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender. |

## Examples

```
switch(config)# ip route 10.0.0.0/24 nullroute
switch(config)# ip route 10.0.1.0/24 reject
switch(config)# ip route 10.0.2.0/24 20.0.0.2
switch(config)# ip route 10.0.3.0/24 1/1/1
switch(config)# ip route 10.0.3.0/24 1/1/1.110
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification               |
|------------------|----------------------------|
| 10.10            | Inclusive language update. |
| 10.07 or earlier | --                         |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## ip route distance



Not supported on the HPE Aruba Networking 6200 Switch Series.

```
ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
no ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
```

## Description

Configures the administrative distance for the IPv4 static route.

The **no** form of this command deletes the static route.

| Parameter        | Description                                                                                                  |
|------------------|--------------------------------------------------------------------------------------------------------------|
| <DEST-IPV4-ADDR> | Specifies an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255. |
| <MASK>           | Specifies the number of bits in the address mask in CIDR format                                              |

| Parameter          | Description                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------|
|                    | ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 32.                                                           |
| <NEXT-HOP-IP-ADDR> | Specifies the next hop IPv4 address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255. |
| <INTERFACE>        | Specifies the next hop as an outgoing interface.                                                                         |
| distance <VALUE>   | Specifies the administrative distance to associate with this static route. Default: 1. Range: 1-255.                     |

## Examples

```
switch(config)# ip route 10.0.2.0/24 20.0.0.2 distance 4
switch(config)# ip route 10.0.3.0/24 1/1/1 distance 6
```

```
switch(config)# no ip route 10.0.3.0/24 1/1/1 distance 6
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms             | Command context | Authority                                                                          |
|-----------------------|-----------------|------------------------------------------------------------------------------------|
| 4100i<br>6000<br>6100 | config          | Administrators or local user group members with execution rights for this command. |

## ip route tag

```
ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |
nullroute} [tag] <1-4294967295>
no ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject
|
nullroute} [tag] <1-4294967295>
```

## Description

Configures tag for IPv4 static route.

The **no** form of this command deletes tag for IPv4 static route.

| Parameter                  | Description                                                                                                                        |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <DEST-IPV4-ADDR>/<NETMASK> | Specifies the IPv4 route destination.                                                                                              |
| <NEXTHOP-ADDR>             | Specifies the next hop address for reaching the destination in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.   |
| <NEXTHOP-PORT-LAG-VLAN>    | Specifies the next hop as an outgoing interface.                                                                                   |
| reject                     | Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.          |
| nullroute                  | Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender. |
| tag                        | Specifies and assigns tag for the route.                                                                                           |

## Examples

```
switch(config)# ip route 10.1.1.1/32 20.1.1.2 tag 10
switch(config)# ip route 10.1.1.5/32 1/1/1 tag 20
```

```
switch(config)# no ip route 10.1.1.1/32 20.1.1.2 tag 10
switch(config)# no route 10.1.1.5/32 1/1/1 tag 20
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification               |
|------------------|----------------------------|
| 10.10            | Inclusive language update. |
| 10.07 or earlier | --                         |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## ipv6 route

```
ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject
| nullroute}
no ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> |
reject | nullroute}
```

## Description

Adds an IPv6 static route.

The **no** form of this command deletes an IPv6 static route on the default VRF.

| Parameter               | Description                                                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <DEST-IPV6-ADDR>        | Specifies the route destination in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where <b>x</b> is a hexadecimal number from 0 to F. |
| <NETMASK>               | Specifies the number of bits in the address mask in CIDR format ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 128.               |
| <NEXTHOP-ADDR>          | Specifies the next hop in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where <b>x</b> is a hexadecimal number from 0 to F.          |
| <NEXTHOP-PORT-LAG-VLAN> | Specifies the next hop as an outgoing interface.                                                                                              |
| reject                  | Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.                     |
| nullroute               | Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.            |

## Examples

```
switch(config)# ipv6 route 120::/124 nullroute
switch(config)# ipv6 route 121::/124 nullroute
switch(config)# ipv6 route 122::/124 1/1/1
switch(config)# ipv6 route 122::/124 1/1/1.110
```

```
switch(config)# no ipv6 route 122::/124 1/1/1.110
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification               |
|------------------|----------------------------|
| 10.10            | Inclusive language update. |
| 10.07 or earlier | --                         |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## ipv6 route distance

```
ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
no ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
```

## Description

Configures the administrative distance for the IPv6 static route

The **no** form of this command deletes the static route.

| Parameter                             | Description                                                                                                                                           |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;DEST-IPV6-ADDR&gt;</code>   | Specifies the route destination address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where <b>x</b> is a hexadecimal number from 0 to F. |
| <code>&lt;MASK&gt;</code>             | Specifies the number of bits in the address mask in CIDR format ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 128.                       |
| <code>&lt;NEXT-HOP-IP-ADDR&gt;</code> | Specifies the next hop in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where <b>x</b> is a hexadecimal number from 0 to F.                  |
| <code>&lt;INTERFACE&gt;</code>        | Specifies the next hop as an outgoing interface.                                                                                                      |
| <code>distance &lt;VALUE&gt;</code>   | Specifies the administrative distance to associate with this static route. Range: 1 to 255. Default: 1.                                               |

## Examples

```
switch(config)# ipv6 route 122::/124 1/1/1 distance 5
switch(config)# ipv6 route 123::/124 120::1 distance 6
```

```
switch(config)# no ipv6 route 123::/124 120::1 distance 6
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## ipv6 route tag



Not supported on the HPE Aruba Networking 6200 Switch Series.

```
ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject
|
nullroute} [tag] <1-4294967295>
no ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> |
```



```
reject | nullroute} [tag] <1-4294967295>
```

## Description

Configures tag for IPv6 static route.

| Parameter               | Description                                                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <DEST-IPV6-ADDR>        | Specifies the route destination in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where <b>x</b> is a hexadecimal number from 0 to F. |
| <NETMASK>               | Specifies the number of bits in the address mask in CIDR format ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 128.               |
| <NEXTHOP-ADDR>          | Specifies the next hop in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where <b>x</b> is a hexadecimal number from 0 to F.          |
| <NEXTHOP-PORT-LAG-VLAN> | Specifies the next hop as an outgoing interface.                                                                                              |
| reject                  | Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.                     |
| nullroute               | Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.            |
| tag                     | Specifies and assigns tag for the route.                                                                                                      |

## Examples

```
switch(config)# ipv6 route 3001::1/128 1/1/1 tag 10
switch(config)# ipv6 route 3002::1/128 1000::2 tag 20
```

```
switch(config)# no ipv6 route 3001::1/128 1/1/1 tag 10
switch(config)# no ipv6 route 3002::1/128 1000::2 tag 20
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification               |
|------------------|----------------------------|
| 10.10            | Inclusive language update. |
| 10.07 or earlier | --                         |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| 4100i<br>5420 | config          | Administrators or local user group members with execution rights for this command. |

| Platforms            | Command context | Authority |
|----------------------|-----------------|-----------|
| 6000<br>6100<br>6200 |                 |           |

## show ip rib

```
show ip rib <FILTER> [vrf <VRF-NAME>]
```

### Description

Shows the IPv4 Routing Information Base (RIB) of VRF with name (<VRF-NAME>). If VRF name is not specified, default VRF routes are displayed.

| Parameter      | Description                        |
|----------------|------------------------------------|
| <FILTER>       | Selects filter, see Usage section. |
| vrf <VRF-NAME> | Specifies the VRF name.            |

### Usage

There are sub-options available within this command:

- **A.B.C.D:** Shows longest prefix match.
- **A.B.C.D/M:** Shows exact route match.
- **all-vrfs:** Shows all VRF information.
- **bgp:** Shows BGP routes only.
- **connected:** Shows connected routes only.
- **connected:** Shows connected routes only.
- **local:** Shows local routes only.
- **ospf:** Shows OSPF routes only.
- **rip:** Shows RIP routes only.
- **static:** Shows static routes only.
- **summary:** Shows aggregate count of routes per routing protocol.
- **vrf:** Specifies the VRF name.
- **selected:** Shows routes selected for forwarding only.
- **non-selected:** Shows routes not selected for forwarding only.

The output of the **show ip rib** commands are not available in information generated by the **show tech files** command. This information is now available in the file `ipv4_rib_dump.gz`, which can be generated using the command **sudo ovs-appctl -t hpe-routing hpe-metaswitch/show\_ip\_rib ipv4 all-vrfs**.

### Examples

Showing IPv4 routes in RIB for HPE Aruba Networking 8325, 10000, 8360, and 9300/9300S Switch series:

```
switch# show ip rib
Origin Codes: R - RIP, O - OSPFv2, B - BGP
               C - connected, S - static, H - host-routes
```

Type Codes: E - External BGP, I - Internal BGP, IA - OSPF inter area  
 E1 - OSPF external type 1, E2 - OSPF external type 2  
 \* indicates selected for forwarding

VRF: default

| Prefix          | Nexthop   | Interface | VRF     | Origin/<br>Type | Distance/<br>Metric | Age |
|-----------------|-----------|-----------|---------|-----------------|---------------------|-----|
| -----           |           |           |         |                 |                     |     |
| --              |           |           |         |                 |                     |     |
| *1.1.1.1/32     | -         | 1/1/33    | -       | H               | [1/0]               |     |
| 00h:00m:07s     |           |           |         |                 |                     |     |
| *10.0.0.0/30    | -         | 1/1/1     | -       | S               | [20/0]              |     |
| 0d:10h:01m:41s  |           |           |         |                 |                     |     |
| *10.0.1.0/30    | -         | 1/1/1     | -       | B/I             | [200/0]             |     |
| 2d:20h:01m:42s  |           |           |         |                 |                     |     |
| *10.1.64.0/18   | -         | loopback2 | -       | C               | [0/0]               | -   |
| *10.2.64.0/18   | 10.0.0.3  | lag1      | -       | O/E1            | [110/25]            |     |
| 1d:05h:03m:43s  |           |           |         |                 |                     |     |
| *10.2.64.0/18   | 20.10.0.1 | vlan100   | -       | O/E1            | [110/25]            |     |
| 0d:05h:03m:43s  |           |           |         |                 |                     |     |
| *20.1.2.3/32    | 2.2.2.2   | 1/1/4     | vrf_red | B/E             | [20/0]              |     |
| 2d:10h:01m:45s  |           |           |         |                 |                     |     |
| *30.1.3.0/24    | -         | reject    | -       | S               | [1/0]               |     |
| 33d:10h:01m:43s |           |           |         |                 |                     |     |
| *50.10.13.0/24  | -         | reject    | -       | S               | [1/0]               |     |
| 12d:10h:01m:44s |           |           |         |                 |                     |     |
| *61.1.1.2/32    | 4.4.4.4   | 1/1/5     | -       | B/I             | [200/0]             |     |
| 1d:11h:01m:45s  |           |           |         |                 |                     |     |
| *62.1.1.3/32    | 5.5.5.5   | 1/1/6     | -       | B/I             | [200/0]             |     |
| 0d:12h:01m:45s  |           |           |         |                 |                     |     |
| *193.0.0.2/32   | 50.0.0.2  | 1/1/2     | -       | S               | [1/0]               |     |
| 0d:04h:01m:43s  |           |           |         |                 |                     |     |
| 193.0.0.2/32    | 56.0.0.3  | 1/1/3     | -       | O/E1            | [110/25]            |     |
| 0d:04h:03m:43s  |           |           |         |                 |                     |     |

Total Route Count : 14

Showing IPv4 routes in RIB for all other switches:

switch# **show ip rib**

Origin Codes: R - RIP, O - OSPFv2, B - BGP

C - connected, S - static, D - DHCP

Type Codes: E - External BGP, I - Internal BGP, IA - OSPF inter area

E1 - OSPF external type 1, E2 - OSPF external type 2

\* indicates selected for forwarding

VRF: default

| Prefix         | Nexthop | Interface | VRF | Origin/<br>Type | Distance/<br>Metric | Age |
|----------------|---------|-----------|-----|-----------------|---------------------|-----|
| -----          |         |           |     |                 |                     |     |
| --             |         |           |     |                 |                     |     |
| *1.1.1.1/32    | -       | 1/1/33    | -   | H               | [1/0]               |     |
| 00h:00m:07s    |         |           |     |                 |                     |     |
| *10.0.0.0/30   | -       | 1/1/1     | -   | S               | [20/0]              |     |
| 0d:10h:01m:41s |         |           |     |                 |                     |     |
| *10.0.1.0/30   | -       | 1/1/1     | -   | B/I             | [200/0]             |     |
| 2d:20h:01m:42s |         |           |     |                 |                     |     |
| *10.1.64.0/18  | -       | loopback2 | -   | C               | [0/0]               | -   |

```

*10.2.64.0/18    10.0.0.3    lag1    -    O/E1    [110/25]
1d:05h:03m:43s
*10.2.64.0/18    20.10.0.1    vlan100    -    O/E1    [110/25]
0d:05h:03m:43s
*20.1.2.3/32     2.2.2.2     1/1/4     vrf_red    B/E     [20/0]
2d:10h:01m:45s
*30.1.3.0/24     -            reject    -            S       [1/0]
33d:10h:01m:43s
*50.10.13.0/24   -            reject    -            S       [1/0]
12d:10h:01m:44s
*61.1.1.2/32     4.4.4.4     1/1/5     -            B/I     [200/0]
1d:11h:01m:45s
*62.1.1.3/32     5.5.5.5     1/1/6     -            B/I     [200/0]
0d:12h:01m:45s
*193.0.0.2/32    50.0.0.2     1/1/2     -            S       [1/0]
0d:04h:01m:43s
 193.0.0.2/32    56.0.0.3     1/1/3     -            O/E1    [110/25]
0d:04h:03m:43s

Total Route Count : 14

```

Showing IPv4 exact route match in RIB:

```

switch# show ip rib 10.0.0.0/30

VRF : default

Prefix      : 10.0.0.0/30          VRF(egress)      : -
Nexthop     : -                  Interface        : 1/1/1
Origin      : Connected          Type             : -
Distance    : 0                  Metric           : 0
Age         : -                  Tag              : 0
Selected    : Yes                Recursive Nexthop : No

```

Showing IPv4 RIB summary:

```

switch# show ip rib summary

IPv4 RIB Table Summary

VRF name :    default
Protocol  RIB Routes
-----
connected 1010
local     1011
static     4
ospfv2     509
bgp        9014
selected  10008
non-selected 1518

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                          |
|---------------|-----------------------------|------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## show ipv6 rib

```
show ipv6 rib <FILTER> [vrf <VRF-NAME>]
```

### Description

Shows the IPv6 Routing Information Base (RIB) of VRF with name (<VRF-NAME>). If VRF name is not specified, default VRF routes are displayed.

| Parameter      | Description                                     |
|----------------|-------------------------------------------------|
| <FILTER>       | Selects filter, see usage section.              |
| vrf <VRF-NAME> | Shows routes in the VRF and specifies VRF name. |

### Usage

There are sub-options available within this command:

- **X:X:X:X:** Shows longest prefix match.
- **X:X:X:X/M:** Shows exact route match.
- **all-vrfs:** Shows all VRF information.
- **bgp:** Shows BGP routes only.
- **connected:** Shows connected routes only.
- **local:** Shows local routes only.
- **ospf:** Shows OSPF routes only.
- **rip:** Shows RIP routes only.
- **static:** Shows static routes only.
- **summary:** Shows aggregate count of routes per routing protocol.
- **vrf:** Specifies the VRF name.
- **selected:** Shows routes selected for forwarding only.
- **non-selected:** Shows routes not selected for forwarding only.

### Examples

Showing IPv6 routes in RIB for HPE Aruba Networking 8325, 10000, 8360, and 9300/9300S Switch series:

```
switch# show ipv6 rib
Origin Codes: R - RIPng, O - OSPFv3, B - BGP
              C - connected, S - static, H - host-routes
```

```

Type Codes:   E - External BGP, I - Internal BGP, IA - OSPF inter area
              E1 - OSPF external type 1, E2 - OSPF external type 2
* indicates selected for forwarding

VRF: default

Prefix          Nexthop    Interface  VRF      Origin/ Distance/ Age
              Type        Metric
-----
--
*1::2/128        1::2        1/1/1      -         H         [1/0]       00h:00m:06s
*1000::/64       -            1/1/1      -         C         [0/0]       -
*1000::8/128     -            1/1/1      -         L         [0/0]       -
*1001:db8::/32   1000::10    1/1/1      -         B/I        [200/0]     1d:20h:01m:42s
*2000::/64       fe80::3182  vlan100    -         S         [1/0]       2d:05h:03m:43s
 2000::/64       fe80::1241  1/1/1      -         O/E1      [110/25]    0d:05h:03m:43s
*2000::2000:0:0:0/67 fe80::1111 lag1        Green      B/E       [20/0]     1d:10h:01m:45s
*3001::0/64      -            vlan100    -         C         [0/0]       -
*3001::1/128     -            vlan100    -         L         [0/0]       -
*6101::0/64      -            nullroute  -         S         [1/0]       -
12d:10h:01m:43s

Total Route Count : 10

```

Showing IPv6 routes in RIB for the other switch series:

```

switch# show ipv6 rib
Origin Codes: R - RIPng, O - OSPFv3, B - BGP
              C - connected, S - static, D- DHCP
Type Codes:   E - External BGP, I - Internal BGP, IA - OSPF inter area
              E1 - OSPF external type 1, E2 - OSPF external type 2
* indicates selected for forwarding

VRF: default

Prefix          Nexthop    Interface  VRF      Origin/ Distance/ Age
              Type        Metric
-----
--
*1000::/64       -            1/1/1      -         C         [0/0]       -
*1000::8/128     -            1/1/1      -         L         [0/0]       -
*1001:db8::/32   1000::10    1/1/1      -         B/I        [200/0]     1d:20h:01m:42s
*2000::/64       fe80::3182  vlan100    -         S         [1/0]       2d:05h:03m:43s
 2000::/64       fe80::1241  1/1/1      -         O/E1      [110/25]    0d:05h:03m:43s
*2000::2000:0:0:0/67 fe80::1111 lag1        Green      B/E       [20/0]     1d:10h:01m:45s
*3001::0/64      -            vlan100    -         C         [0/0]       -
*3001::1/128     -            vlan100    -         L         [0/0]       -
*6101::0/64      -            nullroute  -         S         [1/0]       -
12d:10h:01m:43s

Total Route Count : 10

```

Showing IPv6 exact route match in RIB:

```

switch# show ipv6 rib 2000::2000:0:0:0

VRF : default

```

|          |                       |                   |            |
|----------|-----------------------|-------------------|------------|
| Prefix   | : 2000::2000:0:0:0/67 | VRF (egress)      | : Green    |
| Nexthop  | : fe80::1111          | Interface         | : lag1     |
| Origin   | : BGP                 | Type              | : External |
| Distance | : 20                  | Metric            | : 0        |
| Age      | : 1d:10h:01m:45s      | Tag               | : 20       |
| Selected | : Yes                 | Recursive Nexthop | : Yes      |

Showing IPv6 RIB summary:

```
switch# show ipv6 rib summary

IPv6 RIB Table Summary

VRF name : default
  Protocol      RIB Routes
-----
connected      1009
local           1010
static          3
ospfv3          508
bgp             1013

selected        10004
non-selected    1527
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification               |
|------------------|----------------------------|
| 10.10            | Inclusive language update. |
| 10.07 or earlier | --                         |

## Command Information

| Platforms     | Command context             | Authority                                                                          |
|---------------|-----------------------------|------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |



---

OSPFv2 is not supported on the 6000 and 6100 Switch Series.

---



---

OSPFv2 is not supported on the 4100i Switch Series.

---

Open Shortest Path First version 2 (OSPFv2) is a routing protocol described in RFC2328. It is a Link State-based IGP (Interior Gateway Protocol) routing protocol applied to routers grouped into OSPF areas identified by the routing configuration on each routing switch. It is widely used in medium to large-sized enterprise networks.

## Overview

The characteristics of OSPFv2 are:

- Provides a loop-free topology using SPF algorithm.
- Supports load balancing with equal cost routes for the same destination.
- OSPFv2 is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking) and route summarization.
- Provides authentication of routing messages.
- Provides fast convergence with triggered, incremental updates via LSAs.

Some OSPFv2 configuration is done in the global configuration context, others in the router ospf context, or in the interface configuration context, or in the vlink context. OSPFv2 can be configured on L3 ports, VLAN interfaces, LAG interfaces, and loopback interfaces. All such configurations work in the mentioned interfaces context. OSPFv2 mandates the associated interface to be a routed interface.

### Supported features

- OSPFv2 neighbor adjacency, Hello protocol, multiple areas, Inter-area routing
- AS external routes, stub areas, totally stubby areas, NSSA, ASBR
- Designated Router/Backup Designated Router
- Point-to-point interfaces/broadcast interfaces
- Equal-cost multipath
- Null authentication, Simple password authentication, MD5 authentication
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- SPF throttling
- LSA Throttling
- SPF throttling
- Configuration of interface parameters such as priority, cost, hello-interval, dead-interval, retransmit-interval, transit-delay, etc.



- Configuration of virtual link parameters such as hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Route redistribution
- Passive interfaces
- Multi VRF support with each VRF having up to eight OSPF process instances
- Congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling)
- IP unnumbered interfaces

Listed below are a few recommendations while configuring IP unnumbered interfaces:

- Enable OSPF on the lending interface in the same area as the borrowing interface.
- If different unnumbered interfaces are used for different areas, then configure them to use different loop backs.
- Configure the address on the lending loopback as /32.
- OSPFv3 can also be enabled on an IP unnumbered interface. OSPFv3 neighborship can be established on IP unnumbered interfaces as it works over IPv6 link-local address.

## How OSPFv2 protocol works

The protocol uses Link State Advertisements (LSAs) transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces. Each routing switch in an area also maintains a link-state database (LSDB) that describes the area topology. (All routers in a given OSPF area have identical LSDBs.) The routing switches used to connect areas to each other flood summary link LSAs and external link LSAs to neighboring OSPF areas to update them regarding available routes. Through this means, each OSPF router determines the shortest path between itself and a desired destination router in the same OSPF domain (Autonomous System (AS)).

## OSPFv2 concepts

The following sections describe OSPFv2 Link-State types, router types, and area types.

### OSPFv2 Link-state advertisement (LSA) types

OSPFv2 sends routing information in Link-State Advertisements (LSAs). The switches support the following types of LSAs.

- **Router LSA—Type-1 LSA.** Describes the states of the router interfaces to an area, and is flooded throughout a single area only.
- **Network LSA—Type-2 LSA.** Describes the list of routers connected to the network, and is flooded throughout a single area only.
- **Summary LSA—Type-3 LSA.** Describes the route to networks in another OSPF area of the same Autonomous System (AS). Propagated through backbone area to other areas.
- **ASBR summary LSA—Type-4 LSA.** Describes the route to an ASBR in an OSPF normal or backbone area of the same AS. Propagated through backbone area to other areas.
- **AS external LSA—Type-5 LSA.** Describes the route to a destination in another AS (external route.) Originated by ASBR in normal or backbone areas of an AS and propagates through backbone area to other normal areas.

- **NSSA LSA—Type-7 LSA.** Describes the route to a destination in another AS (external route.) Originated by ASBR in NSSA. ABR converts type-7 LSAs to type-5 LSAs for injection into the backbone area.

## OSPFv2 router types

The following router types are supported

### Internal routers

Internal OSPFv2 routers belong to only one area. Internal routers flood type-1 LSAs to all routers in the same area and maintain identical LSDBs.

### Area border routers (ABRs)

Area border routers have membership in multiple areas. ABRs are used to connect the various areas in an AS to the backbone area for that AS. Multiple ABRs can be used to connect a given area to the backbone, and a given ABR can belong to multiple areas other than the backbone.

An ABR maintains a separate LSDB for each area to which it belongs. (All routers within the same area have identical LSDBs.) The ABR is responsible for flooding summary LSAs between its border areas. You can reduce summary LSA flooding by configuring area ranges. An area range enables you to assign an aggregate address to a range of IP addresses. This aggregate address is advertised instead of all the individual addresses it represents.

### Autonomous system boundary router (ASBR)

Autonomous system boundary routers run multiple interior gateway protocols and serve as a gateway to other autonomous systems operating with interior gateway protocols. The ASBR imports and translates different protocol routes into OSPF through redistribution. ASBRs can be used in backbone areas, normal areas, and NSSAs, but not in stub areas.

### Designated routers (DRs)

In an OSPF network having two or more routers, one router is elected to serve as the DR and another router to act as the Backup Designated Router (BDR). All other routers in the area forward their routing information to the DR and BDR, and the DR forwards this information to all routers in the network. This action minimizes the amount of repetitive information that is forwarded on the network by eliminating the need for each individual router in the area to forward its routing information to all other routers in the network. If the area includes multiple networks, each network elects its own DR and BDR.

In an OSPF network with no DR and no BDR, the neighboring router with the highest priority is elected the DR, and the router with the next highest priority is elected the BDR. If the DR goes off-line, the BDR automatically becomes the DR, and the router with the next highest priority then becomes the new BDR. If multiple routing switches on the same OSPF network are declaring themselves DRs, both priority and router ID are used to select the DR and BDRs.

Priority is configurable using the `ip ospf priority` command at the interface level. If two neighbors share the same priority, the router with the highest router ID is elected as the DR. The router with the next highest router ID is elected as the BDR.

## OSPFv2 area types

OSPFv2 is built upon a hierarchy of network areas. All areas for a given OSPF domain reside in the same AS. An AS is defined as a number of contiguous networks that share the same interior gateway routing protocol.

An AS can be divided into multiple areas. Each area represents a collection of contiguous networks and hosts, and the topology of a given area is not known by the internal routers in any other area. Areas define the boundaries to which types 1 and 2 LSAs are broadcast, which limits the amount of LSA flooding that occurs within the AS and also helps to control the size of the LSDBs maintained in OSPF routers. An area is represented in OSPF by either an IP address or a number. Area types include: Backbone, Normal, Not-so-stubby (NSSA), and stub.

## Backbone area

Every AS must have one (and only one) backbone area (identified as area 0 or 0.0.0.0.) The ABRs of all other areas in the same AS connect to the backbone area, either physically through an ABR or through a configured virtual link. The backbone is a transit area that carries the type-3 summary LSAs, type-5 AS external link LSAs and routed traffic between non-backbone areas, as well as the type-1 and type-2 LSAs and routed traffic internal to the area. ASBRs are allowed in backbone areas.

## Normal area

Normal area connects to the backbone area through one or more ABRs (physically or through a virtual link) and supports type-3 summary LSAs and type-5 external link LSAs to and from the backbone area. ASBRs are allowed in normal areas.

## Stub area

Stub area connects to the AS backbone through one or more ABRs. It does not allow an internal ASBR, and does not allow external (type 5) LSAs. A stub area supports these actions:

- Advertise the area summary routes to the backbone area.
- Advertise summary routes from other areas.
- Use the default summary (type-3) route to advertise both of the following:
  - Summary routes to other areas in the AS
  - External routes to other ASs

You can configure the stub area ABR to do the following:

- Suppress advertising from some or all area summarized internal routes into the backbone area.
- Suppress LSA traffic from other areas in the AS by replacing type-3 summary LSAs and the default external route from the backbone area with the default summary route (0.0.0.0/0.)

Virtual links are not allowed for stub areas.

## Not-so-stubby (NSSA) area

NSSA area connects to the backbone area through one or more ABRs. NSSAs are intended for use where an ASBR exists in an area where you want to control the following:

- Advertising the ASBR external route paths to the backbone area
- Allowing LSAs from the backbone area to advertise in the NSSA:
  - Summary routes (type-3 LSAs) from other areas
  - External routes (type-5 LSAs) from other areas as a default external route (type-7 LSAs)

Virtual links are not allowed for NSSAs.

# OSPFv2 configuration task list

## Tasks at a glance

- [Configuring OSPF on the routing switch](#)
- [Configuring external route redistribution and control](#)
- [Influencing route choice by changing the administrative distance](#)
- [Configuring graceful restart of OSPF routing](#)
- [Configuring OSPF interface settings](#)
- [Configuring OSPF interface authentication](#)
- [Configuring all OSPF interfaces as passive](#)
- [Clearing OSPF statistics on a switch](#)

## Configuring OSPF on the routing switch

Create the OSPF instance and enter the OSPF router configuration context. From this, you can proceed with other OSPF configuration tasks.

### Prerequisites

- You must be in the global configuration context, as indicated by the `switch(config)#` prompt to create the OSPF instance and enter the OSPF router configuration context.
- To configure a router ID, create OSPF network areas, or adjust other global OSPF configuration items, you must be in the router configuration context, as indicated by the `switch(config-ospf)#` prompt.

### Procedure

1. Create the OSPF instance and enter the OSPF router configuration context using the following command. For command details, see [router ospf](#).

```
router ospf <PROCESS-ID>
```

For example, the following command creates OSPF instance 1.

```
switch(config)# router ospf 1
switch(config-ospf-1)#
```

2. Configure a global router ID using the following command. For command details, see [router-id \(ospf\)](#).

```
router-id <ROUTER-ADDRESS>
```

For example, the following command sets the router ID to 1.1.1.1.

```
switch(config-ospf-1)# router-id 1.1.1.1
```

3. Optionally, if the OSPF process was disabled (it is enabled by default), enable the OSPF process using the following command.

```
enable
```

For command details, see [enable \(router ospf\)](#). (Refer to [disable \(router ospf\)](#) for disabling the OSPF process).

## Setting OSPF network for the area

After you define an OSPF area, you can assign one or more networks to it. The OSPF protocol will run on the interface with the configured IPv4 address. The interfaces which have an IP address configured in this network or in a subset of this network, will participate in the OSPF protocol.

## Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

## Procedure

1. Set an OSPF network for the area using the following command. For command details, see [ip ospf area](#).

```
ip ospf <PROCESS-ID> area <AREA-ID>
```

For example, use the following command to assign interface 1/1/1 to OSPF area 1. The area can alternatively be entered as an IPv4 address.

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf 1 area 1
```

2. Optionally, you can disable OSPF on the interface using the following command. For command details, see [ip ospf shutdown](#).

```
ip ospf shutdown
```

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf shutdown
```

## Configuring external route redistribution and control

Configuring route redistribution for OSPFv2 establishes the routing switch as an ASBR for importing and translating different protocol routes into OSPFv2. When you configure redistribution for OSPFv2, you can specify that routes external to the OSPFv2 domain are imported as OSPFv2 routes.

1. Enable route redistribution using the `redistribute` command.  
`redistribute {bgp | connected | local loopback | static | rip | ospf <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]`

For example, setting redistribution of connected routes as OSPFv2 routes:

```
switch(config-ospf-1)# redistribute connected
```

If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv2. For example, setting redistribution of local loopback routes, that only match the routes specified by the route map:

```
switch(config-ospf-1)# redistribute local loopback route-map local_routes
```

2. Optionally, modify the default metric for redistribution using the `default-metric` command.  
`default-metric <METRIC-VALUE>`

For example, setting the default metric for redistribution to 37.

```
switch(config-ospf-1)# default-metric 37
```

3. Optionally, set the cost of default-summary LSAs using the area command.

```
area <AREA-ID> default-metric <COST>
```

For example, setting the cost of default summary LSAs to 2.

```
switch (config-ospf-1)# area 1 default-metric 2
```

4. Optionally, use the `max-metric router-lsa` command to set the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations.

```
max-metric router-lsa [on-startup [<ADVERT-TIME>]]
```

For example, setting advertise max-metric router-lsa on startup.

```
switch(config-ospf-1)# max-metric router-lsa on-startup 3000
```

## Influencing route choice by changing the administrative distance

The administrative distance value can be left in its default configuration setting (110) unless a change is needed to improve OSPF performance for a specific network configuration.

### Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

### Procedure

Reconfigure the administrative distance using the following command. For command details, see [distance \(ospf\)](#) command.

```
distance <distance>
```

For example, use the following command to set administrative distance to 100.

```
switch(config-ospf-1)# distance 100
```

## Configuring graceful restart of OSPF routing

OSPF routing can be gracefully restarted on the switch without losing packets that are in transit. OSPF neighbors are informed that the router is completing a graceful restart, which allows for maintenance on the switch without interrupting traffic in the network. There is no effect on the saved switch configuration.

### Prerequisites



---

Graceful restart is only applicable to the 5420 and 6200-VSF switches.

---

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Manually create a router-id configuration to support graceful restart and management module failover. A dynamic router-id is chosen from the list of configured IPs at the time of the router-id election, and in the event of a graceful restart and management module failover, the router-id should be preserved before and after these events. Graceful restart would fail if there is a change in the router-id, causing traffic loss.

## Procedure

Enable graceful restart of OSPF routing using the following command. For command details, see [graceful-restart \(router ospf\)](#).

```
graceful-restart {restart-interval <seconds> | helper}
```

For example, the following command specifies 50 seconds as the maximum interval another router will wait for this router to gracefully restart.

```
switch(config-ospf-1)# graceful-restart restart-interval 50
```

## Configuring OSPF interface settings

You can optionally adjust the following OSPF interface settings.

### Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

### Procedure

1. Set the interface cost using the following command. For command details, see [ip ospf cost](#).

```
ip ospf cost <interface-cost>
```

For example, the following command sets the cost associated with interface 1/1/1 to 100.

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf cost 100
```

2. Set the time interval between OSPF hello packets for the OSPF interface using the following command. For command details, see [ip ospf hello-interval](#).  

```
ip ospf hello-interval <seconds>
```
3. Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPF interface using the following command. For command details, see [ip ospf dead-interval](#).  

```
ip ospf dead-interval <seconds>
```
4. Set the time between retransmitting lost link state advertisements for the OSPF interface using the following command. For command details, see [ip ospf retransmit-interval](#).  

```
ip ospf retransmit-interval <seconds>
```
5. Sets the transit delay in link state transmission for the OSPF interface using the following command. For command details, see [ip ospf transit-delay](#).  

```
ip ospf transit-delay <seconds>
```
6. Set the OSPF network type for the interface. For command details, see [ip ospf network](#).  

```
ip ospf network {broadcast|point-to-point}
```
7. Set the OSPF priority on the interface using the following command. For command details, see [ip ospf priority](#).  

```
ip ospf priority <number-value>
```

8. Set the OSPF interface as OSPF passive interface. The interface participates in OSPF but does not send or receive packets on that interface. For command details, see [ip ospf passive](#).

```
ip ospf passive
```

## Configuring OSPF interface authentication

Configure authentication on the interface. Only one method of authentication can be active on an interface at a time. If one method is already configured on an interface, configuring an alternative method on the same interface automatically overwrites the first method used.

### Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

### Procedure

1. Set the authentication type that will be used for authentication with the neighbor router using the following command. For command details, see [ip ospf authentication](#).

```
ip ospf authentication {message-digest | simple-text | null | keychain}
```

For example, the following command sets the authentication type to simple-text.

```
switch(config-if)# ip ospf authentication simple-text
```

2. For simple-text authentication, set the password using the following command. For command details, see [ip ospf authentication-key](#).

```
ip ospf authentication-key <PASSWORD>
```

For example:

```
switch(config-if)# ip ospf authentication-key secure
```

3. For MD5 authentication, set the password using the following command. For command details, see [ip ospf message-digest-key md5](#).

```
ip ospf message-digest-key {Key ID} md5 {ciphertext | plaintext} <KEY>
```

For example:

```
switch(config-if)# ip ospf message-digest-key 1 md5 ciphertext 100
```

## Configuring all OSPF interfaces as passive

OSPF sends LSAs to all other routers in the same AS. To limit the flooding of LSAs throughout the AS, you can configure OSPF to be passive.

### Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

### Procedure

Configure all OSPF interfaces as passive using the following command. For command details, see [passive-interface default \(OSPFv2\)](#).

```
passive-interface default
```



```
switch(config-ospf-1)# passive-interface default
```

## Configuring SPF throttling timers

SPF calculation is throttled with default timer values (start-time 200ms, hold-time 1000ms, max-wait-time 5000ms). You can throttle SPF calculation by configuring non-default timers to improve performance of a specific network configuration.

### Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

### Procedure

Reconfigure SPF throttling timers using the following command. For command details, see [timers throttle spf](#).

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds> max-wait-time <milliseconds>
```

For example, use the following command to set start-time to 500ms, hold-time to 3000ms and max-wait-time to 9000ms:

```
switch(config-ospf-1)# timers throttle spf start-time 500 hold-time 3000 max-wait-time 9000
```

## Clearing OSPF statistics on a switch

Clear OSPF event statics using the following command. For command details, see [clear ip ospf statistics](#).

```
clear ip ospf [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-vrfs | vrf <VRF-NAME>]
```

For example, the following command clears OSPF event statistics from interface 1/1/1.

```
switch(config-router)# clear ip ospf statistics interface 1/1/1
```

## An example of the OSPFv2 information in the show running-config command

When OSPFv2 is configured, the output of the `show running-config` command includes OSPFv2 information.

For example:

```
switch# show running-config  
Current configuration:  
!  
!Version AOS-CX 10.0X.XXXX  
!  
lldp enable  
timezone set utc
```

```

mgmt
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf vrf_default
    area 0.0.0.0
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ip address 33.1.1.1/24
    ip ospf 1 area 0.0.0.0
    ip ospf hello-interval 22
interface 1/1/1
    no shutdown
    ip address 33.44.1.1/24
    ip ospf 1 area 0.0.0.0
    ip ospf passive
    ip ospf dead-interval 88
    ip ospf transit-delay 44
    ip ospf priority 88
    ip ospf cost 8
    ip ospf network point-to-point
    ip ospf authentication simple-text
    ip ospf authentication-key kkk
    ip ospf shutdown
interface 1/1/2
interface loopback 2
    ip address 55.55.55.55/32
    ip ospf 1 area 0.0.0.0

```

## OSPFv2 commands

### active-backbone

```

active-backbone stub-default-route
no active-backbone stub-default-route

```

#### Description

This command enables the router to send a default route to stub areas if there is an active loopback link in the backbone area. The configuration is not required if backbone area has neighbors or passive interfaces configured. By default active backbone detection is enabled.

#### Examples

```
switch(config)# router ospf 1  
switch(config-ospf-1)# active-backbone stub-default-route
```

```
switch(config)# no active-backbone stub-default-route
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release    | Modification       |
|------------|--------------------|
| 10.10.1000 | Command Introduced |

## Command Information

| Platforms    | Command context                                                                    | Authority                                                                          |
|--------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf- <i>&lt;PROCESS-ID&gt;</i><br>config-ospfv3- <i>&lt;PROCESS-ID&gt;</i> | Administrators or local user group members with execution rights for this command. |

## area (ospf)

```
area <AREA-ID>  
no area <AREA-ID>
```

## Description

Creates a normal area, with *<AREA-ID>* set if not present. If the area is already present and it is not a normal area, then this command changes the area type to normal.

The **no** form of this command deletes the area with the *<AREA-ID>* specified. Area can be of any type (nssa, nssa no-summary, stub, stub no-summary, and default normal area).

| Parameter              | Description                                                                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>&lt;AREA-ID&gt;</i> | Specifies the area ID in one of the following formats.<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |

## Examples

Creating a normal area:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# area 1  
Switch(config-ospf-1)# show running-config current-context router ospf 1  
router-id 1.1.1.1  
area 0.0.0.1
```

Deleting an area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## area default-metric (ospf)

```
area <AERA-ID> default-metric <COST>
no area <AREA-ID> default-metric
```

## Description

Sets the cost of the default route announced to NSSA or stub areas.

The **no** form of this command resets the cost of the default route announced to NSSA or stub areas, to the default value of 1.

| Parameter             | Description                                                                                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID>             | Specifies area ID in one of the following formats.<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |
| default-metric <COST> | Sets the cost of default-summary LSAs announced to NSSA or stub areas, to the specified value. Default cost: 1. Range: 0 to 16777215.                                                                                              |

## Examples

Setting cost for default LSA summary:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1 default-metric 2
```

Setting cost for default LSA summary to default:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# no area 1 default-metric
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context                        | Authority                                                                          |
|--------------|----------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf- <i>&lt;PROCESS-ID&gt;</i> | Administrators or local user group members with execution rights for this command. |

## area nssa (ospf)

```
area <AREA-ID> nssa [no-summary]  
no area <AREA-ID> nssa [no-summary]
```

## Description

Creates the NSSA area (Not So Stubby Area) with *<AREA-ID>* if not present. If area is present and not NSSA area, this command changes the area type to NSSA area. If **no-summary** is used, area type will be NSSA No-Summary.

The **no** form of this command unsets the area type as NSSA. That is, the configured area will be changed to default normal area. The **no area *<AREA-ID>* nssa no-summary** command enables sending inter-area routes into NSSA, but will not unset the area as NSSA.

| Parameter              | Description                                                                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>&lt;AREA-ID&gt;</i> | Specifies the area ID in one of the following formats.<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295.                          |
| nssa [no-summary]      | Specifies Not So Stubby Area (NSSA) area type. If area is present and not NSSA area, parameter changes the area type to NSSA area. If <b>no-summary</b> is specified, area type will be NSSA No-Summary, which means do not inject inter-area routes into NSSA. |

## Examples

Creating an NSSA area:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# area 1 nssa  
switch(config-ospf-1)# area 1 nssa no-summary
```

Unsetting the area as NSSA

```
switch(config)# router ospf 1  
switch(config-ospf-1)# no area 1 nssa  
switch(config-ospf-1)# no area 1 nssa no-summary
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context                        | Authority                                                                          |
|--------------|----------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf- <i>&lt;PROCESS-ID&gt;</i> | Administrators or local user group members with execution rights for this command. |

## clear ip ospf neighbors

```
clear ip ospf [<PROCESS-ID>] neighbor [<NEIGHBOR>] [interface [<INTERFACE-NAME>]]  
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Resets the neighbor and clears the OSPF neighbor information.

| Parameter                     | Description                                                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------------------|
| <i>&lt;PROCESS-ID&gt;</i>     | Specifies the OSPFv2 process ID to clear the statistics for the particular OSPFv2 process. Range: 1 to 65535. |
| <i>&lt;NEIGHBOR&gt;</i>       | Specifies the router ID of a neighbor.                                                                        |
| <i>&lt;INTERFACE-NAME&gt;</i> | Specifies the OSPFv2 statistics to clear for the specified interface.                                         |
| all-vrfs                      | Select to clear the OSPFv2 statistics for all VRFs.                                                           |
| vrf <i>&lt;VRF-NAME&gt;</i>   | Specifies the name of a VRF.                                                                                  |

## Example

Clearing the OSPFv2 neighbor information:

```
switch# clear ip ospf 1 neighbor 1.1.1.2
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ip ospf 1 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear ip ospf statistics

```
clear ip ospf [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Clear the OSPF event statistics.

| Parameter        | Description                                                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>     | OSPF process ID. Clear the statistics for the particular OSPF process. Range: 1 to 65535.                                                     |
| <INTERFACE-NAME> | Clear the OSPF statistics for the specified interface.                                                                                        |
| all-vrfs         | Optionally select to clear the OSPF statistics for all VRFs.                                                                                  |
| vrf <VRF-NAME>   | Optionally select to clear the OSPF statistics for a particular VRF. If the VRF is not specified, information for the default VRF is cleared. |

### Examples

Clearing the OSPF event statistics:

```
switch# clear ip ospf statistics
switch# clear ip ospf statistics interface 1/1/1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## default-information originate (ospf)

```
default-information originate [metric <METRIC-VALUE>]
no default-information originate [metric <METRIC-VALUE>]
```

### Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors if it is present in the routing table. Optionally, the metric value can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

| Parameter                          | Description                                                                  |
|------------------------------------|------------------------------------------------------------------------------|
| <i>metric &lt;METRIC-VALUE&gt;</i> | Specifies the OSPF metric value for the default route. Optional. Default: 1. |

### Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate
```

Setting advertisement of the default route and specifying an optional metric value of 20:



```
switch(config)# router ospf 1  
switch(config-ospfv3-1)# default-information originate  
switch(config-ospfv3-1)# default-information originate metric 20
```

Disabling advertisement of the default route and setting metric to the default value:

```
switch(config)# router ospf 1  
switch(config-ospfv3-1)# no default-information originate metric
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                  |
|------------------|-----------------------------------------------|
| 10.09            | Added parameter: <b>metric</b> <METRIC-VALUE> |
| 10.07 or earlier | --                                            |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## default-information originate always (ospf)

```
default-information originate always [metric <METRIC-VALUE>]  
no default-information originate always [metric <METRIC-VALUE>]
```

## Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors, regardless if it is present in the routing table or not. Optionally, metric can be set for default route 0.0.0.0/0. The default value is 1. The **no** form of this command disables advertisement of the default route.

| Parameter                    | Description                                                        |
|------------------------------|--------------------------------------------------------------------|
| <i>metric</i> <METRIC-VALUE> | Specifies the OSPF metric value for the default route. Default: 1. |

## Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1  
switch(config-ospfv3-1)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# no default-information originate always
```

Setting advertisement of the default route with metric set to 20:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# default-information originate always metric 20
```

Disabling advertisement of the default route and setting the metric to the default value:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# no default-information originate always metric
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                           |
|------------------|----------------------------------------|
| 10.09            | Added parameter: metric <METRIC-VALUE> |
| 10.07 or earlier | --                                     |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## default-metric (ospf)

```
default-metric <METRIC-VALUE>  
no default-metric
```

## Description

Sets the default metric for redistributed routes in the OSPF.

The **no** form of this command sets the default metric to be used for redistributed routes into OSPF to the default of 25.

| Parameter      | Description                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------|
| <METRIC-VALUE> | Specifies the default metric value to use for redistributed routes. Default: 25. Range: 0-1677214. |

## Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# default-metric 37
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# no default-metric
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## disable (router ospf)

disable

### Description

Disables the OSPF process.



This command does not remove the OSPF configurations.

## Examples

Disabling OSPF process:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## distance (ospf)

```
distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area [<DISTANCE-VAL>] |
external [<DISTANCE-VAL>]]
no distance [intra-area | inter-area | external]
```

### Description

Defines an Administrative Distance (AD) for OSPF. Administrative Distance is used as a criteria to select the best route when multiple routes are present from different routing protocols.

The **no** form of this command sets the OSPF administrative distance to the default of 110. Optionally, administrative distance can be set to default for the specific OSPF route type: intra-area, inter-area, or external type-5 and type-7 routes.

| Parameter      | Description                                                                |
|----------------|----------------------------------------------------------------------------|
| <DISTANCE-VAL> | Specifies the OSPF administrative distance. Range: 1 to 255. Default: 110. |
| intra-area     | Specifies the OSPF distance for intra-area routes.                         |
| inter-area     | Specifies the OSPF distance for inter-area routes.                         |
| external       | Specifies the OSPF distance for external type 5 and type 7 routes.         |

### Usage

Within a given OSPF process, intra-area routes are always given precedence even when distances are configured for inter-area or external type routes.

### Examples

Setting OSPF administrative distance:

```
switch(config)# router ospf 1
switch(config-ospf-1)# distance 100
switch(config-ospf-1)# distance intra-area 24 external 55 inter-area 66
switch(config-ospf-1)# distance intra-area 24 external 55
switch(config-ospf-1)# distance external 55
switch(config-ospf-1)#exit

switch(config)# router ospf 2
switch(config-ospf-2)# distance 200
switch(config-ospf-2)# distance external 60
switch(config-ospf-2)# distance intra-area 24 inter-area 66
```

Setting OSPF administrative distance to the default:

```

switch(config)# router ospf 1
switch(config-ospf-1)# no distance
switch(config-ospf-1)# no distance external
switch(config-ospf-1)# no distance inter-area
switch(config-ospf-1)# no distance intra-area
switch(config-ospf-1)# no distance 100
switch(config-ospf-1)# no distance 220

switch(config)# router ospf 2 vrf blue
switch(config-ospf-2)# no distance 200
switch(config-ospf-2)# no distance external 60
switch(config-ospf-2)# no distance intra-area 24 inter-area 66

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                                                             |
|------------------|------------------------------------------------------------------------------------------|
| 10.14            | Added capability to have individual admin distance for multiple OSPF processes in a VRF. |
| 10.09            | Added parameters: <b>intra-area</b> , <b>inter-area</b> , <b>external</b>                |
| 10.07 or earlier | --                                                                                       |

## Command Information

| Platforms    | Command context                        | Authority                                                                          |
|--------------|----------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf- <i>&lt;PROCESS-ID&gt;</i> | Administrators or local user group members with execution rights for this command. |

## distribute-list prefix (OSPFv2)

```

distribute-list prefix <prefix-list-name> {in | out}
no distribute-list prefix <prefix-list-name> {in | out}

```

### Description

This command uses an existing prefix list to filter routes that are being installed in the routing table or redistributed to another routing protocol.

The **distribute-list prefix** command filters routes in the inbound or the outbound direction. When this command is issued with the **in** parameter, it filters routes from being installed in the routing table, it does not filter LSAs. When this command is issued with the **out** parameter, it filters only the desired redistributed routes from other protocols.



This command requires that your prefix list is already defined using the ip prefix commands. Route-maps are not supported with the distribute-list feature.

| Parameter                 | Description                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prefix <prefix-list-name> | Specify the name of an existing prefix.                                                                                                                                                                                                 |
| {in   out}                | Select one of the following parameters to set the filter direction: <ul style="list-style-type: none"> <li>▪ <b>in</b>: Filter incoming routes into the routing table</li> <li>▪ <b>out</b>: Filter outgoing routing updates</li> </ul> |

## Examples

The following commands enable the filtering of OSPFv2 routes in an IPv4 network, so routes are no longer installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv2 1
switch(config-ospfv2-1)# distribute-list prefix listA in
switch(config-ospfv2-1)# distribute-list prefix listB out
```

The following command disables the filtering of OSPFv2 routes in an IPv4 network, so routes can be installed in the routing table or redistributed from another routing protocol.

```
switch# configure terminal
switch(config)# router ospfv2 1
switch(config-ospfv2-1)# no distribute-list prefix listA in
switch(config-ospfv2-1)# no distribute-list prefix listB out
```

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.13   | Command introduced. |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv2-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## enable (router ospf)

enable

### Description

Enables the OSPF process, if disabled. By default the OSPF process is enabled.

### Examples

Enabling OSPF process:

```
switch(config)# router ospf 1
switch(config-ospf-1)# enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## graceful-restart (router ospf)

```
graceful-restart
  restart-interval <INTERVAL>
  helper [strict-lsa-check]
  ignore-lost-interface
  no...
```

## Description

Configures graceful restart parameters for OSPF.

The **no** form of this command sets the restart interval to the default interval of 120 seconds or disables the helper mode, depending on the parameter specified.

| Parameter                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| restart-interval <INTERVAL> | Specifies the time another router waits for this router to gracefully restart and selects the maximum time to wait in seconds. Range: 5 to 1800. Default: 120.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| helper                      | Specifies that the router will participate in the graceful restart of a neighbor router.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| strict-lsa-check            | (Optional). Use with the <b>helper</b> parameter to enable strict Link state Advertisement (LSA) checking when acting as a restart helper for a restarting peer.<br><br><b>NOTE:</b> OSPF neighbors must disable strict LSA checking. If the local node has fewer OSPF interfaces after restarting, then the neighbors that were adjacent on those interfaces will clear up their adjacencies to the restarting node and will send out link state updates to advertise the dropped adjacency. If strict LSA checking is enabled, the restarting router's neighbors will exit helper mode when they receive the updated LSAs and the graceful restart will still fail. |
| ignore-lost-interface       | Enable the restarting router to ignore lost OSPF interfaces during a graceful restart process. This setting should be enabled on a high availability system to ensure a graceful restart completes                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Parameter | Description                                                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | successfully, even if OSPF-enabled links fail due to High Availability events like a switchover or failover.                                                  |
|           | <b>NOTE:</b> Enabling this setting means that the hitless restart procedures do not strictly follow those defined in <i>RFC 3623, Graceful OSPF Restart</i> . |
| no        | Negate any parameter or return the setting to its default.                                                                                                    |

## Examples

Enabling OSPF graceful restart:

```
switch(config)# router ospf 1
switch(config-ospf-1)# graceful-restart restart-interval 40
switch(config-ospf-1)# graceful-restart helper strict-lsa-check
```

Enabling the switch to ignore lost OSPF interfaces during a graceful restart process:

```
switch(config)# router ospf 1
switch (config-ospf-1)# graceful-restart ignore-lost-interface
```

Setting the restart interval to default, and disabling helper mode:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# no graceful-restart restart-interval
switch(config-ospfv3-1)# no graceful-restart helper
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## ip ospf area

```
ip ospf <PROCESS-ID> area <AREA-ID>
no ip ospf <PROCESS-ID> area <AREA-ID>
```

## Description



Runs the OSPF protocol on the interface with the configured IPv4 address for the area specified. The interfaces which have an IP address configured in this network or in a subset of this network, will participate in the OSPF protocol.

To move an interface to a new area, unmap the existing area and then associate a new area with the interface.

The **no** form of this command disables OSPF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol.

| Parameter    | Description                                                                                                                                                                                                                       |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID> | Specifies the OSPF process Id. Range: 1 to 65535.                                                                                                                                                                                 |
| <AREA-ID>    | Specifies the OSPF area ID in one of the following formats.<br>Area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>Area identifier in decimal format. Range: 0 to 4294967295. |

## Examples

Setting OSPF network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf 1 area 1
switch(config-if-vlan)# ip ospf 1 area 0.0.0.1
```

Disabling OSPF network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf 1 area 1
switch(config-if-vlan)# no ip ospf 1 area 0.0.0.1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf authentication

```
ip ospf authentication {message-digest | simple-text | null | keychain | hmac-sha-1 |  
hmac-sha-256 | hmac-sha-384 | hmac-sha-512}
```

```
no ip ospf authentication
```

## Description

Sets the authentication type that will be used for authentication with the neighbor router.

The **no** form of this command deletes the authentication type used for a particular authentication with the neighbor router and sets to null authentication.

| Parameter      | Description                                        |
|----------------|----------------------------------------------------|
| message-digest | Sets authentication type as message-digest.        |
| simple-text    | Sets authentication type as simple-text.           |
| null           | Sets authentication type as null.                  |
| keychain       | Sets the authentication type to use the key chain. |
| hmac-sha-1     | Sets the authentication type to SHA-1.             |
| hmac-sha-256   | Sets the authentication type to SHA-256.           |
| hmac-sha-384   | Sets the authentication type to SHA-384.           |
| hmac-sha-512   | Sets the authentication type to SHA-512.           |

## Examples

Setting OSPF authentication type on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication simple-text
```

Deleting OSPF authentication type on the interface and sets it to null:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf authentication
```

Setting OSPF authentication type to SHA-384 on the interface:

```
switch(config)# interface vlan 5
switch(config-if-vlan)# ip ospf authentication hmac-sha-384
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf authentication-key

```
ip ospf authentication-key [{ciphertext | plaintext} <PASSWORD>]
no ip ospf authentication-key
```

### Description

Sets the authentication password used for simple-text authentication. If the password is given in ciphertext it will be decrypted and applied to the protocol.

The **no** form of this command deletes the authentication password used for simple-text authentication.

| Parameter                | Description                  |
|--------------------------|------------------------------|
| {ciphertext   plaintext} | Selects the password format. |
| <PASSWORD>               | Specifies the password.      |



When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

### Examples

Setting the OSPF simple-text authentication password in plaintext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key plaintext F82#450b
```

Setting the OSPF simple-text authentication password with a prompted plaintext password:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key
Enter the authentication key: *****
Re-Enter the authentication key: *****
```

Setting the OSPF simple-text authentication password in ciphertext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key ciphertext AQBaz...ecopg=
```

Deleting the OSPF simple-text authentication password:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf authentication-key
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf cost

```
ip ospf cost <INTERFACE-COST>
no ip ospf cost
```

### Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The **no** form of this command sets the cost (metric) associated with a particular interface to the default cost 1.

| Parameter        | Description                                                        |
|------------------|--------------------------------------------------------------------|
| <INTERFACE-COST> | Specifies the interface cost value. Range: 1 to 65535. Default: 1. |

## Examples

Setting OSPF interface cost

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf cost 100
```

Setting the OSPF interface cost to default

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf cost
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf dead-interval

```
ip ospf dead-interval <INTERVAL>
no ip ospf dead-interval
```

### Description

Sets the interval after which a neighbor is declared dead if no hello packet is received on the OSPF interface.

The **no** form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPF interface. The default value is 40 seconds (generally 4 times the hello packet interval).

| Parameter  | Description                                                                                    |
|------------|------------------------------------------------------------------------------------------------|
| <INTERVAL> | Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40. |

### Examples

Setting OSPF dead interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf dead-interval 30
```

Setting OSPF dead interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf dead-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf hello-interval

```
ip ospf hello-interval <INTERVAL>
no ip ospf hello-interval
```

### Description

Sets the time interval between OSPF hello packets for the OSPF interface.

The **no** form of this command sets the time interval OSPF hello packets to the default of 10 seconds for the OSPF interface.

| Parameter  | Description                                                                                     |
|------------|-------------------------------------------------------------------------------------------------|
| <INTERVAL> | Specifies the time interval for the hello interval, in seconds. Range: 1 to 65535. Default: 10. |

### Examples

Setting OSPF hello interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf hello-interval 30
```

Setting OSPF hello interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf hello-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf keychain

```
ip ospf keychain <KEYCHAIN-NAME>
no ip ospf keychain
```

## Description

Sets the key chain for md5 authentication. A key chain configures rotating keys for packet authenticating, reducing the risk of keys being compromised.

The **no** form of this command deletes the key chain used for md5 authentication.

| Parameter                          | Description                                          |
|------------------------------------|------------------------------------------------------|
| <code>&lt;KEYCHAIN-NAME&gt;</code> | Name of key chain to be used for md5 authentication. |

## Examples

Setting OSPFv2 key chain authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf keychain ospf_keys
```

Deleting OSPFv2 key chain authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip ospf keychain
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ip ospf message-digest-key md5

```
ip ospf message-digest-key <KEY-ID> md5 [{ciphertext | plaintext} <KEY>]
no ip ospf message-digest-key <KEY-ID>
```

## Description

Sets the md5 message digest authentication key. If the md5 key is given in ciphertext, it will be decrypted and applied to the protocol.

The **no** form of this command deletes the md5 authentication key.

| Parameter                | Description                                |
|--------------------------|--------------------------------------------|
| <KEY-ID>                 | Specifies the md5 key ID. Range: 1 to 255. |
| {ciphertext   plaintext} | Selects the md5 key format.                |
| <KEY>                    | Specifies the md5 authentication key.      |



When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting the md5 key in plaintext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5 plaintext F82#450b
```

Setting the md5 key with a prompted plaintext key:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5
Enter the MD5 authentication key: *****
Re-Enter the MD5 authentication key: *****
```

Setting the md5 key in ciphertext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5 ciphertext AQ6e...7qEa4=
```

Deleting the md5 key:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf message-digest-key 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information



| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf network

```
ip ospf network {broadcast | point-to-point}
no ip ospf network
```

### Description

Configures the OSPF network type for the interface. Choose one of the following parameters as the interface network type.

The **no** form of this command sets the network type for the interface to the system default which is broadcast network.



For an IP unnumbered interface, the network type should be configured as a **point-to-point**.

| Parameter      | Description                                                          |
|----------------|----------------------------------------------------------------------|
| broadcast      | Specifies the OSPF network type as a broadcast multi-access network. |
| point-to-point | Specifies the OSPF network type as a point-to-point network.         |

### Examples

Setting OSPF network type for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf network broadcast
switch(config-if-vlan)# ip ospf network point-to-point
```

Disabling OSPF network type for the interface to system default of broadcast network:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf network
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf passive

```
ip ospf passive
no ip ospf passive
```

### Description

Configures the interface as an OSPF passive interface. With this setting, the interface participates in OSPF but does not send or receive packets on that interface.

The **no** form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

### Examples

Setting the interface as OSPF passive interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf passive
```

Setting the interface as OSPF active interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf passive
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf priority

```
ip ospf priority <PRIORITY-VALUE>
no ip ospf priority
```

### Description

Sets the OSPF priority for the interface. The larger the numeric value of the priority, the higher the chances for it to become the designated router. Setting a priority of zero makes the router ineligible to become a designated router or back up designated router.

The **no** form of this command sets the OSPF priority for the interface to the default of 1.

| Parameter                           | Description                                                     |
|-------------------------------------|-----------------------------------------------------------------|
| <code>&lt;PRIORITY-VALUE&gt;</code> | Specifies the OSPF priority value. Range: 0 to 255. Default: 1. |

## Examples

Setting OSPF priority for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf priority 50
```

Disabling OSPF priority for the interface to default:

```
switch(config)# interface vlan1
switch(config-if-vlan)# no ip ospf priority
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf retransmit-interval

```
ip ospf retransmit-interval <INTERVAL>
no ip ospf retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for the OSPF interface.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for the OSPF interface.

| Parameter  | Description                                                                  |
|------------|------------------------------------------------------------------------------|
| <INTERVAL> | Specifies the retransmit interval, in seconds. Range: 1 to 3600. Default: 5. |

## Examples

Setting OSPF retransmit interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf retransmit-interval 30
```

Setting OSPF retransmit interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf retransmit-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf sha-key sha

```
ip ospf sha-key <KEY-ID> sha [{ciphertext | plaintext} <KEY>]
no ip ospf sha-key <KEY-ID>
```

## Description

Sets the SHA (secure hash authentication) key for the selected interface. If the SHA key is given in ciphertext, it will be decrypted and applied to the protocol. This command accepts a key of up to 64 characters irrespective of the SHA version configured on the interface. OSPF will internally pad zeros to the key to obtain a 64-byte key. For all types of SHA, key length is adjusted to 64 bytes.

The **no** form of this command deletes the SHA authentication key.

| Parameter | Description                                |
|-----------|--------------------------------------------|
| <KEY-ID>  | Specifies the SHA key ID. Range: 1 to 255. |

| Parameter                | Description                           |
|--------------------------|---------------------------------------|
| {ciphertext   plaintext} | Selects the SHA key format.           |
| <KEY>                    | Specifies the SHA authentication key. |



When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting the SHA authentication key in plaintext format:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf sha-key 1 sha plaintext F82#450b
```

Setting the SHA authentication key in prompted plaintext format:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf sha-key 1 sha
Enter the SHA authentication key: *****
Re-Enter the SHA authentication key: *****
```

Setting the SHA authentication key in ciphertext format:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf sha-key 1 sha ciphertext AQapu...C2K47A=
```

Deleting the SHA authentication key:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip ospf sha-key 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf shutdown

```
ip ospf shutdown
no ip ospf shutdown
```

### Description

Disables OSPF on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command **no ip ospf area**.

The **no** form of this command re-enables OSPF on the interface

### Examples

Disabling OSPF on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf shutdown
```

Re-enabling OSPF on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf shutdown
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf transit-delay

```
ip ospf transit-delay <DELAY>
no ip ospf transit-delay
```

### Description

Sets the time delay in link state transmission for the OSPF interface.

The **no** form of this command sets the delay in link state transmission to the default of 1 second for the OSPF interface.

| Parameter | Description                                                           |
|-----------|-----------------------------------------------------------------------|
| <DELAY>   | Specifies the transit delay in seconds. Range: 1 to 3600. Default: 1. |

## Examples

Setting OSPF transit delay on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf transit-delay 30
```

Setting OSPF transit delay to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf transit-delay
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## keychain (ospf)

```
keychain <KEYCHAIN-NAME>
no keychain
```

### Description

Sets the key chain for md5 authentication. A key chain configures rotating keys for packet authenticating, reducing the risk of keys being compromised.

The **no** form of this command deletes the key chain used for md5 authentication.

| Parameter       | Description                                          |
|-----------------|------------------------------------------------------|
| <KEYCHAIN-NAME> | Name of key chain to be used for md5 authentication. |

## Examples

Setting OSPF virtual link key chain authentication:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# keychain ospf_keys
```

Deleting OSPF virtual link key chain authentication:

```
switch# configure terminal
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# no keychain
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |

## max-metric router-lsa (ospf)

```
max-metric router-lsa [on-startup [<ADVERT-TIME>]]
no max-metric router-lsa [on-startup]
```

### Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations. If the on-startup parameter is used, the router is configured to advertise a maximum metric at startup for the time mentioned in seconds or for a default value of 600 seconds.

The **no** form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

| Parameter                | Description                                                                                                                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| on-startup <ADVERT-TIME> | Specifies the time in seconds to advertise self as stub-router on startup. If no time is specified, the default time of 600 seconds is used. Range: 5 to 86400. Default: 600. |

## Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospf 1
switch(config-ospf-1)# max-metric router-lsa
switch(config-ospf-1)# max-metric router-lsa on-startup
switch(config-ospf-1)# max-metric router-lsa on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:



```
switch(config)# router ospf 1
switch(config-ospf-1)# no max-metric router-lsa
switch(config-ospf-1)# no max-metric router-lsa on-startup
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## passive-interface default (OSPFv2)

```
passive-interface default
no passive-interface
```

### Description

Configures all OSPF interfaces as passive.

The **no** form of this command sets all OSPF interfaces as active.

### Examples

Setting OSPF-enabled interfaces as passive:

```
switch(config)# router ospf 1
switch(config-ospf-1)# passive-interface default
```

Setting OSPF-enabled interfaces as active:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no passive-interface
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## redistribute (ospf)

```
redistribute {bgp | connected | local loopback | static | rip | ospf <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]
no redistribute {bgp | connected | local loopback | static | rip | ospf <PROCESS-ID>}
[route-map <ROUTE-MAP-NAME>]
```

### Description

Redistributes routes originating from other protocols, or from another OSPFv2 process, to the current OSPFv2 process.

If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv2. Configuration is not allowed if the referenced route map has not yet been configured.

If you try to redistribute routes from an OSPFv2 process which is not created, you are prompted to allow the OSPFv2 process to be auto-created before proceeding with redistribution. If you confirm at the prompt, the OSPFv2 process is created with defaults and redistribution configuration applied. If you deny at the prompt, redistribution configuration is skipped.

If command **route-redistribute active-routes-only** has been issued, only the routes from other protocols which are selected for forwarding are considered for redistribution into OSPFv2.

The **no** form of this command disables redistribution of routes to the current OSPFv2 process.

| Parameter                  | Description                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------------------|
| bgp                        | Specifies redistributing BGP (Border Gateway Protocol) routes.                                         |
| connected                  | Specifies redistributing connected (directly attached subnet or host).                                 |
| local loopback             | Specifies redistributing local routes of the loopback interface.                                       |
| static                     | Specifies redistributing static routes.                                                                |
| rip                        | Specifies redistributing RIP routes.                                                                   |
| ospf <PROCESS-ID>          | Specifies redistributing routes from the specified OSPFv2 process ID. Range: 1 to 65535.               |
| route-map <ROUTE-MAP-NAME> | Specifies redistribution filtering by route map. To create a route map, use command <b>route-map</b> . |

### Examples

Redistributing routes to OSPFv2:

```
switch(config)# router ospf 1
switch(config-ospf-1)# redistribute connected
switch(config-ospf-1)# redistribute connected route-map connected_routes
switch(config-ospf-1)# redistribute local loopback
```

```

switch(config-ospf-1)# redistribute local loopback route-map local_routes
switch(config-ospf-1)# redistribute static
switch(config-ospf-1)# redistribute static route-map static_networks
switch(config-ospf-1)# redistribute rip
switch(config-ospf-1)# redistribute rip route-map rip-routes
switch(config-ospf-1)# redistribute ospf 2

```

Disabling redistributing routes to OSPFv2:

```

switch(config)# router ospf 1
switch(config-ospf-1)# no redistribute connected
switch(config-ospf-1)# no redistribute connected route-map connected_routes
switch(config-ospf-1)# no redistribute local loopback
switch(config-ospf-1)# no redistribute local loopback route-map local_routes
switch(config-ospf-1)# no redistribute static
switch(config-ospf-1)# no redistribute static route-map static_networks
switch(config-ospf-1)# no redistribute rip
switch(config-ospf-1)# no redistribute rip route-map rip-routes
switch(config-ospf-1)# no redistribute ospf 2

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                                                                                  |
|------------------|---------------------------------------------------------------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535.                                                     |
| 10.08            | Added <b>route-map</b> support for supported redistribute source-protocols. Updated information and examples. |
| 10.07 or earlier | --                                                                                                            |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## reference-bandwidth (OSPFv2)

```

reference-bandwidth <BANDWIDTH>
no reference-bandwidth

```

### Description

Sets the reference bandwidth for OSPFv2. If the OSPFv2 interface cost is not explicitly set, then the cost of all the OSPFv2 interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the link speed value is taken as 1 Gbps (if the OSPFv2 interface cost is not explicitly set).

The **no** form of this command sets the reference bandwidth for OSPF to the default of 100000 Mbps.

| Parameter   | Description                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| <BANDWIDTH> | Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000. |

## Examples

Setting the reference bandwidth:

```
switch(config)# router ospf 1
switch(config-ospf-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no reference-bandwidth
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## rfc1583-compatibility

```
rfc1583-compatibility
no rfc1583-compatibility
```

## Description

Enables OSPF compatibility with RFC1583 (backward compatibility). If RFC1583 compatibility is enabled, then the route cost calculation follows a different method.

The **no** form of this command disables OSPF compatibility with RFC1583 (backward compatibility). By default the RFC1583 compatibility is disabled.

## Examples

Enabling OSPF RFC1583 compatibility:

```
switch(config)# router ospf 1
switch(config-ospf-1)# rfc1583-compatibility
```

Disabling OSPF RFC1583 compatibility:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# no rfc1583-compatibility
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## router ospf

```
router ospf <PROCESS-ID> [vrf <VRF-NAME>]  
no router ospf <PROCESS-ID> [vrf <VRF-NAME>]
```

## Description

Creates an OSPF process (if not created already) on a VRF, and switches to the OSPF router instance context. Up to eight OSPF processes are supported per VRF.

The **no** form of this command removes the OSPF instance.

| Parameter      | Description                                                  |
|----------------|--------------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPF process ID. Range: 1 to 65535.             |
| vrf <VRF-NAME> | Specifies a VRF name for the OSPF process. Default: default. |

## Examples

```
switch(config)# router ospf 1  
switch(config-ospf-1)#
```

```
switch(config)# router ospf 1 vrf vrf_red
```

```
switch(config)# no router ospf 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## router-id (ospf)

router-id <ROUTER-ADDR>  
no router-id

### Description

Sets an ID for the router in an IPv4 address format.

The **no** form of this command unconfigures the router-id for the instance and sets the router-id to the default as follows: the router-id is selected dynamically as equal to the highest loopback address on the router, or the highest active interface if there are no loopback addresses. If no IP address is configured on any interfaces on the router, OSPF will not form an adjacency.

| Parameter     | Description                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------|
| <ROUTER-ADDR> | Specifies the Router ID in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255. |

### Examples

Setting router-id in the OSPF context:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# router-id 1.1.1.1
```

Unconfiguring router-id:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# no router-id
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## sha-key sha

```
sha-key <KEY-ID> sha [{ciphertext | plaintext} <KEY>]
no sha-key <KEY-ID>
```

### Description

Sets the SHA (secure hash authentication) key for the selected virtual link. If the SHA key is given in ciphertext, it will be decrypted and applied to the protocol. This command accepts a key of up to 64 characters irrespective of the SHA version configured on virtual link. OSPF will internally pad zeros to the key to obtain a 64-byte key. For all types of SHA, key length is adjusted to 64 bytes.

The **no** form of this command deletes the virtual link SHA authentication key.

| Parameter                | Description                                             |
|--------------------------|---------------------------------------------------------|
| <KEY-ID>                 | Specifies the virtual link SHA key ID. Range: 1 to 255. |
| {ciphertext   plaintext} | Selects the virtual link SHA key format.                |
| <KEY>                    | Specifies the virtual link SHA authentication key.      |



When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

### Examples

Setting virtual link SHA authentication key in plaintext format:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# sha-key 1 sha plaintext F82#450b
```

Setting the virtual link SHA authentication key in prompted plaintext format:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# sha-key 1 sha
Enter the SHA authentication key: *****
Re-Enter the SHA authentication key: *****
```

Setting the virtual link SHA authentication key in ciphertext format:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# sha-key 1 sha ciphertext AQapu...C2K47A=
```

Deleting the virtual link SHA authentication key:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# no sha-key 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |

## show ip ospf

show ip ospf [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]

## Description

Displays general OSPF, area, state, and configuration information.

| Parameter      | Description                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535. |
| all-vrfs       | Optionally select to display general OSPF information for all VRFs.                                            |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.                                                                 |

## Examples

Showing general OSPF configurations:

```
switch# show ip ospf 200
VRF : Default                               Process : 200
-----
Router ID           : 1.1.1.1                OSPFv2           : Enabled
```



```

BFD : Disabled
SPF Hold Interval : 1000 ms
LSA Start Interval : 5000 ms
LSA Max Wait Interval : 1000 ms
External LSAs : 4
ECMP : 4
Area Border : Yes
GR Status : Disabled
GR State : Inactive
GR Helper : Enabled
GR Ignore Lost I/F : Disabled
Summary address:
  prefix 10.1.1.0/24, advertise, tag 10

Area      Total      Active
-----
Normal    2          1
Stub      2          1
NSSA      0          0

Area : 0.0.0.1
-----
Area Type : Normal      Status : Active
Total Interfaces : 100   Active Interfaces : 10
Passive Interfaces : 5   Loopback Interfaces : 85
SPF Calculation Count : 1500
Area ranges:
  ip-prefix 10.1.1.1/24, inter-area, advertise
Number of LSAs : 5000      Checksum Sum : 99122
SPF Start Interval : 5000 ms
SPF Max Wait Interval : 1000 ms
LSA Hold Interval : 1000 ms
LSA Arrival Interval : 1000 ms
Checksum Sum : 133302
Reference Bandwidth : 100000 Mbps
AS Border : No
GR Interval : 120 sec
GR Exit Status : None
GR Strict LSA Check : Enabled
Internal Process ID : 1

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf border-routers

```

show ip ospf [<PROCESS-ID>]
  border-routers [all-vrfs | vrf <VRF-NAME>]

```

## Description

Displays the OSPF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

| Parameter      | Description                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535. |
| all-vrfs       | Optionally select to display general OSPF information for all VRFs.                                            |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.                                                                 |

## Examples

Showing OSPF border routers information:

```
switch# show ip ospf border-routers
VRF : default                      Process : 1
Internal Routing Table
-----

Codes: i - Intra-area route, I - Inter-area route

  Router-ID    Cost Type Area      SPF  Nexthop  Interface
i 40.40.40.40  10  ABR  0.0.0.0  71    192.0.2.1 1/1/1
i 60.60.60.60  20  ABR  0.0.0.0  71    192.0.2.1 1/1/1
i 40.40.40.40  10  ABR  0.0.0.1  71    192.0.2.1 1/1/2
i 60.60.60.60  20  ABR  0.0.0.1  71    192.0.2.1 1/1/2
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf interface

```
show ip ospf [<PROCESS-ID>] interface [<interface-name>] [brief]
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays general OSPF, area, state, and configuration information.



For an IP unnumbered enabled OSPF interface, the output of the command displays the IP address borrowed from the donor interface.

| Parameter         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>      | Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <interface-name>] | Specify the name of an OSPF interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| brief             | Include this parameter to display a brief overview of the following OSPF configuration information. <ul style="list-style-type: none"><li>Interface: OSPF interface name.</li><li>Area: OSPF area ID.</li><li>Cost: The metric OSPF uses to judge a path's feasibility, calculated as (reference bandwidth / interface bandwidth).</li><li>State: Indicates if the interface is a designated router (<b>Dr</b>) or a backup designated router (<b>Backup-dr</b>).</li><li>Status: Indicates if the interface is <b>up</b> or <b>down</b>.</li><li>Flags: P - Passive A - Active.</li></ul> |
| all-vrfs          | Optionally select to display general OSPF information for all VRFs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| vrf <VRF-NAME>    | Specifies the name of a VRF. Default: default.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Examples

Showing general OSPF configuration settings for the default VRF:

```
switch (config-if)# show ip ospf 1
VRF : default                               Process : 1
-----

RouterID          : 20.0.0.1                OSPFv2          : Enabled
SPF Start Interval : 200 ms
SPF Hold Interval  : 1000 ms
LSA Start Time     : 5000 ms
LSA Max Wait Time  : 0 ms
External LSAs      : 0
ECMP               : 4
Area Border        : false
GR Status          : Enabled
GR State           : inactive
GR Helper          : Enabled
GR Ignore Lost I/F : Disabled
Summary address:

Area      Total      Active
-----
Normal    1          1
Stub      0          0
NSSA      0          0

Area : 0.0.0.0
-----
Area Type          : Normal                Status          : Active
```

```

Total Interfaces      : 1                Active Interfaces   : 1
Passive Interfaces    : 0                Loopback Interfaces  : 0
SPF Calculation Count : 4
Area ranges          :
Number of LSAs       : 3                Checksum Sum         : 82420

```

Showing OSPF configuration settings for all interfaces:

```

switch(config)# show ip ospf interface
Codes: DR - Designated router BDR - Backup designated router

Interface 1/1/1 is up, line protocol is up
-----
VRF          : default                Process          : 1
IP Address    : 10.10.10.1/24         Area             : 0.0.0.0
Status        : Up                   Network Type     : Broadcast
Hello Interval : 10 sec               Dead Interval    : 40 sec
Transit Delay  : 1 sec                Retransmit Interval : 5 sec
Link Speed     : 1000 Mbps
Cost Configured : 1                   Cost Calculated  : 1
State/Type     : BDR                  Router Priority   : 1
DR             : 10.10.10.2           BDR              : 10.10.10.1
Link LSAs      : 0                   Checksum Sum     : 0
Authentication : Md5                  Passive          : Yes
Interface 1/1/2 is up, line protocol is up
-----
VRF          : default                Process          : 1
IP Address    : 10.10.10.1/24         Area             : 0.0.0.0
Status        : Up                   Network Type     : Broadcast
Hello Interval : 10 sec               Dead Interval    : 40 sec
Transit Delay  : 1 sec                Retransmit Interval : 5 sec
Link Speed     : 1000 Mbps
Cost Configured : 1                   Cost Calculated  : 1
State/Type     : BDR                  Router Priority   : 1
DR             : 20.10.10.2           BDR              : 20.10.10.1
Link LSAs      : 0                   Checksum Sum     : 0
Authentication : Simple               Passive          : No

```

Displaying brief OSPF information

```

switch(config-if)# show ip ospf interface brief
VRF : default   Process : 1
=====
Total Number of Interfaces: 2
Flags: P - Passive A - Active
Interface      Area                IP Address/Mask      Cost  State   Status  Flags
-----
1/1/1          0.0.0.0                10.10.10.1/24       40    DR      Up      P
1/1/2          255.255.255.255       200.200.200.123/24  4      Waiting Up      A

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                                                                                                       |
|------------------|------------------------------------------------------------------------------------------------------------------------------------|
| 10.14.1000       | , The output of the command displays the borrowed IP address for an IP unnumbered enabled OSPF interface.                          |
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535.                                                                          |
| 10.09            | Output of the <b>show ip ospf interface</b> command includes flags to indicate whether the interface is in passive or active mode. |
| 10.07 or earlier | --                                                                                                                                 |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf lsdb

```
show ip ospf [<process-id>] lsdb
  adv-router {<ROUTER-ID> | self}
  area <AREA-ID>
  lsid <link-state-id>
  all-vrfs [{vrf <VRF-NAME>}
    asbr-summary
    database-summary
    external
    lsid <LINK-STATE-ID>
    network
    nssa-external
    router
    summary
```

## Description

Shows the OSPF link state database summary for different OSPF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

| Parameter                     | Description                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>                  | Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535.                                                                                                                                                                 |
| adv-router {<ROUTER-ID> self} | Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify <b>self</b> to show self-originated link states.                                                                                            |
| area <AREA-ID>                | Select to display information filtered for the specified area in one of the following formats.<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Value: 0 to 4294967295. |

| Parameter                                    | Description                                                                                                                                                                                                                                                                             |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>lsid &lt;LINK-STATE-ID&gt;</code>      | Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D).                                                                                                                                                                             |
| <code>all-vrfs {vrf &lt;vrf-name&gt;}</code> | Select <b>all-vrfs</b> to display general OSPF information for all VRFs, or use the <b>vrf &lt;VRF-NAME&gt;</b> option to display information for a specific VRF.<br><br>Optionally select one of the following parameters to filter the link state database information.               |
| <code>asbr-summary</code>                    | Show ASBR summary link states (LSA type 4).                                                                                                                                                                                                                                             |
| <code>database-summary</code>                | Select to display the count of each type of LSA and each area in the database.<br><br><b>NOTE:</b> The <b>database-summary</b> parameter does not support the <b>area &lt;area-id&gt;</b> , <b>lsid &lt;link-state-id&gt;</b> or <b>adv-router {&lt;router-id&gt; self}</b> parameters. |
| <code>external</code>                        | Show external link states (LSA type 5).<br><br><b>NOTE:</b> The <b>external</b> parameter does not support the area <b>area &lt;area-id&gt;</b> parameter.                                                                                                                              |
| <code>network</code>                         | Show network LSAs (LSA type 2).                                                                                                                                                                                                                                                         |
| <code>nssa-external</code>                   | Show NSSA external link states (LSA type 7).                                                                                                                                                                                                                                            |
| <code>router</code>                          | Show router LSAs (LSA type 1).                                                                                                                                                                                                                                                          |
| <code>summary</code>                         | Show network-summary link states (LSA type 3).                                                                                                                                                                                                                                          |

## Examples

Showing OSPF link state database (LSDB) general information:

```
switch# show ip ospf lsdb
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
=====

Router Link State Advertisements (Area 0.0.0.0)
-----
ADV Router    Age      Seq#      Checksum    LSID        Link Count  Bits
-----
40.40.40.40   930     0x80000004 0x2ea1      0           3           None
50.50.50.50   935     0x80000002 0x8b52      0           1           E
60.60.60.60   943     0x800003c5 0x9854      0           2           None

Network Link State Advertisements (Area 0.0.0.0)
-----
ADV Router    Age      Seq#      Checksum    LSID        Router Count
-----
60.60.60.60   944     0x80000001 0x7179      1360007168  2
50.50.50.50   935     0x80000001 0x516a      19          1
```

```

Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Prefix
-----
40.40.40.40    929     0x80000001 0x2498     131072    FEC0:3344::/32
50.50.50.50    928     0x80000001 0x5b2f     65536     111::/64

Inter Area Router Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Destination Router ID
-----
40.40.40.40    929     0x80000001 0x2498     1         33.33.33.33

AS External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Prefix
-----
40.40.40.40    264     0x80000001 0x24cc4     1         10::/64
40.40.40.40    675     0x80000001 0x5b00f     2         11::/64

NSSA External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Prefix
-----
3.3.3.3        264     0x80000001 0x24ac2     1         200::/64

Link-local Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Interface
-----
50.50.50.50    264     0x80000001 0x653c4     19        1/1/1

Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Referenced LS Type
Referenced LSID
-----
50.50.50.50    263     0x80000001 0x1da34     1         0x2001          0
50.50.50.50    264     0x80000001 0x2a45d     1         0x2002          19

```

Showing ASBR summary link states:

```

switch# show ip ospf lsdbr asbr-summary
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====

ASBR Summary Link State Advertisements (Area 0.0.0.0)
-----
LSID            ADV Router      Age      Seq#      Checksum
-----
209.165.201.3   60.60.60.60    944     0x80000001 0x7179
192.0.2.1       50.50.50.50    935     0x80000001 0x516a

```

Showing external link states:

```

switch# show ip ospf lsdb external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====

AS External Link State Advertisements
-----

LSID                ADV Router    Age      Seq#          Checksum
-----
209.165.201.3      60.60.60.60   944      0x80000001   0x7179
192.0.2.1          50.50.50.50   935      0x80000001   0x516a

```

Showing database summary:

```

switch# show ip ospf lsdb database-summary
OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
=====

Area 0.0.0.0 database summary
-----

LSA Type            Count
-----
Router              2
Network             1
Inter-area Summary  1
ASBR Summary        0
NSSA External       0
Subtotal            4

Process 1 database summary
-----

LSA Type            Count
-----
Router              2
Network             1
Inter-area Summary  1
ASBR Summary        0
NSSA External       0
AS External         0
Total               4

```

Showing router LSAs:

```

switch# show ip ospf lsdb router
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====

Router Link State Advertisements (Area 0.0.0.0)
-----

LSID                ADV Router    Age      Seq#          Checksum Link Count
-----
1.1.1.2             1.1.1.2       15       0x80000004   0xf526    1
2.2.2.1             2.2.2.1       14       0x80000005   0x6c5e    2
2.2.2.2             2.2.2.2       104      0x80000004   0xf51a    1
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====

```



# Router Link State Advertisements (Area 0.0.0.0)

| LSID    | ADV Router | Age | Seq#       | Checksum | Link Count |
|---------|------------|-----|------------|----------|------------|
| 1.1.1.2 | 1.1.1.2    | 15  | 0x80000004 | 0xf526   | 1          |
| 2.2.2.1 | 2.2.2.1    | 14  | 0x80000005 | 0x6c5e   | 2          |
| 2.2.2.2 | 2.2.2.2    | 104 | 0x80000004 | 0xf51a   | 1          |

Showing network LSAs:

```
switch# show ip ospf lsdb network
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====
```

# Network Link State Advertisements (Area 0.0.0.0)

| LSID    | ADV Router | Age | Seq#       | Checksum |
|---------|------------|-----|------------|----------|
| 1.1.1.2 | 1.1.1.2    | 141 | 0x80000001 | 0xc55e   |
| 2.2.2.2 | 2.2.2.2    | 230 | 0x80000001 | 0xa179   |

Showing network-summary link states:

```
switch# show ip ospf lsdb summary
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====
```

# Inter-area Summary Link State Advertisements (Area 0.0.0.0)

| LSID    | ADV Router | Age | Seq#       | Checksum |
|---------|------------|-----|------------|----------|
| 1.1.1.0 | 2.2.2.1    | 133 | 0x80000002 | 0xa089   |

# Inter-area Summary Link State Advertisements (Area 0.0.0.1)

| LSID    | ADV Router | Age | Seq#       | Checksum |
|---------|------------|-----|------------|----------|
| 2.2.2.0 | 2.2.2.1    | 133 | 0x80000002 | 0x7caa   |

Showing NSSA external link states:

```
switch(config-ospf-1)# show ip ospf lsdb nssa-external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====
```

# NSSA External Link State Advertisements (Area 0.0.0.1)

| LSID    | ADV Router | Age | Seq#       | Checksum |
|---------|------------|-----|------------|----------|
| 8.8.8.0 | 1.1.1.2    | 162 | 0x80000003 | 0xc7b2   |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf neighbors

```
show ip ospf [<PROCESS-ID>] neighbors [<NEIGHBOR-ID>]  
    [interface <INTERFACE-NAME>] [detail | summary]  
    [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays information about OSPF neighbors.

| Parameter                  | Description                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Enter an OSPF process ID to display OSPF neighbor information for the particular OSPF process. Range: 1 to 65535. |
| neighbors <NEIGHBOR-ID>    | Select to display information about a particular neighbor, specified in IPv4 format (A.B.C.D).                    |
| interface <INTERFACE-NAME> | Select to display neighbor information only for the specified interface.                                          |
| detail                     | Select to display detailed information for all the neighbors.                                                     |
| summary                    | Select to display summary information for the neighbors.                                                          |
| all-vrfs                   | Select to display neighbor information for all VRFs.                                                              |
| vrf <VRF-NAME>             | Specify the name of a VRF. Default: default.                                                                      |

## Examples

Showing OSPF neighbors information for the default VRF:

```
switch# show ip ospf neighbors  
OSPF Process ID 1 VRF default  
=====
```

Total Number of Neighbors: 1

| Neighbor ID | Priority | State   | Nbr Address | Interface |
|-------------|----------|---------|-------------|-----------|
| 2.2.2.2     | 1        | FULL/DR | 10.1.1.2    | 1/1/1     |



The **Nbr Address** displays the borrowed IP address for IP unnumbered interfaces.

Showing OSPF neighbors information for a specific neighbor:

```
switch# show ip ospf neighbors 2.2.2.2
switch# show ip ospf neighbors 2.2.2.2
VRF : default                               Process : 1
-----
Router-Id      : 2.2.2.2                     Area Id       : 0.0.0.0
Interface      : 1/1/1                       Address       : 10.10.10.2
State          : FULL                        Neighbor Priority : 1
Dead Timer Due : 00:00:36                     Options      : 0x42
```

Showing OSPF neighbors information for a specific neighbor and interface:

```
switch# show ip ospf neighbors 2.2.2.2 interface 1/1/1
VRF : default                               Process : 1
-----
Router-Id      : 2.2.2.2                     Area Id       : 0.0.0.0
Interface      : 1/1/1                       Address       : 10.10.10.2
State          : FULL                        Neighbor Priority : 1
Dead Timer Due : 00:00:36                     Options      : 0x42
```

Showing detail information for OSPF neighbors:

```
switch# show ip ospf neighbors detail
VRF : default                               Process : 1
-----
Router-Id      : 2.2.2.2                     Area Id       : 0.0.0.0
Interface      : 1/1/1                       Address       : 10.10.10.2
State          : FULL                        Neighbor Priority : 1
DR             : 10.10.10.2                   BDR          : 10.10.10.1
Dead Timer Due : 00:00:38                     Options      : 0x42
Retransmission Queue Length : 0
Time Since Last State Change : 00h:11m:37s
```

Showing summary information for OSPF neighbors in the **default** VRF:

```
switch# show ip ospf neighbors summary
OSPF Process ID 1 VRF default, Neighbor Summary
=====
Interface  Down Attempt Init TwoWay ExStart Exchange Loading Full Total
-----
1/1/1      0    0      0    0      0      0      0      1    1
```

|       |   |   |   |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|---|---|---|
| 1/1/2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Total | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                                                                            |
|------------------|---------------------------------------------------------------------------------------------------------|
| 10.14.1000       | The output parameter, <b>Nbr Address</b> displays the borrowed IP address for IP unnumbered interfaces. |
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535.                                               |
| 10.07 or earlier | --                                                                                                      |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf routes

```
show ip ospf [<PROCESS-ID>] routes
            [<IPv4-ADDR>/<MASK>] protected [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays OSPF routing information.

| Parameter      | Description                                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| <IPv4-ADDR>    | Specify an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.                     |
| <MASK>         | Specifies the number of bits in the address mask in CIDR format ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 32. |
| <PROCESS-ID>   | Enter an OSPF process ID to display OSPF neighbor information for the particular OSPF process. Range: 1 to 65535.              |
| all-vrfs       | Select to display neighbor information for all VRFs.                                                                           |
| vrf <VRF-NAME> | Specify the name of a VRF. Default: default.                                                                                   |
| protected      | Selects a protected route. Specifying an IPv4 address will select protected route for that prefix.                             |

## Examples

Showing OSPF routing table information:

```

switch# show ip ospf routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1 VRF default, Routing Table
-----

Total Number of Routes : 2

10.1.1.0/24      (i) area: 0.0.0.0
                 directly attached to interface 1/1/1, cost 1 distance 110
20.1.1.0/24      (I)
                 via 10.1.1.2 interface 1/1/1, cost 2 distance 110

```

Showing OSPF routing table information for a specific subnet:

```

switch# show ip ospf routes 10.1.1.0/2
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1 VRF default, Routing Table for prefixes 10.1.1.0/24
-----

Total Number of Routes : 1

10.1.1.0/24      (i) area: 0.0.0.0
                 directly attached to interface 1/1/1, cost 1 distance 110

```

Showing OSPF routing table information for a specific protected subnet on the default VRF:

```

switch# show ip ospf routes 211.0.0.0/24 protected vrf default
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1, VRF default, Protected Routes for prefixes 211.0.0.0/24
-----

211.0.0.0/24      (i) area: 0.0.0.0
                 via 110.110.110.1, interface vlan20, cost 122
                 bkp 130.130.130.1, interface vlan22, cost 132

Total Number of Routes : 1

```

Showing OSPF routing table information for a specific protected subnet on all VRFs

```

switch# show ip ospf routes 211.0.0.0/24 protected all-vrfs
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1, VRF default, Protected Routes for prefixes 211.0.0.0/24
-----

211.0.0.0/24      (i) area: 0.0.0.0
                 via 110.110.110.1, interface vlan20, cost 122
                 bkp 130.130.130.1, interface vlan22, cost 132

```

```
Total Number of Routes : 1
```

Showing OSPF routing table information for a specific protected subnet:

```
switch# show ip ospf routes 211.0.0.0/24 protected
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1, VRF default, Protected Routes for prefixes 211.0.0.0/24
-----

211.0.0.0/24      (i) area: 0.0.0.0
                  via 110.110.110.1, interface vlan20, cost 122
                  bkp 130.130.130.1, interface vlan22, cost 132

Total Number of Routes : 1
```

Showing OSPF routing table information for all protected routes:

```
switch# show ip ospf routes 211.0.0.0/24 protected
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1, VRF default, Protected Routes
-----

120.120.120.0/24  (i) area: 0.0.0.0
                  via 110.110.110.1, interface vlan20, cost 120
                  bkp 130.130.130.1, interface vlan22, cost 230

140.140.140.0/24  (i) area: 0.0.0.0
                  via 110.110.110.1, interface vlan20, cost 130
                  bkp 130.130.130.1, interface vlan22, cost 220

211.0.0.0/24      (i) area: 0.0.0.0
                  via 110.110.110.1, interface vlan20, cost 122
                  bkp 130.130.130.1, interface vlan22, cost 132

Total Number of Routes : 3

OSPFs Process ID 1, VRF red, Protected Routes
-----

Total Number of Protected Routes: 0
```

Showing OSPF routing table information for protected routes on the default VRF:

```
switch# show ip ospf routes protected vrf default
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1, VRF default, Protected Routes
-----

140.140.140.0/24  (i) area: 0.0.0.0
```

```
via 130.130.130.1, interface vlan22, cost 130
bkp 110.110.110.1, interface vlan20, cost 220

211.0.0.0/24      (i) area: 0.0.0.0
via 110.110.110.1, interface vlan20, cost 122
bkp 130.130.130.1, interface vlan22, cost 132

120.120.120.0/24 (i) area: 0.0.0.0
via 110.110.110.1, interface vlan20, cost 120
bkp 130.130.130.1, interface vlan22, cost 230

Total Number of Routes : 3
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.15            | Introduces the <b>protected</b> parameter.                |
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf statistics

```
show ip ospf [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays OSPF statistics.

| Parameter      | Description                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Enter an OSPF process ID to display OSPF neighbor information for the particular OSPF process. Range: 1 to 65535. |
| all-vrfs       | Select to display OSPF statistics information for all VRFs.                                                       |
| vrf <VRF-NAME> | Specify the name of a VRF. Default: default.                                                                      |

## Examples

Showing OSPF statistics:

```
switch# show ip ospf statistics
OSPF Process ID 1 VRF default, Statistics (cleared 1h 16m 24s ago)
-----

Unknown Interface Drops           : 0
Unknown Virtual Interface Drops   : 0
Bad Instance ID Drops             : 0
Bad IP Header Length Drops        : 0
Wrong OSPF Version Drops          : 0
Bad Source IP Drops               : 0
Resource Failure Drops            : 0
Bad Header Length Drops           : 0
Total Drops                       : 0
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf statistics interface

```
show ip ospf [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>]
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays OSPF statistics for the OSPF-enabled interfaces.

| Parameter        | Description                                                                                                                         |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>     | Enter an OSPF process ID to display OSPF-enabled interface statistics information on the specified OSPF process. Range: 1 to 65535. |
| <INTERFACE-NAME> | Select to display information only for the specified interface.                                                                     |
| all-vrfs         | Select to display OSPF-enabled interface statistics information for all VRFs.                                                       |
| vrf <VRF-NAME>   | Specify the name of a VRF. Default: default.                                                                                        |

## Examples



Showing OSPF-enabled interfaces information:

```
switch# show ip ospf statistics interface 1/1/1
OSPF Process ID 1 VRF default, interface 1/1/1 statistics (cleared 0h 30m 28s ago)
=====

Tx Hello Packets      : 101          Rx Hello Packets      : 99
Tx Hello Bytes       : 101          Rx Hello Bytes       : 99
Tx DD Packets        : 101          Rx DD Packets        : 99
Tx DD Bytes          : 101          Rx DD Bytes          : 99
Tx LS Request Packets : 101          Rx LS Requests Packets : 99
Tx LS Request Bytes  : 101          Rx LS Request Bytes  : 99
Tx LS Update Packets : 101          Rx LS Update Packets : 99
Tx LS Update Bytes   : 101          Rx LS Update Bytes   : 99
Tx LS Ack Packets    : 101          Rx LS Ack Packets    : 99
Tx LS Ack Bytes      : 101          Rx LS Ack Bytes      : 99

Total Number of State Changes : 8
Number of LSAs                : 29
LSA Checksum Sum              : 2345
Total Transmit Failures       : 29
Total OSPF Packets Discarded  : 999

Reason                        Packets Dropped
-----
Invalid type                  19
Invalid length                9
Invalid checksum              0
Invalid version               23
Bad or unknown source         67
Area mismatch                 1
Self-originated               19
Duplicate router ID           9
Interface standby             0
Total Hello packets dropped   60
  Network Mask mismatch       10
  Hello interval mismatch     10
  Dead interval mismatch      10
  Options mismatch            10
  MTU mismatch                10
  Neighbor ignored            10
Authentication errors         12
  Type mismatch               6
  Authentication failures     6
Wrong protocol                 0
Resource failures              0
Bad LSA length                 0
Others                         0

Total LSAs Ignored : 176
Bad Type           : 10
Bad Length         : 56
Invalid Data       : 55
Invalid Checksum   : 55
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## summary-address (ospf)

```
summary-address <IPv4-ADDR>/<MASK> [no-advertise | tag <TAG-VALUE>]
no summary-address <prefix/length> [no-advertise | tag <tag-value>]
```

## Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The **no** form of this command disables route summarization.



This command only works for an ASBR (Autonomous System Boundary Router).

| Parameter       | Description                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <IPv4-ADDR>     | Specifies an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.                                     |
| <MASK>          | Specifies the number of bits in the address mask in CIDR format ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 32.                   |
| no-advertise    | Do not advertise the aggregate route. Suppress routes that match the specified prefix/mask pair.                                                 |
| tag <TAG-VALUE> | Specify the tag for the aggregate route. The summary prefix will be advertised along with the tag value in External LSAs. Range: 0 to 4294967295 |

## Examples

Setting OSPF route summarization:

```
switch(config)# router ospf 1
switch(config-ospf-1)# summary-address 10.1.0.0/16
```

Disabling route summarization:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no summary-address 10.1.0.0/16
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context                        | Authority                                                                          |
|--------------|----------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf- <i>&lt;PROCESS-ID&gt;</i> | Administrators or local user group members with execution rights for this command. |

## timers lsa-arrival (OSPFv2)

```
timers lsa-arrival <DELAY>
no timers lsa-arrival
```

### Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives sooner before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command **timers throttle lsa start** on the neighbor.

The **no** form of this command sets the LSA timers to default values.

| Parameter | Description                                                             |
|-----------|-------------------------------------------------------------------------|
| <DELAY>   | Specifies the delay in milliseconds. Range: 0 to 600000. Default: 1000. |

### Examples

Setting the LSA arrival timer:

```
switch(config)# router ospf 1
switch(config-ospf-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no timers lsa-arrival
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## timers throttle lsa (OSPFv2)

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-time <WAIT-TIME>
```

```
no timers throttle lsa
```

## Description

Configures the timers for LSA generation.

The **no** form of this command sets the LSA timers to default values.

| Parameter                                    | Description                                                                                                                                                                                                                                                                    |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>start-time &lt;START-TIME&gt;</code>   | Specifies the initial wait time in milliseconds after which LSAs are generated. When set to 0, the LSAs are generated without any delay. Range: 0 to 600000. Default: 5000.                                                                                                    |
| <code>hold-time &lt;HOLD-TIME&gt;</code>     | Specifies the amount of time, in milliseconds, between regeneration of an LSA. The hold time doubles each time the same LSA must be regenerated, until <b>max-wait-time</b> is reached. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0. |
| <code>max-wait-time &lt;WAIT-TIME&gt;</code> | Specifies the maximum wait time, in milliseconds, for regeneration of the same LSA. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.                                                                                                     |

## Examples

Setting the LSA timers:

```
switch(config)# router ospf 1
switch(config-ospf-1)# timers throttle lsa start-time 100 hold-time 1000 max-wait-time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no timers throttle lsa
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## timers throttle spf

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds>  
max-wait-time <milliseconds>  
no timers throttle spf
```

### Description

Configures timers for SPF calculation. There are three timers:

- **start-time:** Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- **hold-time:** Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the *max-wait-time*.
- **max-wait-time:** Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The **no** form of this command sets all the configured non-default timers to default value.



The configuration is supported globally as well as per address-families ipv4-unicast or ipv6-unicast. If address-family specific timers is configured, it takes precedence over global configuration.

| Parameter    | Description                                                                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <START-TIME> | Time in milliseconds to set timer for initial SPF delay.<br>Default: 200.<br>Range: 1-600000<br><i>Required</i>                                     |
| <HOLD-TIME>  | Time in milliseconds to set the minimum hold time between two consecutive SPF calculations.<br>Default: 1000.<br>Range: 1-600000<br><i>Required</i> |
| <WAIT-TIME>  | Time in milliseconds to set the maximum wait time between two consecutive SPF calculations.<br>Default: 5000.<br>Range: 1-600000<br><i>Required</i> |

## Examples

Setting non-default timer values for SPF throttling:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# timers throttle spf start-time 100 hold-time 1000 max-
wait-time 10000
```

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# no throttle spf
switch(config-ospfv3-af-1)# no timers throttle spf start-time 100 hold-time 1000
max-wait-time 10000
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context               | Authority                                                                          |
|--------------|-------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-af-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## trap-enable (ospf)

```
trap-enable
no trap-enable
```

### Description

Enables the notification of the events to be sent as traps to the SNMP management stations for OSPF. The **no** form of this command disables the notification of the events to be sent as traps to the SNMP management stations for OSPF.

## Examples

Enabling sending notification of events as traps:

```
switch(config)# router ospf 1
switch(config-ospf-1)# trap-enable
```

Disabling sending notification of events as traps:

```
switch(config)# router ospf 1  
switch(config-ospf-1)# no trap-enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |



---

OSPFv2 is not supported on the 6000 and 6100 Switch Series.

---



---

OSPFv2 is not supported on the 4100i Switch Series.

---

OSPFv3 is the IPv6 implementation of Open Shortest Path First protocol (OSPFv2 is the IPv4 implementation of this protocol). OSPFv3 is a routing protocol which is described in RFC5340 entitled OSPF for IPv6. It is a link-state based IGP (Interior Gateway Protocol) routing protocol. It is widely used with medium to large-sized enterprise networks.

## Overview

The characteristics of OSPFv3 are:

- Provides a loop-free topology using SPF algorithm.
- Allows hierarchical routing using area 0 (backbone area) as the top of the hierarchy.
- Supports load balancing with equal cost routes for same destination.
- OSPFv3 is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking) and route summarization.
- Scales enterprise size network easily with area concept.
- Provides fast convergence with triggered, incremental updates through Link State Advertisements (LSAs).

Some OSPFv3 configuration is done in the configuration context, others in the OSPFv3 router context, or in the interface context. OSPFv3 can be configured on routed ports, VLAN interfaces, LAG interfaces, and loopback interfaces. All such configurations work in the mentioned interfaces context. OSPFv3 mandates the associated interface to be routing interface.

The switch supports congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling).

### Supported features

- OSPFv3 neighbor adjacency, Hello protocol, multiple areas, Inter-area routing
- AS external routes, stub areas, totally stubby areas, NSSA, ASBR
- Designated Router/Backup Designated Router
- Point-to-point interfaces/broadcast interfaces
- Equal-cost multipath
- Null authentication, Simple password authentication, MD5 authentication
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- SPF throttling



- LSA Throttling
- SPF throttling
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- Configuration of interface parameters such as priority, cost, hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Configuration of virtual link parameters such as hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Route redistribution
- Passive interfaces
- Multi VRF support with each VRF having up to eight OSPF process instances
- Congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling)




---

An OSPFv3 session will not come up in 6in6 tunnel.

---

## How OSPFv3 protocol works

### OSPFv3 protocol

The protocol uses Link State Advertisements (LSAs) transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces. Each routing switch in an area also maintains a link-state database (LSDB) that describes the area topology. (All routers in a given OSPF area have identical LSDBs.) The routing switches used to connect areas to each other flood summary link LSAs and external link LSAs to neighboring OSPF areas to update them regarding available routes. Through this means, each OSPF router determines the shortest path between itself and a desired destination router in the same OSPF domain (Autonomous System (AS)).

### OSPFv3 concepts

OSPFv3 is a link-state routing protocol applied to IPv6 routers grouped into OSPFv3 areas identified by the IPv6 routing configuration on each routing switch. Each OSPFv3 area includes one or more networks. OSPFv3 routers use hello packets and LSAs to maintain OSPFv3 operation across networks within an area and between areas within an OSPFv3 domain.

OSPFv3 uses hello packets to initiate and preserve relationships between neighboring routers on the same interface.

### OSPFv3 Link-state advertisement (LSA) types

OSPFv3 uses LSAs transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces.

Each routing switch in an area also maintains an LSDB that describes the area topology. (All routers in a given OSPFv3 area have identical LSDBs, and each router uses the LSDB to build its own shortest-path tree.)

The routing switches used to connect areas to each other flood inter-area-prefix-LSAs, inter-area-router-LSAs, and AS-external-LSAs to backbone area to update it regarding available routes. Each OSPFv3 router determines the shortest path between itself and a desired destination router in the same OSPFv3 domain (AS). Routed traffic in an OSPFv3 AS is classified as one of the following:

- Intra-area traffic
- Inter-area traffic
- External traffic

The routing switches support the LSAs listed in the following table.

| Link-state type | Description           | Use                                                                                                                                                                                                                                                                                                                                                                                                                                        | Flood scope |
|-----------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 0x2001          | Router-LSA            | Describes the state of each active interface on a router for Area a given area. (Excludes loopback interfaces and interfaces that have not achieved full adjacency.)                                                                                                                                                                                                                                                                       | Area        |
| 0x2002          | Network-LSA           | Describes the OSPFv3 routers in a given network.                                                                                                                                                                                                                                                                                                                                                                                           | Area        |
| 0x2003          | Inter-area-prefix-LSA | Describes the route to a prefix in another OSPFv3 area of Area the same AS. (Excludes prefixes for link-local addresses.)<br>Propagated through backbone area to other areas.                                                                                                                                                                                                                                                              | Area        |
| 0x2004          | Inter-area-router-LSA | Describes the route to an ASBR in another OSPFv3 normal area (including the backbone area) of the same AS.<br>Propagated through backbone area to other areas. (Excludes any ASBR in the same area as the router sending the LSA.)                                                                                                                                                                                                         | Area        |
| 0x4005          | AS-external-LSA       | Describes the route to a destination prefix in another AS (external route). (Excludes prefixes for link-local addresses.)<br>Originated by ASBR in normal or backbone areas of an AS and propagates through backbone area to other normal areas.<br>Does not flood over virtual links and is not summarized in virtual links. For injection into an NSSA, an NSSA ABR generates a type-7-default-LSA advertising the default route (::/0). | AS          |

| Link-state type | Description           | Use                                                                                                                                                                                                           | Flood scope |
|-----------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 0x2007          | NSSA-LSA              | Describes the route to a destination in another AS (external route). Originated by ASBR in NSSA. ABR translates type-7 LSAs to AS-external-LSAs for injection into the backbone area.                         | NSSA        |
| 0x0008          | Link-LSA              | For other routers on the same VLAN interface (except virtual links), describes the router link-local address and any other IPv6 prefixes reachable on the VLAN. Link LSAs are not flooded over virtual links. | Link-local  |
| 0x2009          | Intra-area-prefix-LSA | Generated on transit links within an area by the DR operating on those links. Also, every OSPFv3 router generates this LSA to refer to stub and loopback prefixes on the router.                              | Area        |

## OSPFv3 area types

OSPFv3 is built upon a hierarchy of network areas. All areas for a given OSPFv3 domain reside in the same AS. An AS is defined as a number of contiguous networks that share an interior gateway routing protocol.

An AS can be divided into multiple areas, including the backbone (area 0). Because each area represents a collection of contiguous networks and hosts, the topology of a given area is not known by the internal routers in any other area. Areas define the boundaries to which router-LSAs and network-LSAs are broadcast. This functionality limits the amount of LSA flooding that occurs within the AS. It also helps to control the size of the link-state databases (LSDBs) maintained in OSPFv3 routers.

An area is represented in OSPFv3 by either a 32-bit dotted-decimal address or a number. Area types include: Backbone, Normal, Not-so-stubby (NSSA), and Stub.

### Backbone area

Every AS must have one (and only one) backbone area (identified as area 0 or 0.0.0.0.) The ABRs of all other areas in the same AS connect to the backbone area, either physically through an ABR or through a configured, virtual link. The backbone is a special type of area and serves as a transit area for carrying the inter-area-prefix-LSAs, AS-external-LSAs, and routed traffic between non-backbone areas, as well as the router-LSAs, network-LSAs, and routed traffic internal to the area. ASBRs are allowed in backbone areas.

### Normal area

A normal area allows inter-area-prefix-LSAs and AS-external-LSAs to and from the backbone area. A normal area connects to the AS backbone area through one or more ABRs (physically or through a

virtual link) and allows router-LSAs and network LSAs within this area. ASBRs are allowed in normal areas.

## Stub area

Stub area connects to the AS backbone through one or more ABRs. It does not allow an internal ASBR, and does not allow AS-external-LSAs. A stub area supports these actions:

- Advertise the area summary routes to the backbone area.
- Advertise inter-area routes from other areas.
- Use the inter-area-prefix-LSA default route to advertise routes to an ASBR and to other areas.

You can configure the stub area ABR to do the following:

- Suppress advertising on the area's summarized internal routes into the backbone area.
- Suppress LSA traffic from other areas in the AS by replacing inter-area-prefix-LSAs and the default external route from the backbone area with the default summary route (::/0.). This area is called totally stubby area.

Virtual links are not allowed for stub areas.

## Not-so-stubby (NSSA) area

An NSSA area connects to the backbone area through one or more ABRs. NSSAs are used where an ASBR exists in an area where you want to:

- Block injection of external routes from other areas of the AS.
- Advertise type-7-LSA external routes (learned from the ASBR) to the backbone area as AS-external-LSAs.

NSSAs also support the following:

- Advertise inter-area-prefix-LSAs from the backbone area into the NSSA. (If no-summary is enabled, the NSSA ABR suppresses these LSAs from the backbone and, instead, injects the default route into the NSSA.)
- Advertise NSSA inter-area-prefix-LSAs to the backbone area.

Virtual links are not allowed for NSSAs.

## OSPFv3 router types

### Internal routers

Internal OSPFv3 routers belong to only one area. Internal routers flood router-LSAs to all routers in the same area and maintain identical LSDBs.

### Area border routers (ABRs)

Area border routers have membership in multiple areas. ABRs are used to connect the various areas in an AS to the backbone area for that AS. Multiple ABRs can be used to connect a given area to the backbone, and a given ABR can belong to multiple areas other than the backbone.

An ABR maintains a separate LSDB for each area to which it belongs. (All routers within the same area have identical LSDBs.) The ABR is responsible for flooding inter-area-prefix-LSAs and inter-area router LSAs between its border areas.

You can reduce summary LSA flooding by configuring area ranges. An area range enables you to assign an aggregate address to a range of IPv6 addresses. This aggregate address is advertised instead of all the individual addresses it represents. You can assign up to eight ranges in an OSPFv3 area.

### **Autonomous system boundary router (ASBR)**

Autonomous system boundary routers run one or more interior gateway protocols and serve as a gateway to other autonomous systems operating with interior gateway protocols. The ASBR imports and translates different protocol routes into OSPFv3 through redistribution. ASBRs can be used in backbone areas, normal areas, and NSSAs, but not in stub areas.

### **Designated routers (DRs)**

In an OSPFv3 network having two or more routers, one router is elected to serve as the designated router (DR) and another router to act as the backup designated router (BDR). All other routers in the same network segment forward their routing information to the DR and BDR, and the DR forwards this information to all routers in the network. This functionality minimizes the amount of repetitive information that is forwarded on the network by eliminating the need for each individual router in the area to forward its routing information to all other routers in the network. If the area includes multiple networks, each network elects its own DR and BDR.

In an OSPFv3 network with no DR and no BDR, the neighboring router with the highest priority is elected the DR and the router with the next highest priority is elected the BDR. If the DR goes off-line, the BDR automatically becomes the DR, and the router with the next highest priority then becomes the new BDR. If multiple routing switches on the same OSPFv3 network are declaring themselves DRs, both priority and router ID are used to select the DR and BDR.

Priority is configurable using the `ipv6 ospfv3 priority` command at the interface level. You can use this command to help bias one router as the DR. If two neighbors share the same priority, the router with the highest router ID is designated the DR. The router with the next highest router ID is designated the BDR. Routers retain DR/BDR states throughout the period for which the adjacency is up and running.

## **OSPFv3 configuration task list**

### **Tasks at a glance**

- [Configuring OSPFv3 on the routing switch](#)
- [Setting OSPFv3 network for the area](#)
- [Configuring external route redistribution and control](#)
- [Influencing route choice by changing the administrative distance](#)
- [Configuring graceful restart](#)
- [Configuring OSPFv3 interface settings](#)
- [Configuring all OSPFv3 interfaces as passive](#)
- [Viewing OSPFv3 information](#)
- [Clearing OSPFv3 statistics on a switch](#)

## **Configuring OSPFv3 on the routing switch**

### **Prerequisites**

- You must be in the global configuration context, as indicated by the `switch(config)#` prompt to create the OSPFv3 instance and enter the OSPFv3 configuration context.
- To configure a router ID, and other OSPFv3 configuration you must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

Create the OSPFv3 instance and enter the OSPFv3 configuration context. From this context, you can proceed with other OSPFv3 configuration.

## Procedure

1. Create the OSPFv3 instance and enter the OSPFv3 configuration context using the following command. For command details, see [router ospfv3](#).

```
router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]
```

For example, the following command creates OSPF instance 1.

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
```

2. Configure a global router ID using the following command. For command details, see [router-id \(ospfv3\)](#).

```
router-id <ROUTER-ADDRESS>
```

For example, the following command sets the router ID to 1.1.1.1.

```
switch(config-ospfv3-1)# router-id 1.1.1.1
```

3. Optionally, if the OSPFv3 process was disabled (it is enabled by default), enable the OSPFv3 process using the following command.

```
enable
```

For command details, see [enable \(router ospfv3\)](#). (Refer to [disable \(router ospfv3\)](#) for disabling the OSPFv3 process).

## Creating an OSPFv3 area

### Prerequisites

You must be in the OSPFv3 router configuration context.

Create a Normal, Stub, or Not So Stubby (NSSA) area.

### Procedure

Create an OSPFv3 area for the routing switch using one of the following commands:

- Create a Normal area using the following command: `area <area-id>`. For command details, see [area \(ospfv3\)](#).
- Create a Not So Stubby Area using the command: `area <area-id> nssa`.

For example, the following command creates a normal area with an area identifier of 10.1.1.1. Area identifier could alternatively be entered in decimal format such as 1.

```
switch(config-ospfv3-1) # area 1.1.1.1
```

| Section                                      | Owner-Group |
|----------------------------------------------|-------------|
| Important Information                        | NTL/Dev     |
| Products Supported (Define New HW Platforms) | PLM         |
| Compatibility/interoperability               | NTL/Dev     |
| Transceiver Support                          | PLM         |
| Enhancements and new features                | PLM         |
| Upgrade                                      | NTL         |
| Fixes (Defects)                              |             |
| Define CFDs to include                       | PMO         |
| Define IFDs to include                       | NTL         |
| Issues & Workarounds                         |             |
| Define IFDs to include                       | NTL         |

## Setting OSPFv3 network for the area

### Prerequisites

Routing must be enabled on the interface where you are planning to enable OSPFv3.

You must be in the interface configuration context, as indicated by the `switch(config-if) #` prompt.

After you define an OSPFv3 area, you can assign one or more networks to it. The OSPFv3 protocol will run on the interface with the configured link-local ipv6 address. The interfaces that have an IPv6 address configured in this network or in a subset of this network will participate in the OSPFv3 protocol.

### Procedure

1. Set an OSPFv3 network for the area using the following command. For command details, see [ipv6 ospfv3 area](#).

```
ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>
```

For example, use the following command to assign interface 1/1/1 to OSPFv3 area 1. The area can alternatively be entered in IO address format.

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ospfv3 1 area 1
```

2. Optionally, you can disable OSPFv3 on the interface using the following command. For command details, see [ipv6 ospfv3 area](#).

```
ipv6 ospfv3 shutdown
```

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ospfv3 shutdown
```

## Configuring external route redistribution and control

Configuring route redistribution for OSPFv3 establishes the routing switch as an ASBR for importing and translating different protocol routes from other IGP domains into an OSPFv3 domain. When you configure redistribution for OSPFv3, you can specify that routes external to the OSPFv3 domain are imported as OSPFv3 routes.

1. Enable route redistribution using the `redistribute` command.

```
redistribute {bgp | connected | local loopback | static | ripng | ospf <PROCESS-
ID>}
                [route-map <ROUTE-MAP-NAME>]
```

For example, setting redistribution of connected routes as OSPFv3 routes:

```
switch(config-ospfv3-1)# redistribute connected
```

If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv3. For example, setting redistribution of local loopback routes, that only match the routes specified by the route map:

```
switch(config-ospf-1)# redistribute local loopback route-map local_routes
```

2. Optionally, modify the default metric for redistribution using the `default-metric` command.

```
default-metric <METRIC-VALUE>
```

For example, setting the default metric for redistribution to 37.

```
switch(config-ospfv3-1)# default-metric 37
```

3. Optionally, set the cost of default-summary LSAs using the `area` command.

```
area <AREA-ID> default-metric <COST>
```

For example, setting the cost of default summary LSAs to 2.

```
switch (config-ospfv3-1)# area 1 default-metric 2
```

4. Optionally, use the `max-metric router-lsa` command to set the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations.

```
max-metric router-lsa [on-startup [<ADVERT-TIME>]]
```



For example, setting advertise max-metric router-lsa on startup.

```
switch(config-ospfv3-1) # max-metric router-lsa on-startup 3000
```

## Influencing route choice by changing the administrative distance

### Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1) #` prompt.

The administrative distance is used to influence the selection of routes learned by different protocols.

### Procedure

Reconfigure the administrative distance using the following command. For command details, see [distance \(ospfv3\)](#).

```
distance <distance>
```

For example, use the following command to set administrative distance to 100.

```
switch(config-ospfv3-1) # distance 100
```

## Configuring graceful restart

### Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1) #` prompt.

OSPFv3 routing can be gracefully restarted on switches without losing packets that are in transit. There is no effect on the saved switch configuration.

Manually create a router-id configuration to support graceful restart and management module failover. A dynamic router-id is chosen from the list of configured IPs at the time of the router-id election, and in the event of a graceful restart and management module failover, the router-id should be preserved before and after these events. Graceful restart would fail if there is a change in the router-id, causing traffic loss.

### Procedure

Configure graceful restart of using the following command. For command details, see [graceful-restart \(router ospfv3\)](#).

```
graceful-restart [restart-interval <seconds> | helper]
```

For example, the following command specifies 50 seconds as the maximum interval another router will wait for this router to gracefully restart.

```
switch(config-ospfv3-1) # graceful-restart restart-interval 50
```

## Configuring OSPFv3 interface settings

## Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt. You can optionally adjust the following OSPFv3 interface settings.

1. Set the interface cost using the following command. For command details, see [ipv6 ospfv3 cost](#).  
`ipv6 ospfv3 cost <interface-cost>`

For example, the following command sets the cost associated with interface 1/1/1 to 100.

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ospfv3 cost 100
```

2. Set the time interval between OSPFv3 hello packets for the OSPFv3 interface using the following command. For command details, see [ipv6 ospfv3 hello-interval](#).  
`ipv6 ospfv3 hello-interval <seconds>`
3. Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPFv3 interface using the following command. For command details, see [ipv6 ospfv3 dead-interval](#).  
`ipv6 ospfv3 dead-interval <seconds>`
4. Set the time between re-transmitting lost link state advertisements for the OSPFv3 interface using the following command. For command details, see [ipv6 ospfv3 af retransmit-interval](#).  
`ipv6 ospfv3 retransmit-interval <seconds>`
5. Sets the time delay in Link state transmission for the OSPFv3 interface using the following command. For command details, see [ipv6 ospfv3 transit-delay](#).  
`ipv6 ospfv3 transit-delay <seconds>`
6. Set the OSPFv3 network type for the interface. For command details, see [ipv6 ospfv3 network](#).  
`ipv6 ospfv3 network {broadcast|point-to-point}`
7. Set the OSPFv3 priority on the interface using the following command. For command details, see [ipv6 ospfv3 priority](#).  
`ipv6 ospfv3 priority <number-value>`
8. Set the OSPFv3 interface as OSPFv3 passive interface. With this setting the interface participates in OSPFv3 but does not send or receive packets on that interface. For command details, see [ipv6 ospfv3 passive](#).  
`ipv6 ospfv3 passive`

## Configuring all OSPFv3 interfaces as passive

### Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

OSPFv3 sends LSAs to all other routers in the same AS. To limit the flooding of LSAs throughout the AS, you can configure OSPFv3 to be passive.

### Procedure

Configure all OSPFv3 interfaces as passive using the following command. For command details, see [passive-interface default \(OSPFv3\)](#).

```
passive-interface default
```

```
switch(config-ospfv3-1)# passive-interface default
```

## Configuring SPF throttling timers

SPF calculation is throttled with default timer values (start-time 200ms, hold-time 1000ms, max-wait-time 5000ms). You can throttle SPF calculation by configuring non-default timers to improve performance of a specific network configuration.

### Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

### Procedure

Reconfigure SPF throttling timers using the following command. For command details, see [timers throttle spf](#).

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds> max-wait-time <milliseconds>
```

For example, use the following command to set start-time to 500ms, hold-time to 3000ms and max-wait-time to 9000ms:

```
switch(config-ospf-1)# timers throttle spf start-time 500 hold-time 3000 max-wait-time 9000
```

## Viewing OSPFv3 information

### Prerequisites

Use these show commands from the Manager context, as indicated by the `switch#` prompt.

### Procedure

To view OSPFv3 information, use the following commands. For command details and examples, click the links.

- To view general OSPFv3, area, state and configuration information use: [show ipv6 ospfv3](#).
- To view information about OSPFv3 interfaces use: [show ipv6 ospfv3 interface](#).
- To view information about OSPFv3 neighbors use: [show ipv6 ospfv3 neighbors](#).
- To view OSPFv3 routing table information use: [show ipv6 ospfv3 routes](#).
- To view OSPFv3 statistics for the OSPFv3 interfaces use [show ipv6 ospfv3 statistics interface](#).

## Clearing OSPFv3 statistics on a switch

Clear OSPFv3 event statics using the following command. For command details, see [clear ipv6 ospfv3 statistics](#)

```
clear ipv6 ospfv3 [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-vrfs | vrf <VRF-NAME>]
```

For example, the following command clears OSPFv3 event statistics from interface 1/1/1.

```
switch# clear ipv6 ospfv3 statistics interface 1/1/1
```

## OSPFv3 commands

### active-backbone

```
active-backbone stub-default-route  
no active-backbone stub-default-route
```

#### Description

This command enables the router to send a default route to stub areas if there is an active loopback link in the backbone area. The configuration is not required if backbone area has neighbors or passive interfaces configured. By default active backbone detection is enabled.

#### Examples

```
switch(config)# router ospf 1  
switch(config-ospf-1)# active-backbone stub-default-route
```

```
switch(config)# no active-backbone stub-default-route
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

#### Command History

| Release    | Modification       |
|------------|--------------------|
| 10.10.1000 | Command Introduced |

#### Command Information

| Platforms    | Command context                                                                    | Authority                                                                          |
|--------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf- <i>&lt;PROCESS-ID&gt;</i><br>config-ospfv3- <i>&lt;PROCESS-ID&gt;</i> | Administrators or local user group members with execution rights for this command. |

### area (ospfv3)

```
area <AREA-ID>  
no area <AREA-ID>
```

#### Description

Creates a normal area with **<AREA-ID>** set if not present. If area is present and is not the normal area, this command changes the area type to normal area.

The **no** form of this command deletes the area with the **<AREA-ID>** specified. The area can be of any type (stub, stub no-summary, and default normal area).

| Parameter | Description                                                                                                                                                                                                                            |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID> | Specifies the area ID is one of the following formats.<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |

## Examples

Creating a normal area:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 1
```

Deleting an area:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## area authentication ipsec

```
area <AREA-ID> authentication ipsec spi <SPI-INDEX> <AUTH-TYPE> [<KEY-TYPE> <AUTH-KEY>]
no area <AREA-ID> authentication
```

### Description

Configures IPsec AH authentication for the specified area. OSPFv3 interfaces which have IPsec configured at the interface context will not use area level IPsec.

The **no** form of this command removes IPsec AH authentication for the specified area.



IPsec is not supported for 6in6 tunnel interfaces.

| Parameter       | Description                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID>       | Specifies the area ID is one of the following formats:<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295.                                                          |
| spi <SPI-INDEX> | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| <AUTH-TYPE>     | Specifies the authentication type: <b>md5</b> or <b>sha1</b> .                                                                                                                                                                                                                                  |
| <KEY-TYPE>      | Specifies the key type to use: <b>plaintext</b> (unencrypted), <b>hex-string</b> (encrypted) or <b>ciphertext</b> (encrypted).                                                                                                                                                                  |
| <AUTH-KEY>      | Specifies the authentication key.                                                                                                                                                                                                                                                               |



When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting area 0 to use IPsec authentication with a provided plaintext authentication key:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 authentication ipsec spi 256 sha1 plaintext
F82#450
```

Setting area 5 to use IPsec authentication with a prompted plaintext authentication key:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 5 authentication ipsec spi 256 sha1
Enter the IPsec authentication key: *****
Re-Enter the IPsec authentication key: *****
```

Removing IPsec authentication from area 1:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 1 authentication
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## area encryption ipsec

```
area <AREA-ID> encryption ipsec spi <SPI-INDEX> <AUTH-TYPE>
    [<KEY-TYPE> <AUTH-KEY> <ENCR-TYPE> [<KEY-TYPE> <ENCR-KEY>]]
no area <AREA-ID> encryption
```

### Description

Configures IPsec ESP with the authentication and encryption algorithm types and keys for the specified area. OSPFv3 interfaces with IPsec configured at the interface context will not use area level IPsec ESP configuration.

The **no** form of this command removes IPsec ESP from the specified area.



IPsec is not supported for 6in6 tunnel interfaces.

| Parameter       | Description                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID>       | Specifies the area ID is one of the following formats.<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295.                                                          |
| spi <SPI-INDEX> | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| <AUTH-TYPE>     | Specifies the authentication type: <b>md5</b> or <b>sha1</b> .                                                                                                                                                                                                                                  |
| <KEY-TYPE>      | Specifies the key type to use: <b>plaintext</b> (unencrypted), <b>hex-string</b> (encrypted) or <b>ciphertext</b> (encrypted).                                                                                                                                                                  |
| <AUTH-KEY>      | Specifies the authentication key.                                                                                                                                                                                                                                                               |
| <ENCR-TYPE>     | Specifies the encryption type: <b>des</b> , <b>3des</b> , <b>aes</b> , or <b>null</b> .<br><br><b>NOTE:</b> Encryption type <b>aes</b> is considered to be AES128, AES192 or AES256 based on key length.                                                                                        |
| <ENCR-KEY>      | Specifies the encryption key.                                                                                                                                                                                                                                                                   |



When the authentication key is not provided on the command line, plaintext authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally plaintext encryption key prompting. The entered key characters are masked with asterisks.



When the authentication key and encryption type are provided on the command line but the encryption key is not provided, plaintext encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting area 0 to use IPsec ESP with provided authentication and encryption keys:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 encryption ipsec spi 256 md5 plaintext F824eva
des plaintext F82#450b
```

Setting area 5 to use IPsec ESP with prompted authentication and encryption keys:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 5 encryption ipsec spi 256 md5
Enter the IPsec authentication key: *****
Re-Enter the IPsec authentication key: *****

Enter the IPsec encryption type (3des/aes/des/null)? des

Enter the IPsec encryption key: *****
Re-Enter the IPsec encryption key: *****
```

Setting area 2 to use IPsec ESP with provided authentication password and encryption type but a prompted encryption key:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 2 encryption ipsec spi 256 md5 plaintext F82# des
Enter the IPsec encryption key: *****
Re-Enter the IPsec encryption key: *****
```

Setting area 0 to use IPsec ESP with provided plaintext authentication key and null encryption:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 encryption ipsec spi 256 md5 plaintext axtw null
```

Removing IPsec ESP from area 0:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 0 encryption
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |



## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## clear ipv6 ospfv3 neighbors

```
clear ipv6 ospfv3 [<PROCESS-ID>] neighbor [<NEIGHBOR>] [interface [<INTERFACE-NAME>]]  
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Resets the neighbor and clears the OSPF neighbor information.

| Parameter        | Description                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>     | Specifies the OSPFv3 process ID to clear the statistics for the particular OSPFv3 process. Range: 1 to 65535. |
| <NEIGHBOR>       | Specifies the router ID of a neighbor.                                                                        |
| <INTERFACE-NAME> | Specifies the OSPFv3 statistics to clear for the specified interface.                                         |
| all-vrfs         | Select to clear the OSPFv3 statistics for all VRFs.                                                           |
| vrf <VRF-NAME>   | Specifies the name of a VRF.                                                                                  |

### Example

Clearing the OSPFv3 neighbor information:

```
switch# clear ipv6 ospfv3 1 neighbor 3.3.3.3  
Performing clear ospf neighbor may result in traffic disruption.  
Do you want to continue (y/n)? y  
switch# clear ipv6 ospfv3 1 neighbor interface 1/1/1  
Performing clear ospf neighbor may result in traffic disruption.  
Do you want to continue (y/n)? y
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context         | Authority                                                    |
|-----------|-------------------------|--------------------------------------------------------------|
| 5420      | Operator (>) or Manager | Operators or Administrators or local user group members with |

| Platforms | Command context | Authority                                                                                                 |
|-----------|-----------------|-----------------------------------------------------------------------------------------------------------|
| 6200      | (#)             | execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear ipv6 ospfv3 statistics

```
clear ipv6 ospfv3 [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Clears the OSPFv3 event statistics.

| Parameter        | Description                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>     | Specifies the OSPFv3 process ID to clear the statistics for the particular OSPFv3 process. Range: 1 to 65535. |
| <INTERFACE-NAME> | Specifies the OSPFv3 statistics to clear for the specified interface.                                         |
| all-vrfs         | Select to clear the OSPFv3 statistics for all VRFs.                                                           |
| vrf <VRF-NAME>   | Specifies the name of a VRF.                                                                                  |

### Example

Clearing the OSPFv3 event statistics:

```
switch# clear ipv6 ospfv3 statistics
switch# clear ipv6 ospfv3 statistics interface 1/1/1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear ospfv3-af ipv4 neighbor

```
clear ospfv3-af [<PROCESS-ID>] ipv4 neighbor [<NEIGHBOR-ID>] [interface <INTERFACE-NAME>
| vlan <VLAN-ID> | lag <LAG-ID> | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>] [all-vrfs
| vrf <VRF-NAME>]
```

## Description

Resets the neighbor and clears the OSPFv3 AF neighbor information.

| Parameter                  | Description                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies the OSPFv3 AF process ID to clear the statistics for the particular OSPFv3 AF process. Range: 1 to 65535. |
| neighbor <NEIGHBOR-ID>     | Specifies the router ID of a neighbor.                                                                              |
| interface <INTERFACE-NAME> | Specifies the OSPFv3 AF statistics to clear for the specified interface.                                            |
| vlan <VLAN-ID>             | Specifies the vlan interface name.                                                                                  |
| lag <LAG-ID>               | Specifies the lag interface name.                                                                                   |
| loopback <LOOPBACK-ID>     | Specifies the loopback interface name.                                                                              |
| tunnel <TUNNEL-ID>         | Specifies the tunnel interface name.                                                                                |
| all-vrfs                   | Select to clear the OSPFv3 AF statistics for all VRFs.                                                              |
| vrf <VRF-NAME>             | Specifies the name of a VRF.                                                                                        |

## Example

Clearing the OSPFv3 AF neighbor information:

```
switch# clear ospfv3-af 1 ipv4 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv4 neighbor 3.3.3.3
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv4 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv4 neighbor 3.3.3.3 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor 5.5.5.5
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor 5.5.5.5 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear ospfv3-af ipv4 statistics

```
clear ospfv3-af [<PROCESS-ID>] ipv4 statistics [interface [interface <INTERFACE-NAME> |
vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>] [all-vrfs |
vrf <VRF-NAME>]
```

## Description

Clears the OSPFv3 AF event statistics.

| Parameter                  | Description                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies the OSPFv3 AF process ID to clear the statistics for the particular OSPFv3 AF process. Range: 1 to 65535. |
| interface <INTERFACE-NAME> | Specifies the OSPFv3 AF statistics to clear for the specified interface.                                            |
| vlan <VLAN-ID>             | Specifies the vlan interface name.                                                                                  |
| lag <LAG-ID>               | Specifies the lag interface name.                                                                                   |
| loopback <LOOPBACK-ID>     | Specifies the loopback interface name.                                                                              |
| tunnel <TUNNEL-ID>         | Specifies the tunnel interface name.                                                                                |
| all-vrfs                   | Select to clear the OSPFv3 AF statistics for all VRFs.                                                              |
| vrf <VRF-NAME>             | Specifies the name of a VRF.                                                                                        |

## Example

Clearing the OSPFv3 AF event statistics:

```
switch# clear ospfv3-af ipv4 statistics
switch# clear ospfv3-af ipv4 statistics interface 1/1/1
switch# clear ospfv3-af ipv4 statistics interface 1/1/1 vrf vrf_red
switch# clear ospfv3-af ipv4 statistics interface tunnell
switch# clear ospfv3-af ipv4 statistics interface tunnell vrf vrf_red
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear ospfv3-af ipv6 neighbor

```
clear ospfv3-af [<PROCESS-ID>] ipv6 neighbor [<NEIGHBOR-ID>] [interface <INTERFACE-NAME>
| vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>] [all-vrfs
| vrf <VRF-NAME>]
```

## Description

Resets the neighbor and clears the OSPFv3 AF neighbor information.

| Parameter                  | Description                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies the OSPFv3 AF process ID to clear the statistics for the particular OSPFv3 AF process. Range: 1 to 65535. |
| neighbor <NEIGHBOR-ID>     | Specifies the router ID of a neighbor.                                                                              |
| interface <INTERFACE-NAME> | Specifies the OSPFv3 AF statistics to clear for the specified interface.                                            |
| vlan <VLAN-ID>             | Specifies the vlan interface name.                                                                                  |
| lag <LAG-ID>               | Specifies the lag interface name.                                                                                   |
| loopback <LOOPBACK-ID>     | Specifies the loopback interface name.                                                                              |
| tunnel <TUNNEL-ID>         | Specifies the tunnel interface name.                                                                                |
| all-vrfs                   | Select to clear the OSPFv3 AF statistics for all VRFs.                                                              |
| vrf <VRF-NAME>             | Specifies the name of a VRF.                                                                                        |

## Example

Clearing the OSPFv3 AF neighbor information:

```
switch# clear ospfv3-af 1 ipv6 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv6 neighbor 3.3.3.3
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv6 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
```

```

switch# clear ospfv3-af 1 ipv6 neighbor 3.3.3.3 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor 5.5.5.5
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor 5.5.5.5 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear ospfv3-af ipv6 statistics

```

clear ospfv3-af [<PROCESS-ID>] ipv6 statistics [interface [interface <INTERFACE-NAME> |
vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>] [all-vrfs |
vrf <VRF-NAME>]

```

## Description

Clears the OSPFv3 AF event statistics.

| Parameter                  | Description                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies the OSPFv3 AF process ID to clear the statistics for the particular OSPFv3 AF process. Range: 1 to 65535. |
| interface <INTERFACE-NAME> | Specifies the OSPFv3 AF statistics to clear for the specified interface.                                            |
| vlan <VLAN-ID>             | Specifies the vlan interface name.                                                                                  |
| lag <LAG-ID>               | Specifies the lag interface name.                                                                                   |

| Parameter              | Description                                            |
|------------------------|--------------------------------------------------------|
| loopback <LOOPBACK-ID> | Specifies the loopback interface name.                 |
| tunnel <TUNNEL-ID>     | Specifies the tunnel interface name.                   |
| all-vrfs               | Select to clear the OSPFv3 AF statistics for all VRFs. |
| vrf <VRF-NAME>         | Specifies the name of a VRF.                           |

## Example

Clearing the OSPFv3 AF event statistics:

```
switch# clear ospfv3-af ipv6 statistics
switch# clear ospfv3-af ipv6 statistics interface 1/1/1
switch# clear ospfv3-af ipv6 statistics interface 1/1/1 vrf vrf_red
switch# clear ospfv3-af ipv6 statistics interface tunnel1
switch# clear ospfv3-af ipv6 statistics interface tunnel1 vrf vrf_red
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## default-metric (ospfv3)

```
default-metric <METRIC-VALUE>
no default-metric
```

### Description

Sets the default metric for redistributed routes in the OSPFv3.

The **no** form of this command sets the default metric to be used for redistributed routes into OSPFv3 to the default of 25.

| Parameter      | Description                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------|
| <METRIC-VALUE> | Specifies the default metric value to use for redistributed routes. Range: 1 to 1677214. Default: 25. |

## Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# default-metric 36
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# no default-metric
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## disable (router ospfv3)

disable

### Description

Disables the OSPFv3 process. By default OSPFv3 process is enabled.  
This command does not remove the OSPFv3 configurations.

### Example

Disabling OSPFv3 process:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information



| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## distance (ospfv3)

```
distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area [<DISTANCE-VAL>] |
external [<DISTANCE-VAL>]]
no distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area [<DISTANCE-VAL>] |
external [<DISTANCE-VAL>]]
```

### Description

Defines an administrative distance for OSPFv3. Administrative distance is used as a criteria to select the best route when the same route is learned by multiple routing protocols.

The **no** form of this command sets the OSPFv3 administrative distance to the default value of 110. Optionally, administrative distance can be set to default for the specific OSPF route type: intra-area, inter-area, or external type-5 and type-7 routes.

| Parameter      | Description                                                                  |
|----------------|------------------------------------------------------------------------------|
| <DISTANCE-VAL> | Specifies the OSPFv3 administrative distance. Range: 1 to 255. Default: 110. |
| intra-area     | Specifies the OSPFv3 distance for intra-area routes.                         |
| inter-area     | Specifies the OSPFv3 distance for inter-area routes.                         |
| external       | Specifies the OSPFv3 distance for external type 5 and type 7 routes.         |

### Usage

Within a given OSPF process, intra-area routes are always given precedence even when distances are configured for inter-area or external type routes.

### Examples

Setting OSPFv3 administrative distance:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# distance 100
switch(config-ospfv3-1)# distance intra-area 24 external 55 inter-area 66
switch(config-ospfv3-1)# distance intra-area 24 external 55
switch(config-ospfv3-1)# distance external 55
switch(config-ospfv3-1)#exit

switch(config)# router ospfv3 2
switch(config-ospfv3-2)# distance 200
switch(config-ospfv3-2)# distance external 60
switch(config-ospfv3-2)# distance intra-area 24 inter-area 66
```

Setting OSPFv3 administrative distance to the default:

```

switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no distance
switch(config-ospfv3-1)# no distance external
switch(config-ospfv3-1)# no distance intra-area
switch(config-ospfv3-1)# no distance inter-area
switch(config-ospfv3-1) # no distance 1
switch(config-ospfv3-1) #exit

switch(config)# router ospfv3 2 vrf blue
switch(config-ospfv3-2) # no distance 200
switch(config-ospfv3-2) # no distance external 60
switch(config-ospfv3-2) # no distance intra-area 24 inter-area 66

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                                                             |
|------------------|------------------------------------------------------------------------------------------|
| 10.14            | Added capability to have individual admin distance for multiple OSPF processes in a VRF. |
| 10.09            | Added parameters: <b>intra-area</b> , <b>inter-area</b> , <b>external</b>                |
| 10.07 or earlier | --                                                                                       |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## distribute-list prefix (OSPFv3)

```

distribute-list prefix <prefix-list-name> {in | out}
no distribute-list prefix <prefix-list-name> {in | out}

```

### Description

This command uses an existing prefix list to filter routes that are being installed in the routing table or redistributed to another routing protocol.

The **distribute-list prefix** command filters routes in the inbound or the outbound direction. When this command is issued with the **in** parameter, it filters routes from being installed in the routing table, it does not filter LSAs. When this command is issued with the **out** parameter, it filters only the desired redistributed routes from other protocols.



This command requires that your prefix list is already defined using the ipv6 prefix commands. Route-maps are not supported with the distribute-list feature.

| Parameter                 | Description                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prefix <prefix-list-name> | Specify the name of an existing prefix.                                                                                                                                                                                                 |
| {in   out}                | Select one of the following parameters to set the filter direction: <ul style="list-style-type: none"> <li>▪ <b>in</b>: Filter incoming routes into the routing table</li> <li>▪ <b>out</b>: Filter outgoing routing updates</li> </ul> |

## Examples

The following commands enable the filtering of OSPFv3 routes in an IPv6 network, so routes are no longer installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# distribute-list prefix listA in
switch(config-ospfv3-1)# distribute-list prefix listB out
```

The following command disables the filtering of OSPFv3 routes in an IPv6 network, so routes can be installed in the routing table or redistributed from another routing protocol.

```
switch# configure terminal
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no distribute-list prefix listA in
switch(config-ospfv3-1)# no distribute-list prefix listB out
```

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.13   | Command introduced. |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## enable (router ospfv3)

enable

### Description

Enables OSPFv3 process when disabled. By default OSPFv3 process is enabled.

### Example

Enabling OSPFv3 process:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## default-information originate (ospfv3)

```
default-information originate [metric <METRIC-VALUE>]
no default-information originate [metric <METRIC-VALUE>]
```

### Description

Configures OSPFv3 to advertise the default route (::/0) to its neighbors if it is present in the routing table. Optionally, the metric value can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

| Parameter                          | Description                                                                  |
|------------------------------------|------------------------------------------------------------------------------|
| <i>metric &lt;METRIC-VALUE&gt;</i> | Specifies the OSPF metric value for the default route. Optional. Default: 1. |

## Examples

Setting advertisement of the default route:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-information originate
```

Setting advertisement of the default route and specifying an optional metric value of 20:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-information originate
switch(config-ospfv3-1)# default-information originate metric 20
```

Disabling advertisement of the default route and setting metric to the default value:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# no default-information originate metric
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                        |
|------------------|-----------------------------------------------------|
| 10.09            | Added parameter: <b>metric &lt;METRIC-VALUE&gt;</b> |
| 10.07 or earlier | --                                                  |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## default-information originate always (ospfv3)

```
default-information originate always [metric <METRIC-VALUE>]  
no default-information originate always [metric <METRIC-VALUE>]
```

## Description

Configures OSPFv3 to advertise the default route (::/0) to its neighbors, regardless if it is present in the routing table or not. Optionally, metric can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

| Parameter                          | Description                                                          |
|------------------------------------|----------------------------------------------------------------------|
| <i>metric &lt;METRIC-VALUE&gt;</i> | Specifies the OSPFv3 metric value for the default route. Default: 1. |

## Examples

Setting advertisement of the default route:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# no default-information originate always
```

Setting advertisement of the default route with metric set to 20:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# default-information originate always metric 20
```

Disabling advertisement of the default route and setting the metric to the default value:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# no default-information originate always metric
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                        |
|------------------|-----------------------------------------------------|
| 10.09            | Added parameter: <b>metric &lt;METRIC-VALUE&gt;</b> |
| 10.07 or earlier | --                                                  |

## Command Information

| Platforms    | Command context          | Authority                                                                          |
|--------------|--------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospf-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## graceful-restart (router ospfv3)

```
graceful-restart  
  restart-interval <INTERVAL>  
  helper [strict-lsa-check]  
  ignore-lost-interface  
  no...
```

### Description

Configures graceful restart for OSPFv3. By default graceful restart is enabled on the OSPFv3 router. The **no** form of this command sets the restart interval to the default of 120 seconds or disables helper mode depending on the specified parameters.

| Parameter                   | Description                                                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| restart-interval <INTERVAL> | Specifies the time another router waits for this router to gracefully restart and selects the maximum time to wait in seconds. Range: 5 to 1800. Default: 120.   |
| helper                      | Specifies that the router will participate in the graceful restart of a neighbor router.                                                                         |
| strict-lsa-check            | (Optional). Use with the <b>helper</b> parameter to enable strict Link state Advertisement (LSA) checking when acting as a restart helper for a restarting peer. |

| Parameter             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p><b>NOTE:</b> OSPF neighbors must disable strict LSA checking. If the local node has fewer OSPF interfaces after restarting, then the neighbors that were adjacent on those interfaces will clear up their adjacencies to the restarting node and will send out link state updates to advertise the dropped adjacency. If strict LSA checking is enabled, the restarting router's neighbors will exit helper mode when they receive the updated LSAs and the graceful restart will still fail.</p> |
| ignore-lost-interface | <p>Enable the restarting router to ignore lost OSPF interfaces during a graceful restart process. This setting should be enabled on a high availability system to ensure a graceful restart completes successfully, even if OSPF-enabled links fail due to High Availability events like a switchover or failover.</p> <p><b>NOTE:</b> Enabling this setting means that the hitless restart procedures do not strictly follow those defined in <i>RFC 3623, Graceful OSPF Restart</i>.</p>           |
| no                    | Negate any parameter or return the setting to its default..                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Examples

Enabling OSPF graceful restart:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# graceful-restart restart-interval 40
switch(config-ospfv3-1)# graceful-restart helper strict-lsa-check
```

Enabling the switch to ignore lost OSPF interfaces during a graceful restart process:

```
switch(config)# router ospfv3 1
switch (config-ospfv3-1)# graceful-restart ignore-lost-interface
```

Setting the restart interval to default, and disabling helper mode:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no graceful-restart restart-interval
switch(config-ospfv3-1)# no graceful-restart helper
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 area

```
ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>
no ipv6 ospfv3 <PROCESS-ID> area <area-id>
```

### Description

Runs the OSPFv3 protocol on the interface for the area specified.

To move an interface to a new area, unmap the existing area and then associate a new area with the interface.

The **no** form of this command disables OSPF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol

| Parameter    | Description                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID>    | Specifies the area ID is one of the following formats.<br>Area ID in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>Area ID as a decimal value. Range: 0-4294967295. |
| <PROCESS-ID> | Specifies the OSPFv3 process ID. Range: 1 to 65535.                                                                                                                                                        |

### Examples

Setting OSPFv3 network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 1 area 1
switch(config-if-vlan)# ipv6 ospfv3 1 area 0.0.0.1
```

Disabling OSPFv3 network for the area:

```
switch(config)# interface 1/1/1
switch(config-if-vlan)# no ipv6 ospfv3 1 area 1
switch(config-if-vlan)# no ipv6 ospfv3 1 area 0.0.0.1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information



| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 authentication null

ipv6 ospfv3 authentication null

### Description

Configures null authentication on an interface which disables IPsec authentication.

### Examples

Disabling IPsec on interface VLAN 1:

```
switch(config)# interface van 1
switch(config-if-vlan)# ipv6 ospfv3 authentication null
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 authentication ipsec

ipv6 ospfv3 authentication ipsec spi <SPI-INDEX> <AUTH-TYPE> [<KEY-TYPE> <AUTH-KEY>]  
no ipv6 ospfv3 authentication

### Description

Configures IPsec AH authentication. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec.

The **no** form of this command removes IPsec AH authentication for the specified area.

| Parameter       | Description                                                                                                                                                                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spi <SPI-INDEX> | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. |

| Parameter   | Description                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
|             | Range: 256 to 4294967295.                                                                                                      |
| <AUTH-TYPE> | Specifies the authentication type: <b>md5</b> or <b>sha1</b> .                                                                 |
| <KEY-TYPE>  | Specifies the key type to use: <b>plaintext</b> (unencrypted), <b>hex-string</b> (encrypted) or <b>ciphertext</b> (encrypted). |
| <AUTH-KEY>  | Specifies the authentication key.                                                                                              |



When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting interface VLAN **1** to use IPsec authentication with a provided plaintext authentication key:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 authentication ipsec spi 256 md5 plaintext
F82#
```

Setting interface VLAN **4** to use IPsec authentication with a prompted plaintext authentication key:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# ipv6 ospfv3 authentication ipsec spi 256 md5
Enter the IPsec authentication key: *****
Re-Enter the IPsec authentication key: *****
```

Removing IPsec authentication from interface VLAN **1**:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 authentication
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 cost

```
ipv6 ospfv3 cost <INTERFACE-COST>
no ipv6 ospfv3 cost
```

### Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The **no** form of this command sets the cost (metric) associated with a particular interface to the default of 1.

| Parameter        | Description                                                        |
|------------------|--------------------------------------------------------------------|
| <INTERFACE-COST> | Specifies the interface cost value. Range: 1 to 65535. Default: 1. |

### Examples

Setting OSPFv3 interface cost:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 cost 100
```

Setting the OSPFv3 interface cost to default:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 cost
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 dead-interval

```
ipv6 ospfv3 dead-interval <INTERVAL>
no ipv6 ospfv3 dead-interval
```

### Description

Sets the interval after a neighbor is declared dead when no hello packet is received on the OSPFv3 interface.

The **no** form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPFv3 interface. The default value is 40 seconds (generally four times the hello packet interval).

| Parameter                     | Description                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------|
| <code>&lt;INTERVAL&gt;</code> | Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40. |

## Examples

Setting OSPFv3 dead interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 dead-interval 30
```

Setting OSPFv3 dead interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 dead-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if-vlan  | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 encryption ipsec

```
ipv6 ospfv3 encryption ipsec spi <SPI-INDEX> <AUTH-TYPE>
[<KEY-TYPE> <AUTH-KEY> <ENCR-TYPE> [<KEY-TYPE> <ENCR-KEY>]]
no ipv6 ospfv3 encryption
```

## Description

Configures IPsec ESP authentication. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec ESP.

The **no** form of this command removes IPsec ESP for the specified area.

| Parameter                          | Description                                                                                                                                                                                                                                                                                     |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>spi &lt;SPI-INDEX&gt;</code> | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| <code>&lt;AUTH-TYPE&gt;</code>     | Specifies the authentication type: <b>md5</b> or <b>sha1</b> .                                                                                                                                                                                                                                  |
| <code>&lt;KEY-TYPE&gt;</code>      | Specifies the key type to use: <b>plaintext</b> (unencrypted), <b>hex-string</b> (encrypted) or <b>ciphertext</b> (encrypted).                                                                                                                                                                  |
| <code>&lt;AUTH-KEY&gt;</code>      | Specifies the authentication key.                                                                                                                                                                                                                                                               |
| <code>&lt;ENCR-TYPE&gt;</code>     | Specifies the encryption type: <b>des</b> , <b>3des</b> , <b>aes</b> , or <b>null</b> .<br><br><b>NOTE:</b> Encryption type <b>aes</b> is considered to be AES128, AES192, or AES256 based on key length.                                                                                       |
| <code>&lt;ENCR-KEY&gt;</code>      | Specifies the encryption key.                                                                                                                                                                                                                                                                   |



When the authentication key is not provided on the command line, plaintext authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally plaintext encryption key prompting. The entered key characters are masked with asterisks.



When the authentication key and encryption type are provided on the command line but the encryption key is not provided, plaintext encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting interface VLAN 1 to use IPsec ESP with provided authentication and encryption keys:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext F82
des plaintext F82#450b
```

Setting interface VLAN 3 to use IPsec ESP with prompted authentication and encryption keys:

```
switch(config)# interface vlan 3
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1
Enter the IPsec authentication key: *****
Re-Enter the IPsec authentication key: *****

Enter the IPsec encryption type (3des/aes/des/null)? des

Enter the IPsec encryption key: *****
Re-Enter the IPsec encryption key: *****
```

Setting interface VLAN 4 to use IPsec ESP with provided authentication password and encryption type but a prompted encryption key:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext F82
des
Enter the IPsec encryption key: *****
Re-Enter the IPsec encryption key: *****
```

Setting interface VLAN 1 to use IPsec ESP with provided plaintext authentication key and null encryption:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext F82
null
```

Removing IPsec from interface VLAN 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 encryption
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 encryption null

ipv6 ospfv3 encryption null

### Description

Configures NULL ESP on an interface which disables IPsec ESP.

### Examples

Disable IPsec ESP on interface VLAN 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption null
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 hello-interval

```
ipv6 ospfv3 hello-interval <INTERVAL>
no ipv6 ospfv3 hello-interval
```

### Description

Sets the time interval between OSPFv3 hello packets for the OSPFv3 interface.

The **no** form of this command sets the time interval between OSPFv3 hello packets to the default for the OSPFv3 interface of 10 seconds.

| Parameter  | Description                                                                                    |
|------------|------------------------------------------------------------------------------------------------|
| <INTERVAL> | Specifies the time interval between hello packets, in seconds. Range: 1 to 65535. Default: 10. |

### Examples

Setting OSPFv3 hello interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 hello-interval 30
```

Setting OSPFv3 hello interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 hello-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 network

```
ipv6 ospfv3 network {broadcast|point-to-point}
no ipv6 ospfv3 network
```

### Description

Configures the network type for the interface. By default the network type is broadcast network.

The **no** form of this command sets the network type for the interface to the system default of broadcast network.

| Parameter      | Description                                                           |
|----------------|-----------------------------------------------------------------------|
| broadcast      | Specifies the OSPFv3 network type as a broadcast multiaccess network. |
| point-to-point | Specifies the OSPFv3 network type as a point-to-point network.        |

### Examples

Setting OSPFv3 network type for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 network broadcast
switch(config-if-vlan)# ipv6 ospfv3 network point-to-point
```

Disabling OSPFv3 network type for the interface to system default of broadcast network:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 network
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |



## ipv6 ospfv3 passive

```
ipv6 ospfv3 passive
no ipv6 ospfv3 passive
```

### Description

Configures the interface as an OSPFv3 passive interface. With this setting, the interface participates in the OSPF, but does not send or receive OSPF packets on that interface.

The **no** form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

### Examples

Setting the interface as OSPFv3 passive interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 passive
```

Setting the interface as OSPFv3 active interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 passive
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 priority

```
ipv6 ospfv3 priority <number-value>
no ipv6 ospfv3 priority
```

### Description

Sets the OSPFv3 priority for the interface. The larger the numeric value of the priority, the higher the chance it will become the designated router. Setting a priority of 0 makes the router ineligible to become a designated router or back up designated router.

The **no** form of this command sets the OSPFv3 priority for the interface to the default of 1.

| Parameter                   | Description                                                       |
|-----------------------------|-------------------------------------------------------------------|
| <i>&lt;number-value&gt;</i> | Specifies the OSPFv3 priority value. Default: 1. Range: 0 to 255. |

## Examples

Setting the OSPFv3 priority for the interface:

```
switch(config)# interface vlan /1
switch(config-if-vlan)# ipv6 ospfv3 priority 50
```

Setting the OSPFv3 priority for the interface to the default of 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 priority
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 af retransmit-interval

```
ipv6 ospfv3 af retransmit-interval <seconds>
no ipv6 ospfv3 af retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for the OSPFv3 interface.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default 5 seconds.

| Parameter              | Description                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------|
| <i>&lt;seconds&gt;</i> | Specifies the time interval for the retransmit interval, in seconds. Range: 1 to 3600. Default: 5 |

## Examples

Setting OSPFv3 retransmit interval on the interface:

```
switch# configure terminal
switch(config)# interface 1
switch(config-if)# ospfv3-af ipv6 retransmit-interval 30
```

Setting OSPFv3 retransmit interval to the default on the interface:

```
switch# configure terminal
switch(config)# interface 1
switch(config-if)# no ospfv3-af ipv6 retransmit-interval
switch(config-if)# no ospfv3-af ipv6 retransmit-interval 30
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 shutdown

```
ipv6 ospfv3 shutdown
no ipv6 ospfv3 shutdown
```

### Description

Disables OSPFv3 on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command **no ip ospf area**.

The **no** form of this command re-enables OSPFv3 on the interface.

### Examples

Disabling OSPFv3 on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 shutdown
```

Re-enabling OSPFv3 on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 shutdown
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 transit-delay

ipv6 ospfv3 transit-delay <DELAY>  
no ipv6 ospfv3 transit-delay

### Description

Sets the time delay in Link state transmission for the OSPFv3 interface.

The **no** form of this command sets the transit delay in Link state transmission to the default of 1 second.

| Parameter | Description                                                                               |
|-----------|-------------------------------------------------------------------------------------------|
| <DELAY>   | Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1. |

## Examples

Setting OSPFv3 transit delay on the interface

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 transit-delay 30
```

Setting OSPFv3 transit delay to default on the interface

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 transit-delay
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## maximum-paths (ospfv3)

maximum-paths <MAXIMUM>  
no maximum-paths

### Description

Sets the maximum number of ECMP routes that OSPFv3 can support.

The **no** form of this command sets the maximum number of ECMP routes that OSPFv3 can support to the default value of 4.

| Parameter | Description                                                              |
|-----------|--------------------------------------------------------------------------|
| <MAXIMUM> | Specifies the maximum number of ECMP routes. Range: 1 to 32. Default: 4. |

### Examples

Setting maximum number of parallel routes:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# maximum-paths 32
```

Setting maximum number of parallel paths to the default value of 4:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# no maximum-paths
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification                                                 |
|------------------|--------------------------------------------------------------|
| 10.10            | Increased upper limit of range of <MAXIMUM> parameter to 32. |
| 10.07 or earlier | --                                                           |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## max-metric router-lsa (ospfv3)

```
max-metric router-lsa [on-startup <INTERVAL>]
no max-metric router-lsa [on-startup]
```

### Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer the router as an intermediate hop in their shortest path first (SPF) calculations. If the on-startup parameter is used, the router is configured to advertise a maximum metric at startup. That is, for the time specified in seconds, or the default value of 600 seconds.

To disable advertisement of the maximum metric, use the **no** form of the command.

The **no** form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

| Parameter             | Description                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| on-startup <INTERVAL> | Automatically advertises the stub Router-LSA (or maximize the router-LSA cost metric) for a specified time interval upon OSPFv3 startup.<br>Specifies the time in seconds. Range: 5 to 86400. Default: 600. |

### Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# max-metric router-lsa
switch(config-ospfv3-1)# max-metric router-lsa on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no max-metric router-lsa
switch(config-ospfv3-1)# no max-metric router-lsa on-startup
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## passive-interface default (OSPFv3)

```
passive-interface default
no passive-interface default
```

### Description

Sets all OSPFv3 interfaces as passive.

The **no** form of this command sets all the OSPFv3 interfaces as active.

### Examples

Setting OSPFv3-enabled interfaces as passive:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# passive-interface default
```

Setting OSPFv3-enabled interfaces as active:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no passive-interface default
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## redistribute (ospfv3)

```
redistribute {bgp | connected | local loopback | static | ripng | ospf <PROCESS-ID>}
[route-map <ROUTE-MAP-NAME>]
no redistribute {bgp | connected | local loopback | static | ripng | ospf <PROCESS-ID>}
[route-map <ROUTE-MAP-NAME>]
```

### Description

Redistributes routes originating from other protocols, or from another OSPFv3 process, to the current OSPFv3 process.

If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv3. Configuration is not allowed if the referenced route map has not yet been configured.

If you try to redistribute routes from an OSPFv3 process which is not created, you are prompted to allow the OSPFv3 process to be auto-created before proceeding with redistribution. If you confirm at the

prompt, the OSPFv3 process is created with defaults and redistribution configuration applied. If you deny at the prompt, redistribution configuration is skipped.

If command **route-redistribute active-routes-only** has been issued, only the routes from other protocols which are selected for forwarding are considered for redistribution into OSPFv3.

The **no** form of this command disables redistribution of routes to the current OSPFv3 process.

| Parameter                  | Description                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------------------|
| bgp                        | Specifies redistributing BGP (Border Gateway Protocol) routes.                                         |
| connected                  | Specifies redistributing connected (directly attached subnet or host).                                 |
| local loopback             | Specifies redistributing local routes of the loopback interface.                                       |
| static                     | Specifies redistributing static routes.                                                                |
| ripng                      | Specifies redistributing RIPv3 routes.                                                                 |
| ospf <PROCESS-ID>          | Specifies redistributing routes from the specified OSPFv3 process ID. Range: 1 to 65535.               |
| route-map <ROUTE-MAP-NAME> | Specifies redistribution filtering by route map. To create a route map, use command <b>route-map</b> . |

## Examples

Redistributing routes to OSPFv3:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# redistribute connected
switch(config-ospfv3-1)# redistribute connected route-map connected_routes
switch(config-ospfv3-1)# redistribute local loopback
switch(config-ospfv3-1)# redistribute local loopback route-map local_routes
switch(config-ospfv3-1)# redistribute static
switch(config-ospfv3-1)# redistribute static route-map static_networks
switch(config-ospfv3-1)# redistribute ripng
switch(config-ospfv3-1)# redistribute ripng route-map rip-routes
switch(config-ospfv3-1)# redistribute ospf 2
```

Disabling redistributing routes to OSPFv3:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no redistribute connected
switch(config-ospfv3-1)# no redistribute connected route-map connected_routes
switch(config-ospfv3-1)# no redistribute local loopback
switch(config-ospfv3-1)# no redistribute local loopback route-map local_routes
switch(config-ospfv3-1)# no redistribute static
switch(config-ospfv3-1)# no redistribute static route-map static_networks
switch(config-ospfv3-1)#
switch(config-ospfv3-1)# no redistribute ripng
switch(config-ospfv3-1)# no redistribute ripng route-map rip-routes
switch(config-ospfv3-1)# no redistribute ospf 2
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.



## Command History

| Release          | Modification                                                                                                  |
|------------------|---------------------------------------------------------------------------------------------------------------|
| 10.08            | Added <b>route-map</b> support for supported redistribute source-protocols. Updated information and examples. |
| 10.07 or earlier | --                                                                                                            |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## reference-bandwidth (OSPFv3)

```
reference-bandwidth <BANDWIDTH>
no reference-bandwidth
```

### Description

Sets the reference bandwidth for OSPFv3. If the OSPFv3 interface cost is not explicitly set, then the cost of all the OSPFv3 interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the calculated link speed value is 1 Gbps (if the OSPFv3 interface cost is not explicitly set).

The **no** form of this command sets the reference bandwidth for OSPF to the default of 100000 Mbps.

| Parameter   | Description                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| <BANDWIDTH> | Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000. |

## Examples

Setting the reference bandwidth:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# routerv3 ospf 1
switch(config-ospfv3-1)# no reference-bandwidth
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## retransmit-interval (OSPFv3)

```
retransmit-interval <INTERVAL>
no retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for virtual links.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for virtual links.

| Parameter  | Description                                                                                        |
|------------|----------------------------------------------------------------------------------------------------|
| <INTERVAL> | Specifies the time interval for the retransmit interval, in seconds. Range: 1 to 3600. Default: 5. |

### Examples

Setting OSPFv3 virtual links retransmit interval:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# retransmit-interval 30
```

Setting OSPFv3 virtual links retransmit interval to default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no retransmit-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |

## router-id (ospfv3)

```
router-id <ROUTER-ADDRESS>
no router-id
```

### Description

Sets an ID for the router in an IPv4 address format.

The **no** form of this command unconfigures the router-id for the instance and sets the router-id to the default. The router-id is changed to the dynamically selected router-id. The default router-id 0.0.0.0 updates to the routing stack that triggers to auto-elect a router-id based on the highest IP address of loopback interface, or the highest IP address of interfaces.

| Parameter        | Description                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------|
| <ROUTER-ADDRESS> | Specifies the router address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. |

### Examples

Setting router-id in the OSPFv3 context:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1) # router-id 1.1.1.1
```

Unconfiguring router-id:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1) # no router-id
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## router ospfv3

```
router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]
no router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]
```

## Description

Creates the OSPFv3 process (if not created already) and enters the router OSPFv3 instance context. Optionally if specified, you can specify a named VRF, or the default VRF if the <vrf-name> is not specified. Only one OSPFv3 process is allowed per VRF.

The **no** form of this command removes the OSPFv3 instance. If a VRF is specified, it removes the OSPF instance from the named VRF, or the default VRF if the <var-name> is not specified.

| Parameter      | Description                                         |
|----------------|-----------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 process ID. Length: 1 to 65535. |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.      |

## Examples

Entering the router OSPFv3 instance:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
```

Setting the router OSPFv3 VRF instance:

```
switch(config)# router ospfv3 1 vrf vrf_red
```

Removing the router OSPFv3 instance:

```
switch#(config)# no router ospfv3 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## show ipv6 ospfv3

```
show ipv6 ospfv3 [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows OSPFv3 information including area, state, and configuration information.

| Parameter      | Description                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 process ID optionally to show OSPFv3 information for a particular OSPFv3 process. Range: 1 to 65535. |
| all-vrfs       | Select to show OSPFv3 information for all VRFs.                                                                          |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.                                                                           |

## Example

Showing general OSPFv3 configurations:

```
switch# show ipv6 ospfv3 200
VRF : default                                Process : 200
-----

Router ID          : 1.1.1.1                OSPFv3           : Enabled
BFD                : Disabled               SPF Start Interval : 200 ms
SPF Hold Interval  : 1000 ms                SPF Max Wait Interval : 5000 ms
LSA Start Interval : 5000 ms                LSA Hold Interval   : 1000 ms
LSA Max Wait Interval : 1000 ms             LSA Arrival Interval : 1000 ms
External LSAs      : 0                     Checksum Sum       : 0
ECMP               : 4                     Reference Bandwidth : 100000 Mbps
Area Border        : Yes                   AS Border          : No
GR Status          : Enabled               GR Interval        : 120 sec
GR State           : Inactive              GR Exit Status     : None
GR Helper          : Enabled               GR Strict LSA Check : Enabled
GR Ignore Lost I/F : Disabled              Internal Process ID : 1
Summary address:
  prefix fd00::1/64, advertise, tag 10

Area      Total Active
-----
Normal    2          2
Stub      0          0

Area : 0.0.0.0
-----
Area Type          : Normal                Status           : Active
Total Interfaces   : 1                    Active Interfaces : 1
Passive Interfaces : 0                    Loopback Interfaces : 0
SPF calculation count : 4
Area ranges:
  fd00::1/64, inter-area, no-advertise
AH Authentication   : SHA1, SPI 256
Number of LSAs      : 5                    Checksum Sum      : 99122

Area : 0.0.0.1
-----
Area Type          : Normal                Status           : Active
Total Interfaces   : 1                    Active Interfaces : 1
Passive Interfaces : 0                    Loopback Interfaces : 0

SPF Calculation Count : 4
```

```
Area ranges:
  fd00::1/64, inter-area, no-advertise
ESP Authentication      : SHA1
Encryption              : 3DES, SPI 256
Number of LSAs         : 5           Checksum Sum           : 99122
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 border-routers

```
show ipv6 ospfv3 [<PROCESS-ID>] border-routers [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows the OSPFv3 routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

| Parameter      | Description                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 process ID to show the OSPFv3 routing table entries for ABR and ASBR for the particular OSPFv3 process. Range: 1 to 65535. |
| all-vrfs       | Select to show OSPFv3 border router information for all VRFs.                                                                                  |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.                                                                                                 |

### Example

Showing OSPFv3 border routers information for VRF **vrf\_red**:

```
switch# show ipv6 ospfv3 border-routers vrf vrf_red
VRF : vrf_red          Process ID : 1
          Internal Routing Table
-----
```

```

Codes: i - Intra-area route, I - Inter-area route
Router-ID Cost  Type   Area   SPF  Nexthop          Interface
i  1.1.1.1    1    ASBR   0.0.0.0  9    fe80::7272:cfff:fe9a:a15d  1/1/2
i  3.3.3.3    1    ASBR   1.1.1.1  9    fe80::7272:cfff:fe1f:d80   ** tunnel1

```

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 interface

```

show ipv6 ospfv3 [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief]
[all-vrfs | vrf <VRF-NAME>]

```

## Description

Shows information about OSPFv3 enabled interfaces.

| Parameter        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>     | Specifies an OSPFv3 process ID optionally to show the OSPFv3 enabled interfaces for the particular OSPFv3 process. Range: 1 to 65535.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <INTERFACE-NAME> | Selects to show information only for the specified OSPFv3-enabled interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| brief            | Include this parameter to display a brief overview of the following OSPF configuration information. <ul style="list-style-type: none"> <li>Interface: OSPF interface name.</li> <li>Area: OSPF area ID.</li> <li>Cost: The metric OSPF uses to judge a path's feasibility, calculated as (reference bandwidth / interface bandwidth).</li> <li>State: Indicates if the interface is a designated router (<b>Dr</b>) or a backup designated router (<b>Backup-dr</b>).</li> <li>Status: Indicates if the interface is <b>up</b> or <b>down</b>.</li> <li>Flags: P - Passive A - Active.</li> </ul> |
| all-vrfs         | Select to show OSPF-enabled interface information for all VRFs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| vrf <VRF-NAME>   | Specifies the name of a VRF. Default: default.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Examples

Showing OSPFv3 information for all interfaces in default VRF:

```
switch# show ipv6 ospfv3 interface
Codes: DR - Designated router  BDR - Backup Designated router

Interface 1/3/10 is Up, Line Protocol is Up
-----
VRF                : default                               Process          : 1
IPv6 address       : fe80::9020:c203:280a:e800             Area              :
0.0.0.0
Status             : Up                                     Network Type     :
Broadcast
Hello Interval     : 10      sec                             Dead Interval    : 40
sec
Transit Delay      : 1      sec                             Retransmit Interval : 5
sec
BFD                : Disabled                               Link Speed       : 1000
Mbps
Cost Configured    : NA                                     Cost Calculated  : 100
State/Type         : DR                                     Router Priority   : 2
DR                 : 1.1.1.1                               BDR              :
2.2.2.2
Link LSAs          : 2                                     Checksum Sum     :
39245
Authentication     : no                                    Passive          : No

Codes: DR - Designated router  BDR - Backup Designated router

Interface 1/3/11 is Up, Line Protocol is Up
-----
VRF                : default                               Process          : 1
IPv6 address       : fe80::9020:c203:2c0a:e800             Area              :
0.0.0.1
Status             : Up                                     Network Type     :
Broadcast
Hello Interval     : 10      sec                             Dead Interval    : 40
sec
Transit Delay      : 1      sec                             Retransmit Interval : 5
sec
BFD                : Disabled                               Link Speed       : 1000
Mbps
Cost Configured    : NA                                     Cost Calculated  : 100
State/Type         : BDR                                     Router Priority   : 1
DR                 : 3.3.3.3                               BDR              :
1.1.1.1
Link LSAs          : 2                                     Checksum Sum     :
83119
Authentication     : no                                    Passive          : No
```

Showing overview information for OSPFv3 enabled interfaces for all VRFs in brief:

```
switch# show ipv6 ospfv3 interface brief all-vrfs
VRF : default                               Process : 1
=====

Total Number of Interfaces: 2

Flags: P - Passive  A - Active

Interface      Area              Cost      State      Status      Flags
```



| Interface | Area    | Cost | State | Status | Flags |
|-----------|---------|------|-------|--------|-------|
| 1/3/10    | 0.0.0.0 | 100  | DR    | Up     | A     |
| 1/3/11    | 0.0.0.1 | 100  | BDR   | Up     | A     |

Showing overview information for OSPFv3 enabled interfaces for all VRFs:

```
6200(config)# show ipv6 ospfv3 interface br all-vrfs
VRF : default                               Process : 1
=====
```

Total Number of Interfaces: 1

Flags: P - Passive A - Active

| Interface | Area    | Cost | State | Status | Flags |
|-----------|---------|------|-------|--------|-------|
| 1/1/1     | 0.0.0.0 | 100  | BDR   | Up     | A     |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                                                                                                           |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535.                                                                              |
| 10.09            | Output of the <b>show ipv6 ospfv3 interface</b> command includes flags to indicate whether the interface is in passive or active mode. |
| 10.07 or earlier | --                                                                                                                                     |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 lsdb

```
show ipv6 ospfv3 [<process-id>] lsdb
  adv-router {<ROUTER-ID>|self}
  area <AREA-ID>
  lsid <link-state-id>
  all-vrfs|vrf <VRF-NAME>
  as-external
  asbr-summary
  database-summary
  inter-area-prefix
  inter-area-router
  intra-area-prefix
  link
```

```
network
nssa-external
router
summary
```

## Description

Shows the OSPFv3 link state database summary for different OSPF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

| Parameter                                        | Description                                                                                                                                                                                                                                                                             |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;PROCESS-ID&gt;</code>                  | Enter an OSPFv3 process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535.                                                                                                                                                                        |
| <code>adv-router {&lt;ROUTER-ID&gt; self}</code> | Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify <b>self</b> to show self-originated link states.                                                                                                     |
| <code>area &lt;AREA-ID&gt;</code>                | Select to display information filtered for the specified area in one of the following formats.<br>OSPFv3 area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPFv3 area identifier in decimal format. Value: 0 to 4294967295.      |
| <code>lsid &lt;LINK-STATE-ID&gt;</code>          | Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D).                                                                                                                                                                             |
| <code>all-vrfs {vrf &lt;vrf-name&gt;}</code>     | Select <b>all-vrfs</b> to display general OSPF information for all VRFs, or use the <b>vrf &lt;VRF-NAME&gt;</b> option to display information for a specific VRF.<br><br>Optionally select one of the following parameters to filter the link state database information.               |
| <code>as-external</code>                         | Show external link states (LSA type 5)                                                                                                                                                                                                                                                  |
| <code>database-summary</code>                    | Select to display the count of each type of LSA and each area in the database.<br><br><b>NOTE:</b> The <b>database-summary</b> parameter does not support the <b>area &lt;area-id&gt;</b> , <b>lsid &lt;link-state-id&gt;</b> or <b>adv-router {&lt;router-id&gt; self}</b> parameters. |
| <code>inter-area-prefix</code>                   | Show inter-area prefix link states (LSA type 3)                                                                                                                                                                                                                                         |
| <code>inter-area-router</code>                   | Show inter-area router link states (LSA type 4)                                                                                                                                                                                                                                         |
| <code>intra-area-prefix</code>                   | Show intra-area prefix link states (LSA type 9)                                                                                                                                                                                                                                         |
| <code>link</code>                                | Show link states (LSA type 8)                                                                                                                                                                                                                                                           |
| <code>network</code>                             | Show network LSAs (LSA type 2).                                                                                                                                                                                                                                                         |
| <code>nssa-external</code>                       | Show NSSA external link states (LSA type 7).                                                                                                                                                                                                                                            |
| <code>router</code>                              | Show router LSAs (LSA type 1).                                                                                                                                                                                                                                                          |

## Examples

## Showing OSPFv3 link state database (LSDB) general information:

```
switch# show ipv6 ospfv3 lsd
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
=====

Router Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Link Count  Bits
-----
40.40.40.40     930     0x80000004 0x2ea1      0           3           None
50.50.50.50     935     0x80000002 0x8b52      0           1           E
60.60.60.60     943     0x800003c5 0x9854      0           2           None

Network Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Router Count
-----
60.60.60.60     944     0x80000001 0x7179      1360007168 2
50.50.50.50     935     0x80000001 0x516a      19          1

Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Prefix
-----
40.40.40.40     929     0x80000001 0x2498      131072      FEC0:3344::/32
50.50.50.50     928     0x80000001 0x5b2f      65536       111::/64

Inter Area Router Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Destination Router ID
-----
40.40.40.40     929     0x80000001 0x2498      1           33.33.33.33

AS External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Prefix
-----
40.40.40.40     264     0x80000001 0x24cc4     1           10::/64
40.40.40.40     675     0x80000001 0x5b00f     2           11::/64

NSSA External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Prefix
-----
3.3.3.3         264     0x80000001 0x24ac2     1           200::/64

Link-local Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Interface
-----
50.50.50.50     264     0x80000001 0x653c4     19         1/1/1

Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Referenced LS Type
Referenced LSID
-----
50.50.50.50     263     0x80000001 0x1da34     1           0x2001          0
50.50.50.50     264     0x80000001 0x2a45d     1           0x2002          19
```

## Showing AS external link states:

```
switch# show ipv6 ospfv3 lsdb as-external
OSPF Router with ID (60.60.60.60) (Process ID 1 VRF default)
=====

AS External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Prefix
-----
40.40.40.40     264     0x80000001 0x24cc4     1         10::/64
40.40.40.40     675     0x80000001 0x5b00f     2         11::/64
```

Showing the LSDB database summary:

```
switch# show ipv6 ospfv3 lsdb database-summary
OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
=====

Area 0.0.0.0 database summary
-----
LSA Type      Count
-----
Router        2
Network       1
Inter Area Prefix  1
Inter Area Router  0
NSSA External  0
Link          3
Intra Area Prefix  3
-----
Total         10

Process 1 database summary
-----
LSA Type      Count
-----
Router        2
Network       1
Inter Area Prefix  1
Inter Area Router  0
AS External    2
NSSA External  0
Link          3
Intra Area Prefix  3
-----
Total         12
```

Showing inter-area prefix LSAs:

```
switch# show ipv6 ospfv3 lsdb inter-area-prefix
OSPF Router with ID (6.6.6.6) (Process ID 1 VRF default)
=====

Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Prefix
-----
40.40.40.40     929     0x80000001 0x2498     131072    FEC0:3344::/32
50.50.50.50     928     0x80000001 0x5b2f     65536     111::/64
```

Showing network LSAs:

```
switch# show ipv6 ospfv3 lsdbs network
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
=====

Network Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Router Count
-----
60.60.60.60    944     0x80000001 0x7179     1360007168 2
50.50.50.50    935     0x80000001 0x516a     19          1
```

Showing NSSA external link states:

```
switch# show ipv6 ospfv3 lsdbs nssa-external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====

NSSA External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Prefix
-----
3.3.3.3         264     0x80000001 0x24ac2     1           200::/64
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 neighbors

```
show ipv6 ospfv3 [<PROCESS-ID>] neighbors [<NEIGHBOR-ID>]
[interface <INTERFACE-NAME>] [detail | summary]
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows information about OSPFv3 neighbors.

| Parameter                  | Description                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies an OSPFv3 process ID to show OSPFv3 neighbor information for the particular OSPFv3 process. Range: 1 to 65535. |
| neighbors <NEIGHBOR-ID>    | Shows information about a particular neighbor, specified in IPv4 format (A.B.C.D).                                       |
| interface <INTERFACE-NAME> | Shows neighbor information only for the specified interface.                                                             |
| detail                     | Shows detailed information for the neighbors.                                                                            |
| summary                    | Shows summary information for the neighbors.                                                                             |
| all-vrfs                   | Shows neighbor information for all VRFs.                                                                                 |
| vrf <VRF-NAME>             | Specifies the name of a VRF. Default: default.                                                                           |

## Examples

Showing OSPFv3 neighbors information:

Showing OSPFv3 neighbors information for a specific neighbor:

```
switch# show ipv6 ospfv3 neighbors 3.3.3.3
VRF : default                      Process : 1
-----
Router-Id      : 3.3.3.3           Area          : 0.0.0.0
Interface     : 1/1/1             Address         :
fe80::7272:cfff:fe79:7510
State          : FULL              Neighbor Priority : 1
Dead Timer Due : 00:00:36          Options         : 0x13
Time since last state change : 00h:14m:45s
```

Showing detail OSPFv3 neighbors information for a specific neighbor:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2 detail
VRF : default                      Process : 1
-----
Router-Id      : 3.3.3.3           Area          : 0.0.0.0
Interface     : 1/1/1             Address         :
fe80::7272:cfff:fe79:7510
State          : FULL              Neighbor Priority : 1
DR             : 3.3.3.3           BDR            : 1.1.1.3
Dead Timer Due : 00:00:36          Options         : 0x13
Retransmission Queue Length : 0
Time Since Last State Change : 00h:14m:45s
```

Showing OSPFv3 neighbors information for interface **1/1/1**:

```
switch# show ipv6 ospfv3 neighbors 3.3.3.3 interface 1/1/1
VRF : default                      Process : 1
-----
Router-Id      : 3.3.3.3           Area          : 0.0.0.0
Interface     : 1/1/1             Address         :
fe80::7272:cfff:fe79:7510
```

```
State           : FULL           Neighbor Priority : 1
Dead Timer Due   : 00:00:36       Options           : 0x13
Time Since Last State Change : 00h:14m:45s
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 routes

```
show ipv6 ospfv3 [<PROCESS-ID>] routes [<PREFIX/LENGTH>]
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows the OSPFv3 routing table information.

| Parameter       | Description                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>    | Specifies an OSPFv3 process ID that shows information from the OSPFv3 routing table for the particular OSPFv3 process. Range: 1 to 65535. |
| <PREFIX/LENGTH> | Specifies the IPv6 destination prefix showing information about a particular destination prefix. For example, 2010:bd9::/32.              |
| all-vrfs        | Select to show OSPFv3 routing table information for all VRFs.                                                                             |
| vrf <VRF-NAME>  | Specifies the name of a VRF. Default: default.                                                                                            |

## Examples

Showing OSPFv3 routing table information:

```
switch# show ipv6 ospfv3 routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 1 VRF default, Routing Table
```

```

-----

Total Number of OSPFv3 Routes : 2

111::/64          (i) area:0.0.0.0
    directly attached to interface 1/1/1, cost 1 distance 110
fd00::/64         (i) area:0.0.0.1
    directly attached to interface vlan10, cost 1 distance 110

```

Showing OSPFv3 routing table information for VRF red:

```

switch# show ipv6 ospfv3 routes vrf red

Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 2 VRF red, Routing Table
-----

Total Number of OSPFv3 Routes : 1

222::/64          (i) area:0.0.0.1
    directly attached to interface 1/1/2, cost 1 distance 110

```

Showing OSPFv3 routing table information for destination prefix fd00::/64:

```

switch# show ipv6 ospfv3 1 routes fd00::/64

Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 1 VRF default, Routing Table for prefixes fd00::/64
-----

Total Number of OSPFv3 Routes : 1

fd00::/64          (i) area:0.0.0.1
    directly attached to interface vlan10, cost 1 distance 110

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |



# show ipv6 ospfv3 statistics

```
show ipv6 ospfv3 [<PROCESS-ID>] statistics
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows OSPFv3 statistics.

| Parameter      | Description                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 process ID that shows information on the OSPFv3 SPF statistics for the particular OSPFv3 process. Range: 1 to 65535. |
| all-vrfs       | Select to show OSPFv3 SPF statistics information for all VRFs.                                                                           |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.                                                                                           |

## Examples

Showing OSPFv3 statistics information:

```
switch# show ipv6 ospfv3 statistics
OSPFv3 Process ID 1 VRF default, Statistics (cleared 3h 2m 21s ago)
-----

Unknown Interface Drops           : 0
Unknown Virtual Interface Drops   : 0
Bad IPv6 Header Length Drops      : 0
Wrong OSPFv3 Version Drops        : 0
Bad Source IPv6 Drops             : 0
Resource Failure Drops            : 0
Bad Header Length Drops           : 0
Total Drops                       : 0
```

Showing OSPFv3 statistics information for VRF red:

```
switch# show ipv6 ospfv3 2 statistics vrf red
OSPFv3 Process ID 2 VRF red, Statistics (cleared 3h 2m 30s ago)
-----

Unknown Interface Drops           : 0
Unknown Virtual Interface Drops   : 0
Bad IPv6 Header Length Drops      : 0
Wrong OSPFv3 Version Drops        : 0
Bad Source IPv6 Drops             : 0
Resource Failure Drops            : 0
Bad Header Length Drops           : 0
Total Drops                       : 0
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 statistics interface

```
show ipv6 ospfv3 [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>]
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows the OSPFv3 statistics for the OSPFv3-enabled interfaces.

| Parameter        | Description                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>     | Specifies an OSPFv3 process ID to show OSPF-enabled interface statistics information on the specified OSPFv3 process. Range: 1 to 65535. |
| <INTERFACE-NAME> | Selects to show information only for the specified interface.                                                                            |
| all-vrfs         | Select to show OSPF-enabled interface statistics information for all VRFs.                                                               |
| vrf <VRF-NAME>   | Specifies the name of a VRF. Default: default.                                                                                           |

## Example

Showing OSPFv3-enabled interfaces information for interface 1/1/1:

```
switch# show ipv6 ospfv3 statistics interface all-vrfs
OSPFv3 Process ID 1 VRF default, Interface vlan2000 Statistics (cleared 0h 2m 52s
ago)

=====
====

Tx Hello packets      : 0          Rx Hello packets      : 0
Tx Hello bytes        : 0          Rx Hello bytes        : 0
Tx DD packets         : 0          Rx DD packets         : 0
Tx DD bytes           : 0          Rx DD bytes           : 0
Tx LS request packets : 0          Rx LS request packets : 0
Tx LS request bytes   : 0          Rx LS request bytes   : 0
Tx LS update packets  : 0          Rx LS update packets  : 0
Tx LS update bytes    : 0          Rx LS update bytes    : 0
Tx LS ack packets     : 0          Rx LS ack packets     : 0
Tx LS ack bytes       : 0          Rx LS ack bytes       : 0
```

```
Total IPsec packets processed : 0
Total IPsec bytes processed   : 0
Total Number of State Changes : 0
Number of LSAs                : 0
LSA Checksum Sum              : 0
```

```
Total OSPFv3 Packets Discarded: 0
-----
```

| Reason                      | Packets Dropped |
|-----------------------------|-----------------|
| Invalid Type                | 0               |
| Invalid Length              | 0               |
| Invalid Version             | 0               |
| Bad or Unknown Source       | 0               |
| Area Mismatch               | 0               |
| Self-originated             | 0               |
| Duplicate Router ID         | 0               |
| Interface Standby           | 0               |
| Total Hello Packets Dropped | 0               |
| Hello Interval Mismatch     | 0               |
| Dead Interval Mismatch      | 0               |
| Options Mismatch            | 0               |
| MTU Mismatch                | 0               |
| Neighbor Ignored            | 0               |
| Resource Failures           | 0               |
| Bad LSA Length              | 0               |
| Others                      | 0               |
| IPsec Authentication Errors | 0               |
| IPsec ESP Errors            | 0               |

```
Total LSAs Ignored : 0
-----
```

```
Bad Type      : 0
Bad Length    : 0
Invalid Data  : 0
Invalid Checksum : 0
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## summary-address (ospfv3)

```
summary-address <IPV6-ADDR>/<MASK> [no-advertise | tag <TAG-VALUE>]  
no summary-address <prefix/length> [no-advertise | tag <tag-value>]
```

### Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The **no** form of this command disables route summarization.



---

This command only works for an ASBR (Autonomous System Boundary Router).

---

| Parameter       | Description                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <IPV6-ADDR>     | Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where <b>x</b> is a hexadecimal number from 0 to F.            |
| <MASK>          | Specifies the number of bits in the address mask in CIDR format ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 128.                  |
| no-advertise    | Do not advertise the aggregate route. Suppress routes that match the specified prefix/mask pair.                                                 |
| tag <TAG-VALUE> | Specify the tag for the aggregate route. The summary prefix will be advertised along with the tag value in External LSAs. Range: 0 to 4294967295 |

### Examples

Setting OSPF route summarization:

```
switch(config) # router ospfv3 1  
switch(config-ospfv3-1) # summary-address 2001:DB8::1/32
```

Disabling route summarization:

```
switch(config) # router ospfv3 1  
switch(config-ospfv3-1) # no summary-address 2001:DB8::1/32
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## timers lsa-arrival (OSPFv3)

```
timers lsa-arrival <DELAY>
no timers lsa-arrival
```

### Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives sooner before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command **timers throttle lsa start** on the neighbor.

The **no** form of this command sets the LSA timers to default values.

| Parameter | Description                                                             |
|-----------|-------------------------------------------------------------------------|
| <DELAY>   | Specifies the delay in milliseconds. Range: 0 to 600000. Default: 1000. |

### Examples

Setting the LSA arrival timer:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# no timers lsa-arrival
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## timers throttle lsa (OSPFv3)

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-time <WAIT-TIME>
no timers throttle lsa
```

### Description

Configures the timers for LSA generation.

The **no** form of this command sets the LSA timers to default values.

| Parameter                 | Description                                                                                                                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| start-time <START-TIME>   | Specifies the initial wait time in milliseconds after which LSAs are generated. When set to 0, the LSAs are generated without any delay. Range: 0 to 600000. Default: 5000.                                                                                                    |
| hold-time <HOLD-TIME>     | Specifies the amount of time, in milliseconds, between regeneration of an LSA. The hold time doubles each time the same LSA must be regenerated, until <b>max-wait-time</b> is reached. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0. |
| max-wait-time <WAIT-TIME> | Specifies the maximum wait time, in milliseconds, for regeneration of the same LSA. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.                                                                                                     |

### Examples

Setting the LSA timers:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttle lsa start-time 100 hold-time 1000 max-wait-time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttle lsa
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## timers throttle spf

```
timers throttle spf start-time <START-TIME> hold-time <HOLD-TIME>
max-wait-time <WAIT-TIME>
no timers throttle spf
```

### Description

Configures timers for SPF calculation. There are three timers:

- **start-time** Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- **hold-time** Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the *max-wait-time*.
- **max-wait-time** Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The **no** form of this command sets all the configured non-default timers to default value.

| Parameter    | Description                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------|
| <START-TIME> | Time in milliseconds to set timer for initial SPF delay. Default: 200.                                     |
| <HOLD-TIME>  | Time in milliseconds to set the minimum hold time between two consecutive SPF calculations. Default: 1000. |
| <WAIT-TIME>  | Time in milliseconds to set the maximum wait time between two consecutive SPF calculations. Default: 5000. |

### Examples

Setting non-default timer values for SPF throttling:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttling spf start-time 500 hold-time 3000 max-
wait-time 9000
Switch(config-ospfv3-1)# show running-config current-context
router ospfv3 1
    area 0.0.0.0
```

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttling spf
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## transit-delay (OSPFv3)

```
transit-delay <DELAY>
no transit-delay
```

### Description

Sets the time delay in Link state transmission for virtual links.

The **no** form of this command sets the delay in Link state transmission to the default of 1 second for virtual links.

| Parameter | Description                                                                               |
|-----------|-------------------------------------------------------------------------------------------|
| <DELAY>   | Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1. |

### Examples

Setting OSPFv3 virtual links transit delay:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# transit-delay 30
```

Setting OSPFv3 virtual links transit delay to default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no transit-delay
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |



## trap-enable (ospfv3)

trap-enable  
no trap-enable

### Description

Enables the notification of the events to be sent as traps to the SNMP management stations for OSPFv3. The **no** form of this command disables the notification of the events to be sent as traps to the SNMP management stations for OSPFv3.

### Examples

Enabling sending notification of events as traps:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# trap-enable
```

Disabling sending notification of events as traps:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-1)# no trap-enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## Open Shortest Path First version 3 Address Family (OSPFv3 AF)

OSPFv3 AF extends the existing OSPFv3 protocol by adding support for multiple unicast address families as described in RFC5838. As an extension of OSPFv3, OSPFv3 AF retains all the functionalities, LSA types, and concepts of the original OSPFv3 protocol.

### Overview

The characteristics of OSPFv3 AF are:

- Provides a loop-free topology using SPF algorithm.
- Allows hierarchical routing using area 0 (backbone area) as the top of the hierarchy.
- Supports load balancing with equal cost routes for same destination.
- OSPFv3 AF is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking) and route summarization.
- Scales enterprise size network easily with area concept.
- Provides fast convergence with triggered, incremental updates through Link State Advertisements (LSAs).

OSPFv3 AF configurations are supported in global router context, address-family context, and interface context. Some OSPFv3 AF router configurations are supported only in address-family context, while others are supported in both AF router and global router context. OSPFv3 AF can be configured on routed ports, VLAN interfaces, LAG interfaces, and loopback interfaces. All such configurations work in the mentioned interfaces context. OSPFv3 AF mandates the associated interface to be routing interface. The switch supports congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling).

### Supported features

- OSPFv3 neighbor adjacency, Hello protocol, multiple areas, Inter-area routing
- AS external routes, stub areas, totally stubby areas, NSSA, ABR, ASBR
- Designated Router/Backup Designated Router
- Point-to-point interfaces/broadcast interfaces
- Virtual Links (Supported only by IPv6 AF)
- Bidirectional Forwarding Detection (BFD) - refer to the *High Availability Guide* for additional information
- Equal-cost multipath
- Null authentication, Simple password authentication, MD5 authentication
- Area range aggregation - Type-3/Type-7
- External route aggregation - Type-5 LSAs
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- SPF throttling
- LSA Throttling
- Configuration of interface parameters such as priority, cost, hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Configuration of virtual link parameters such as hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Route redistribution
- Passive interfaces
- Multi VRF support with each VRF having up to eight OSPF process instances
- Congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling)



---

An OSPFv3 AF session will not come up in 6in6 tunnel.

---

## Limitations

- The following configurations are not supported in router context:
  - area
  - bfd all-interfaces
  - default-metric
  - passive-interface-default
  - redistribute
  - maximum-paths
  - traps
  - default-information originate
  - summary-address
  - distribute-list
- Only one instance per address-family can be configured on the interface.
- Multicast address-families are not supported.
- OSPFv3 AF processes cannot establish neighborships with peers that do not support OSPFv3 AF or OSPFv3 Instance-IDs.
- OSPFv3 AF area-level IPsec authentication is not supported.
- Instance ID mismatch counter is not supported per interface. To identify a mismatch in instance IDs, the user must review the configuration of that particular interface. To help identify debug logs for mismatched instance IDs, the user must enable debug ospfv3-af and monitor debug logs for the specific interface.
- The Options field in the show ospfv3-af neighbor detail output does not display the address-family information.
- Simultaneous configurations of OSPFv3-AF with OSPFv2 or OSPFv3 on the same interface are not supported.

## How OSPFv3 AF protocol works

### OSPFv3 AF protocol

The protocol uses Link State Advertisements (LSAs) transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces. Each routing switch in an area also maintains a link-state database (LSDB) that describes the area topology. (All routers in a given OSPF area have identical LSDBs). The routing switches used to connect areas to each other flood summary link LSAs and external link LSAs to neighboring OSPF areas to update them regarding available routes. Through this means, each OSPF router determines the shortest path between itself and a desired destination router in the same OSPF domain (Autonomous System (AS)). OSPFv3 AF supports unicast address-families by mapping each AF to a separate Instance ID and OSPFv3 AF instance.

The following ranges are defined for different AFs. The first value of each range is the default value for the corresponding AF.

| Instance ID | Description     |
|-------------|-----------------|
| 0 - 31      | IPv6 unicast AF |
| 64 - 95     | IPv4 unicast AF |

## OSPFv3 AF concepts

OSPFv3 Address Family is a link-state routing protocol applied to routers grouped into OSPFv3 AF areas identified by the IPv4 and IPv6 routing configuration on each routing switch. Each OSPFv3 AF area includes one or more networks. OSPFv3 AF routers use hello packets and LSAs to maintain OSPFv3 AF operation across networks within an area and between areas within an OSPFv3 AF domain.

OSPFv3 AF uses hello packets to establish and maintain connections between neighboring routers on the same interface. Because neighborship and LSA exchanges are conducted using IPv6 link-local addresses, OSPFv3 AF requires an IPv6 address, even when operating with the IPv4 unicast address family.

## OSPFv3 AF Link-state advertisement (LSA) types

OSPFv3 AF uses LSAs transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces.

Each routing switch in an area also maintains an LSDB that describes the area topology. (All routers in a given OSPFv3 AF area have identical LSDBs, and each router uses the LSDB to build its own shortest-path tree.)

The routing switches used to connect areas to each other flood inter-area-prefix-LSAs, inter-area-router-LSAs, and AS-external-LSAs to backbone area to update it regarding available routes. Each OSPFv3 AF router determines the shortest path between itself and a desired destination router in the same OSPFv3 AF domain (AS). Routed traffic in an OSPFv3 AF AS is classified as one of the following:

- Intra-area traffic
- Inter-area traffic
- External traffic

The routing switches support the LSAs listed in the following table.

| Link-state type | Description           | Use                                                                                                                                                                  | Flood scope |
|-----------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 0x2001          | Router-LSA            | Describes the state of each active interface on a router for Area a given area. (Excludes loopback interfaces and interfaces that have not achieved full adjacency.) | Area        |
| 0x2002          | Network-LSA           | Describes the OSPFv3 AF routers in a given network.                                                                                                                  | Area        |
| 0x2003          | Inter-area-prefix-LSA | Describes the route to a prefix in another OSPFv3 AF area of Area the same AS. (Excludes prefixes for                                                                | Area        |

| Link-state type | Description           | Use                                                                                                                                                                                                                                                                                                                                                                                                                                            | Flood scope |
|-----------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                 |                       | link-local addresses.)<br>Propagated through backbone area to other areas.                                                                                                                                                                                                                                                                                                                                                                     |             |
| 0x2004          | Inter-area-router-LSA | Describes the route to an ASBR in another OSPFv3 AF normal area (including the backbone area) of the same AS.<br><br>Propagated through backbone area to other areas. (Excludes any ASBR in the same area as the router sending the LSA.)                                                                                                                                                                                                      | Area        |
| 0x4005          | AS-external-LSA       | Describes the route to a destination prefix in another AS (external route). (Excludes prefixes for link-local addresses.)<br>Originated by ASBR in normal or backbone areas of an AS and propagates through backbone area to other normal areas.<br><br>Does not flood over virtual links and is not summarized in virtual links. For injection into an NSSA, an NSSA ABR generates a type-7-default-LSA advertising the default route (::/0). | AS          |
| 0x2007          | NSSA-LSA              | Describes the route to a destination in another AS (external route). Originated by ASBR in NSSA. ABR translates type-7 LSAs to AS-external-LSAs for injection into the backbone area.                                                                                                                                                                                                                                                          | NSSA        |
| 0x0008          | Link-LSA              | For other routers on the same VLAN interface (except virtual links), describes the router link-local address and any other IPv6 prefixes reachable on the VLAN. Link LSAs are not flooded over virtual links.                                                                                                                                                                                                                                  | Link-local  |
| 0x2009          | Intra-area-prefix-LSA | Generated on transit links within an area by the DR operating on those links. Also, every OSPFv3 AF                                                                                                                                                                                                                                                                                                                                            | Area        |

| Link-state type | Description | Use                                                                             | Flood scope |
|-----------------|-------------|---------------------------------------------------------------------------------|-------------|
|                 |             | router generates this LSA to refer to stub and loopback prefixes on the router. |             |

## OSPFv3 AF area types

OSPFv3 AF is built upon a hierarchy of network areas. All areas for a given OSPFv3 AF domain reside in the same AS. An AS is defined as a number of contiguous networks that share an interior gateway routing protocol.

An AS can be divided into multiple areas, including the backbone (area 0). Because each area represents a collection of contiguous networks and hosts, the topology of a given area is not known by the internal routers in any other area. Areas define the boundaries to which router-LSAs and network-LSAs are broadcast. This functionality limits the amount of LSA flooding that occurs within the AS. It also helps to control the size of the link-state databases (LSDBs) maintained in OSPFv3 AF routers.

An area is represented in OSPFv3 AF by either a 32-bit dotted-decimal address or a number. Area types include: Backbone, Normal, Not-so-stubby (NSSA), and Stub.

### Backbone area

Every AS must have one (and only one) backbone area (identified as area 0 or 0.0.0.0.) The ABRs of all other areas in the same AS connect to the backbone area, either physically through an ABR or through a configured, virtual link. The backbone is a special type of area and serves as a transit area for carrying the inter-area-prefix-LSAs, AS-external-LSAs, and routed traffic between non-backbone areas, as well as the router-LSAs, network-LSAs, and routed traffic internal to the area. ASBRs are allowed in backbone areas.

### Normal area

A normal area allows inter-area-prefix-LSAs and AS-external-LSAs to and from the backbone area. A normal area connects to the AS backbone area through one or more ABRs (physically or through a virtual link) and allows router-LSAs and network LSAs within this area. ASBRs are allowed in normal areas.

### Stub area

Stub area connects to the AS backbone through one or more ABRs. It does not allow an internal ASBR, and does not allow AS-external-LSAs. A stub area supports these actions:

- Advertise the area summary routes to the backbone area.
- Advertise inter-area routes from other areas.
- Use the inter-area-prefix-LSA default route to advertise routes to an ASBR and to other areas.

You can configure the stub area ABR to do the following:

- Suppress advertising on the area's summarized internal routes into the backbone area.
- Suppress LSA traffic from other areas in the AS by replacing inter-area-prefix-LSAs and the default external route from the backbone area with the default summary route (::/0.). This area is called totally stubby area.

Virtual links are not allowed for stub areas.

## Not-so-stubby (NSSA) area

An NSSA area connects to the backbone area through one or more ABRs. NSSAs are used where an ASBR exists in an area where you want to:

- Block injection of external routes from other areas of the AS.
- Advertise type-7-LSA external routes (learned from the ASBR) to the backbone area as AS-external-LSAs.

NSSAs also support the following:

- Advertise inter-area-prefix-LSAs from the backbone area into the NSSA. (If no-summary is enabled, the NSSA ABR suppresses these LSAs from the backbone and, instead, injects the default route into the NSSA.)
- Advertise NSSA inter-area-prefix-LSAs to the backbone area.

Virtual links are not allowed for NSSAs.

## OSPFv3 AF router types

### Internal routers

Internal OSPFv3 AF routers belong to only one area. Internal routers flood router-LSAs to all routers in the same area and maintain identical LSDBs.

### Area border routers (ABRs)

Area border routers have membership in multiple areas. ABRs are used to connect the various areas in an AS to the backbone area for that AS. Multiple ABRs can be used to connect a given area to the backbone, and a given ABR can belong to multiple areas other than the backbone.

An ABR maintains a separate LSDB for each area to which it belongs. (All routers within the same area have identical LSDBs.) The ABR is responsible for flooding inter-area-prefix-LSAs and inter-area router LSAs between its border areas.

You can reduce summary LSA flooding by configuring area ranges. An area range enables you to assign an aggregate address to a range of IPv6 addresses. This aggregate address is advertised instead of all the individual addresses it represents. You can assign up to eight ranges in an OSPFv3 AF area.

### Autonomous system boundary router (ASBR)

Autonomous system boundary routers run one or more interior gateway protocols and serve as a gateway to other autonomous systems operating with interior gateway protocols. The ASBR imports and translates different protocol routes into OSPFv3 AF through redistribution. ASBRs can be used in backbone areas, normal areas, and NSSAs, but not in stub areas.

### Designated routers (DRs)

In an OSPFv3 AF network having two or more routers, one router is elected to serve as the designated router (DR) and another router to act as the backup designated router (BDR). All other routers in the same network segment forward their routing information to the DR and BDR, and the DR forwards this information to all routers in the network. This functionality minimizes the amount of repetitive information that is forwarded on the network by eliminating the need for each individual router in the area to forward its routing information to all other routers in the network. If the area includes multiple networks, each network elects its own DR and BDR.

In an OSPFv3 AF network with no DR and no BDR, the neighboring router with the highest priority is elected the DR and the router with the next highest priority is elected the BDR. If the DR goes off-line,

the BDR automatically becomes the DR, and the router with the next highest priority then becomes the new BDR. If multiple routing switches on the same OSPFv3 AF network are declaring themselves DRs, both priority and router ID are used to select the DR and BDR.

Priority is configurable using the `ospfv3-af ipv4 priority` or `ospfv3-af ipv6 priority` command at the interface level. You can use this command to help bias one router as the DR. If two neighbors share the same priority, the router with the highest router ID is designated the DR. The router with the next highest router ID is designated the BDR. Routers retain DR/BDR states throughout the period for which the adjacency is up and running.

## OSPFv3 AF configuration task list

### Tasks at a glance

- [Configuring OSPFv3 AF on the routing switch](#)
- [Assigning the routing switch to an OSPFv3 AF area](#)
- [Setting OSPFv3 AF network for the area](#)
- [Configuring external route redistribution and control](#)
- [Configuring area ranges on an ABR to reduce advertisements to the backbone](#)
- [Influencing route choice by changing the administrative distance](#)
- [Configuring graceful restart](#)
- [Configuring OSPFv3 AF virtual link settings](#)
- [Configuring OSPFv3 AF interface settings](#)
- [Configuring BFD for OSPFv3 AF](#)
- [Configuring all OSPFv3 AF interfaces as passive](#)
- [Viewing OSPFv3 AF information](#)
- [Clearing OSPFv3 AF statistics on a switch](#)

## Configuring OSPFv3 AF on the routing switch

### Prerequisites

- You must be in the global configuration context, as indicated by the `switch(config)#` prompt to create the OSPFv3 AF instance and enter the OSPFv3 AF configuration context.
- To configure a router ID, and other OSPFv3 AF configuration you must be in the OSPFv3 AF router configuration context, as indicated by the `switch(config-ospfv3-af-1)#` prompt.

Create the OSPFv3 AF instance and enter the OSPFv3 AF configuration context. From this context, you can proceed with other OSPFv3 AF configuration.

### Procedure

1. Create the OSPFv3 AF instance and enter the OSPFv3 AF configuration context using the following command. For command details, see [router ospfv3-af](#).  
`router ospfv3-af <PROCESS-ID> [vrf <VRF-NAME>]`



For example, the following command creates OSPF instance 1.

```
switch(config)# router ospfv3 af 1  
switch(config-ospfv3-af-1)#
```

For example, the following commands enter the IPv4, or IPv6 AF context respectively.

```
switch(config-ospfv3-af-1)# address-family ipv4 unicast  
switch(config-ospfv3-ipv4-uc)#  
  
switch(config-ospfv3-af-1)# address-family ipv6 unicast  
switch(config-ospfv3-ipv6-uc)#
```

2. Configure a global router ID using the following command. For command details, see [router-id](#).

```
router-id <ROUTER-ADDRESS>
```

For example, the following command sets the router ID to 1.1.1.1.

```
switch(config-ospfv3-af-1)# router-id 1.1.1.1
```

3. Optionally, if the OSPFv3 AF process was disabled (it is enabled by default), enable the OSPFv3 AF process using the following command.

```
enable
```

For command details, see [enable](#). (Refer to [disable](#) for disabling the OSPFv3 AF process).

## Assigning the routing switch to an OSPFv3 AF area

Create a Normal, Stub, or Not So Stubby (NSSA) area.

### Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

### Procedure

Create an OSPFv3 AF area for the routing switch using one of the following commands:

- Create a normal area using the following command: `area <AREA-ID>`. For command details, see [area](#).
- Create a stub area using the following command: `area <AREA-ID> stub`. For command details, see [area stub](#).
- Create a Not-So-Stubby-Area using the command: `area <AREA-ID> nssa`. For command details, see [area nssa](#).

For example, the following commands create either an IPv4 or IPv6 normal area with an area identifier of 1.1.1.1. Area identifier could alternatively be entered in decimal format such as 1.

```
switch(config-ospfv3-af-1)# address-family ipv4 unicast  
switch(config-ospfv3-ipv4-uc)# area 1  
switch(config-ospfv3-ipv4-uc)# area 1.1.1.1
```

```
switch(config-ospfv3-af-1)# address-family ipv6 unicast
```

```
switch(config-ospfv3-ipv6-uc) # area 1
switch(config-ospfv3-ipv6-uc) # area 1.1.1.1
```

## Setting OSPFv3 AF network for the area

### Prerequisites

Routing must be enabled on the interface where you are planning to enable OSPFv3 AF.

You must be in the interface configuration context, as indicated by the `switch(config-if) #` prompt.

After you define an OSPFv3 AF area, you can assign one or more networks to it. The OSPFv3 AF protocol will run on the interface with the configured link-local ipv6 address. The interfaces that have an IPv6 address configured in this network or in a subset of this network will participate in the OSPFv3 AF protocol.

### Procedure

1. Set an OSPFv3 AF network for the area using any of the following commands. For command details, see [ospfv3-af ipv4 area](#) or [ospfv3-af ipv6 area](#).

```
ospfv3-af <PROCESS-ID> ipv4 area <AREA-ID>
ospfv3-af <PROCESS-ID> ipv6 area <AREA-ID>
```

For example, use the following command to assign interface 1/1/1 to OSPFv3 AF area 1. The area can alternatively be entered in IO address format.

```
switch(config) # interface 1/1/1
switch(config-if) # ospfv3-af 1 ipv4 area 1
```

```
switch(config) # interface 1/1/1
switch(config-if) # ospfv3-af 1 ipv6 area 1
```

2. Optionally, you can disable OSPFv3 AF on the interface using any of the following commands. For command details, see [ospfv3-af ipv4 shutdown](#) or [ospfv3-af ipv6 shutdown](#).

```
ospfv3-af ipv4 shutdown
ospfv3-af ipv6 shutdown
```

```
switch(config) # interface 1
switch(config-if) # ospfv3-af ipv4 shutdown
```

```
switch(config) # interface 1
switch(config-if) # ospfv3-af ipv6 shutdown
```

## Configuring external route redistribution and control

Configuring route redistribution for OSPFv3 AF establishes the routing switch as an ASBR for importing and translating different protocol routes from other IGP domains into an OSPFv3 AF domain. When you configure redistribution for OSPFv3 AF, you can specify that routes external to the OSPFv3 AF domain are imported as OSPFv3 AF routes.

1. Enable route redistribution using the `redistribute` command.

```
redistribute {bgp | connected | host-routes | local loopback | static | ripng |  
ospfv3-af <PROCESS-ID>}  
[route-map <ROUTE-MAP-NAME>]
```

For example, setting redistribution of connected routes as OSPFv3 AF routes:

```
switch(config-ospfv3-ipv4-uc) # redistribute connected
```

If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv3 AF. For example, setting redistribution of local loopback routes, that only match the routes specified by the route map:

```
switch(config-ospfv3-ipv4-uc) # redistribute local loopback route-map local_  
routes
```

2. Optionally, modify the default metric for redistribution using the `default-metric` command.  
`default-metric <METRIC-VALUE>`

For example, setting the default metric for redistribution to 37.

```
switch(config-ospfv3-ipv4-uc) # default-metric 37
```

3. Optionally, set the cost of default-summary LSAs using the `area` command.

```
area <AREA-ID> default-metric <COST>
```

For example, setting the cost of default summary LSAs to 2.

```
switch(config-ospfv3-ipv4-uc) # area 1 default-metric 2
```

4. Optionally, use the `max-metric router-lsa` command to set the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations.

```
max-metric router-lsa [include-stub] | [on-startup <INTERVAL>]
```

For example, setting advertise max-metric router-lsa on startup.

```
switch(config-ospfv3-af-1) # max-metric router-lsa on-startup 3000
```

5. Optionally set the maximum number of ECMP routes that OSPFv3 can support using the `maximum-paths` command.

```
maximum-paths <MAXIMUM>
```

For example, setting the maximum number of ECMP routes to 8.

```
switch(config-ospfv3-ipv4-uc) # maximum-paths 8
```

## Configuring area ranges on an ABR to reduce advertisements to the backbone

You can configure area ranges to reduce inter-area advertisements by summarizing a range of IP addresses into a single route advertisement. This action prevents an ABR from advertising specific networks or subnets to the backbone area.

## Prerequisites

You must be in the address-family configuration context, as indicated by the `switch(config-ospfv3-ipv4-uc)#` prompt.

## Procedure

Summarize inter-area or NSSA paths using the following command. For command details, see [area range](#).

```
area <AREA-ID> range <IP-PREFIX> type {inter-area | nssa} [no-advertise]
```

For example, use any of the following commands to summarize routes matching the area range fd00::64 using inter-area as the type of address aggregation.

```
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 range fd00::/64 type inter-area
```

```
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 range fd00::/64 type inter-area
```

In another example, use the following command to specify DoNotAdvertise status for routes matching the area range 10.1.0.0/16. Use nssa as the type of address aggregation.

```
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 range 10.1.0.0/16 type inter-area no
advertise
```

## Influencing route choice by changing the administrative distance

### Prerequisites

You must be in the OSPFv3 AF router configuration context, as indicated by the `switch(config-ospfv3-af-1)#` prompt.

The administrative distance is used to influence the selection of routes learned by different protocols.

### Procedure

Reconfigure the administrative distance using the following command. For command details, see [distance](#).

```
distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area [<DISTANCE-VAL>] |
external [<DISTANCE-VAL>]]
```

For example, use the following command to set administrative distance to 100.

```
switch(config-ospfv3-ipv4-uc)# distance 100
```

# Configuring graceful restart

## Prerequisites

You must be in the OSPFv3 address-family configuration context, as indicated by the `switch(config-ospfv3-ipv4-uc)#` prompt.

OSPFv3 AF routing can be gracefully restarted on switches without losing packets that are in transit. There is no effect on the saved switch configuration.

Manually create a router-id configuration to support graceful restart and management module failover. A dynamic router-id is chosen from the list of configured IPs at the time of the router-id election, and in the event of a graceful restart and management module failover, the router-id should be preserved before and after these events. Graceful restart would fail if there is a change in the router-id, causing traffic loss.

## Procedure

Configure graceful restart of using the following command. For command details, see [graceful-restart](#).  
`graceful-restart {restart-interval <INTERVAL> | helper [strict-lsa-check]}`

For example, the following command specifies 50 seconds as the maximum interval another router will wait for this router to gracefully restart.

```
switch(config-ospfv3-ipv4-uc)# graceful-restart restart-interval 50
```

# Configuring OSPFv3 AF virtual link settings

You must be in the router vlink configuration context, as indicated by the `switch(config-router-vlink)#` prompt.

The OSPFv3 AF interface parameters for this process are automatically set to their default values for virtual links. You can optionally adjust the following OSPFv3 AF virtual link settings.

1. Set the time interval between OSPF hello packets for the OSPFv3 AF virtual links using the following command. For command details, see [hello-interval](#).

```
hello-interval <seconds>
```

For example:

```
switch(config)# router ospfv3-af 1
switch(config)# address-family ipv6 unicast
switch(config-ospfv3-ipv4-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# hello-interval 30
```

2. Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPFv3 AF virtual links. Use the following command. For command details, see [dead-interval](#).  
`dead-interval <seconds>`
3. Set the time between retransmitting lost link state advertisements for the OSPFv3 AF virtual links using the following command. For command details, see [retransmit-interval](#).  
`retransmit-interval <seconds>`
4. Sets the transit delay in Link state transmission for the OSPFv3 AF virtual links using the following command. For command details, see [transit-delay](#).  
`transit-delay <seconds>`

# Configuring OSPFv3 AF interface settings

## Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt. You can optionally adjust the following OSPFv3 AF interface settings.

1. Set the interface cost using any of the following commands. For command details, see [ospfv3-af ipv4 cost](#) or [ospfv3-af ipv6 cost](#).

```
ospfv3-af ipv4 cost <INTERFACE-COST>
ospfv3-af ipv6 cost <INTERFACE-COST>
```

For example, the following command sets the cost associated with interface 1/1/1 to 100.

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 cost 100
```

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv6 cost 100
```

2. Set the time interval between OSPFv3 AF hello packets for the OSPFv3 AF interface using the following command. For command details, see [ospfv3-af ipv6 hello-interval](#).  
`ospfv3-af ipv6 hello-interval <INTERVAL>`
3. Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPFv3 AF interface using any of the following command. For command details, see [ospfv3-af ipv4 dead-interval](#) or [ospfv3-af ipv6 dead-interval](#).  
`ospfv3-af ipv4 dead-interval <INTERVAL>`  
`ospfv3-af ipv6 dead-interval <INTERVAL>`
4. Set the time between re-transmitting lost link state advertisements for the OSPFv3 AF interface using any of the following command. For command details, see [ospfv3-af ipv4 retransmit-interval](#) or [ospfv3-af ipv6 retransmit-interval](#).  
`ospfv3-af ipv4 retransmit-interval <INTERVAL>`  
`ospfv3-af ipv6 retransmit-interval <INTERVAL>`
5. Sets the time delay in Link state transmission for the OSPFv3 AF interface using any of the following command. For command details, see [ospfv3-af ipv4 transit-delay](#) or [ospfv3-af ipv6 transit-delay](#).  
`ospfv3-af ipv4 transit-delay <INTERVAL>`  
`ospfv3-af ipv6 transit-delay <INTERVAL>`
6. Set the OSPFv3 AF network type for the interface. For command details, see [ospfv3-af ipv4 network](#) or [ospfv3-af ipv6 network](#).  
`ospfv3-af ipv4 network {broadcast|point-to-point}`  
`ospfv3-af ipv6 network {broadcast|point-to-point}`
7. Set the OSPFv3 AF priority on the interface using any of the following command. For command details, see [ospfv3-af ipv4 priority](#) or [ospfv3-af ipv6 priority](#).  
`ospfv3-af ipv4 priority <number-value>`  
`ospfv3-af ipv6 priority <number-value>`
8. Set the OSPFv3 AF interface as OSPFv3 AF passive interface. With this setting the interface participates in OSPFv3 AF but does not send or receive packets on that interface. For command details, see [ospfv3-af ipv4 passive](#) or [ospfv3-af ipv6 passive](#).  
`ospfv3-af ipv4 passive`  
`ospfv3-af ipv6 passive`

# Configuring BFD for OSPFv3 AF

Enabling BFD Support for OSPFv3 AF over IPv6 enables OSPFv3 AF to register with Bidirectional Forwarding Detection (BFD) to receive forwarding path detection failure messages. You can either configure BFD for OSPFv3 AF globally on all interfaces or configure it selectively on one or more interfaces. BFD creates a session in asynchronous mode as soon as the switch reaches the 2-Way state with a neighbor. Once the session is established BFD will commence sending echos for path failure detection.

OSPFv3 AF uses IPv6 for neighborship in both IPv4 and IPv6 unicast address-families. Therefore, only a single BFD session is created to monitor the IPv6 link-local address, even when both address-families are enabled on the same interface.

There are two methods for enabling BFD for OSPFv3 AF:

1. Enable BFD for all interfaces enabled for OSPFv3 AF by using the BFD all-interfaces command in address-family configuration mode
2. Enable BFD for a subset of interfaces that have OSPFv3 AF enabled by using the `ospfv3-af bfd` command in interface configuration mode



---

OSPFv3 AF needs to be enabled prior to enabling BFD on one or more interfaces.

---

## Examples

### Enable BFD on all OSPFv3 AF Interfaces

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# bfd all-interfaces
```

### Disable BFD on all OSPFv3 AF Interfaces

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no bfd all-interfaces
```

### Enable BFD on a single OSPFv3 AF interface

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 bfd
switch(config-if)# ospfv3-af ipv6 bfd
```

### Disable BFD on a single OSPFv3 AF interface

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 bfd disable
switch(config-if)# ospfv3-af ipv6 bfd disable
```

### Set BFD as default on an OSPFv3 AF interface

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv4 bfd
switch(config-if)# no ospfv3-af ipv6 bfd
```

## Configuring all OSPFv3 AF interfaces as passive

### Prerequisites

You must be in the OSPFv3 address-family configuration context, as indicated by the `switch(config-ospfv3-ipv4-uc)#` prompt.

OSPFv3 AF sends LSAs to all other routers in the same AS. To limit the flooding of LSAs throughout the AS, you can configure OSPFv3 AF to be passive.

### Procedure

Configure all OSPFv3 AF interfaces as passive using the following command. For command details, see [passive-interface default](#).

```
passive-interface default
```

```
switch(config-ospfv3-ipv4-uc)# passive-interface default
switch(config-ospfv3-ipv6-uc)# passive-interface default
```

## Configuring SPF throttling timers

SPF calculation is throttled with default timer values (start-time 200ms, hold-time 1000ms, max-wait-time 5000ms). You can throttle SPF calculation by configuring non-default timers to improve performance of a specific network configuration.

### Prerequisites

SPF throttling timers can be configured on any of the following contexts:

**Address-family configuration context**, as indicated by the `switch(config-ospfv3-ipv4-uc)#` or `switch(config-ospfv3-ipv6-uc)#` prompt.

**Router configuration context**, as indicated by the `switch(config-router)#` prompt.

### Procedure

Reconfigure SPF throttling timers using the following command. For command details, see [timers throttle spf](#).

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds> max-wait-time <milliseconds>
```

For example, use the following command to set start-time to 500ms, hold-time to 3000ms and max-wait-time to 9000ms:

```
switch(config-ospfv3-ipv4-uc)# timers throttle spf start-time 500 hold-time 3000 max-wait-time 9000
```

## Viewing OSPFv3 AF information



## Prerequisites

Use these show commands from the Manager context, as indicated by the `switch#` prompt.

## Procedure

To view OSPFv3 AF information, use the following commands. For command details and examples, click the links.

- To view general OSPFv3 AF, area, state and configuration information use: [show ospfv3-af ipv4](#) or [show ospfv3-af ipv6](#).
- To view information about OSPFv3 AF interfaces use: [show ospfv3-af ipv4 interface](#) or [show ospfv3-af ipv6 interface](#).
- To view information about OSPFv3 AF neighbors use: [show ospfv3-af ipv4 neighbors](#) or [show ospfv3-af ipv6 neighbors](#).
- To view OSPFv3 AF routing table information use: [show ospfv3-af ipv4 routes](#) or [show ospfv3-af ipv6 routes](#).
- To view OSPFv3 AF statistics for the OSPFv3 AF interfaces use [show ospfv3-af ipv4 statistics interface](#) or [show ospfv3-af ipv6 statistics interface](#).

## Clearing OSPFv3 AF statistics on a switch

Clear OSPFv3 AF event statics using the following command. For command details, see [clear ospfv3-af ipv4 statistics](#) or [clear ospfv3-af ipv6 statistics](#). [clear ipv6 ospfv3 statistics](#)

```
clear ospfv3-af [<PROCESS-ID>] <address-family> statistics [interface [interface  
<INTERFACE-NAME> | vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID> | tunnel  
<TUNNEL-ID>] [all-vrfs | vrf <VRF-NAME>]
```

For example, the following command clears OSPFv3 AF event statistics from interface 1/1/1.

```
switch# clear ospfv3-af ipv4 statistics interface 1/1/1
```

```
switch# clear ospfv3-af ipv6 statistics interface 1/1/1
```

## OSPFv3 AF commands

### active-backbone stub-default-route

`active-backbone stub-default-route`

#### Description

This command enables the router to send a default route to stub areas if there is an active loopback link in the backbone area. The configuration is not required if backbone area has neighbors or passive interfaces configured. By default, active backbone detection is disabled.

| Parameter                       | Description                                                   |
|---------------------------------|---------------------------------------------------------------|
| <code>stub-default-route</code> | Advertise default route to stub if loopback in backbone area. |

## Examples

Enabling the router to send an IPv4 default route to stub areas:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-af-1)# address-family ipv4 unicast  
switch(config-ospfv3-ipv4-uc)# active-backbone stub-default-route
```

Enabling the router to send an IPv6 default route to stub areas:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-af-1)# address-family ipv6 unicast  
switch(config-ospfv3-ipv6-uc)# active-backbone stub-default-route
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## address-family unicast (OSPFv3-AF)

```
address-family { <ipv4> | <ipv6> } unicast
```

### Description

This command specifies the IPv4/IPv6, address-family and enters address family configuration mode.

## Examples

Specifying an IPv4 address-family:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-af-1)# address-family ipv4 unicast
```

Specifying an IPv6 address-family:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-af-1)# address-family ipv6 unicast
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context               | Authority                                                                          |
|--------------|-------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-af-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## area default-metric (OSPFv3 AF)

```
area <AREA-ID> default-metric <METRIC>
no area <AREA-ID> default-metric
```

### Description

Sets the cost of default-summary LSAs announced to the stub/nssa areas.

The **no** form of this command resets the cost of the default-summary LSAs announced to stub/nssa areas to the default of 1.

| Parameter               | Description                                                                                                                                                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID>               | Specifies the area ID as one of the following formats.<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |
| default-metric <METRIC> | Specifies the default metric of default-summary LSAs announced to the stub/nssa areas, to the specified value. Default: 1. Range: 0 to 16777215.                                                                                       |

### Examples

Setting cost for default IPv4 LSA summary:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 default-metric 2
switch(config-ospfv3-ipv4-uc)# area 1.1.1.1 default-metric 2
```

Setting cost for default IPv4 LSA summary to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no area 1 default-metric
switch(config-ospfv3-ipv4-uc)# no area 0.0.0.1 default-metric
switch(config-ospfv3-ipv4-uc)# no area 0.0.0.1 default-metric 2
```

Setting cost for default IPv6 LSA summary:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 default-metric 2
switch(config-ospfv3-ipv6-uc)# area 1.1.1.1 default-metric 2
```

Setting cost for default IPv6 LSA summary to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# no area 1 default-metric
switch(config-ospfv3-ipv6-uc)# no area 0.0.0.1 default-metric
switch(config-ospfv3-ipv6-uc)# no area 0.0.0.1 default-metric 2
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## area nssa

```
area <AREA-ID> nssa [no-summary]
no area <AREA-ID> nssa [no-summary]
```

## Description

This command sets the area type as NSSA. If area is present and not NSSA area, changes the area type to NSSA area. If **no-summary** is configured the area type will be NSSA No-Summary.

| Parameter  | Description                                                                                                                                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID>  | Specifies the area ID is one of the following formats.<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295.<br><i>Required</i> |
| nssa       | Configure OSPFv3 AF area as NSSA.<br><i>Required</i>                                                                                                                                                                                                      |
| no-summary | Do not inject inter-area routes into NSSA.<br><i>Optional</i>                                                                                                                                                                                             |

| Parameter | Description                                                                                                                                                                                                                                                                                          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| no        | Unsets the area type as NSSA. This means that the configured area becomes an area type default. The <b>no area area-id nssa no-summary</b> command will stop sending Area Border Router(ABR) summary link advertisements into the NSSA area, but will not unset the area as NSSA.<br><i>Optional</i> |

## Examples

Creating an NSSA area for OSPFv3 AF:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 nssa
switch(config-ospfv3-ipv4-uc)# area 1 nssa no-summary
```

Clearing the NSSA area for OSPFv3 AF:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# no area 1 nssa
switch(config-ospfv3-ipv6-uc)# no area 1 nssa no-summary
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## area (OSPFv3 AF)

```
area <AREA-ID>
no area <AREA-ID>
```

### Description

Creates a normal area with <AREA-ID> set if not present. If area is present and is not the normal area, this command changes the area type to normal area.

The **no** form of this command deletes the area with the <AREA-ID> specified. The area can be of any type (stub, stub no-summary, and default normal area).

| Parameter | Description                                                                                                                                                                                                                                      |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID> | Specifies the area ID is one of the following formats.<br>OSPFv3 AF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPFv3 AF area identifier in decimal format. Range: 0 to 4294967295. |

## Examples

Creating an IPv4 normal area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1
switch(config-ospfv3-ipv4-uc)# area 1.1.1.1
```

Creating an IPv6 normal area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1.1.1.1
```

Deleting an IPv4 area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no area 1
switch(config-ospfv3-ipv4-uc)# no area 0.0.0.1
```

Deleting an IPv6 area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# no area 1
switch(config-ospfv3-ipv6-uc)# no area 1.1.1.1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## area range (OSPFv3 AF)

```
area <AREA-ID> range <IP-PREFIX> type {inter-area | nssa} [no-advertise]
no area <AREA-ID> range <IP-PREFIX> type {inter-area | nssa} [no-advertise]
```

### Description

Summarizes the routes with the matching address or masks for OSPFv3 AF. This command only works for border routers.

The **no** form of this command unsets the route summarization for the configured IP prefix address on the ABR. When using the **no** form of the command with the **no-advertise** option, enables advertising this range to other areas.

| Parameter                | Description                                                                                                                                                                                                                                              |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID>                | Specifies the area identifier as one of the following formats.<br>OSPFv3 AF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPFv3 AF area identifier in decimal format. Range: 0 to 4294967295. |
| range <IP-PREFIX>        | Specifies summarizing routes matching the area range prefix/mask in IPv4 or IPv6 format.                                                                                                                                                                 |
| type {inter-area   nssa} | Specifies the type this address aggregation applies to as either inter-area range prefix or NSSA range prefix.                                                                                                                                           |
| no-advertise             | Specifies the address range status as <b>DoNotAdvertise</b> (do not advertise this range to other areas).                                                                                                                                                |

### Examples

Summarizing inter-area or NSSA paths on OSPFv3 AF for IPv4:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 range 10.1.0.0/16 type inter-area
switch(config-ospfv3-ipv4-uc)# area 1 range 10.1.0.0/16 type inter-area no
advertise
switch(config-ospfv3-ipv4-uc)# area 1 range 10.1.0.0/16 type nssa
switch(config-ospfv3-ipv4-uc)# area 1 range 10.1.0.0/16 type nssa no-advertise
```

Unsetting summarization on OSPFv3 AF for IPv4:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no area 1 range 10.1.0.0/16 type inter-area
switch(config-ospfv3-ipv4-uc)# no area 1 range 10.1.0.0/16 type inter-area no
advertise
switch(config-ospfv3-ipv4-uc)# no area 1 range 10.1.0.0/16 type nssa
```

```
switch(config-ospfv3-ipv4-uc) # no area 1 range 10.1.0.0/16 type nssa no-advertise
```

Summarizing inter-area or NSSA paths on OSPFv3 AF for IPv6:

```
switch(config) # router ospfv3-af 1
switch(config-ospfv3-af-1) # address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc) # area 1 range fd00::1/64 type inter-area
switch(config-ospfv3-ipv6-uc) # area 1 range fd00::1/64 type inter-area no
advertise
switch(config-ospfv3-ipv6-uc) # area 1 range fd00::1/64 type nssa
switch(config-ospfv3-ipv6-uc) # area 1 range fd00::1/64 type nssa no-advertise
```

Unsetting summarization on OSPFv3 AF for IPv6:

```
switch(config) # router ospfv3-af 1
switch(config-ospfv3-af-1) # address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc) # no area 1 range fd00::1/64 type inter-area
switch(config-ospfv3-ipv6-uc) # no area 1 range fd00::1/64 type inter-area no
advertise
switch(config-ospfv3-ipv6-uc) # no area 1 range fd00::1/64 type nssa
switch(config-ospfv3-ipv6-uc) # no area 1 range fd00::1/64 type nssa no-advertise
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms | Command context                   | Authority                                                                          |
|-----------|-----------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-ospfv3-ipv4-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |
| 6200      | config-ospfv3-ipv6-uc<PROCESS-ID> |                                                                                    |

## area stub (OSPFv3 AF)

```
area <AREA-ID> stub [no-summary]
no area <AREA-ID> stub [no-summary]
```

## Description

Creates the stub area with <AREA-ID> if not present. If area is present and is not the stub area, this command changes the area type to stub area. If **no-summary** is used, area type will be totally stubby area.

The **no** form of this command unsets the area type as stub. The configured area will be changed to the default normal area. The **no area <AREA-ID> stub no-summary** command will start sending Area Border Router (ABR) summary link advertisements into the stub area, but will not unset the stub area.





ABR does not inject the default route in a Totally Stubby Area with loopback in Area 0.0.0.0. As a workaround, configure a passive interface or active neighbors in the backbone area.

| Parameter                      | Description                                                                                                                                                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;AREA-ID&gt;</code>   | Specifies the area ID as one of the following formats.<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295.              |
| <code>stub [no-summary]</code> | Specifies stub area type. If area is present and not stub area, this parameter changes the area type to stub area. If <b>no-summary</b> is specified, area type will be totally stubby area, which means do not inject inter-area routes into stub. |

## Examples

Creating an IPv4 stub area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 stub
switch(config-ospfv3-ipv4-uc)# area 1 stub no-summary
```

Unsetting the IPv4 stub area type:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc) # no area 1 stub
switch(config-ospfv3-ipv4-uc) # no area 1 sub no-summary
```

Creating an IPv6 stub area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 stub
switch(config-ospfv3-ipv6-uc)# area 1 stub no-summary
```

Unsetting the IPv6 stub area type:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc) # no area 1 stub
switch(config-ospfv3-ipv6-uc) # no area 1 sub no-summary
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms | Command context                   | Authority                                                                          |
|-----------|-----------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-ospfv3-ipv4-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |
| 6200      | config-ospfv3-ipv6-uc<PROCESS-ID> |                                                                                    |

## area virtual-link (OSPFv3 AF)

```
area <AREA-ID> virtual-link <ROUTER-ID>
no area <AREA-ID> virtual-link <ROUTER-ID>
```

### Description

Creates an OSPF virtual link with remote ABR (if not created already) and enters the vlink context. The configuration is supported only in ipv6-unicast address-family.

The **no** form of this command deletes an OSPF virtual link with the specified router ID of the remote ABR. If no **<ROUTER-ID>** is specified, the **no** form of the command sets the virtual link to the default settings.

| Parameter                | Description                                                                                                                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID>                | Specifies the area ID is one of the following formats.<br>OSPF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |
| virtual-link <ROUTER-ID> | Configures a virtual link with the specified router ID of the remote ABR.                                                                                                                                                              |

### Examples

Configuring OSPFv3 AF virtual links:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
```

Deleting OSPFv3 AF virtual links:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# no area 100 virtual-link 100.0.1.1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## authentication ipsec spi

authentication ipsec spi <SPI-INDEX> <AUTH-TYPE> [<KEY-TYPE> <AUTH-KEY>]  
no authentication

### Description

Configures IPsec AH authentication for the selected Vlink.

The **no** form of this command removes IPsec AH authentication for the selected Vlink.

| Parameter       | Description                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spi <SPI-INDEX> | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| <AUTH-TYPE>     | Specifies the authentication type: <b>md5</b> or <b>sha1</b> .                                                                                                                                                                                                                                  |
| <KEY-TYPE>      | Specifies the key type to use: <b>plaintext</b> (unencrypted), <b>hex-string</b> (encrypted) or <b>ciphertext</b> (encrypted).                                                                                                                                                                  |
| <AUTH-KEY>      | Specifies the authentication key.                                                                                                                                                                                                                                                               |



When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

### Examples

Setting area 1 to use IPsec AH authentication for Vlink with provided plaintext key:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# authentication ipsec spi 256 sha1 plaintext abcd
```

Setting area 1 to use IPsec AH authentication for Vlink with prompted plaintext key:

```
switch(config)# router ospfv3-af 1
```

```

switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# authentication ipsec spi 256 sha1
Enter the IPsec authentication key: *****
Re-Enter the IPsec authentication key: *****

```

Removing IPsec AH authentication for Vlink on area 1:

```

switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# no authentication
switch(config-router-vlink6)# no authentication ipsec spi 1234 md5 ciphertext
qwertyuioplkjhgfsazxcvbnmkjhgfsa
switch(config-router-vlink6)# no authentication ipsec spi 1234 sha1 ciphertext
qwertyuioplkjhgfsazxcvbnmkjhgfsa

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |

## clear ospfv3-af ipv4 neighbor

```

clear ospfv3-af [<PROCESS-ID>] ipv4 neighbor [<NEIGHBOR-ID>] [interface <INTERFACE-NAME>
| vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>] [all-vrfs
| vrf <VRF-NAME>]

```

## Description

Resets the neighbor and clears the OSPFv3 AF neighbor information.

| Parameter                  | Description                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies the OSPFv3 AF process ID to clear the statistics for the particular OSPFv3 AF process. Range: 1 to 65535. |
| neighbor <NEIGHBOR-ID>     | Specifies the router ID of a neighbor.                                                                              |
| interface <INTERFACE-NAME> | Specifies the OSPFv3 AF statistics to clear for the specified interface.                                            |

| Parameter              | Description                                            |
|------------------------|--------------------------------------------------------|
| vlan <VLAN-ID>         | Specifies the vlan interface name.                     |
| lag <LAG-ID>           | Specifies the lag interface name.                      |
| loopback <LOOPBACK-ID> | Specifies the loopback interface name.                 |
| tunnel <TUNNEL-ID>     | Specifies the tunnel interface name.                   |
| all-vrfs               | Select to clear the OSPFv3 AF statistics for all VRFs. |
| vrf <VRF-NAME>         | Specifies the name of a VRF.                           |

## Example

Clearing the OSPFv3 AF neighbor information:

```
switch# clear ospfv3-af 1 ipv4 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv4 neighbor 3.3.3.3
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv4 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv4 neighbor 3.3.3.3 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor 5.5.5.5
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor 5.5.5.5 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear ospfv3-af ipv4 statistics

```
clear ospfv3-af [<PROCESS-ID>] ipv4 statistics [interface [interface <INTERFACE-NAME> |
vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>] [all-vrfs |
vrf <VRF-NAME>]
```

### Description

Clears the OSPFv3 AF event statistics.

| Parameter                  | Description                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies the OSPFv3 AF process ID to clear the statistics for the particular OSPFv3 AF process. Range: 1 to 65535. |
| interface <INTERFACE-NAME> | Specifies the OSPFv3 AF statistics to clear for the specified interface.                                            |
| vlan <VLAN-ID>             | Specifies the vlan interface name.                                                                                  |
| lag <LAG-ID>               | Specifies the lag interface name.                                                                                   |
| loopback <LOOPBACK-ID>     | Specifies the loopback interface name.                                                                              |
| tunnel <TUNNEL-ID>         | Specifies the tunnel interface name.                                                                                |
| all-vrfs                   | Select to clear the OSPFv3 AF statistics for all VRFs.                                                              |
| vrf <VRF-NAME>             | Specifies the name of a VRF.                                                                                        |

### Example

Clearing the OSPFv3 AF event statistics:

```
switch# clear ospfv3-af ipv4 statistics
switch# clear ospfv3-af ipv4 statistics interface 1/1/1
switch# clear ospfv3-af ipv4 statistics interface 1/1/1 vrf vrf_red
switch# clear ospfv3-af ipv4 statistics interface tunnell
switch# clear ospfv3-af ipv4 statistics interface tunnell vrf vrf_red
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

### Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear ospfv3-af ipv6 neighbor

```
clear ospfv3-af [<PROCESS-ID>] ipv6 neighbor [<NEIGHBOR-ID>] [interface <INTERFACE-NAME>
| vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>] [all-vrfs
| vrf <VRF-NAME>]
```

### Description

Resets the neighbor and clears the OSPFv3 AF neighbor information.

| Parameter                  | Description                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies the OSPFv3 AF process ID to clear the statistics for the particular OSPFv3 AF process. Range: 1 to 65535. |
| neighbor <NEIGHBOR-ID>     | Specifies the router ID of a neighbor.                                                                              |
| interface <INTERFACE-NAME> | Specifies the OSPFv3 AF statistics to clear for the specified interface.                                            |
| vlan <VLAN-ID>             | Specifies the vlan interface name.                                                                                  |
| lag <LAG-ID>               | Specifies the lag interface name.                                                                                   |
| loopback <LOOPBACK-ID>     | Specifies the loopback interface name.                                                                              |
| tunnel <TUNNEL-ID>         | Specifies the tunnel interface name.                                                                                |
| all-vrfs                   | Select to clear the OSPFv3 AF statistics for all VRFs.                                                              |
| vrf <VRF-NAME>             | Specifies the name of a VRF.                                                                                        |

### Example

Clearing the OSPFv3 AF neighbor information:

```
switch# clear ospfv3-af 1 ipv6 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv6 neighbor 3.3.3.3
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv6 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv6 neighbor 3.3.3.3 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor 5.5.5.5
```

```

Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor 5.5.5.5 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear ospfv3-af ipv6 statistics

clear ospfv3-af [<PROCESS-ID>] ipv6 statistics [interface [interface <INTERFACE-NAME> | vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>] [all-vrfs | vrf <VRF-NAME>]

## Description

Clears the OSPFv3 AF event statistics.

| Parameter                  | Description                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies the OSPFv3 AF process ID to clear the statistics for the particular OSPFv3 AF process. Range: 1 to 65535. |
| interface <INTERFACE-NAME> | Specifies the OSPFv3 AF statistics to clear for the specified interface.                                            |
| vlan <VLAN-ID>             | Specifies the vlan interface name.                                                                                  |
| lag <LAG-ID>               | Specifies the lag interface name.                                                                                   |
| loopback <LOOPBACK-ID>     | Specifies the loopback interface name.                                                                              |
| tunnel <TUNNEL-ID>         | Specifies the tunnel interface name.                                                                                |
| all-vrfs                   | Select to clear the OSPFv3 AF statistics for all VRFs.                                                              |
| vrf <VRF-NAME>             | Specifies the name of a VRF.                                                                                        |



## Example

Clearing the OSPFv3 AF event statistics:

```
switch# clear ospfv3-af ipv6 statistics
switch# clear ospfv3-af ipv6 statistics interface 1/1/1
switch# clear ospfv3-af ipv6 statistics interface 1/1/1 vrf vrf_red
switch# clear ospfv3-af ipv6 statistics interface tunnell
switch# clear ospfv3-af ipv6 statistics interface tunnell vrf vrf_red
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## dead-interval (OSPFv3 AF)

```
dead-interval <INTERVAL>
no dead-interval
```

### Description

Sets the interval after which a neighbor is declared dead if no hello packet comes in for virtual links.

The **no** form of this command sets the dead interval to default for virtual links. The default value is 40 seconds (generally four times the hello packet interval).



For proper operation, set the dead interval higher than the hello interval.

| Parameter  | Description                                                                                    |
|------------|------------------------------------------------------------------------------------------------|
| <INTERVAL> | Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40. |

## Examples

Setting OSPFv3 AF virtual links dead interval:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
```

```
switch(config-router-vlink6)# dead-interval 30
```

Setting OSPFv3 AF virtual links dead interval to default:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-af-1)# address-family ipv6 unicast  
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1  
switch(config-router-vlink6)# no dead-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |

## debug ospfv3-af

```
debug ospfv3-af <SUBMODULE-NAME> [instance <PROCESS-ID>] [ip <IP-ADDRESS>] [port <IF-NAME>] [vrf <VRF-NAME>] [severity {alert | crit | debug | emer | err | info | notice | warning}]  
no debug ospfv3-af <SUBMODULE-NAME> [instance <PROCESS-ID>] [ip <IP-ADDRESS>] [port <IF-NAME>] [vrf <VRF-NAME>] [severity {alert | crit | debug | emer | err | info | notice | warning}]
```

## Description

Enables problem logs to the debug logs. If <SEVERITY> is omitted, all severities are logged.

The **no** form of this command disables problem logs to the debug logs.

| Parameter             | Description                                                                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <SUBMODULE-NAME>      | Enables debug logging for a specific submodule. For a list of supported submodules, enter the debug <MODULE-NAME> command followed by a space and a question mark (?). |
| instance <PROCESS-ID> | Displays debug logs for the specified OSPFv3 AF process ID. Range: 1 to 65535.                                                                                         |
| ip <IP-ADDRESS>       | Displays debug logs for the specified IP Address.                                                                                                                      |

| Parameter                                                | Description                                                                                             |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| port <INTERFACE-NAME>                                    | Displays debug logs for the OSPFv3 AF specified interfaces.                                             |
| vrf <VRF-NAME>                                           | Displays debug logs for the specified VRF name.                                                         |
| severity {alert crit debug emer err info notice warning} | Enables the debug logs for OSPv3 AF based on the required severity level for the destination. Optional. |

## Examples

Enables debug logging of event submodules for port 1/1/1 and debug severity:

```
switch# debug ospfv3-af ipv4 event port 1/1/1 severity debug
```

Enables all debug logging for VRF red:

```
switch# debug ospfv3-af ipv4 all vrf vrf_red
```

Disables all OSPFv3 AF debug logging:

```
switch# no debug ospfv3-af ipv4 all
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## default-information originate always (OSPFv3 AF)

```
default-information originate always [metric <METRIC-VALUE>]
no default-information originate always [metric <METRIC-VALUE>]
```

### Description

Configures OSPFv3 AF to advertise the default route (::/0) to its neighbors, regardless if it is present in the routing table or not. Optionally, metric can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

| Parameter                    | Description                                                        |
|------------------------------|--------------------------------------------------------------------|
| <i>metric</i> <METRIC-VALUE> | Specifies the OSPF metric value for the default route. Default: 1. |

## Examples

Setting advertisement of the default route:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no default-information originate always
```

Setting advertisement of the default route with metric set to 20:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# default-information originate always metric 20
```

Disabling advertisement of the default route and setting the metric to the default value:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no default-information originate always metric
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                   | Authority                                                                          |
|--------------|-----------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## default-information originate (OSPFv3 AF)

```
default-information originate [metric <METRIC-VALUE>]
no default-information originate [metric <METRIC-VALUE>]
```

## Description

Configures OSPFv3 AF to advertise the default route (::/0) to its neighbors if it is present in the routing table. Optionally, the metric value can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

| Parameter                                | Description                                                                  |
|------------------------------------------|------------------------------------------------------------------------------|
| <code>metric &lt;METRIC-VALUE&gt;</code> | Specifies the OSPF metric value for the default route. Optional. Default: 1. |

## Examples

Setting advertisement of the default route and specifying an optional metric value of 20:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# default-information originate
switch(config-ospfv3-ipv4-uc)# default-information originate metric 20
```

Disabling advertisement of the default route:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no default-information originate metric
switch(config-ospfv3-ipv4-uc)# no default-information originate
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                   | Authority                                                                          |
|--------------|-----------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## default-metric (OSPFv3 AF)

```
default-metric <METRIC-VALUE>
no default-metric
```

## Description

Sets the default metric for redistributed routes in the OSPFv3 AF.

The **no** form of this command sets the default metric to be used for redistributed routes into OSPFv3 AF to the default of 25.

| Parameter                         | Description                                                                                           |
|-----------------------------------|-------------------------------------------------------------------------------------------------------|
| <code>&lt;METRIC-VALUE&gt;</code> | Specifies the default metric value to use for redistributed routes. Range: 1 to 1677214. Default: 25. |

## Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# default-metric 36
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1) # address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc) # no default-metric
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                   | Authority                                                                          |
|--------------|-----------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## diag ospfv3 lsdb dump

diag ospfv3 lsdb dump

## Description

This command dumps the OSPFv3 AF protocol LSA information.

## Examples

```
switch# diag ospfv3 lsdb dump
ospfv3PmAreaLsdbTable row:
ospfv3PmAreaLsdbApplIndex:      537265152
ospfv3PmAreaLsdbAreaId:        0x00000000
ospfv3PmAreaLsdbType:          ospfv3RouterLsa
ospfv3PmAreaLsdbRouterId:      0x01010103
ospfv3PmAreaLsdbLsid:          0x00000000
```

```

ospfv3PmAreaLsdbSequence:      -2147483646
ospfv3PmAreaLsdbAge:           872
ospfv3PmAreaLsdbChecksum:      12468
ospfv3PmAreaLsdbAdvertisement:  0x00 0x01 0x20 0x01 0x00 0x00 0x00 0x00
                                0x01 0x01 0x01 0x03 0x80 0x00 0x00 0x02
                                0x30 0xB4 0x00 0x28 0x01 0x00 0x00 0x13
                                0x02 0x00 0x00 0x01 0x10 0x01 0x01 0x01
                                0x10 0x01 0x01 0x01 0x03 0x03 0x03 0x03
ospfv3PmAreaLsdbTypeKnown:     true

ospfv3PmAreaLsdbTable row:
ospfv3PmAreaLsdbApplIndex:      537265152
ospfv3PmAreaLsdbAreaId:         0x00000000
ospfv3PmAreaLsdbType:           ospfv3RouterLsa
ospfv3PmAreaLsdbRouterId:       0x03030303
ospfv3PmAreaLsdbLsid:           0x00000000
ospfv3PmAreaLsdbSequence:      -2147483646
ospfv3PmAreaLsdbAge:           873
ospfv3PmAreaLsdbChecksum:       64738
ospfv3PmAreaLsdbAdvertisement:  0x00 0x01 0x20 0x01 0x00 0x00 0x00 0x00
                                0x03 0x03 0x03 0x03 0x80 0x00 0x00 0x02
                                0xFC 0xE2 0x00 0x28 0x00 0x00 0x00 0x13
                                0x02 0x00 0x00 0x01 0x10 0x01 0x01 0x01
                                0x10 0x01 0x01 0x01 0x03 0x03 0x03 0x03
ospfv3PmAreaLsdbTypeKnown:     true

ospfv3PmAreaLsdbTable row:
ospfv3PmAreaLsdbApplIndex:      537265152
ospfv3PmAreaLsdbAreaId:         0x00000000
ospfv3PmAreaLsdbType:           ospfv3NetworkLsa
ospfv3PmAreaLsdbRouterId:       0x03030303
ospfv3PmAreaLsdbLsid:           0x10010101
ospfv3PmAreaLsdbSequence:      -2147483647
ospfv3PmAreaLsdbAge:           878
ospfv3PmAreaLsdbChecksum:       45127
ospfv3PmAreaLsdbAdvertisement:  0x00 0x01 0x20 0x02 0x10 0x01 0x01 0x01
                                0x03 0x03 0x03 0x03 0x80 0x00 0x00 0x01
                                0xB0 0x47 0x00 0x20 0x00 0x00 0x00 0x13
                                0x01 0x01 0x01 0x03 0x03 0x03 0x03 0x03
ospfv3PmAreaLsdbTypeKnown:     true

ospfv3PmAreaLsdbTable row:
ospfv3PmAreaLsdbApplIndex:      537265152
ospfv3PmAreaLsdbAreaId:         0x00000000
ospfv3PmAreaLsdbType:           ospfv3InterAreaPrfxLsa
ospfv3PmAreaLsdbRouterId:       0x01010103
ospfv3PmAreaLsdbLsid:           0x00000001
ospfv3PmAreaLsdbSequence:      -2147483647
ospfv3PmAreaLsdbAge:           913
ospfv3PmAreaLsdbChecksum:       13755
ospfv3PmAreaLsdbAdvertisement:  0x00 0x24 0x20 0x03 0x00 0x00 0x00 0x01
                                0x01 0x01 0x01 0x03 0x80 0x00 0x00 0x01
                                0x35 0xBB 0x00 0x24 0x00 0x00 0x00 0x01
                                0x40 0x00 0x00 0x00 0xFD 0x00 0x00 0x00
                                0x00 0x00 0x00 0x00
ospfv3PmAreaLsdbTypeKnown:     true

ospfv3PmAreaLsdbTable row:
ospfv3PmAreaLsdbApplIndex:      537265152
ospfv3PmAreaLsdbAreaId:         0x00000000
ospfv3PmAreaLsdbType:           ospfv3IntraAreaPrfxLsa
ospfv3PmAreaLsdbRouterId:       0x03030303

```

```

ospfv3PmAreaLsdbLsid:      0x00010000
ospfv3PmAreaLsdbSequence:  -2147483645
ospfv3PmAreaLsdbAge:       873
ospfv3PmAreaLsdbChecksum:  2173
ospfv3PmAreaLsdbAdvertisement: 0x00 0x01 0x20 0x09 0x00 0x01 0x00 0x00
                                0x03 0x03 0x03 0x03 0x80 0x00 0x00 0x03
                                0x08 0x7D 0x00 0x2C 0x00 0x01 0x20 0x02
                                0x10 0x01 0x01 0x01 0x03 0x03 0x03 0x03
                                0x40 0x00 0x00 0x00 0x01 0x11 0x00 0x00
                                0x00 0x00 0x00 0x00
ospfv3PmAreaLsdbTypeKnown: true

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## disable (OSPFv3 AF)

disable

### Description

Disables the OSPFv3 AF process. By default OSPFv3 AF process is enabled.  
This command does not remove the OSPFv3 AF configurations.



The configuration is supported globally as well as per address-families ipv4-unicast or ipv6-unicast. If address-family specific timers is configured, it takes precedence over global configuration.

### Example

Disabling OSPFv3 AF process:

```

switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1) # disable

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History



| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## distance (OSPFv3 AF)

```
distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area [<DISTANCE-VAL>] |
external [<DISTANCE-VAL>]]
no distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area [<DISTANCE-VAL>] |
external [<DISTANCE-VAL>]]
```

## Description

Defines an administrative distance for OSPFv3 AF. Administrative distance is used as a criteria to select the best route when the same route is learned by multiple routing protocols.

The **no** form of this command sets the OSPFv3 AF administrative distance to the default value of 110. Optionally, administrative distance can be set to default for the specific OSPF route type: intra-area, inter-area, or external type-5 and type-7 routes.

| Parameter      | Description                                                                     |
|----------------|---------------------------------------------------------------------------------|
| <DISTANCE-VAL> | Specifies the OSPFv3 AF administrative distance. Range: 1 to 255. Default: 110. |
| intra-area     | Specifies the OSPFv3 AF distance for intra-area routes.                         |
| inter-area     | Specifies the OSPFv3 AF distance for inter-area routes.                         |
| external       | Specifies the OSPFv3 AF distance for external type 5 and type 7 routes.         |

## Usage

Within a given OSPF process, intra-area routes are always given precedence even when distances are configured for inter-area or external type routes.

## Examples

Setting OSPFv3 AF administrative distance for address-family ipv6-unicast:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# distance 100
switch(config-ospfv3-ipv6-uc)# distance intra-area 24 external 55 inter-area 66
switch(config-ospfv3-ipv6-uc)# distance intra-area 24 external 55
switch(config-ospfv3-ipv6-uc)# distance external 55
switch(config-ospfv3-ipv6-uc)#exit
```

Setting OSPFv3 AF administrative distance for address-family ipv4-unicast to default:

```

switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no distance
switch(config-ospfv3-ipv4-uc)# no distance external
switch(config-ospfv3-ipv4-uc)# no distance intra-area
switch(config-ospfv3-ipv4-uc)# no distance inter-area
switch(config-ospfv3-ipv4-uc)# no distance 1
switch(config-ospfv3-ipv4-uc)#exit

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## distribute-list prefix (ospfv3 AF)

```

distribute-list prefix <PREFIX-LIST-NAME> {in | out}
no distribute-list prefix <PREFIX-LIST-NAME> {in | out}

```

## Description

This command uses an existing prefix list to filter routes that are being installed in the routing table or redistributed to another routing protocol.

The **distribute-list prefix** command filters routes in the inbound or the outbound direction. When this command is issued with the **in** parameter, it filters routes from being installed in the routing table, it does not filter LSAs. When this command is issued with the **out** parameter, it filters only the desired redistributed routes from other protocols.

| Parameter                 | Description                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prefix <PREFIX-LIST-NAME> | Specify the name of an existing prefix.                                                                                                                                                                                                 |
| {in   out}                | Select one of the following parameters to set the filter direction: <ul style="list-style-type: none"> <li>▪ <b>in</b>: Filter incoming routes into the routing table</li> <li>▪ <b>out</b>: Filter outgoing routing updates</li> </ul> |

## Examples

Enabling the filtering of OSPFv3 AF routes for an IPv4 network, preventing routes from being installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# distribute-list prefix list1 in
switch(config-ospfv3-ipv4-uc)# distribute-list prefix list2 out
```

Disabling the filtering of OSPFv3 AF routes for an IPv4 network, allowing routes to be installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no distribute-list prefix list1 in
switch(config-ospfv3-ipv4-uc)# no distribute-list prefix list2 out
```

Enabling the filtering of OSPFv3 AF routes for an IPv6 network, preventing routes from being installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# distribute-list prefix list1 in
switch(config-ospfv3-ipv6-uc)# distribute-list prefix list2 out
```

Disabling the filtering of OSPFv3 AF routes for an IPv6 network, allowing routes to be installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# no distribute-list prefix list1 in
switch(config-ospfv3-ipv6-uc)# no distribute-list prefix list2 out
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms | Command context                   | Authority                                                                          |
|-----------|-----------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-ospfv3-ipv4-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |
| 6200      | config-ospfv3-ipv6-uc<PROCESS-ID> |                                                                                    |

## enable (OSPFv3 AF)

enable

## Description

Enables OSPFv3 AF process when disabled. By default OSPFv3 AF process is enabled.



Configuration is supported both globally and in the address-family for ipv4-unicast and ipv6-unicast. If enable is configured for the address-family, it will override the global configuration.

## Example

Enabling OSPFv3 AF process:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-af-1)# enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                             | Authority                                                                          |
|--------------|---------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-af- <i>&lt;PROCESS-ID&gt;</i> | Administrators or local user group members with execution rights for this command. |

## encryption ipsec

```
encryption ipsec spi <SPI-INDEX> <AUTH-TYPE> [<KEY-TYPE> <AUTH-KEY>  
    {<ENCR-TYPE> [<KEY-TYPE> <ENCR-KEY>] | null}]  
no encryption
```

## Description

Configures IPsec ESP authentication and encryption for the selected Vlink.

The **no** form of this command removes IPsec ESP authentication and encryption for the selected Vlink.

| Parameter                    | Description                                                                                                                                                                                                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spi <i>&lt;SPI-INDEX&gt;</i> | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPFv3 AF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| <i>&lt;AUTH-TYPE&gt;</i>     | Specifies the authentication type: <b>md5</b> or <b>sha1</b> .                                                                                                                                                                                                                                       |
| <i>&lt;KEY-TYPE&gt;</i>      | Specifies the key type to use: <b>plaintext</b> (unencrypted), <b>hex-string</b> (encrypted) or <b>ciphertext</b> (encrypted).                                                                                                                                                                       |
| <i>&lt;AUTH-KEY&gt;</i>      | Specifies the authentication key.                                                                                                                                                                                                                                                                    |

| Parameter   | Description                                                                                                                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ENCR-TYPE> | Specifies the encryption type: <b>des</b> , <b>3des</b> , <b>aes</b> , or <b>null</b> .<br><br><b>NOTE:</b> Encryption type <b>aes</b> is considered to be AES128, AES192, or AES256 based on key length. |
| <ENCR-KEY>  | Specifies the encryption key.                                                                                                                                                                             |



When the authentication key is not provided on the command line, plaintext authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally plaintext encryption key prompting. The entered key characters are masked with asterisks.



When the authentication key and encryption type are provided on the command line but the encryption key is not provided, plaintext encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting area 1 to use IPsec ESP authentication and encryption for Vlink with provided plaintext keys:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# encryption ipsec spi 256 md5 plaintext abcd
des plaintext abcdefab
```

Setting area 1 to use IPsec md5 authentication and encryption for Vlink with a prompted plaintext authentication key and encryption type:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# encryption ipsec spi 256 md5
Enter the IPsec authentication key: *****
Re-Enter the IPsec authentication key: *****

Enter the IPsec encryption type (3des/aes/des/null)? des

Enter the IPsec encryption key: *****
Re-Enter the IPsec encryption key: *****
```

Setting area 1 to use IPsec sha1 authentication and encryption for Vlink with a provided plaintext authentication key and prompted encryption type:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
```

```
switch(config-router-vlink6)# encryption ipsec spi 256 sha1 plaintext pass des
ciphertext Configure ciphertext key.
hex-string Configure hex-string key. Key length must be 16 Hex digits.
plaintext Configure plaintext key. Key length must be 8 characters.

switch(config-router-vlink6)# encryption ipsec spi 256 sha1 plaintext pass des
Enter the IPsec encryption key: *****
Re-Enter the IPsec encryption key: *****
```

Setting area 1 to use IPsec ESP authentication for Vlink with a provided plaintext authentication key and null encryption:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-1)# area 1
switch(config-ospfv3-1)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# encryption ipsec spi 256 md5 plaintext abcd null
```

Removing IPsec ESP authentication and encryption for Vlink on area 1:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# no encryption
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |

## graceful-restart ignore-lost-interface

```
graceful-restart ignore-lost-interface
no graceful-restart ignore-lost-interface
```

### Description

Enables the restarting router to ignore lost OSPFv3 AF interfaces during a graceful restart process. It is recommended to enable the ignore-lost-interface parameter on the VSF device configured with OSPF enabled on ROPs to ensure the graceful restart completes successfully, as a VSF switchover or failover can render a ROP non-operational, leading to a Router LSA change published to neighbors.

The **no** form of this commands disables the restarting router to ignore OSPFv3 AF interfaces during a graceful restart process.



Configuration is supported both globally and in the address-family for ipv4-unicast and ipv6-unicast. If disable is configured for the address-family, it will override the global configuration.

| Parameter                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ignore-lost-interface</code> | <p>Enable the restarting router to ignore lost OSPFv3 AF interfaces during a graceful restart process. This setting should be enabled on a high availability system to ensure a graceful restart completes successfully, even if OSPFv3 AF-enabled links fail due to High Availability events like a switchover or failover.</p> <p><b>NOTE:</b> Enabling this setting means that the hitless restart procedures do not strictly follow those defined in <i>RFC 3623, Graceful OSPF Restart</i>.</p> <p><b>NOTE:</b> OSPFv3 AF neighbors must disable strict LSA checking. If the local node has fewer OSPFv3 AF interfaces after restarting, then the neighbors that were adjacent on those interfaces will clear up their adjacencies to the restarting node and will send out link state updates to advertise the dropped adjacency. If strict LSA checking is enabled, the restarting router's neighbors will exit helper mode when they receive the updated LSAs and the graceful restart will still fail.</p> |

## Examples

Enabling the restarting router to ignore lost OSPFv3 AF interfaces during a graceful restart process:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc) # graceful-restart ignore-lost-interface
```

Disabling the restarting router:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc) # no graceful-restart ignore-lost-interface
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                      | Authority                                                                          |
|--------------|------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | <code>config-ospfv3-ipv4-uc&lt;PROCESS-ID&gt;</code> | Administrators or local user group members with execution rights for this command. |

## graceful-restart (OSPFv3 AF)

```
graceful-restart {restart-interval <INTERVAL> | helper [strict-lsa-check]}  
no graceful-restart {restart-interval <INTERVAL> | helper [strict-lsa-check]}
```

### Description

Configures graceful restart for OSPFv3 AF. By default graceful restart is enabled on the OSPFv3 AF router.

The **no** form of this command sets the restart interval to the default of 120 seconds or disables helper mode depending on the specified parameters.



Configuration is supported both globally and in the address-family for ipv4-unicast and ipv6-unicast. If disable is configured for the address-family, it will override the global configuration.

| Parameter                                      | Description                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>restart-interval &lt;INTERVAL&gt;</code> | Specifies the time another router waits for this router to gracefully restart and selects the maximum time to wait in seconds. Range: 5 to 1800. Default: 120.                                                                                                                                                                                               |
| <code>helper</code>                            | Specifies that the router will participate in the graceful restart of a neighbor router.                                                                                                                                                                                                                                                                     |
| <code>strict-lsa-check</code>                  | (Optional). Use with the <b>helper</b> parameter to enable strict Link state Advertisement (LSA) checking when acting as a restart helper for a restarting peer.<br><br><b>NOTE:</b> If strict LSA checking is enabled, the restarting router's neighbors will exit helper mode when they receive the updated LSAs and the graceful restart will still fail. |

### Examples

Enabling OSPFv3 AF graceful restart:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-ipv4-uc) # graceful-restart helper  
switch(config-ospfv3-ipv4-uc) # graceful-restart helper strict-lsa-check  
switch(config-ospfv3-ipv4-uc) # graceful-restart restart-interval 40
```

Setting the restart interval to default, and disabling helper mode:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-ipv4-uc) # no graceful-restart  
switch(config-ospfv3-ipv4-uc) # no graceful-restart helper  
switch(config-ospfv3-ipv4-uc) # no graceful-restart helper strict-lsa-check  
switch(config-ospfv3-ipv4-uc) # no graceful-restart restart-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History



| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                   | Authority                                                                          |
|--------------|-----------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## hello-interval (OSPFv3 AF)

```
hello-interval <INTERVAL>
no hello-interval
```

### Description

Sets the time interval between OSPFv3 AF hello packets for virtual links.

The **no** form of this command sets the hello interval to the default value of 10 seconds for virtual links.

For proper operation, set the hello interval lower than the dead interval.

| Parameter  | Description                                                                                     |
|------------|-------------------------------------------------------------------------------------------------|
| <INTERVAL> | Specifies the time interval for the hello interval, in seconds. Range: 1 to 65535. Default: 10. |

### Examples

Setting OSPFv3 AF virtual links hello interval:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# hello-interval 30
```

Setting OSPFv3 AF virtual links hello interval to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no hello-interval
```

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv4 retransmit-interval

```
ospfv3-af ipv4 retransmit-interval <INTERVAL>  
no ospfv3-af ipv4 retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for the OSPFv3 AF interface.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for the OSPFv3 AF interface.

| Parameter  | Description                                                                  |
|------------|------------------------------------------------------------------------------|
| <INTERVAL> | Specifies the retransmit interval, in seconds. Range: 1 to 3600. Default: 5. |

### Examples

Setting OSPFv3 AF retransmit interval on the interface:

```
switch(config)# interface 1  
switch(config-if)# ospfv3-af ipv4 retransmit-interval 30
```

Setting OSPFv3 AF retransmit interval to the default on the interface:

```
switch(config)# interface 1  
switch(config-if)# no ospfv3-af ipv4 retransmit-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv4 cost

```
ospfv3-af ipv4 cost <INTERFACE-COST>
no ospfv3-af ipv4 cost
```

## Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The **no** form of this command sets the cost (metric) associated with a particular interface to the default of 1. The default cost is calculated automatically depending on the bandwidth of the interface.

| Parameter        | Description                                                        |
|------------------|--------------------------------------------------------------------|
| <INTERFACE-COST> | Specifies the interface cost value. Range: 1 to 65535. Default: 1. |

## Examples

Setting OSPFv3 AF interface cost:

```
switch(config)# interface 1
switch(config-if)# ospfv3-af ipv4 cost 100
```

Setting the OSPFv3 AF interface cost to default:

```
switch(config)# interface 1
switch(config-if)# no ospfv3-af ipv4 cost
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv4 hello-interval

```
ospfv3-af ipv4 hello-interval <INTERVAL>
no ospfv3-af ipv4 hello-interval
```

## Description

Sets the time interval between OSPFv3 AF hello packets for the OSPFv3 AF interface.

The **no** form of this command sets the time interval between OSPFv3 AF hello packets to the default for the OSPFv3 AF interface of 10 seconds.



For proper operation, set the hello interval lower than the dead interval.

| Parameter                     | Description                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------|
| <code>&lt;INTERVAL&gt;</code> | Specifies the time interval between hello packets, in seconds. Range: 1 to 65535. Default: 10. |

## Examples

Setting OSPFv3 AF hello interval on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 hello-interval 30
```

Setting OSPFv3 AF hello interval to default on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv4 hello-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv4 network

```
ospfv3-af ipv4 network {broadcast|point-to-point}
no ospfv3-af ipv4 network {broadcast|point-to-point}
```

## Description

Configures the network type for the interface. By default the network type is broadcast network.

The **no** form of this command sets the network type for the interface to the system default as broadcast network for ethernet interface and point-to-point network for tunnel interface.

| Parameter              | Description                                                     |
|------------------------|-----------------------------------------------------------------|
| <code>broadcast</code> | Specifies the OSPFv3 AF network type as a broadcast multiaccess |

| Parameter      | Description                                                       |
|----------------|-------------------------------------------------------------------|
|                | network.                                                          |
| point-to-point | Specifies the OSPFv3 AF network type as a point-to-point network. |

## Examples

Setting OSPFv3 AF network type for the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 network broadcast
switch(config-if)# ospfv3-af ipv4 network point-to-point
```

Disabling OSPFv3 AF network type for the interface to system default of broadcast network:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv4 network
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv4 priority

```
ospfv3-af ipv4 priority <number-value>
no ospfv3-af ipv4 priority
```

### Description

Sets the OSPFv3 AF priority for the interface. The larger the numeric value of the priority, the higher the chance it will become the designated router. Setting a priority of 0 makes the router ineligible to become a designated router or back up designated router.

The **no** form of this command sets the OSPFv3 AF priority for the interface to the default of 1.

| Parameter      | Description                                                |
|----------------|------------------------------------------------------------|
| <number-value> | Specifies the priority value. Default: 1. Range: 0 to 255. |

## Examples

Setting the OSPFv3 AF priority for the interface:

```
switch(config)# interface 1  
switch(config-if)# ospfv3-af ipv4 priority 50
```

Setting the OSPFv3 AF priority for the interface to the default of 1:

```
switch(config)# interface 1  
switch(config-if)# no ospfv3-af ipv4 priority
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv4 shutdown

```
ospfv3-af ipv4 shutdown  
no ospfv3-af ipv4 shutdown
```

### Description

Disables OSPFv3 AF on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command **no ospfv3-af ipv4 area**. The **no** form of this command re-enables OSPFv3 AF on the interface.

### Examples

Disabling OSPFv3 AF on the interface:

```
switch(config)# interface 1/1/1  
switch(config-if)# ospfv3-af ipv4 shutdown
```

Re-enabling OSPFv3 AF on the interface:

```
switch(config)# interface 1/1/1  
switch(config-if)# no ospfv3-af ipv4 shutdown
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv4 transit-delay

```
ospfv3-af ipv4 transit-delay <seconds>  
no ospfv3-af ipv4 transit-delay <seconds>
```

### Description

Sets the time delay in Link state transmission for the OSPFv3 AF interface.

The **no** form of this command sets the transit delay in Link state transmission to the default of 1 second.

| Parameter | Description                                                                               |
|-----------|-------------------------------------------------------------------------------------------|
| <DELAY>   | Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1. |
| no        | Sets the delay in Link state transmission to the default of 1 for virtual links.          |

### Examples

Setting OSPFv3 AF transit delay on the interface

```
switch(config)# interface 1/1/1  
switch(config-if)# ospfv3-af ipv4 transit-delay 30
```

Setting OSPFv3 AF transit delay to default on the interface

```
switch(config)# interface 1/1/1  
switch(config-if)# no ospfv3-af ipv4 transit-delay
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv6 area

```
ospfv3-af <PROCESS-ID> ipv6 area <AREA-ID>  
no ospfv3-af <PROCESS-ID> ipv6 area <area-id>
```

### Description

This command runs the OSPFv3 AF IPv6 on the interface with the configured link-local IPv6 address for the area specified. To move an interface to a new area, unmap the existing area and then associate a new area with the interface. If the OSPFv3 AF process or area is not created, user confirmation is requested to create the OSPFv3 AF process and area.

The **no** form of this command disables OSPFv3 AF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol

| Parameter    | Description                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AREA-ID>    | Specifies the area ID is one of the following formats.<br>Area ID in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>Area ID as a decimal value. Range: 0-4294967295. |
| <PROCESS-ID> | Specifies the OSPFv3 AF process ID. Range: 1 to 65535.                                                                                                                                                     |

### Examples

Setting OSPFv3 AF network for the area:

```
switch(config)# interface 1  
switch(config-if)# ospfv3-af 1 ipv6 area 1  
switch(config-if)# ospfv3-af 1 ipv6 area 0.0.0.1
```

Disabling OSPFv3 AF network for the area:

```
switch(config)# interface 1  
switch(config-if)# no ospfv3-af 1 ipv6 area 1  
switch(config-if)# no ospfv3-af 1 ipv6 area 0.0.0.1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |



## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv6 cost

```
ospfv3-af ipv6 cost <INTERFACE-COST>  
no ospfv3-af ipv6 cost
```

### Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The **no** form of this command sets the cost (metric) associated with a particular interface to the default which is calculated automatically depending on the bandwidth of the interface.

| Parameter        | Description                                                        |
|------------------|--------------------------------------------------------------------|
| <INTERFACE-COST> | Specifies the interface cost value. Range: 1 to 65535. Default: 1. |

### Examples

Setting OSPFv3 AF interface cost:

```
switch(config)# interface vlan 1  
switch(config-if)# ospfv3-af ipv6 cost 20
```

Setting the OSPFv3 AF interface cost to default:

```
switch(config)# interface vlan 1  
switch(config-if)# no ospfv3-af ipv6 cost
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3 af ipv6 dead-interval

```
ospfv3 af ipv6 dead-interval <seconds>  
no ospfv3 af ipv6 dead-interval
```

### Description

Sets the interval after a neighbor is declared dead when no hello packet is received on the OSPFv3 AF interface. Dead interval must be greater than hello interval, otherwise, the OSPF protocol will have an undefined behavior.

The **no** form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPFv3 AF interface. The default value is 40 seconds (generally four times the hello packet interval).

| Parameter | Description                                                                                    |
|-----------|------------------------------------------------------------------------------------------------|
| <seconds> | Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40. |

### Examples

Setting OSPFv3 AF dead interval on the interface:

```
switch(config)# interface 1  
switch(config-if)# ospfv3 af ipv6 dead-interval 30
```

Setting OSPFv3 AF dead interval to default on the interface:

```
switch(config)# interface 1  
switch(config-if)# no ospfv3 af ipv6 dead-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

### Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3 af ipv6 hello-interval

```
ospfv3 af ipv6 hello-interval <INTERVAL>  
no ospfv3 af ipv6 hello-interval
```

### Description

Sets the time interval between OSPFv3 AF hello packets for the OSPFv3 AF interface.

The **no** form of this command sets the time interval between OSPFv3 AF hello packets to the default for the OSPFv3 AF interface of 10 seconds.



---

For proper operation, set the hello interval lower than the dead interval.

---

| Parameter                     | Description                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------|
| <code>&lt;INTERVAL&gt;</code> | Specifies the time interval between hello packets, in seconds. Range: 1 to 65535. Default: 10. |

## Examples

Setting OSPFv3 AF hello interval on the interface:

```
switch(config)# interface 1
switch(config-if)# ospfv3-af ipv6 hello-interval 30
```

Setting OSPFv3 AF hello interval to default on the interface:

```
switch(config)# interface 1
switch(config-if)# no ospfv3-af ipv6 hello-interval
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv6 network

```
ospfv3-af ipv6 network {broadcast|point-to-point}
no ospfv3-af ipv6 network {broadcast|point-to-point}
```

## Description

Configures the network type for the interface. By default the network type is broadcast network.

The **no** form of this command sets the network type for the interface to the system default as broadcast network for ethernet interface and point-to-point network for tunnel interface.

| Parameter      | Description                                                              |
|----------------|--------------------------------------------------------------------------|
| broadcast      | Specifies the OSPFv3 AF network type as a broadcast multiaccess network. |
| point-to-point | Specifies the OSPFv3 AF network type as a point-to-point network.        |

## Examples

Setting OSPFv3 AF network type for the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv6 network broadcast
switch(config-if)# ospfv3-af ipv6 network point-to-point
```

Disabling OSPFv3 AF network type for the interface to system default of broadcast network:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv6 network
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv6 passive

```
ospfv3-af ipv6 passive
no ospfv3-af ipv6 passive
```

## Description

Configures the interface as an OSPFv3 AF passive interface. With this setting, the interface participates in the OSPF, but does not send or receive OSPF packets on that interface.

The **no** form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

## Examples

Setting the interface as OSPFv3 AF passive interface:

```
switch(config)# interface vlan 1
switch(config-if)# ospfv3-af ipv6 passive
```

Setting the interface as OSPFv3 AF active interface:

```
switch(config)# interface vlan 1
switch(config-if)# no ospfv3-af ipv6 passive
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv6 priority

```
ospfv3-af ipv6 priority <number-value>
no ospfv3-af ipv6 priority
```

### Description

Sets the OSPFv3 AF priority for the interface. The larger the numeric value of the priority, the higher the chance it will become the designated router. Setting a priority of 0 makes the router ineligible to become a designated router or back up designated router.

The **no** form of this command sets the OSPFv3 AF priority for the interface to the default of 1.

| Parameter      | Description                                                          |
|----------------|----------------------------------------------------------------------|
| <number-value> | Specifies the OSPFv3 AF priority value. Default: 1. Range: 0 to 255. |

## Examples

Setting the OSPFv3 AF priority for the interface:

```
switch(config)# interface vlan /1
switch(config-if)# ospfv3-af ipv6 priority 50
```

Setting the OSPFv3 AF priority for the interface to the default of 1:

```
switch(config)# interface vlan 1
switch(config-if)# no ospfv3-af ipv6 priority
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv6 retransmit-interval

```
ospfv3-af ipv6 retransmit-interval <INTERVAL>
no ospfv3-af ipv6 retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for the OSPFv3 AF interface.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default 5 seconds.

| Parameter  | Description                                                                                       |
|------------|---------------------------------------------------------------------------------------------------|
| <INTERVAL> | Specifies the time interval for the retransmit interval, in seconds. Range: 1 to 3600. Default: 5 |

### Examples

Setting OSPFv3 AF retransmit interval on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv6 retransmit-interval 30
```

Setting OSPFv3 AF retransmit interval to the default on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv6 retransmit-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv6 shutdown

```
ospfv3-af ipv6 shutdown
no ospfv3-af ipv6 shutdown
```

### Description

Disables OSPFv3 AF on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command **no ospfv3-af ipv6 area**. The **no** form of this command re-enables OSPFv3 AF on the interface.

### Examples

Disabling OSPFv3 AF on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv6 shutdown
```

Re-enabling OSPFv3 AF on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv6 shutdown
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv6 transit-delay

```
ospfv3-af ipv6 transit-delay <seconds>
```

```
no ospfv3-af ipv6 transit-delay <seconds>
```

## Description

Sets the time delay in Link state transmission for the OSPFv3 AF virtual link.

The **no** form of this command sets the transit delay in Link state transmission to the default of 1 second.

| Parameter | Description                                                                               |
|-----------|-------------------------------------------------------------------------------------------|
| <DELAY>   | Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1. |
| no        | Sets the delay in Link state transmission to the default of 1 for virtual links.          |

## Examples

Setting OSPFv3 AF transit delay on the interface

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# transit-delay 30
```

Setting OSPFv3 AF transit delay to default on the interface

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no transit-delay
switch(config-router-vlink6)# no transit-delay 30
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |

## max-metric router-lsa (OSPFv3 AF)

```
max-metric router-lsa [include-stub] | [on-startup <INTERVAL>]
no max-metric router-lsa [include-stub] | [on-startup]
```

## Description



Sets the protocol to advertise a maximum metric so that other routers do not prefer the router as an intermediate hop in their shortest path first (SPF) calculations. If the on-startup parameter is used, the router is configured to advertise a maximum metric at start up. That is, for the time specified in seconds, or the default value of 600 seconds. The include-stub parameter configures the router to advertise maximum metric for links including stub links.

To disable advertisement of the maximum metric, use the **no** form of the command.

The **no** form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

| Parameter             | Description                                                                                                                                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| on-startup <INTERVAL> | Automatically advertises the stub Router-LSA (or maximize the router-LSA cost metric) for a specified time interval upon OSPFv3 AF startup.<br>Specifies the time in seconds. Range: 5 to 86400. Default: 600. |
| include-stub          | Advertise router-LSAs with maximum metric value for stub links.                                                                                                                                                |

## Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# max-metric router-lsa on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# no max-metric router-lsa
switch(config-ospfv3-ipv6-uc)# no max-metric router-lsa on-startup
```

Setting to advertise Router LSA for stub links:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# max-metric router-lsa
switch(config-ospfv3-ipv6-uc)# max-metric router-lsa include-stub on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric for stub links:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-af-1)# no max-metric router-lsa
switch(config-ospfv3-ipv6-uc)# no max-metric router-lsa include-stub on-startup
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## maximum-paths (OSPFv3 AF)

maximum-paths <MAXIMUM>  
no maximum-paths

### Description

Sets the maximum number of ECMP routes that OSPFv3 AF can support.

The **no** form of this command sets the maximum number of ECMP routes that OSPFv3 AF can support to the default value of 4.

| Parameter | Description                                                              |
|-----------|--------------------------------------------------------------------------|
| <MAXIMUM> | Specifies the maximum number of ECMP routes. Range: 1 to 32. Default: 4. |

### Examples

Setting maximum number of parallel routes:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)#address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# maximum-paths 3
```

Setting maximum number of parallel paths to the default value of 4:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)#address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no maximum-paths
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## ospfv3-af encryption ipsec spi

```
ospfv3-af encryption ipsec spi <SPI-INDEX> <AUTH-TYPE>
    [<KEY-TYPE> <AUTH-KEY> <ENCR-TYPE> [<KEY-TYPE> <ENCR-KEY>]]
no ospfv3-af encryption {null| [ipsec spi <SPI-INDEX> <AUTH-TYPE>
    [<KEY-TYPE><AUTH-KEY><ENCR-TYPE> [<KEY-TYPE><ENCR-KEY>]] }
```

### Description

Configures IPsec ESP authentication. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec ESP.

The **no** form of this command removes IPsec ESP for the specified area.

| Parameter       | Description                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spi <SPI-INDEX> | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| <AUTH-TYPE>     | Specifies the authentication type: <b>md5</b> or <b>sha1</b> .                                                                                                                                                                                                                                  |
| <KEY-TYPE>      | Specifies the key type to use: <b>plaintext</b> (unencrypted), <b>hex-string</b> (encrypted) or <b>ciphertext</b> (encrypted).                                                                                                                                                                  |
| <AUTH-KEY>      | Specifies the authentication key.                                                                                                                                                                                                                                                               |
| <ENCR-TYPE>     | Specifies the encryption type: <b>des</b> , <b>3des</b> , <b>aes</b> , or <b>null</b> .<br><br><b>NOTE:</b> Encryption type <b>aes</b> is considered to be AES128, AES192, or AES256 based on key length.                                                                                       |
| <ENCR-KEY>      | Specifies the encryption key.                                                                                                                                                                                                                                                                   |



When the authentication key is not provided on the command line, plaintext authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally plaintext encryption key prompting. The entered key characters are masked with asterisks.



When the authentication key and encryption type are provided on the command line but the encryption key is not provided, plaintext encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

### Examples

Setting interface VLAN 1 to use IPsec ESP with provided authentication and encryption keys:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ospfv3-af encryption ipsec spi 256 sha1 plaintext F82
des plaintext F82#450b
```

Setting interface VLAN 3 to use IPsec ESP with prompted authentication and encryption keys:

```
switch(config)# interface vlan 3
switch(config-if-vlan)# ospfv3-af encryption ipsec spi 256 sha1
Enter the IPsec authentication key: *****
Re-Enter the IPsec authentication key: *****

Enter the IPsec encryption type (3des/aes/des/null)? des

Enter the IPsec encryption key: *****
Re-Enter the IPsec encryption key: *****
```

Setting interface VLAN 4 to use IPsec ESP with provided authentication password and encryption type but a prompted encryption key:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# ospfv3-af encryption ipsec spi 256 sha1 plaintext F82 des
Enter the IPsec encryption key: *****
Re-Enter the IPsec encryption key: *****
```

Setting interface VLAN 1 to use IPsec ESP with provided plaintext authentication key and null encryption:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ospfv3-af encryption ipsec spi 256 sha1 plaintext F82 null
```

Removing IPsec from interface VLAN 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ospfv3-af encryption ipsec spi
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af authentication ipsec spi

```
ospfv3-af authentication ipsec spi <SPI-INDEX> <AUTH-TYPE> [<KEY-TYPE> <AUTH-KEY>]  
no ospfv3-af authentication ipsec spi
```

## Description

Configures IPsec AH authentication. OSPFv3 AF interfaces that have IPsec configured at the interface context will not use area level IPsec. The Security Parameters Index (SPI) is an identification tag carried in IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table.

The **no** form of this command removes IPsec AH authentication for the specified area.



---

IPsec is not supported for 6in6 tunnel interfaces, and IPsec AH SPI must be unique on the router.

---

| Parameter       | Description                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spi <SPI-INDEX> | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| <AUTH-TYPE>     | Specifies the authentication type: <b>md5</b> or <b>sha1</b> .                                                                                                                                                                                                                                  |
| <KEY-TYPE>      | Specifies the key type to use: <b>plaintext</b> (unencrypted), <b>hex-string</b> (encrypted) or <b>ciphertext</b> (encrypted).                                                                                                                                                                  |
| <AUTH-KEY>      | Specifies the authentication key.                                                                                                                                                                                                                                                               |



---

When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

---

## Examples

Setting interface **1/1/1** to use IPsec authentication with a provided plaintext authentication key:

```
switch(config)# interface 1/1/1  
switch(config-if)# ospfv3-af authentication ipsec spi 256 md5 plaintext abcd
```

Setting interface **1/1/1** to use IPsec authentication with a prompted plaintext authentication key:

```
switch(config)# interface 1/1/1  
switch(config-if)# ospfv3-af authentication ipsec spi 256 md5  
Enter the IPsec authentication key: *****  
Re-Enter the IPsec authentication key: *****
```

Removing IPsec authentication from interface **1/1/1**:

```
switch(config)# interface 1/1/1  
switch(config-if)# no ospfv3-af authentication  
switch(config-if)# no ospfv3-af authentication null  
switch(config-if)# no ospfv3-af authentication ipsec spi 300 md5 ciphertext
```

```
qwertyuioplkjhgfdsxcvbnm
switch(config-if)# no ospfv3-af authentication ipsec spi 300 sha1 ciphertext
qwertyuioplkjhgfdsxcvbnm
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af encryption ipsec

```
ospfv3-af encryption ipsec spi <SPI-INDEX> <AUTH-TYPE> [<KEY-TYPE> <AUTH-KEY> {<ENCR-
TYPE> [<KEY-TYPE> <ENCR-KEY>] | null}]
no ospfv3-af encryption {null | [ipsec spi <SPI-INDEX> <AUTH-TYPE> ciphertext <AUTH-KEY>
[<ENCR-TYPE> ciphertext <ENCR-KEY>] | null]}
```

## Description

Configures IPsec ESP authentication. OSPFv3 AF interfaces that have IPsec configured at the interface context will not use area level IPsec ESP.

The **no** form of this command removes IPsec ESP for the specified area.



IPsec is not supported in 6in6 tunnel interfaces.

| Parameter       | Description                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spi <SPI-INDEX> | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| <AUTH-TYPE>     | Specifies the authentication type: <b>md5</b> or <b>sha1</b> .                                                                                                                                                                                                                                  |
| <KEY-TYPE>      | Specifies the key type to use: <b>plaintext</b> (unencrypted), <b>hex-string</b> (encrypted) or <b>ciphertext</b> (encrypted).                                                                                                                                                                  |
| <AUTH-KEY>      | Specifies the authentication key.                                                                                                                                                                                                                                                               |
| <ENCR-TYPE>     | Specifies the encryption type: <b>des</b> , <b>3des</b> , <b>aes</b> , or <b>null</b> .                                                                                                                                                                                                         |

| Parameter  | Description                                                                                                |
|------------|------------------------------------------------------------------------------------------------------------|
|            | <b>NOTE:</b> Encryption type <b>aes</b> is considered to be AES128, AES192, or AES256 based on key length. |
| <ENCR-KEY> | Specifies the encryption key.                                                                              |



When the authentication key is not provided on the command line, **plaintext** authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally **plaintext** encryption key prompting. The entered key characters are masked with asterisks.



When the authentication key and encryption type are provided on the command line but the encryption key is not provided, **plaintext** encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting interface **1/1/1** to use IPsec ESP with provided authentication and encryption keys:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af encryption ipsec spi 256 sha1 plaintext abcdef
aes plaintext abcdefabcdefabcdefab
```

Setting interface **1/1/1** to use IPsec ESP with provided plaintext authentication key and null encryption:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af encryption ipsec spi 256 sha1 plaintext abcdef null
```

Setting interface **1/1/1** to use IPsec ESP with prompted authentication and encryption keys:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af encryption ipsec spi 256 sha1
Enter the IPsec authentication key: *****
Re-Enter the IPsec authentication key: *****

Enter the IPsec encryption type (3des/aes/des/null)? des

Enter the IPsec encryption key: *****
Re-Enter the IPsec encryption key: *****
```

Setting interface **1/1/1** to use IPsec ESP with provided authentication password and encryption type but a prompted encryption key:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af encryption ipsec spi 256 sha1 plaintext abcdef des
ciphertext    Configure ciphertext key.
hex-string    Configure hex-string key. Key length must be 16 Hex digits.
plaintext     Configure plaintext key. Key length must be 8 characters.
Enter the IPsec encryption key: *****
```

Re-Enter the IPsec encryption key: \*\*\*\*\*

Removing IPsec from interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af encryption
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv4 area

```
ospfv3-af <process-id> ipv4 area <area-id>
no ospfv3-af <process-id> ipv4 area <area-id>
```

## Description

This command will run the OSPFv3 AF protocol on the interface with the configured link-local IPv6 address for the area mentioned. For moving an interface to a new area, you need to unmap the existing area and then associate a new area to the interface. If the OSPFv3 AF process or area is not created, user confirmation is requested to create the OSPFv3 AF process and area.

The **no** form of this command disables OSPFv3 AF on the interface.



An IPv6 link-local address must be enabled on the interface for an OSPFv3 AF adjacency to be formed for the IPv4 unicast OSPFv3 AF.

| Parameter    | Description                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <process-id> | Specify the OSPFv3 AF Process ID.<br>Range: 1-65535                                                                                          |
| area-id      | Area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255 or using a decimal value (0-4294967295). |

## Examples

Enabling OSPFv3 AF on an interface using IPv4 address:



```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af 1 ipv4 area 1
switch(config-if)# ospfv3-af 1 ipv4 area 0.0.0.1
```

Disabling OSPFv3 AF on an interface using IPv4 address:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af 1 ipv4 area 1
switch(config-if)# no ospfv3-af 1 ipv4 area 0.0.0.1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af ipv4 dead-interval

ospfv3-af ipv4 dead-interval <seconds>

### Description

This command sets the interval after which a neighbor is declared dead if no hello packet is received for the OSPFv3 AF interface. Dead interval must always be greater than hello-interval, otherwise the OSPFv3 AF protocol will have undefined behavior.

The **no** form of this command sets the interval to the default value of 40 seconds.

| Parameter      | Description                                                                  |
|----------------|------------------------------------------------------------------------------|
| <i>seconds</i> | Specifies Dead interval value in second.<br>Default is 40.<br>Range: 1-65535 |

## Examples

Configuring an interval of 30 to consider an unresponsive neighbor as down:

```
switch(config)# interface 1
switch(config-if)# ospfv3-af ipv4 dead-interval 30
```

Setting the dead interval to the default value:

```
switch(config)# interface 1
switch(config-if)# no ospfv3-af ipv4 dead-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ospfv3-af authentication null

ospfv3-af authentication null

### Description

This command configures NULL authentication on an interface which disables IPsec AH authentication.



Ipsec is not supported on 6in6 Tunnel Interfaces.

## Examples

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af authentication null
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                          |
|--------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

# ospfv3-af encryption null

ospfv3-af encryption null

## Description

This command configures NULL ESP on an interface, which disables IPsec ESP.



IPsec is not supported on 6in6 Tunnel Interfaces.

## Example

Configuring null ESP:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af encryption null
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# ospfv3-af ipv4 passive

ospfv3-af ipv4 passive

## Description

This command configures the interface as an OSPFv3 AF passive interface. With this setting the interface participates in OSPF but does not receive or send OSPF packets on that interface.

## Example

Configuring interface 1/1/1 as an OSPFv3 AF passive interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 passive
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## passive-interface default (OSPFv3 AF)

```
passive-interface default
no passive-interface default
```

### Description

Sets all OSPFv3 AF interfaces as passive.

The **no** form of this command sets all the OSPFv3 AF interfaces as active.

### Examples

Setting OSPFv3 AF enabled interfaces as passive:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc) # passive-interface default
```

Setting OSPFv3 AF enabled interfaces as active:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc) # no passive-interface default
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                   | Authority                                                                          |
|--------------|-----------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## redistribute (OSPFv3 AF)

```
redistribute {bgp | connected | host-routes | local loopback | static | ripng |
```

```
ospf <PROCESS-ID> | ospfv3-af <PROCESS-ID> [route-map <ROUTE-MAP-NAME>]
no redistribute {bgp | connected | host-routes | local loopback | static | ripng |
ospf <PROCESS-ID> | ospfv3-af <PROCESS-ID> [route-map <ROUTE-MAP-NAME>]
```

## Description

This command redistributes routes originating from other protocols to OSPFv3 AF. This command allows to redistribute routes from other OSPFv3 AF instances. IPv4 address family allows redistribution from other OSPFv3 Ipv4 AFs. IPv6 address family allows redistribution from other OSPFv3 Ipv6 AFs. In addition, if a route-map is specified, then routes that pass the match clause specified in the route-map are only redistributed to OSPFv3 AF. Configuration is not allowed if the referenced route-map has not yet been configured.

Only BGP leaked routes will be given to OSPFv3 AF for redistribution. If you try to redistribute routes from an OSPF process which is not created, user confirmation is asked to create the process before proceeding with redistribution. If accepted, the process is created with defaults and redistribution configuration is applied. If denied, redistribution configuration is skipped.




---

All the routes added by a routing protocol into RIB are redistributed into the target routing protocol OSPF.

---

If command **route-redistribute active-routes-only** has been issued, only the routes from other protocols which are selected for forwarding are considered for redistribution into OSPFv3 AF. With the change in redistribute behavior, additional explicit redistribution of connected routes may be needed if any of the deployment assumes redistribution of connected networks will happen via redistribution of dynamic protocol routes into OSPF.

The **no** form of this command disables redistribution of routes to the current OSPFv3 AF process.

| Parameter              | Description                                                                                                                                                                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bgp                    | Specifies redistributing BGP (Border Gateway Protocol) routes into OSPFv3 AF.                                                                                                                                                                                      |
| connected              | Specifies redistributing connected (directly attached subnet or host).                                                                                                                                                                                             |
| host-routes            | Redistribute dynamic, IRL and static ARP/ND entries into OSPFv3 AF. ARP/ND entries learnt from EVPN will not be redistributed.<br><br>The <b>host-routes</b> parameter is applicable to HPE Aruba Networking 8325, 10000, 8360, and 9300/9300S Switch series only. |
| local loopback         | Specifies redistributing local routes of the loopback interface.                                                                                                                                                                                                   |
| static                 | Specifies redistributing static routes into OSPFv3 AF.                                                                                                                                                                                                             |
| ripng                  | Specifies redistributing RIPng routes into OSPFv3 AF.                                                                                                                                                                                                              |
| ospf <PROCESS-ID>      | Specifies redistributing routes from the specified OSPF process ID. Range: 1 to 65535.                                                                                                                                                                             |
| ospfv3-af <PROCESS-ID> | Specifies redistributing routes from the specified OSPFv3 AF process ID. Range: 1 to 65535.                                                                                                                                                                        |

| Parameter                                     | Description                                                                                            |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>route-map &lt;ROUTE-MAP-NAME&gt;</code> | Specifies redistribution filtering by route map. To create a route map, use command <b>route-map</b> . |

## Examples

Redistributing routes to OSPFv3 AF:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc) # redistribute bgp
switch(config-ospfv3-ipv4-uc) # redistribute connected
switch(config-ospfv3-ipv4-uc) # redistribute host-routes
switch(config-ospfv3-ipv4-uc) # redistribute static
switch(config-ospfv3-ipv4-uc) # redistribute ripng
switch(config-ospfv3-ipv4-uc) # redistribute ospfv3-af 2
switch(config-ospfv3-ipv4-uc) # redistribute local loopback
switch(config-ospfv3-ipv4-uc) # redistribute static route-map
% Command incomplete.
switch(config-ospfv3-ipv4-uc) # redistribute static route-map static_networks
switch(config-ospfv3-af-1)# redistribute static route-map static_networks1
Route Map static_networks1 not found
switch(config-ospfv3-af-1)# redistribute static route-map static_networks12
Redistribute with route-map config exists
switch(config-ospfv3-ipv4-uc) # redistribute ripng route-map
% Command incomplete.
switch(config-ospfv3-ipv4-uc) # redistribute ripng route-map ripng_networks
switch(config-ospfv3-af-1)# redistribute ripng route-map ripng_networks1
Route Map ripng_networks1 not found
switch(config-ospfv3-af-1)# redistribute ripng route-map ripng_networks12
Redistribute with route-map config exists
switch(config-ospfv3-ipv4-uc) # redistribute ospfv3-af 2 route-map
% Command incomplete.
switch(config-ospfv3-ipv4-uc) # redistribute ospfv3-af 2 route-map tagged_routes
switch(config-ospfv3-ipv4-uc)# redistribute ospfv3-af 2 route-map tagged_routes1
Route Map tagged_routes1 not found
switch(config-ospfv3-ipv4-uc)# redistribute ospfv3-af 2 route-map tagged_routes12
Redistribute with route-map config exists
switch(config-ospfv3-ipv4-uc)# redistribute local loopback route-map
% Command incomplete.
switch(config-ospfv3-ipv4-uc)# redistribute local loopback route-map local_routes
switch(config-ospfv3-af-1)# redistribute local loopback route-map local_routes1
Route Map local_routes1 not found
switch(config-ospfv3-ipv4-uc)# redistribute local loopback route-map local_
routes12
Redistribute with route-map config exists
switch(config-ospfv3-ipv4-uc)# redistribute host-routes route-map
% Command incomplete.
switch(config-ospfv3-ipv4-uc)# redistribute host-routes route-map host_routes
switch(config-ospfv3-ipv4-uc)# redistribute host-routes route-map host_routes1
Route Map local_routes1 not found
switch(config-ospfv3-ipv4-uc)# redistribute host-routes route-map host_routes12
Redistribute with route-map config exists
```

Disabling redistributing routes to OSPFv3 AF:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv6-uc) # no redistribute bgp
```

```

switch(config-ospfv3-ipv6-uc) # no redistribute connected
switch(config-ospfv3-ipv6-uc) # no redistribute host-routes
switch(config-ospfv3-ipv6-uc) # no redistribute static
switch(config-ospfv3-ipv6-uc) # no redistribute ripng
switch(config-ospfv3-ipv6-uc) # no redistribute ospfv3-af 2
switch(config-ospfv3-ipv6-uc) # no redistribute local loopback
switch(config-ospfv3-ipv6-uc) # no redistribute static route-map
% Command incomplete.
switch(config-ospfv3-ipv6-uc) # no redistribute static route-map static_networks
switch(config-ospfv3-ipv6-uc) # no redistribute ripng route-map
% Command incomplete.
switch(config-ospfv3-ipv6-uc) # no redistribute ripng route-map ripng_networks
switch(config-ospfv3-ipv6-uc) # no redistribute ospfv3-af 2 route-map
% Command incomplete.
switch(config-ospfv3-ipv6-uc) # no redistribute ospfv3-af 2 route-map tagged_
routes
switch(config-ospfv3-ipv6-uc) # no redistribute local loopback route-map
% Command incomplete.
switch(config-ospfv3-ipv6-uc) # no redistribute local loopback route-map local_
routes
switch(config-ospfv3-ipv6-uc) # no redistribute host-routes route-map
% Command incomplete.
switch(config-ospfv3-ipv6-uc) # no redistribute host-routes route-map host_routes

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## reference-bandwidth (OSPFv3 AF)

```

reference-bandwidth <BANDWIDTH-VALUE>
no reference-bandwidth

```

### Description

Sets the reference bandwidth for OSPFv3 AF. If the OSPFv3 AF interface cost is not explicitly set, then the cost of all the OSPFv3 AF interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the calculated link speed value is 1 Gbps (if the OSPFv3 AF interface cost is not explicitly set).

The **no** form of this command sets the reference bandwidth for OSPFv3 AF to the default of 100000 Mbps.

| Parameter         | Description                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <BANDWIDTH-VALUE> | Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000. |

## Examples

Setting the reference bandwidth:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-1)# no reference-bandwidth
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context            | Authority                                                                          |
|--------------|----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## retransmit-interval (OSPFv3 AF)

```
retransmit-interval <INTERVAL>
no retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for virtual links.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for virtual links.

| Parameter  | Description                                                                                        |
|------------|----------------------------------------------------------------------------------------------------|
| <INTERVAL> | Specifies the time interval for the retransmit interval, in seconds. Range: 1 to 3600. Default: 5. |

## Examples

Setting OSPFv3 AF virtual links retransmit interval:



```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# retransmit-interval 30
```

Setting OSPFv3 AF virtual links retransmit interval to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no retransmit-interval
switch(config-router-vlink6)# no retransmit-interval 30
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |

## router-id (OSPFv3 AF)

```
router-id <ROUTER-ADDRESS>
no router-id
```

## Description

Sets an ID for the router in an IPv4 address format.

| Parameter        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ROUTER-ADDRESS> | Specifies the router address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>Required                                                                                                                                                                                                                                                                                                                                                                                        |
| no               | Unconfigures the router-id for the instance. The Router-id is changed to the dynamically selected router-id. The default router-id 0.0.0.0 is updated to the routing stack that triggers to auto-elect a router-id based on the highest ip-address of loopback interface or highest ip-address of interfaces. The configuration is supported globally as well as per address-families ipv4-unicast, ipv6-unicast. If address-family specific router-id is configured, it takes precedence over global configuration. |



In case there are duplicate loopback IPs within an OSPF domain then the recommendation is to explicitly configure router-id to prevent an unintended IP Address from being selected as router-id. Virtual VTEP is one such example which uses same loopback IP on 2 switches. Even if the IP is not the highest address to be selected as router-id, there switches. Dynamic election of router-id is not supported when multiple OSPF processes are configured on the same VRF. Explicit router-id configuration is required on each OSPF process.



Dynamic router-id is chosen from the list of configured IPs at the time of the router-id election. In Graceful Restart and Management Module failover scenarios, the manual router-id configuration is highly recommended as the router-id should be preserved before and after these events. Graceful Restart would fail if there is a change in the router-id causing traffic loss. A few scenarios where the dynamically elected router-id after a restart could be different from what was selected before daemon restart are mentioned below:

- OSPF selects the router-id. Later on, new interface IPs are configured. OSPF doesn't re-calculate the router-id to consider the newer IP additions. However, after the restart, router-id is again calculated which considers all the IPs (including the ones which were later added).
- After a daemon restart, the router-id is chosen based on the order in which the interface IP notifications are received by the OSPF process. OSPF considers the IP notifications received until it starts the router-id selection. The IP notifications received post the router-id selection are not considered.

## Examples

Setting router-id in the OSPFv3 AF context:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1) # router-id 1.1.1.1
switch(config-ospfv3-af-1) # address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc) # router-id 2.2.2.2
switch(config-ospfv3-af-1) # address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc) # router-id 3.3.3.3
```

Unconfiguring router-id:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc) # no router-id
switch(config-ospfv3-ipv4-uc) # no router-id 1.1.1.1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## router ospfv3-af

```
router ospfv3-af <PROCESS-ID> [vrf <VRF-NAME>]
no router ospfv3-af <PROCESS-ID> [vrf <VRF-NAME>]
```

### Description

Creates the OSPFv3 AF process (if not created already) and enters the router OSPFv3 AF instance context. Optionally if specified, you can specify a named VRF, or the default VRF if the <vrf-name> is not specified. Maximum of eight OSPFv3 processes are allowed per VRF.

The switch allows configuration of OSPFv2, OSPFv3, OSPFv3 IPv4 AF and OSPFv3 IPv6 AF. But at any point only 8 OSPF instances can run per address-family, 8 instances for IPv4 (OSPFv2 + OSPFv3 IPv4 AF) and 8 instances for IPv6 (OSPFv3 + OSPFv3 IPv6 AF).

The **no** form of this command removes the OSPFv3 AF instance. If a VRF is specified, it removes the OSPF instance from the named VRF, or the default VRF if the <var-name> is not specified.

| Parameter      | Description                                            |
|----------------|--------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 AF process ID. Length: 1 to 65535. |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.         |

### Examples

Entering the router OSPFv3 AF instance:

```
switch(config)# router ospfv3-af 1
```

Setting the router OSPFv3 VRF instance:

```
switch(config)# router ospfv3-af 1 vrf vrf_red
```

Removing the router OSPFv3 AF instance:

```
switch#(config)# no router ospfv3-af 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

### Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## show ospfv3-af ipv4 statistics

show ospfv3-af [<process-id>] ipv4 statistics [all-vrfs | vrf <vrf-name>]

### Description

This command displays the OSPFv3 AF shortest path first (SPF) calculation statistics.

| Parameter      | Description                                                    |
|----------------|----------------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 AF process ID.<br>Range: 1 to 65535        |
| all-vrfs       | Display the OSPFv3 AF statistics information for all the VRFs. |
| vrf <VRF-NAME> | Specify the VRF name.                                          |

### Example

```
switch# show ospfv3-af ipv4 statistics
Address-family : IPv4 Unicast
-----
OSPFv3 AF Process ID 1 VRF default, Statistics (cleared 3h 2m 21s ago)
-----
Unknown Interface Drops           : 0
Unknown Virtual Interface Drops   : 0
Bad IPv6 Header Length Drops      : 0
Wrong OSPFv3 Version Drops        : 0
Bad Source IPv6 Drops             : 0
Resource Failure Drops            : 0
Bad Header Length Drops           : 0
Total Drops                       : 0
switch# show ospfv3-af 2 ipv4 statistics vrf red
Address-family : IPv4 Unicast
-----
OSPFv3 AF Process ID 2 VRF red, Statistics (cleared 3h 2m 30s ago)
-----
Unknown Interface Drops           : 0
Unknown Virtual Interface Drops   : 0
Bad IPv6 Header Length Drops      : 0
Wrong OSPFv3 Version Drops        : 0
Bad Source IPv6 Drops             : 0
Resource Failure Drops            : 0
Bad Header Length Drops           : 0
Total Drops                       : 0
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show debug buffer module ospfv3-af

```
show debug buffer module ospfv3-af
[severity (alert|crit|debug|emer|err|info|notice|warn)]
```

## Description

Displays debug logs for OSPFv3 AF stored in the specified debug buffer.

| Parameter                                             | Description                                                                                             |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <MODULE-NAME>                                         | Filters debug logs displayed by the specified module name.                                              |
| severity (alert crit debug emer err info notice warn) | Displays debug logs for OSPFv3 AF with a specified severity level. Defaults to <b>debug</b> . Optional. |
| alert                                                 | Displays debug logs with a severity level of <b>alert (6)</b> and above.                                |
| crit                                                  | Displays debug logs with a severity level of <b>critical (5)</b> and above.                             |
| debug                                                 | Displays debug logs with all severities.                                                                |
| emer                                                  | Displays debug logs with a severity level of <b>emergency (7)</b> only.                                 |
| err                                                   | Specifies storage of debug logs with severity level of <b>error (4)</b> and above.                      |
| info                                                  | Displays debug logs with a severity level of <b>info (1)</b> and above.                                 |
| notice                                                | Specifies storage of debug logs with severity level of <b>notice (2)</b> and above.                     |
| warning                                               | Displays debug logs with a severity level of <b>warning (3)</b> and above.                              |

## Examples

```
switch(config)# show debug
```

```
-----
-----
module      sub_module  severity  vlan  port  ip      mac      instance
  vrf
-----
```

```

-----
ospfv3-af    all                debug

switch(config)# show debug buffer module ospfv3-af
-----
show debug buffer
-----
2017-09-18:10:35:38.621611|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|OSPF
537265152 hitless restart state changed (None -> None),
reason: None.
2017-09-18:10:35:38.621626|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF| (OSPFv3
process Id = 1, VRF Name = default)
2017-09-18:10:35:38.621836|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-
AF|0X01111000 Protocol Manager has initialized successfully.
2017-09-18:10:35:38.621846|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|Entity
index                = OSPF 537265152
2017-09-18:10:35:38.621854|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF| (OSPFv3
process Id = 1, VRF Name = default)
2017-09-18:10:35:38.622188|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|OSPF
537265152 hitless restart state changed (None -> None),
reason: None.
2017-09-18:10:35:38.622197|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF| (OSPFv3
process Id = 1, VRF Name = default)
2017-09-18:10:35:38.622423|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|OSPF
537265152 created.
2017-09-18:10:35:38.622433|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF| (OSPFv3
process Id = 0, VRF Name = default)
2017-09-18:10:35:38.622516|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|A master
join has been successfully registered.
2017-09-18:10:35:38.622524|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|Join
interface type       = 2
2017-09-18:10:35:38.622536|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|Join
interface ID         = 0X00000001
2017-09-18:10:35:38.622544|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF| (OSPFv3
process Id = 1, VRF Name = default)
2017-09-18:10:35:38.622717|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|OSPF
537265152 i/f idx 0X10010101: OSPF has been told over
the I3 interface about a new address on an
2017-09-18:10:35:38.622728|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-
AF|interface, or about updated attributes of an existing address.
2017-09-18:10:35:38.622738|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|Address
= 111::1
2017-09-18:10:35:38.622746|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|Scope ID
= 268501249
2017-09-18:10:35:38.622754|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|Prefix
length              = 64
2017-09-18:10:35:38.622762|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|Remote
router address      = ::
2017-09-18:10:35:38.622771|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|Address
use flags           = 0X00
2017-09-18:10:35:38.622780|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF| (OSPFv3
process Id = 1, VRF Name = default)
2017-09-18:10:35:38.625212|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|OSPF
537265152 activated successfully
2017-09-18:10:35:38.625225|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|Router
ID                  rtr ID 1.1.1.1
2017-09-18:10:35:38.625235|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF| (OSPFv3
process Id = 1, VRF Name = default)
2017-09-18:10:35:38.625286|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|OSPF
537265152 was Down and is now Going Up.

```

```

2017-09-18:10:35:38.625295|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|(OSPFv3
process Id = 1, VRF Name = default)
2017-09-18:10:35:38.625309|hpe-routing|LOG_ERR|AMM|-|OSPFV3-AF|OSPFv3_AF_
EVENT|Failed to configure interface cost for stub link port 1/1/1.
2017-09-18:10:35:38.625312|hpe-routing|LOG_ERR|AMM|-|OSPFV3-AF|OSPFv3_AF_
EVENT|Could not send ospfv3VirtNbrStateChange Trap
2020-07-21:11:36:47.324342|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|OSPF 537265152 NM has sent a Hello packet.
2020-07-21:11:36:47.324717|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Sending interface index = i/f idx 0X51101000.
2020-07-21:11:36:47.324952|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Router ID of target neighbor
(null string if sent per interface) = .
2020-07-21:11:36:47.325217|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Destination IP address = IP addr FF02::5.
2020-07-21:11:36:47.325460|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_PACKET|
(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-07-21:11:36:47.326710|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|OSPF message details:
2020-07-21:11:36:47.326946|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Header: Src IP:FF02::5 Dst IP:FE80::800:901:4C3:C58B
Type: 1 Rtr Id:2.2.2.2 Area:0.0.0.1 Inst ID:0
2020-07-21:11:36:47.327166|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Packet: Hello: I/f 0X51101000 Pri:1 Opt:19 Hello
Int:10 Dead Int:40 DR:2.2.2.2 BDR:1.1.1.1
2020-07-21:11:36:47.327350|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_PACKET|
(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-08-12:11:59:50.183653|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|OSPF message details:
2020-08-12:11:59:50.183666|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Header: Src IP:FE80::800:901:4C1:3751 Dst IP:FF02::5
Type: 4 Rtr Id:2.2.2.2 Area:0.0.0.1 Inst ID:0
2020-08-12:11:59:50.183676|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Packet: LS Upd: Num LSAs 2
2020-08-12:11:59:50.183687|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_PACKET|
(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-08-12:11:59:50.272301|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|OSPF message details:
2020-08-12:11:59:50.272313|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Header: Src IP:FE80::800:901:4C1:3751 Dst IP:FF02::5
Type: 5 Rtr Id:2.2.2.2 Area:0.0.0.1 Inst ID:0
2020-08-12:11:59:50.272323|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Packet: LS Ack
2020-08-12:11:59:50.272332|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_PACKET|
(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-08-12:11:59:45.072407|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|OSPF message details:
2020-08-12:11:59:45.072424|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Header: Src IP:FE80::800:901:4C1:3751
Dst IP:FE80::800:901:40D:A274 Type: 2 Rtr Id:2.2.2.2 Area:0.0.0.1 Inst ID:0
2020-08-12:11:59:45.072443|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Packet: DB Desc: If MTU:1500 Opt:19 I,M,MS
bits:0X01 Seq No:1020993
2020-08-12:11:59:45.072460|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_PACKET|
(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-08-12:11:59:45.073077|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|OSPF message details:
2020-08-12:11:59:45.073096|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Header: Src IP:FE80::800:901:4C1:3751
Dst IP:FE80::800:901:40D:A274 Type: 3 Rtr Id:1.1.1.1 Area:0.0.0.1 Inst ID:0
2020-08-12:11:59:45.073114|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Packet: LS Req: First LS Type:8193 LS id:0.0.0.0

```

```

Adv Router:2.2.2.2
2020-08-12:11:59:45.073130|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_PACKET|
(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-08-12:12:11:22.344623|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|OSPF message details:
2020-08-12:12:11:22.344635|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Header: Src IP:FE80::800:901:4C1:3751
Dst IP:FE80::800:901:40D:A274 Type: 2 Rtr Id:2.2.2.2 Area:0.0.0.1 Inst ID:0
2020-08-12:12:11:22.344646|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_
PACKET|Packet: DB Desc: If MTU:1500 Opt:19 I,M,MS bits:0X07
Seq No:1728268
2020-08-12:12:11:22.344663|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|OSPFv3_AF_PACKET|
(End of packet, OSPFv3 process Id = 1, VRF Name = default)

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv4

```
show ospfv3-af [<PROCESS-ID>] ipv4 [all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows OSPFv3 AF information including area, state, and configuration information.

| Parameter      | Description                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 AF process ID optionally to show OSPFv3 AF information for a particular OSPF process. Range: 1 to 65535.       |
| all-vrfs       | Select to show OSPFv3 AF information for all VRFs.                                                                                 |
| vrf <VRF-NAME> | Specifies the name of a VRF. The default VRF OSPFv3 AF process information is displayed if VRF is not mentioned. Default: default. |

## Example

Showing general OSPFv3 AF configurations:



```

switch# show ospfv3-af 200 ipv4
Address-family : IPv4 Unicast
-----

VRF : default                                Process : 20200
-----

Router ID          : 1.1.1.1                OSPFv3-AF          : Enabled
BFD                : Disabled                SPF Start Interval : 200 ms
SPF Hold Interval  : 1000 ms                 SPF Max Wait Interval : 5000 ms
LSA Start Interval : 5000 ms                 LSA Hold Interval   : 1000 ms
LSA Max Wait Interval : 1000 ms              LSA Arrival Interval : 1000 ms
External LSAs      : 0                      Checksum Sum        : 0
ECMP                : 4                      Reference Bandwidth  : 100000 Mbps
Area Border         : Yes                    AS Border           : No
GR Status           : Enabled                GR Interval         : 120 sec
GR State            : Inactive                GR Exit Status      : None
GR Helper           : Enabled                GR Strict LSA Check  : Enabled
GR Ignore Lost I/F  : Disabled                Internal Process ID  : 1
Summary address:
  prefix fd00::1/64, advertise, tag 10

Area      Total Active
-----
Normal    2          2
Stub      0          0

Area : 0.0.0.0
-----
Area Type          : Normal                Status              : Active
Total Interfaces   : 1                    Active Interfaces    : 1
Passive Interfaces  : 0                    Loopback Interfaces   : 0
SPF calculation count : 4
Area ranges:
  fd00::1/64, inter-area, no-advertise
AH Authentication   : SHA1, SPI 256
Number of LSAs      : 5                    Checksum Sum         : 99122

Area : 0.0.0.1
-----
Area Type          : Normal                Status              : Active
Total Interfaces   : 1                    Active Interfaces    : 1
Passive Interfaces  : 0                    Loopback Interfaces   : 0

SPF Calculation Count : 4
Area ranges:
  fd00::1/64, inter-area, no-advertise
ESP Authentication   : SHA1
Encryption           : 3DES, SPI 256
Number of LSAs      : 5                    Checksum Sum         : 99122

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv4 border-routers

show ospfv3-af [<PROCESS-ID>] ipv4 border-routers | [all-vrfs | vrf <VRF-NAME>]

### Description

Shows the OSPFv3 AF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

| Parameter      | Description                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 AF process ID to show the OSPFv3 AF routing table entries for ABR and ASBR for the particular OSPFv3 AF process. Range: 1 to 65535. |
| all-vrfs       | Select to show OSPFv3 AF border router information for all VRFs.                                                                                        |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.                                                                                                          |

### Example

Showing OSPFv3 AF border routers information for VRF **vrf\_red**:

```
switch# show ospfv3-af ipv4 border-routers vrf vrf_red

Address-family : IPv4 Unicast
-----

VRF : vrf_red      Process ID : 1
      Internal Routing Table
-----

Codes: i - Intra-area route, I - Inter-area route
Router-ID Cost  Type   Area   SPF   Nexthop   Interface
i  1.1.1.1    1    ASBR   0.0.0.0  9     10.10.10.1  1/1/2
i  3.3.3.3    1    ASBR   1.1.1.1  9     20.20.20.1  ** tunnel1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv4 interface

```
show ospfv3-af [<PROCESS-ID>] ipv4 interface [<INTERFACE-NAME>] | vlan <VLAN-ID> | lag
<LAG-ID> | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>]
[brief] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows information about OSPFv3 AF enabled IPv4 interfaces.

| Parameter              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>           | Specifies an OSPFv3 AF process ID optionally to show the OSPFv3 AF enabled interfaces for the particular OSPFv3 AF process. Range: 1 to 65535.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <INTERFACE-NAME>       | Selects to show information only for the specified OSPFv3 AF enabled interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| vlan <VLAN-ID>         | Specifies the vlan interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| lag <LAG-ID>           | Specifies the lag interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| loopback <LOOPBACK-ID> | Specifies the loopback interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| tunnel <TUNNEL-ID>     | Specifies the tunnel interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| brief                  | Include this parameter to display a brief overview of the following OSPFv3 AF configuration information. <ul style="list-style-type: none"> <li>Interface: OSPFv3 AF interface name.</li> <li>Area: OSPFv3 AF area ID.</li> <li>Cost: The metric OSPFv3 AF uses to judge a path's feasibility, calculated as (reference bandwidth / interface bandwidth).</li> <li>State: Indicates if the interface is a designated router (<b>Dr</b>) or a backup designated router (<b>Backup-dr</b>).</li> <li>Status: Indicates if the interface is <b>up</b> or <b>down</b>.</li> <li>Flags: P - Passive A - Active.</li> </ul> |
| all-vrfs               | Select to show OSPFv3 AF-enabled interface information for all VRFs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| vrf <VRF-NAME>         | Specifies the name of a VRF. Default: default.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Examples

Showing OSPFv3 AF information for all interfaces in default VRF:

```
switch# show ospfv3-af ipv4 interface
Codes: DR - Designated router BDR - Backup Designated router
```

Address-family : IPv4 Unicast

Interface 1/1/1 is up, Line Protocol is up, Instance Id 64

```
-----
VRF                : default                               Process          : 1
IP Address          : 1.1.1.2                               Area              :
0.0.0.0
Status              : Up                                   Network Type     :
Broadcast
Hello Interval      : 10      sec                          Dead Interval     : 40
sec
Transit Delay       : 1      sec                          Retransmit Interval : 5
sec
BFD                 : Disabled                             Link Speed        :
1000 Mbps
Cost Configured     : 1                                    Cost Calculated   : 4
State/Type          : BDR                                  Router Priority    : 1
DR                  : 3.3.3.3                              BDR               :
1.1.1.3
Link LSAs           : 2                                    Checksum Sum      :
26868
AH Authentication   : SHA1, SPI 256 (Area)                 Passive          : Yes
-----
```

Interface 1/1/2 is up, Line Protocol is up, Instance Id 64

```
-----
VRF                : default                               Process          : 1
IP Address          : 10.10.10.2                           Area              :
0.0.0.0
Status              : Up                                   Network Type     :
Broadcast
Hello Interval      : 10      sec                          Dead Interval     : 40
sec
Transit Delay       : 1      sec                          Retransmit Interval : 5
sec
Link Speed          : 1000 Mbps
Cost Configured     : 1                                    Cost Calculated   : 4
State/Type          : BDR                                  Router Priority    : 1
DR                  : 10.10.10.2                           BDR               :
10.10.10.1
Link LSAs           : 0                                    Checksum Sum      : 0
AH Authentication   : SHA1, SPI 256 (Area)                 Passive          : No
-----
```

Interface tunnelli is up, Line Protocol is up, Instance Id 64

```
-----
VRF                : default                               Process          : 1
IPv6 Address        : 10.10.10.1                           Area              :
0.0.0.1
Status              : Up                                   Network Type     :
Broadcast
Hello Interval      : 10      sec                          Dead Interval     : 40
sec
Transit Delay       : 1      sec                          Retransmit Interval : 5
sec
BFD                 : Disabled                             Link Speed        :
1000 Mbps
Cost Configured     : 1                                    Cost Calculated   : 4
State/Type          : DR                                  Router Priority    : 1
DR                  : 10.10.10.2                           BDR               :
10.10.10.1
Link LSAs           : 1                                    Checksum Sum      :
9168
-----
```

```

AH Authentication      : SHA1, SPI 256 (Area)          Passive      : Yes

Interface vlan10 is up, Line Protocol is up, Instance Id 64
-----
VRF                    : default                      Process      : 1
IPv6 Address           : 10.10.10.3                  Area         :
0.0.0.1
Status                : Up                          Network Type :
Broadcast
Hello Interval        : 10      sec                  Dead Interval : 40
sec
Transit Delay         : 1      sec                  Retransmit Interval : 5
sec
BFD                   : Disabled                      Link Speed   :
1000 Mbps
Cost Configured       : 1                          Cost Calculated : 4
State/Type            : DR                          Router Priority : 1
DR                    : 10.10.10.2                  BDR          :
10.10.10.1
Link LSAs             : 1                          Checksum Sum  :
9168
AH Authentication     : MD5, SPI 257                Passive      : No

Interface vlan20 is up, Line Protocol is up, Instance Id 64
-----
VRF                    : default                      Process      : 1
IPv6 Address           : 1.1.1.1                    Area         :
0.0.0.1
Status                : Up                          Network Type :
Broadcast
Hello Interval        : 10      sec                  Dead Interval : 40
sec
Transit Delay         : 1      sec                  Retransmit Interval : 5
sec
Link Speed            : 1000 Mbps
Cost Configured       : 1                          Cost Calculated : 4
State/Type            : DR                          Router Priority : 1
DR                    : 1.1.1.3                    BDR          : NA
Link LSAs             : 1                          Checksum Sum  :
9168
AH Authentication     : MD5, SPI 257                Passive      : No

```

Showing overview information for OSPFv3 AF enabled interfaces for all VRFs in brief:

```

switch# show ospfv3-af ipv4 interface brief all-vrfs
Address-family : IPv4 Unicast
-----

OSPFv3 AF Process ID 1 VRF default
=====

Total Number of Interfaces: 2

Flags: P - Passive  A - Active

Interface      Area          Cost      State      Status      Flags
-----
1/1/1          0.0.0.0       1         BDR        Up          P
vlan10         0.0.0.1       1         DR         Up          A

```

```
OSPFv3 AF Process ID 2 VRF red
```

```
=====
```

```
Total Number of Interfaces: 1
```

```
Flags: P - Passive  A - Active
```

| Interface | Area    | Cost | State | Status | Flags |
|-----------|---------|------|-------|--------|-------|
| 1/1/2     | 0.0.0.1 | 1    | BDR   | Up     | P     |
| tunnel1   | 0.0.0.0 | 100  | DR    | Up     | A     |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv4 lsdb

```
show ospfv3-af [<PROCESS-ID>] ipv4 lsdb [all-vrfs | vrf <VRF-NAME>] [router | network |  
inter-area-prefix | inter-area-router | as-external | nssa-external | link | intra-area-  
prefix]
```

```
[area <AREA-ID>] [lsid <LINK-STATE-ID>] [adv-router <ROUTER-ID> | self]
```

```
show ospfv3-af [<PROCESS-ID>] ipv4 lsdb [all-vrfs | vrf <VRF-NAME>] database-summary
```

## Description

Shows the OSPFv3 AF link state database summary for different OSPFv3 AF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

| Parameter               | Description                                                                                                                                                                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>            | Enter an OSPFv3 AF process ID to display general OSPF information for a particular OSPFv3 AF process. Range: 1 to 65535.                                                                                                                                                  |
| all-vrfs vrf <vrf-name> | Select <b>all-vrfs</b> to display general OSPF information for all VRFs, or use the <b>vrf &lt;VRF-NAME&gt;</b> option to display information for a specific VRF.<br><br>Optionally select one of the following parameters to filter the link state database information. |

| Parameter                     | Description                                                                                                                                                                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| as-external                   | Show external link states (LSA type 5)                                                                                                                                                                                                                                                   |
| database-summary              | Select to display the count of each type of LSA and each area in the database.<br><br><b>NOTE:</b> The <b>database-summary</b> parameter does not support the <b>area &lt;area-id&gt;</b> , <b>lsid &lt;link-state-id&gt;</b> or <b>adv-router {&lt;router-id&gt; self}</b> parameters.  |
| inter-area-prefix             | Show inter-area prefix link states (LSA type 3)                                                                                                                                                                                                                                          |
| inter-area-router             | Show inter-area router link states (LSA type 4)                                                                                                                                                                                                                                          |
| intra-area-prefix             | Show intra-area prefix link states (LSA type 9)                                                                                                                                                                                                                                          |
| link                          | Show link states (LSA type 8)                                                                                                                                                                                                                                                            |
| network                       | Show network LSAs (LSA type 2).                                                                                                                                                                                                                                                          |
| nssa-external                 | Show NSSA external link states (LSA type 7).                                                                                                                                                                                                                                             |
| router                        | Show router LSAs (LSA type 1).                                                                                                                                                                                                                                                           |
| area <AREA-ID>                | Select to display information filtered for the specified area in one of the following formats.<br>OSPFv3 AF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPFv3 AF area identifier in decimal format. Value: 0 to 4294967295. |
| lsid <LINK-STATE-ID>          | Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D).                                                                                                                                                                              |
| adv-router {<ROUTER-ID> self} | Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify <b>self</b> to show self-originated link states.                                                                                                      |

## Examples

Showing OSPFv3 AF link state database (LSDB) general information:

```
switch# show ospfv3-af ipv4 lsdb
Address-family : IPv4 Unicast
-----

OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
=====

Router Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Link Count  Bits
-----
40.40.40.40     930     0x80000004 0x2ea1      0           3           None
50.50.50.50     935     0x80000002 0x8b52      0           1           E
60.60.60.60     943     0x800003c5 0x9854      0           2           None

Network Link State Advertisements (Area 0.0.0.0)
```

```

-----
ADV Router      Age      Seq#      Checksum    LSID      Router Count
-----
60.60.60.60    944      0x80000001 0x7179      1360007168 2
50.50.50.50    935      0x80000001 0x516a      19          1

Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Prefix
-----
40.40.40.40    929      0x80000001 0x2498      131072     11.0.0.0/8
50.50.50.50    928      0x80000001 0x5b2f      65536     100.100.0.0/16

Inter Area Router Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Destination Router ID
-----
40.40.40.40    929      0x80000001 0x2498      1          33.33.33.33

AS External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Prefix
-----
40.40.40.40    264      0x80000001 0x24cc4     1          10.10.0.0/16
40.40.40.40    675      0x80000001 0x5b00f     2          11.11.0.0/16

NSSA External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Prefix
-----
3.3.3.3        264      0x80000001 0x24ac2     1          200.200.0.0/16

Link-local Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Interface
-----
50.50.50.50    264      0x80000001 0x653c4     19         1/1/1

Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
-----
-----
ADV Router      Age      Seq#      Checksum    LSID      Referenced LS Type
Referenced LSID
-----
50.50.50.50    263      0x80000001 0x1da34     1          0x2001          0
50.50.50.50    264      0x80000001 0x2a45d     1          0x2002          19

```

Showing AS external link states:

```

switch# show ospfv3-af ipv4 lsdb as-external
Address-family : IPv4 Unicast
-----

OSPF Router with ID (60.60.60.60) (Process ID 1 VRF default)
=====

AS External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Prefix
-----

```



|             |     |            |         |   |              |
|-------------|-----|------------|---------|---|--------------|
| 40.40.40.40 | 264 | 0x80000001 | 0x24cc4 | 1 | 10.10.0.0/16 |
| 40.40.40.40 | 675 | 0x80000001 | 0x5b00f | 2 | 11.11.0.0/16 |

Showing the LSDB database summary:

```
switch# show ospfv3-af ipv4 lsdb database-summary
Address-family : IPv4 Unicast
-----

OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
=====

Area 0.0.0.0 database summary
-----
LSA Type                Count
-----
Router                  2
Network                 1
Inter Area Prefix       1
Inter Area Router       0
NSSA External           0
Link                    3
Intra Area Prefix       3
-----
Total                    10

Process 1 database summary
-----
LSA Type                Count
-----
Router                  2
Network                 1
Inter Area Prefix       1
Inter Area Router       0
AS External             2
NSSA External           0
Link                    3
Intra Area Prefix       3
-----
Total                    12
```

Showing inter-area prefix LSAs:

```
switch# show ospfv3-af ipv4 lsdb inter-area-prefix
Address-family : IPv4 Unicast
-----

OSPF Router with ID (6.6.6.6) (Process ID 1 VRF default)
=====

Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
-----
ADV Router    Age    Seq#    Checksum    LSID    Prefix
-----
40.40.40.40   929    0x80000001  0x2498     131072  1.0.0.0/8
50.50.50.50   928    0x80000001  0x5b2f     65536   11.11.0.0./16
```

Showing inter-area router LSAs:

```

switch# show ospfv3-af ipv4 lsdB inter-area-router
Address-family : IPv4 Unicast
-----

OSPF Router with ID (60.60.60.60) (Process ID 1 VRF default)
=====

Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
-----

```

| ADV Router  | Age | Seq#       | Checksum | LSID | Destination Router ID |
|-------------|-----|------------|----------|------|-----------------------|
| 40.40.40.40 | 929 | 0x80000001 | 0x2498   | 1    | 33.33.33.33           |

Showing intra-area prefix LSAs:

```

switch# show ospfv3-af ipv4 lsdB intra-area-prefix
Address-family : IPv4 Unicast
-----

OSPF Router with ID (0.0.0.0) (Process ID 1 VRF default)
=====

Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
-----

```

| ADV Router<br>Referenced LSID | Age | Seq#       | Checksum | LSID | Referenced LS Type |
|-------------------------------|-----|------------|----------|------|--------------------|
| 50.50.50.50<br>0              | 263 | 0x80000001 | 0x1da34  | 1    | 0x2001             |
| 50.50.50.50<br>19             | 264 | 0x80000001 | 0x2a45d  | 1    | 0x2002             |

Showing link LSAs:

```

switch# show ospfv3-af ipv4 lsdB link
Address-family : IPv4 Unicast
-----

OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====

Link-local Link State Advertisements (Area 0.0.0.0)
-----

```

| ADV Router  | Age | Seq#       | Checksum | LSID | Interface |
|-------------|-----|------------|----------|------|-----------|
| 50.50.50.50 | 264 | 0x80000001 | 0x653c4  | 19   | 1/1/1     |

```

Link-local Link State Advertisements (Area 0.0.0.0)
-----

```

| ADV Router  | Age | Seq#       | Checksum | LSID | Interface |
|-------------|-----|------------|----------|------|-----------|
| 50.50.50.50 | 264 | 0x80000001 | 0xad324  | 20   | 1/1/2     |

Showing network LSAs:

```
switch# show ospfv3-af ipv4 lsdB network
Address-family : IPv4 Unicast
-----

OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
=====

Network Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Router Count
-----
60.60.60.60     944     0x80000001 0x7179      1360007168 2
50.50.50.50     935     0x80000001 0x516a      19          1
```

Showing NSSA external link states:

```
switch# show ospfv3-af ipv4 lsdB nssa-external
Address-family : IPv4 Unicast
-----

OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====

NSSA External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Prefix
-----
3.3.3.3         264     0x80000001 0x24ac2     1           200.200.0.0/16
```

Showing router LSAs:

```
switch# show ospfv3-af ipv4 lsdB router
Address-family : IPv4 Unicast
-----

OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
=====

Router Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Link Count  Bits
-----
40.40.40.40     930     0x80000004 0x2ea1      0           3           None
50.50.50.50     935     0x80000002 0x8b52      0           1           E
60.60.60.60     943     0x800003c5 0x9854      0           2           None
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv4 neighbors

```
show ospfv3-af [<PROCESS-ID>] ipv4 neighbors [<NEIGHBOR-ID>]
[interface <INTERFACE-NAME> | vlan <VLAN-ID> | lag <LAG-ID> | loopback <LOOPBACK-ID> |
tunnel <TUNNEL-ID>]
[detail | summary] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows information about OSPFv3 AF neighbors.

| Parameter                  | Description                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies an OSPFv3 AF process ID to show OSPFv3 AF neighbor information for the particular OSPFv3 AF process. Range: 1 to 65535. |
| neighbors <NEIGHBOR-ID>    | Shows information about a particular neighbor, specified in IPv4 format (A.B.C.D).                                                |
| interface <INTERFACE-NAME> | Shows neighbor information only for the specified interface.                                                                      |
| vlan <VLAN-ID>             | Specifies the vlan interface name.                                                                                                |
| lag <LAG-ID>               | Specifies the lag interface name.                                                                                                 |
| loopback <LOOPBACK-ID>     | Specifies the loopback interface name.                                                                                            |
| tunnel <TUNNEL-ID>         | Specifies the tunnel interface name.                                                                                              |
| detail                     | Shows detailed information for the neighbors.                                                                                     |
| summary                    | Shows summary information for the neighbors.                                                                                      |
| all-vrfs                   | Shows neighbor information for all VRFs.                                                                                          |
| vrf <VRF-NAME>             | Specifies the name of a VRF. Default: default.                                                                                    |

### Examples

Showing OSPFv3 AF neighbor information for the specified interface:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3
Address-family : IPv4 Unicast
-----

VRF : default                                Process : 1
-----
Router-Id      : 3.3.3.3                      Area          : 0.0.0.0
Interface      : 1/1/1                        Address        : 1.1.1.1
State          : FULL                          Neighbor Priority : 1
```

```
Dead Timer Due      : 00:00:36      Options      : 0x13
Time since last state change : 00h:14m:45s
```

Showing detail OSPFv3 AF neighbor information:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3 detail
Address-family : IPv4 Unicast
-----
VRF : default                                Process : 1
-----
Router-Id      : 3.3.3.3                    Area          : 0.0.0.0
Interface      : 1/1/1                      Address        : 1.1.1.1
State          : FULL                       Neighbor Priority : 1
DR             : 3.3.3.3                    BDR           : 1.1.1.3
Dead Timer Due : 00:00:36                  Options       : 0x13
Retransmission Queue Length : 0
Time Since Last State Change : 00h:14m:45s
```

Showing OSPFv3 AF neighbor information for interface **1/1/1**:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3 interface 1/1/1
Address-family : IPv4 Unicast
-----
VRF : default                                Process : 1
-----
Router-Id      : 3.3.3.3                    Area          : 0.0.0.0
Interface      : 1/1/1                      Address        : 1.1.1.1
State          : FULL                       Neighbor Priority : 1
Dead Timer Due : 00:00:36                  Options       : 0x13
Time Since Last State Change : 00h:14m:45s
```

Showing OSPFv3 AF neighbor information for tunnel interface:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3
Address-family : IPv4 Unicast
-----
VRF : default                                Process : 1
-----
Router-Id      : 3.3.3.3                    Area          : 0.0.0.0
Interface      : tunnel1                    Address        : 1.1.1.1
State          : FULL                       Neighbor Priority : 1
Dead Timer Due : 00:00:36                  Options       : 0x13
Time since last state change : 00h:14m:45s
```

Showing detail OSPFv3 AF neighbor information for tunnel interface:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3 detail
Address-family : IPv4 Unicast
-----
VRF : default                                Process : 1
-----
Router-Id      : 3.3.3.3                    Area          : 0.0.0.0
Interface      : tunnel1                    Address        : 1.1.1.1
```

```

State           : FULL           Neighbor Priority : 1
DR              : 3.3.3.3        BDR               : 1.1.1.3
Dead Timer Due  : 00:00:36       Options          : 0x13
Retransmission Queue Length : 0
Time Since Last State Change : 00h:14m:45s

```

Showing summary OSPFv3 AF neighbors information for a specific neighbor for all VRFs:

```

switch# show ospfv3-af ipv4 neighbors 3.3.3.3 summary all-vrfs
Address-family : IPv4 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Neighbor Summary
=====

Interface      Down Attempt Init TwoWay ExStart Exchange Loading Full Total
-----
1/1/1          0   0       0   0     0       0       0       1   1
Total          0   0       0   0     0       0       0       1   1

OSPFv3 AF Process ID 1 VRF red, Neighbor Summary
=====

Interface      Down Attempt Init TwoWay ExStart Exchange Loading Full Total
-----
1/1/2          0   0       0   0     0       0       0       1   1
Total          0   0       0   0     0       0       0       1   1

```

Showing OSPFv3 AF neighbors information for VRF red:

```

switch# show ospfv3-af ipv4 neighbors vrf red
Address-family : IPv4 Unicast
-----

OSPFv3 AF Process ID 2 VRF red
=====

Total Number of Neighbors: 1

Neighbor ID      Priority  State           Interface
-----
4.4.4.4          1       FULL/DR         1/1/2
Neighbor address 1.1.1.1

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv4 routes

```
show ospfv3-af [<PROCESS-ID>] ipv4 routes [<PREFIX/LENGTH>]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows the OSPFv3 AF routing table information.

| Parameter       | Description                                                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>    | Specifies an OSPFv3 AF process ID that shows information from the OSPFv3 AF routing table for the particular OSPFv3 AF process. Range: 1 to 65535. |
| <PREFIX/LENGTH> | Specifies an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.                                       |
| all-vrfs        | Select to show OSPFv3 AF routing table information for all VRFs.                                                                                   |
| vrf <VRF-NAME>  | Specifies the name of a VRF. Default: default.                                                                                                     |

### Examples

Showing OSPFv3 AF routing table information:

```
switch# show ospfv3-af ipv4 routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

Address-family : IPv4 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Routing Table
-----

Total Number of OSPFv3 AF Routes : 3

11.11.0.0/16      (i) area:0.0.0.0
    directly attached to interface 1/1/1, cost 1 distance 110
200.200.0.0/16   (i) area:0.0.0.0
    directly attached to interface tunnel1, cost 100 distance 110
1.1.1.1/132      (i) area:0.0.0.1
    directly attached to interface vlan10, cost 1 distance 110
```

Showing OSPFv3 AF routing table information for VRF red:

```
switch# show ospfv3-af ipv4 routes vrf red

Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
```

```

Address-family : IPv4 Unicast
-----

OSPFv3 AF Process ID 2 VRF red, Routing Table
-----

Total Number of OSPFv3 AF Routes : 1

2.2.2.1/24          (i) area:0.0.0.1
    directly attached to interface 1/1/2, cost 1 distance 110

```

Showing OSPFv3 AF routing table information for destination prefix 2.2.2.1/24:

```

switch# show ospfv3-af 1 ipv4 routes 2.2.2.1/24
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

Address-family : IPv4 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Routing Table for prefixes 2.2.2.1/24
-----

Total Number of OSPFv3 AF Routes : 1

2.2.2.1/24          (i) area:0.0.0.1
    directly attached to interface vlan10, cost 1 distance 110

```

Showing OSPFv3 AF routing table information for destination prefix 200.0.0.0/8:

```

switch# show ospfv3-af 1 ipv4 routes 200.0.0.0/8
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

Address-family : IPv4 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Routing Table for prefixes 200.0.0.0/8
-----

Total Number of OSPFv3 AF Routes : 1

200.0.0.0/8          (i) area:0.0.0.0
    directly attached to interface tunnell1, cost 100 distance 110

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information



| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv4 statistics interface

```
show ospfv3-af [<PROCESS-ID>] ipv4 statistics interface [<INTERFACE-NAME>] | vlan <VLAN-ID> | lag <LAG-ID> | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>
[brief] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows information about OSPFv3 AF enabled IPv4 interfaces.

| Parameter              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>           | Specifies an OSPFv3 AF process ID optionally to show the OSPFv3 AF enabled interfaces for the particular OSPFv3 AF process.<br>Range: 1 to 65535.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <INTERFACE-NAME>       | Selects to show information only for the specified OSPFv3 AF enabled interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| vlan <VLAN-ID>         | Specifies the vlan interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| lag <LAG-ID>           | Specifies the lag interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| loopback <LOOPBACK-ID> | Specifies the loopback interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| tunnel <TUNNEL-ID>     | Specifies the tunnel interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| brief                  | Include this parameter to display a brief overview of the following OSPFv3 AF configuration information. <ul style="list-style-type: none"> <li>Interface: OSPFv3 AF interface name.</li> <li>Area: OSPFv3 AF area ID.</li> <li>Cost: The metric OSPFv3 AF uses to judge a path's feasibility, calculated as (reference bandwidth / interface bandwidth).</li> <li>State: Indicates if the interface is a designated router (<b>Dr</b>) or a backup designated router (<b>Backup-dr</b>).</li> <li>Status: Indicates if the interface is <b>up</b> or <b>down</b>.</li> <li>Flags: P - Passive A - Active.</li> </ul> |
| all-vrfs               | Select to show OSPFv3 AF-enabled interface information for all VRFs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| vrf <VRF-NAME>         | Specifies the name of a VRF. Default: default.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Examples

Showing OSPFv3 AF information for all interfaces in default VRF:

```
switch# show ospfv3-af ipv4 interface
Codes: DR - Designated router BDR - Backup Designated router
```

Address-family : IPv4 Unicast

Interface 1/1/1 is up, Line Protocol is up, Instance Id 64

```
-----
VRF                : default                      Process                : 1
IP Address          : 1.1.1.2                      Area                  :
0.0.0.0
Status             : Up                          Network Type          :
Broadcast
Hello Interval      : 10      sec                  Dead Interval          : 40
sec
Transit Delay       : 1      sec                  Retransmit Interval    : 5
sec
BFD                 : Disabled                    Link Speed             :
1000 Mbps
Cost Configured     : 1                          Cost Calculated        : 4
State/Type          : BDR                        Router Priority         : 1
DR                  : 3.3.3.3                    BDR                    :
1.1.1.3
Link LSAs           : 2                          Checksum Sum           :
26868
AH Authentication   : SHA1, SPI 256 (Area)          Passive                : Yes
-----
```

Interface 1/1/2 is up, Line Protocol is up, Instance Id 64

```
-----
VRF                : default                      Process                : 1
IP Address          : 10.10.10.2                   Area                  :
0.0.0.0
Status             : Up                          Network Type          :
Broadcast
Hello Interval      : 10      sec                  Dead Interval          : 40
sec
Transit Delay       : 1      sec                  Retransmit Interval    : 5
sec
Link Speed          : 1000 Mbps
Cost Configured     : 1                          Cost Calculated        : 4
State/Type          : BDR                        Router Priority         : 1
DR                  : 10.10.10.2                   BDR                    :
10.10.10.1
Link LSAs           : 0                          Checksum Sum           : 0
AH Authentication   : SHA1, SPI 256 (Area)          Passive                : No
-----
```

Interface tunnell is up, Line Protocol is up, Instance Id 64

```
-----
VRF                : default                      Process                : 1
IPv6 Address        : 10.10.10.1                   Area                  :
0.0.0.1
Status             : Up                          Network Type          :
Broadcast
Hello Interval      : 10      sec                  Dead Interval          : 40
sec
Transit Delay       : 1      sec                  Retransmit Interval    : 5
sec
BFD                 : Disabled                    Link Speed             :
1000 Mbps
Cost Configured     : 1                          Cost Calculated        : 4
State/Type          : DR                        Router Priority         : 1
DR                  : 10.10.10.2                   BDR                    :
10.10.10.1
Link LSAs           : 1                          Checksum Sum           :
9168
-----
```

```

AH Authentication      : SHA1, SPI 256 (Area)          Passive      : Yes

Interface vlan10 is up, Line Protocol is up, Instance Id 64
-----
VRF                    : default                      Process      : 1
IPv6 Address          : 10.10.10.3                   Area         :
0.0.0.1
Status                : Up                          Network Type :
Broadcast
Hello Interval        : 10      sec                  Dead Interval : 40
sec
Transit Delay         : 1      sec                  Retransmit Interval : 5
sec
BFD                   : Disabled                    Link Speed   :
1000 Mbps
Cost Configured       : 1                          Cost Calculated : 4
State/Type            : DR                          Router Priority : 1
DR                    : 10.10.10.2                  BDR          :
10.10.10.1
Link LSAs              : 1                          Checksum Sum  :
9168
AH Authentication     : MD5, SPI 257                Passive      : No

Interface vlan20 is up, Line Protocol is up, Instance Id 64
-----
VRF                    : default                      Process      : 1
IPv6 Address          : 1.1.1.1                     Area         :
0.0.0.1
Status                : Up                          Network Type :
Broadcast
Hello Interval        : 10      sec                  Dead Interval : 40
sec
Transit Delay         : 1      sec                  Retransmit Interval : 5
sec
Link Speed            : 1000 Mbps
Cost Configured       : 1                          Cost Calculated : 4
State/Type            : DR                          Router Priority : 1
DR                    : 1.1.1.3                    BDR          : NA
Link LSAs              : 1                          Checksum Sum  :
9168
AH Authentication     : MD5, SPI 257                Passive      : No

```

Showing overview information for OSPFv3 AF enabled interfaces for all VRFs in brief:

```
switch# show ospfv3-af ipv4 interface
```

```
Address-family : IPv4 Unicast
```

```
-----
```

```
OSPFv3 AF Process ID 1 VRF default
```

```
=====
```

```
Total Number of Interfaces: 2
```

```
Flags: P - Passive  A - Active
```

| Interface | Area    | Cost | State | Status | Flags |
|-----------|---------|------|-------|--------|-------|
| 1/1/1     | 0.0.0.0 | 1    | BDR   | Up     | P     |
| vlan10    | 0.0.0.1 | 1    | DR    | Up     | A     |

```
OSPFv3 AF Process ID 2 VRF red
=====
```

```
Total Number of Interfaces: 1
```

```
Flags: P - Passive  A - Active
```

| Interface | Area    | Cost | State | Status | Flags |
|-----------|---------|------|-------|--------|-------|
| 1/1/2     | 0.0.0.1 | 1    | BDR   | Up     | P     |
| tunnell   | 0.0.0.0 | 100  | DR    | Up     | A     |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv6

```
show ospfv3-af [<PROCESS-ID>] ipv6 [all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows OSPFv3 AF information including area, state, and configuration information.

| Parameter      | Description                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 AF process ID optionally to show OSPFv3 AF information for a particular OSPF process. Range: 1 to 65535.       |
| all-vrfs       | Select to show OSPFv3 AF information for all VRFs.                                                                                 |
| vrf <VRF-NAME> | Specifies the name of a VRF. The default VRF OSPFv3 AF process information is displayed if VRF is not mentioned. Default: default. |

## Example

Showing general OSPFv3 AF configurations:

```
switch# show ospfv3-af 200 ipv6
Address-family : IPv6 Unicast
-----
```

```

VRF : default                                Process : 20200
-----

Router ID          : 1.1.1.1                OSPFv3-AF          : Enabled
BFD                : Disabled                SPF Start Interval : 200 ms
SPF Hold Interval  : 1000 ms                 SPF Max Wait Interval : 5000 ms
LSA Start Interval : 5000 ms                 LSA Hold Interval   : 1000 ms
LSA Max Wait Interval : 1000 ms              LSA Arrival Interval : 1000 ms
External LSAs      : 0                      Checksum Sum        : 0
ECMP               : 4                      Reference Bandwidth  : 100000 Mbps
Area Border        : Yes                    AS Border           : No
GR Status           : Enabled                GR Interval          : 120 sec
GR State           : Inactive                 GR Exit Status       : None
GR Helper          : Enabled                 GR Strict LSA Check  : Enabled
GR Ignore Lost I/F : Disabled                Internal Process ID  : 1
Summary address:
  prefix fd00::1/64, advertise, tag 10

Area      Total Active
-----
Normal    2          2
Stub      0          0

Area : 0.0.0.0
-----
Area Type          : Normal                Status              : Active
Total Interfaces   : 1                    Active Interfaces    : 1
Passive Interfaces : 0                    Loopback Interfaces  : 0
SPF calculation count : 4
Area ranges:
  fd00::1/64, inter-area, no-advertise
AH Authentication   : SHA1, SPI 256
Number of LSAs      : 5                    Checksum Sum         : 99122

Area : 0.0.0.1
-----
Area Type          : Normal                Status              : Active
Total Interfaces   : 1                    Active Interfaces    : 1
Passive Interfaces : 0                    Loopback Interfaces  : 0

SPF Calculation Count : 4
Area ranges:
  fd00::1/64, inter-area, no-advertise
ESP Authentication  : SHA1
Encryption          : 3DES, SPI 256
Number of LSAs      : 5                    Checksum Sum         : 99122

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv6 border-routers

show ospfv3-af [<PROCESS-ID>] ipv6 border-routers | [all-vrfs | vrf <VRF-NAME>]

### Description

Shows the OSPFv3 AF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

| Parameter      | Description                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 AF process ID to show the OSPFv3 AF routing table entries for ABR and ASBR for the particular OSPFv3 AF process. Range: 1 to 65535. |
| all-vrfs       | Select to show OSPFv3 AF border router information for all VRFs.                                                                                        |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.                                                                                                          |

### Example

Showing OSPFv3 AF border routers information for VRF **vrf\_red**:

```
switch# show ospfv3-af ipv6 border-routers vrf vrf_red

Address-family : IPv6 Unicast
-----

VRF : vrf_red          Process ID : 1
          Internal Routing Table
-----

Codes: i - Intra-area route, I - Inter-area route
Router-ID Cost  Type   Area   SPF  Nexthop                                Interface
i  1.1.1.1     1    ASBR   0.0.0.0  9    fe80::7272:cfff:fe9a:a15d             1/1/2
i  3.3.3.3     1    ASBR   1.1.1.1  9    fe80::7272:cfff:fe1f:d80              ** tunnel1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

### Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv6 interface

```
show ospfv3-af [<PROCESS-ID>] ipv6 interface [<INTERFACE-NAME>] | vlan <vlan id> |
lag <lag id> | loopback <loopback id> | tunnel <tunnel id>] [brief]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows information about OSPFv3 AF enabled interfaces.

| Parameter              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>           | Specifies an OSPFv3 AF process ID optionally to show the OSPFv3 AF enabled interfaces for the particular OSPFv3 AF process. Range: 1 to 65535.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <INTERFACE-NAME>       | Selects to show information only for the specified OSPFv3 AF enabled interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| vlan <VLAN-ID>         | Specifies the vlan interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| lag <LAG-ID>           | Specifies the lag interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| loopback <LOOPBACK-ID> | Specifies the loopback interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| tunnel <TUNNEL-ID>     | Specifies the tunnel interface name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| brief                  | Include this parameter to display a brief overview of the following OSPFv3 AF configuration information. <ul style="list-style-type: none"> <li>Interface: OSPFv3 AF interface name.</li> <li>Area: OSPFv3 AF area ID.</li> <li>Cost: The metric OSPFv3 AF uses to judge a path's feasibility, calculated as (reference bandwidth / interface bandwidth).</li> <li>State: Indicates if the interface is a designated router (<b>Dr</b>) or a backup designated router (<b>Backup-dr</b>).</li> <li>Status: Indicates if the interface is <b>up</b> or <b>down</b>.</li> <li>Flags: P - Passive A - Active.</li> </ul> |
| all-vrfs               | Select to show OSPFv3 AF-enabled interface information for all VRFs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| vrf <VRF-NAME>         | Specifies the name of a VRF. Default: default.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Examples

Showing OSPFv3 AF information for all interfaces in default VRF:

```
switch# show ospfv3-af ipv6 interface
Codes: DR - Designated router  BDR - Backup Designated router
```

Interface 1/1/1 is Up, Line Protocol is Up, Instance ID 10

```
-----
VRF                : default                Process                : 1
IPv6 address       : fe80::7272:cfff:fe70:67ae Area                :
0.0.0.0
Status             : Up                    Network Type          :
Broadcast
Hello Interval     : 10      sec            Dead Interval         : 40
sec
Transit Delay      : 1      sec            Retransmit Interval    : 5
sec
BFD                : Disabled              Link Speed             : 1000
Mbps
Cost Configured    : 1                    Cost Calculated        : 4
State/Type         : BDR                  Router Priority        : 1
DR                 : 3.3.3.3              BDR                   :
1.1.1.3
Link LSAs          : 2                    Checksum Sum           :
26868
AH Authentication  : SHA1, SPI 256 (Area)  Passive                : Yes
```

Codes: DR - Designated router BDR - Backup Designated router

Interface 1/1/2 is Up, Line Protocol is Up, Instance ID 70

```
-----
VRF                : default                Process                : 1
IPv6 address       : fe80::7272:cfff:fe70:67ae Area                :
0.0.0.0
Status             : Up                    Network Type          :
Broadcast
Hello Interval     : 10      sec            Dead Interval         : 40
sec
Transit Delay      : 1      sec            Retransmit Interval    : 5
sec
BFD                : Disabled              Link Speed             : 1000
Mbps
Cost Configured    : 1                    Cost Calculated        : 4
State/Type         : BDR                  Router Priority        : 1
DR                 : 10.10.10.2           BDR                   :
10.10.10.1
Link LSAs          : 0                    Checksum Sum           : 0
AH Authentication  : SHA1, SPI 256 (Area)  Passive                : No
```

Codes: DR - Designated router BDR - Backup Designated router

Interface tunnell is Up, Line Protocol is Up, Instance ID 70

```
-----
VRF                : default                Process                : 1
IPv6 address       : fe80::7272:cfff:fe70:679ae Area                :
0.0.0.1
Status             : Up                    Network Type          :
Broadcast
Hello Interval     : 10      sec            Dead Interval         : 40
sec
Transit Delay      : 1      sec            Retransmit Interval    : 5
sec
BFD                : Disabled              Link Speed             : 1000
Mbps
Cost Configured    : 1                    Cost Calculated        : 4
State/Type         : DR                  Router Priority        : 1
DR                 : 10.10.10.2           BDR                   :
10.10.10.1
```



```

Link LSAs          : 1                      Checksum Sum      : 9168
AH Authentication  : SHA1, SPI 256 (Area)    Passive           : Yes

```

Codes: DR - Designated router BDR - Backup Designated router

Interface vlan10 is Up, Line Protocol is Up, Instance ID 10

```

-----
VRF          : default                      Process          : 1
IPv6 address  : fe80::7272:cfff:fe70:67ae   Area            :
0.0.0.1
Status       : Up                          Network Type     :
Broadcast
Hello Interval : 10      sec                Dead Interval    : 40
sec
Transit Delay : 1      sec                Retransmit Interval : 5
sec
BFD          : Disabled                    Link Speed       : 1000
Mbps
Cost Configured : 1                      Cost Calculated  : 4
State/Type     : DR                      Router Priority   : 1
DR             : 10.10.10.2                BDR              :
10.10.10.1
Link LSAs     : 1                      Checksum Sum     : 9168
AH Authentication : MD5, SPI 257          Passive         : No

```

Codes: DR - Designated router BDR - Backup Designated router

Interface vlan20 is Up, Line Protocol is Up, Instance ID 10

```

-----
VRF          : default                      Process          : 1
IPv6 address  : fe80::7272:cfff:fe70:67ae   Area            :
0.0.0.1
Status       : Up                          Network Type     :
Broadcast
Hello Interval : 10      sec                Dead Interval    : 40
sec
Transit Delay : 1      sec                Retransmit Interval : 5
sec
BFD          : Disabled                    Link Speed       : 1000
Mbps
Cost Configured : 1                      Cost Calculated  : 4
State/Type     : DR                      Router Priority   : 1
DR             : 1.1.1.3                  BDR              : NA
Link LSAs     : 1                      Checksum Sum     : 9168
AH Authentication : MD5, SPI 257          Passive         : No

```

Showing overview information for OSPFv3 AF enabled interfaces for all VRFs in brief:

```

switch# show ospfv3-af ipv6 interface brief all-vrfs

```

```

VRF : default                      Process : 1
=====

```

Total Number of Interfaces: 2

Flags: P - Passive A - Active

| Interface | Area    | Cost | State | Status | Flags |
|-----------|---------|------|-------|--------|-------|
| 1/1/1     | 0.0.0.0 | 1    | BDR   | Up     | P     |
| vlan10    | 0.0.0.1 | 1    | DR    | Up     | A     |

```
VRF : red                                Process : 1
```

```
=====
```

```
Total Number of Interfaces: 1
```

```
Flags: P - Passive  A - Active
```

| Interface | Area    | Cost | State | Status | Flags |
|-----------|---------|------|-------|--------|-------|
| 1/1/2     | 0.0.0.1 | 1    | BDR   | Up     | P     |
| tunnel1   | 0.0.0.0 | 100  | DR    | Up     | A     |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv6 lsdb

```
show ospfv3-af [<process-id>] ipv6 lsdb
  adv-router {<ROUTER-ID>|self}
  area <AREA-ID>
  lsid <link-state-id>
  all-vrfs|vrf <VRF-NAME>
    as-external
    asbr-summary
    database-summary
    inter-area-prefix
    inter-area-router
    intra-area-prefix
    link
    network
    nssa-external
    router
    summary
```

## Description

Shows the OSPFv3 AF link state database summary for different OSPF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

| Parameter                                        | Description                                                                                                                                                                                                                                                                              |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;PROCESS-ID&gt;</code>                  | Enter an OSPFv3 AF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535.                                                                                                                                                                      |
| <code>adv-router {&lt;ROUTER-ID&gt; self}</code> | Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify <b>self</b> to show self-originated link states.                                                                                                      |
| <code>area &lt;AREA-ID&gt;</code>                | Select to display information filtered for the specified area in one of the following formats.<br>OSPFv3 AF area identifier in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>OSPFv3 AF area identifier in decimal format. Value: 0 to 4294967295. |
| <code>lsid &lt;LINK-STATE-ID&gt;</code>          | Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D).                                                                                                                                                                              |
| <code>all-vrfs {vrf &lt;vrf-name&gt;}</code>     | Select <b>all-vrfs</b> to display general OSPF information for all VRFs, or use the <b>vrf &lt;VRF-NAME&gt;</b> option to display information for a specific VRF.<br><br>Optionally select one of the following parameters to filter the link state database information.                |
| <code>as-external</code>                         | Show external link states (LSA type 5)                                                                                                                                                                                                                                                   |
| <code>database-summary</code>                    | Select to display the count of each type of LSA and each area in the database.<br><br><b>NOTE:</b> The <b>database-summary</b> parameter does not support the <b>area &lt;area-id&gt;</b> , <b>lsid &lt;link-state-id&gt;</b> or <b>adv-router {&lt;router-id&gt; self}</b> parameters.  |
| <code>inter-area-prefix</code>                   | Show inter-area prefix link states (LSA type 3)                                                                                                                                                                                                                                          |
| <code>inter-area-router</code>                   | Show inter-area router link states (LSA type 4)                                                                                                                                                                                                                                          |
| <code>intra-area-prefix</code>                   | Show intra-area prefix link states (LSA type 9)                                                                                                                                                                                                                                          |
| <code>link</code>                                | Show link states (LSA type 8)                                                                                                                                                                                                                                                            |
| <code>network</code>                             | Show network LSAs (LSA type 2).                                                                                                                                                                                                                                                          |
| <code>nssa-external</code>                       | Show NSSA external link states (LSA type 7).                                                                                                                                                                                                                                             |
| <code>router</code>                              | Show router LSAs (LSA type 1).                                                                                                                                                                                                                                                           |

## Examples

Showing OSPFv3 AF link state database (LSDB) general information:

```
switch# show ospfv3-af ipv6 lsdb
Address-family : IPv6 Unicast
-----
```

OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)

=====

Router Link State Advertisements (Area 0.0.0.0)

| ADV Router  | Age | Seq#       | Checksum | LSID | Link Count | Bits |
|-------------|-----|------------|----------|------|------------|------|
| 40.40.40.40 | 930 | 0x80000004 | 0x2ea1   | 0    | 3          | None |
| 50.50.50.50 | 935 | 0x80000002 | 0x8b52   | 0    | 1          | E    |
| 60.60.60.60 | 943 | 0x800003c5 | 0x9854   | 0    | 2          | None |

Network Link State Advertisements (Area 0.0.0.0)

| ADV Router  | Age | Seq#       | Checksum | LSID       | Router Count |
|-------------|-----|------------|----------|------------|--------------|
| 60.60.60.60 | 944 | 0x80000001 | 0x7179   | 1360007168 | 2            |
| 50.50.50.50 | 935 | 0x80000001 | 0x516a   | 19         | 1            |

Inter Area Prefix Link State Advertisements (Area 0.0.0.0)

| ADV Router  | Age | Seq#       | Checksum | LSID   | Prefix         |
|-------------|-----|------------|----------|--------|----------------|
| 40.40.40.40 | 929 | 0x80000001 | 0x2498   | 131072 | FEC0:3344::/32 |
| 50.50.50.50 | 928 | 0x80000001 | 0x5b2f   | 65536  | 111::/64       |

Inter Area Router Link State Advertisements (Area 0.0.0.0)

| ADV Router  | Age | Seq#       | Checksum | LSID | Destination Router ID |
|-------------|-----|------------|----------|------|-----------------------|
| 40.40.40.40 | 929 | 0x80000001 | 0x2498   | 1    | 33.33.33.33           |

AS External Link State Advertisements (Area 0.0.0.0)

| ADV Router  | Age | Seq#       | Checksum | LSID | Prefix  |
|-------------|-----|------------|----------|------|---------|
| 40.40.40.40 | 264 | 0x80000001 | 0x24cc4  | 1    | 10::/64 |
| 40.40.40.40 | 675 | 0x80000001 | 0x5b00f  | 2    | 11::/64 |

NSSA External Link State Advertisements (Area 0.0.0.0)

| ADV Router | Age | Seq#       | Checksum | LSID | Prefix   |
|------------|-----|------------|----------|------|----------|
| 3.3.3.3    | 264 | 0x80000001 | 0x24ac2  | 1    | 200::/64 |

Link-local Link State Advertisements (Area 0.0.0.0)

| ADV Router  | Age | Seq#       | Checksum | LSID | Interface |
|-------------|-----|------------|----------|------|-----------|
| 50.50.50.50 | 264 | 0x80000001 | 0x653c4  | 19   | 1/1/1     |

Intra Area Prefix Link State Advertisements (Area 0.0.0.0)

| ADV Router      | Age | Seq#       | Checksum | LSID | Referenced LS Type |    |
|-----------------|-----|------------|----------|------|--------------------|----|
| Referenced LSID |     |            |          |      |                    |    |
| 50.50.50.50     | 263 | 0x80000001 | 0x1da34  | 1    | 0x2001             | 0  |
| 50.50.50.50     | 264 | 0x80000001 | 0x2a45d  | 1    | 0x2002             | 19 |

Showing AS external link states:

```
switch# show ospfv3-af ipv6 lsd as-external

Address-family : IPv6 Unicast
-----

OSPF Router with ID (60.60.60.60) (Process ID 1 VRF default)
=====

AS External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Prefix
-----
40.40.40.40    264     0x80000001 0x24cc4     1         10::/64
40.40.40.40    675     0x80000001 0x5b00f     2         11::/64
```

Showing the LSDB database summary:

```
switch# show ospfv3-af ipv6 lsd database-summary

Address-family : IPv6 Unicast
-----

OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
=====

Area 0.0.0.0 database summary
-----
LSA Type      Count
-----
Router        2
Network       1
Inter Area Prefix 1
Inter Area Router 0
NSSA External 0
Link          3
Intra Area Prefix 3
-----
Total         10

Process 1 database summary
-----
LSA Type      Count
-----
Router        2
Network       1
Inter Area Prefix 1
Inter Area Router 0
AS External   2
NSSA External 0
Link          3
Intra Area Prefix 3
-----
Total         12
```

Showing inter-area prefix LSAs:

```
switch# show ospfv3-af ipv6 lsd inter-area-prefix

Address-family : IPv6 Unicast
```

```

-----
OSPF Router with ID (6.6.6.6) (Process ID 1 VRF default)
=====

Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Prefix
-----
40.40.40.40     929     0x80000001 0x2498      131072      FEC0:3344::/32
50.50.50.50     928     0x80000001 0x5b2f      65536       111::/64

```

Showing network LSAs:

```

switch# show ospfv3-af ipv6 lsd network

Address-family : IPv6 Unicast
-----

OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
=====

Network Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Router Count
-----
60.60.60.60     944     0x80000001 0x7179      1360007168  2
50.50.50.50     935     0x80000001 0x516a      19          1

```

Showing NSSA external link states:

```

switch# show ospfv3-af ipv6 lsd nssa-external

Address-family : IPv6 Unicast
-----

OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====

NSSA External Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID        Prefix
-----
3.3.3.3         264     0x80000001 0x24ac2     1           200::/64

```

Showing router LSAs:

```

switch# show ospfv3-af ipv6 lsd router

Address-family : IPv6 Unicast
-----

OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
=====

Ruter Link State Advertisements (Area 0.0.0.0)
-----

```

| ADV Router  | Age | Seq#       | Checksum | LSID | Link Count | Bits |
|-------------|-----|------------|----------|------|------------|------|
| 40.40.40.40 | 930 | 0x80000004 | 0x2ea1   | 0    | 3          | None |
| 50.50.50.50 | 935 | 0x80000002 | 0x8b52   | 0    | 1          | E    |
| 60.60.60.60 | 943 | 0x800003c5 | 0x9854   | 0    | 2          | None |

Showing inter-area router LSAs:

```
switch# show ospfv3-af ipv6 lsdB inter-area-router

Address-family : IPv6 Unicast
-----

OSPF Router with ID (60.60.60.60) (Process ID 1 VRF default)
=====

Inter Area Router Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Destination Router ID
-----
40.40.40.40    929     0x80000001 0x2498      1         33.33.33.33
```

Showing link states:

```
switch# show ospfv3-af ipv6 lsdB link

Address-family : IPv6 Unicast
-----

OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====

Link-local Link State Advertisements (Area 0.0.0.0)
-----
ADV Router      Age      Seq#      Checksum    LSID      Interface
-----
50.50.50.50    264     0x80000001 0x653c4     19        1/1/1

Link-local Link State Advertisements (Area 0.0.0.1)
-----
ADV Router      Age      Seq#      Checksum    LSID      Interface
-----
50.50.50.50    264     0x80000001 0xad324     20        1/1/2
```

Showing lintra-area prefix LSAs:

```
switch# show ospfv3-af ipv6 lsdB intra-area-prefix

Address-family : IPv6 Unicast
-----

OSPF Router with ID (0.0.0.0) (Process ID 1 VRF default)
=====

Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
-----
-----
```

| ADV Router<br>Referenced LSID | Age | Seq#       | Checksum | LSID | Referenced LS Type |
|-------------------------------|-----|------------|----------|------|--------------------|
| -----                         |     |            |          |      |                    |
| 50.50.50.50                   | 263 | 0x80000001 | 0x1da34  | 1    | 0x2001 0           |
| 50.50.50.50                   | 264 | 0x80000001 | 0x2a45d  | 1    | 0x2002 19          |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv6 neighbors

```
show ospfv3-af [<PROCESS-ID>] ipv6 neighbors [<NEIGHBOR-ID>]
  [interface <INTERFACE-NAME>] | vlan <VLAN ID> | lag <LAG ID> | loopback <LOOPBACK ID>
  | tunnel<tunnel id>][detail | summary]
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows information about OSPFv3 AF neighbors.

| Parameter                  | Description                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>               | Specifies an OSPFv3 AF process ID to show OSPFv3 AF neighbor information for the particular OSPFv3 AF process. Range: 1 to 65535. |
| neighbors <NEIGHBOR-ID>    | Shows information about a particular neighbor, specified in IPv4 format (A.B.C.D).                                                |
| interface <INTERFACE-NAME> | Shows neighbor information only for the specified interface.                                                                      |
| detail                     | Shows detailed information for the neighbors.                                                                                     |
| summary                    | Shows summary information for the neighbors.                                                                                      |
| lag <LAG ID>               | Specifies lag interface name.                                                                                                     |
| loopback <LOOPBACK ID>     | Specifies the vlan interface name.                                                                                                |



| Parameter      | Description                                    |
|----------------|------------------------------------------------|
| all-vrfs       | Shows neighbor information for all VRFs.       |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default. |

## Examples

Showing OSPFv3 AF neighbors information:

```
switch# show ospfv3-af ipv6 neighbors
Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 1 VRF default
=====

Total Number of Neighbors: 1

Neighbor ID      Priority  State          Interface
-----
3.3.3.3          1        FULL/DR        1/1/1
Neighbor address fe80::7272:cfff:fe79:7510
```

Showing OSPFv3 AF neighbors information for neighbor 3.3.3.3:

```
switch# show ospfv3-af ipv6 neighbors 3.3.3.3
Address-family : IPv6 Unicast
-----

VRF : default                                Process : 1
-----
Router-Id      : 3.3.3.3                      Area          : 0.0.0.0
Interface      : 1/1/1                        Address       :
fe80::7272:cfff:fe79:7510
State          : FULL                          Neighbor Priority : 1
Dead Timer Due : 00:00:36                      Options       : 0x13
Time since last state change : 00h:14m:45s
```

Showing detail OSPFv3 AF neighbors information for neighbor 3.3.3.3:

```
switch# show ospfv3-af ipv6 neighbors 3.3.3.3 detail
Address-family : IPv6 Unicast
-----

VRF : default                                Process : 1
-----
Router-Id      : 3.3.3.3                      Area          : 0.0.0.0
Interface      : 1/1/1                        Address       :
fe80::7272:cfff:fe79:7510
State          : FULL                          Neighbor Priority : 1
DR             : 3.3.3.3                      BDR           : 1.1.1.3
Dead Timer Due : 00:00:36                      Options       : 0x13
Retransmission Queue Length : 0
Time Since Last State Change : 00h:14m:45s
```

Showing OSPFv3 AF neighbors information for interface 1/1/1:

```
switch# show ospfv3-af ipv6 neighbors 3.3.3.3 interface 1/1/1

Address-family : IPv6 Unicast
-----

VRF : default                                Process : 1
-----
Router-Id      : 3.3.3.3                    Area          : 0.0.0.0
Interface      : 1/1/1                      Address         :
fe80::7272:cfff:fe79:7510
State          : FULL                      Neighbor Priority : 1
Dead Timer Due : 00:00:36                  Options         : 0x13
Time Since Last State Change : 00h:14m:45s
```

Showing summary OSPFv3 AF neighbors information for a specific neighbor for all VRFs:

```
switch# show ospfv3-af ipv6 neighbors 3.3.3.3 summary all-vrfs

Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Neighbor Summary
=====

Interface      Down Attempt Init TwoWay ExStart Exchange Loading Full Total
-----
1/1/1          0    0      0    0      0      0      0      1    1
Total          0    0      0    0      0      0      0      1    1

OSPFv3 AF Process ID 1 VRF red, Neighbor Summary
=====

Interface      Down Attempt Init TwoWay ExStart Exchange Loading Full Total
-----
1/1/2          0    0      0    0      0      0      0      1    1
Total          0    0      0    0      0      0      0      1    1
```

Showing OSPFv3 AF neighbors information for VRF red:

```
switch# show ospfv3-af ipv6 neighbors vrf red

Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 2 VRF red
=====

Total Number of Neighbors: 1

Neighbor ID      Priority  State          Interface
-----
4.4.4.4          1      FULL/DR        1/1/2
Neighbor address fe80::7272:cfff:fe79:7510
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv6 routes

```
show ospfv3-af [<PROCESS-ID>] ipv6 routes [<PREFIX/LENGTH>]
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows the OSPFv3 AF routing table information.

| Parameter       | Description                                                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>    | Specifies an OSPFv3 AF process ID that shows information from the OSPFv3 AF routing table for the particular OSPFv3 AF process. Range: 1 to 65535. |
| <PREFIX/LENGTH> | Specifies the IPv6 destination prefix showing information about a particular destination prefix. For example, 2010:bd9::/32.                       |
| all-vrfs        | Select to show OSPFv3 AF routing table information for all VRFs.                                                                                   |
| vrf <VRF-NAME>  | Specifies the name of a VRF. Default: default.                                                                                                     |

## Examples

Showing OSPFv3 AF routing table information:

```
switch# show ospfv3-af ipv6 routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Routing Table
-----

Total Number of OSPFv3 AF Routes : 3

111::/64          (i) area:0.0.0.0
    directly attached to interface 1/1/1, cost 1 distance 110
2000::/64         (i) area:0.0.0.0
    directly attached to interface tunnel1, cost 100 distance 110
fd00::/64        (i) area:0.0.0.1
    directly attached to interface vlan10, cost 1 distance 110
```

Showing OSPFv3 AF routing table information for VRF red:

```

switch# show ospfv3-af ipv6 routes vrf red

Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 2 VRF red, Routing Table
-----

Total Number of OSPFv3 AF Routes : 1

222::/64          (i) area:0.0.0.1
    directly attached to interface 1/1/2, cost 1 distance 110

```

Showing OSPFv3 AF routing table information for destination prefix fd00::/64:

```

switch# show ospfv3-af 1 ipv6 routes fd00::/64

Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Routing Table for prefixes fd00::/64
-----

Total Number of OSPFv3 AF Routes : 1

fd00::/64          (i) area:0.0.0.1
    directly attached to interface vlan10, cost 1 distance 110

```

Showing OSPFv3 AF routing table information for destination prefix 2000::/64:

```

switch# show ospfv3-af 1 ipv6 routes 2000::/64

Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Routing Table for prefixes 2000::/64
-----

Total Number of OSPFv3 AF Routes : 1

2000::/64          (i) area:0.0.0.0
    directly attached to interface tunnell1, cost 100 distance 110

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv6 statistics

```
show ospfv3-af ipv6 [<PROCESS-ID>] statistics  
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows OSPFv3 AF statistics.

| Parameter      | Description                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Specifies an OSPFv3 AF process ID that shows information on the OSPFv3 AF SPF statistics for the particular OSPFv3 AF process. Range: 1 to 65535. |
| all-vrfs       | Select to show OSPFv3 AF SPF statistics information for all VRFs.                                                                                 |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.                                                                                                    |

### Examples

Showing OSPFv3 AF statistics information:

```
switch# show ospfv3-af ipv6 statistics  
  
Address-family : IPv6 Unicast  
-----  
  
OSPFv3 AF Process ID 1 VRF default, Statistics (cleared 3h 2m 21s ago)  
-----  
  
Unknown Interface Drops           : 0  
Unknown Virtual Interface Drops   : 0  
Bad IPv6 Header Length Drops      : 0  
Wrong OSPFv3 Version Drops        : 0  
Bad Source IPv6 Drops             : 0  
Resource Failure Drops            : 0  
Bad Header Length Drops           : 0  
Total Drops                       : 0
```

Showing OSPFv3 AF statistics information for VRF red:

```
switch# show ospfv3-af 2 ipv6 statistics vrf red  
  
Address-family : IPv6 Unicast  
-----  
  
OSPFv3 AF Process ID 2 VRF red, Statistics (cleared 3h 2m 30s ago)  
-----
```

```

Unknown Interface Drops      : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops : 0
Wrong OSPFv3 Version Drops   : 0
Bad Source IPv6 Drops        : 0
Resource Failure Drops        : 0
Bad Header Length Drops      : 0
Total Drops                   : 0

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv6 statistics interface

```

show ospfv3-af ipv6 [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>| vlan <VLAN-ID>
| lag <LAG-ID> | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>]
[all-vrfs | vrf <VRF-NAME>]

```

## Description

Shows the OSPFv3 AF statistics for the enabled OSPFv3 AF interfaces.

| Parameter              | Description                                                                                                                                    |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>           | Specifies an OSPFv3 AF process ID to show OSPF-enabled interface statistics information on the specified OSPFv3 AF process. Range: 1 to 65535. |
| <INTERFACE-NAME>       | Selects to show information only for the specified interface.                                                                                  |
| vlan <VLAN-ID>         | Specifies the vlan interface name.                                                                                                             |
| lag <LAG ID>           | Specifies lag interface name.                                                                                                                  |
| loopback <LOOPBACK ID> | Specifies the vlan interface name.                                                                                                             |
| tunnel <TUNNEL-ID>     | Specifies the tunnel interface name.                                                                                                           |
| all-vrfs               | Select to show OSPF-enabled interface statistics information for all VRFs.                                                                     |

**Parameter****Description**`vrf <VRF-NAME>`

Specifies the name of a VRF. Default: default.

**Example**

Showing enabled OSPFv3 AF interfaces information for interface 1/1/1:

```

switch# show ospfv3-af ipv6 statistics interface 1/1/1

Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Interface 1/1/1 Statistics (cleared 3h 22m
18s ago)

=====
====

Tx Hello packets      : 1213          Rx Hello packets      : 1213
Tx Hello bytes        : 48516         Rx Hello bytes        : 48516
Tx DD packets         : 3             Rx DD packets         : 2
Tx DD bytes           : 164           Rx DD bytes           : 136
Tx LS request packets : 1             Rx LS request packets : 1
Tx LS request bytes   : 64            Rx LS request bytes   : 64
Tx LS update packets  : 22            Rx LS update packets  : 22
Tx LS update bytes    : 1520          Rx LS update bytes    : 1844
Tx LS ack packets     : 21            Rx LS ack packets     : 20
Tx LS ack bytes       : 976           Rx LS ack bytes       : 800

Total IPsec packets processed : 10
Total IPsec bytes processed   : 6400
Total Number of State Changes : 3
Number of LSAs                : 2
LSA Checksum Sum              : 21758

Total OSPFv3-AF Packets Discarded: 0
-----

Reason                      Packets Dropped
-----
Invalid Type                 0
Invalid Length               0
Invalid Version              0
Bad or Unknown Source        0
Area Mismatch                0
Self-originated              0
Duplicate Router ID          0
Interface Standby            0
Total Hello Packets Dropped  0
  Hello Interval Mismatch    0
  Dead Interval Mismatch     0
  Options Mismatch           0
  MTU Mismatch               0
  Neighbor Ignored           0
Resource Failures            0
Bad LSA Length               0
Others                       0
IPsec Authentication Errors  0

Total LSAs Ignored : 0

```

```

-----
Bad Type          : 0
Bad Length        : 0
Invalid Data      : 0
Invalid Checksum  : 0

```

Showing enabled OSPFv3 AF interfaces information for tunnel interface:

```

switch# show ospfv3-af ipv6 statistics interface tunnel1

Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Interface tunnel1 Statistics (cleared 0h16m32s
ago)

=====
===

Tx Hello packets      : 100                Rx Hello packets      : 75
Tx Hello bytes        : 3900                Rx Hello bytes        : 3000
Tx DD packets         : 2                   Rx DD packets         : 2
Tx DD bytes           : 136                 Rx DD bytes           : 116
Tx LS request packets : 1                   Rx LS request packets : 1
Tx LS request bytes   : 52                  Rx LS request bytes   : 64
Tx LS update packets  : 3                   Rx LS update packets  : 5
Tx LS update bytes    : 388                 Rx LS update bytes    : 464
Tx LS ack packets     : 4                   Rx LS ack packets     : 3
Tx LS ack bytes       : 244                 Rx LS ack bytes       : 208

Total IPsec packets processed : 0
Total IPsec bytes processed   : 0
Total Number of State Changes : 2
Number of LSAs                : 2
LSA Checksum Sum              : 76646

Total OSPFv3-AF Packets Discarded: 0
-----

Reason                                Packets Dropped
-----
Invalid Type                          0
Invalid Length                        0
Invalid Version                       0
Bad or Unknown Source                 0
Area Mismatch                         0
Self-originated                      0
Duplicate Router ID                   0
Interface Standby                     0
Total Hello Packets Dropped           0
  Hello Interval Mismatch              0
  Dead Interval Mismatch                0
  Options Mismatch                     0
  MTU Mismatch                         0
  Neighbor Ignored                     0
Resource Failures                     0
Bad LSA Length                        0
Others                                0
IPsec Authentication Errors           0
IPsec ESP Errors                      0

```



```
Total LSAs Ignored : 0
-----
Bad Type           : 0
Bad Length          : 0
Invalid Data        : 0
Invalid Checksum    : 0
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ospfv3-af ipv6 virtual-links

```
show ospfv3-af [<PROCESS-ID>] ipv6 virtual-links [brief]
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays the current state and parameters of the OSPFv3 AF virtual links.

| Parameter      | Description                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Enter an OSPFv3 AF process ID to display information on the OSPFv3 AF virtual links for the particular OSPFv3 AF process. Range: 1 to 65535. |
| brief          | Select to display brief overview information for the OSPFv3 AF virtual links.                                                                |
| all-vrfs       | Select to display OSPFv3 AF virtual links information for all VRFs.                                                                          |
| vrf <VRF-NAME> | Specifies the name of a VRF. Default: default.                                                                                               |

## Examples

Show OSPFv3 AF virtual links information:

```
switch# show ospfv3-af ipv6 virtual-links

Address-family : IPv6 Unicast
-----
```

```
Virtual link to router 40.40.40.40 is up
```

```
-----  
VRF                : default                      Process                : 21  
Transit area       : 0.0.0.1                      Authentication       : No  
Hello Interval     : 10                          Dead Interval        : 40  
Transit Delay      : 1                          Retransmit Interval  : 5  
Number of Link LSAs : 0                        Checksum Sum         : 0  
Authentication     : No  
Number of State Changes : 4
```

Show OSPFv3 AF virtual links information with IPsec AH authentication:

```
switch# show ospfv3-af ipv6 virtual-links
```

```
Address-family : IPv6 Unicast
```

```
-----  
Virtual link to router 40.40.40.40 is up
```

```
-----  
VRF                : default                      Process                : 21  
Transit area       : 0.0.0.1                      Authentication       : No  
Hello Interval     : 10                          Dead Interval        : 40  
Transit Delay      : 1                          Retransmit Interval  : 5  
Number of Link LSAs : 0                        Checksum Sum         : 0  
AH Authentication  : MD5   SPI: 1234  
Number of State Changes : 4
```

Show OSPFv3 AF virtual links information with IPsec ESP authentication:

```
switch# show ospfv3-af ipv6 virtual-links
```

```
Address-family : IPv6 Unicast
```

```
-----  
Virtual link to router 40.40.40.40 is up
```

```
-----  
VRF                : default                      Process                : 21  
Transit area       : 0.0.0.1                      Authentication       : No  
Hello Interval     : 10                          Dead Interval        : 40  
Transit Delay      : 1                          Retransmit Interval  : 5  
Number of Link LSAs : 0                        Checksum Sum         : 0  
ESP Authentication : SHA1 Encryption: NULL   SPI: 1234  
Number of State Changes : 4
```

Show brief overview information for OSPFv3 AF virtual links:

```
switch# show ospfv3-af ipv6 virtual-links brief
```

```
Address-family : IPv6 Unicast
```

```
-----  
OSPFv3 AF Process ID 1 VRF default  
=====
```

```
Total Number of Virtual Links: 1
```

| Neighbor-ID | Transit-Area | Status |
|-------------|--------------|--------|
| 2.1.1.2     | 0.0.0.1      | down   |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show running-config

```
show running-config
```

### Description

This command shows the configurations for all the protocols along with OSPFv3 AF.

### Example

```
switch# show running-config
Current configuration:
!
!Version ArubaOS-CX genericx86-64.01.01.000X
!Schema version 0.1.8
vrf blue
vrf green
vrf red
led locator on
!
!
!
!
!
!
!
vlan 1,10
interface 1/1/1
  no shutdown
  ip address 1.1.1.1/24
  ipv6 address 111::1/64
  ip ospf 1 area 0.0.0.0
```

```

    ospfv3-af 1 ipv4 area 0.0.0.0
    ospfv3-af ipv4 authentication ipsec spi 300 sha1 ciphertext
    AQBapXl/Ow3jK2zC6WCov3mGwmvDQ02L9WKRTxVPtSyXpUNqBQAAACoBmj2c
interface 1/1/2
    no shutdown
    vrf attach red
    ip address 2.2.2.2/24
    ipv6 address 222::2/64
    ip ospf 2 area 0.0.0.1
    ospfv3-af 1 ipv4 area 0.0.0.1
    ospfv3-af ipv4 authentication ipsec spi 300 md5 ciphertext
    AQBapYH+i/6HV2w+yyJCb45XhHPFFlNIt876Mh9EMRyf0XgwBAAAAGE/tKs=
interface 1/1/3
    no shutdown
    ip address 3.3.3.1/24
    ipv6 address 333::3/64
    ospfv3-af 1 ipv6 area 0.0.0.0 instance 1
    no ospfv3-af ipv6 passive
    ospfv3-af 1 ipv4 area 0.0.0.0 instance 65
interface 1/1/4
    no shutdown
    ipv6 address 1000::1/64
interface tunnel 1 mode ip 6in6
    no shutdown
    source ipv6 1000::1
    destination ipv6 1000::2
    ipv6 address 2000::1/64
    ospfv3-af 1 ipv4 area 0.0.0.0
    ospfv3-af ipv4 passive
    ospfv3-af ipv4 hello-interval 22
    ospfv3-af ipv4 dead-interval 88
    ospfv3-af ipv4 transit-delay 44
    ospfv3-af ipv4 priority 88
    ospfv3-af ipv4 cost 8
    ospfv3-af ipv4 network point-to-point
    ospfv3-af ipv4 shutdown
interface vlan10
    ip address 10.1.1.1/24
    ipv6 address fd00::1/64
    ip ospf 1 area 0.0.0.0
    ospfv3-af 1 ipv4 area 0.0.0.1
    ospfv3-af ipv4 encryption ipsec spi 401 sha1 ciphertext
    AQBapd+HuD8Y4kMJkh5dFzG6OTBvW4iY+ueK4xpETxZPsZlFDgAAAE3pkLmvMCtdFk2zz/LS
    aes ciphertext
    AQBapbPnk0Kl9giahf2dWGRHcU+6mTFiWzfxeWUXuWGYQ4teIgAAAOG7kcVrkrKV754nX+Li1ZDYPHG5xs
    apOnyZJMv7cfHEctA=
    ospfv3-af 1 ipv6 area 0.0.0.1
    ospfv3-af ipv4 encryption ipsec spi 401 sha1 ciphertext
    AQBapd+HuD8Y4kMJkh5dFzG6OTBvW4iY+ueK4xpETxZPsZlFDgAAAE3pkLmvMCtdFk2zz/LS
    aes ciphertext
    AQBapbPnk0Kl9giahf2dWGRHcU+6mTFiWzfxeWUXuWGYQ4teIgAAAOG7kcVrkrKV754nX+Li1ZDYPHG5xs
    apOnyZJMv7cfHEctA=
interface loopback 2
    ipv6 address 99::1/64
    ospfv3-af 1 ipv4 area 0.0.0.0
router ospfv3-af 1
    router-id 1.1.1.1
    reference-bandwidth 4000
    address-family ipv4 unicast
        router-id 2.2.2.1
        area 0.0.0.0
        area 0.0.0.1

```

```

        area 0.0.0.2
        area 0.0.0.3
        distribute-list prefix list2 in
        redistribute static
        reference-bandwidth 4000
        active-backbone stub-default-route
    exit-address-family
    address-family ipv6 unicast
        router-id 2.2.2.2
        area 0.0.0.0
        area 0.0.0.1
        area 0.0.0.2
        area 0.0.0.3
        distribute-list prefix list2 in
        passive-interface default
        redistribute static
        reference-bandwidth 4000
        active-backbone stub-default-route
    area 0.0.0.1 virtual-link 1.1.1.1
        hello-interval 15
        dead-interval 60
        authentication ipsec spi 256 md5 ciphertext
AQBapYH+i/6HV2w+yyJCb45XhHPFFlNIt876Mh9EMRyf0XgwBAAAAGE/tKs=
        exit
        area 0.0.0.1 authentication ipsec spi 256 md5 ciphertext
AQBapYH+i/6HV2w+yyJCb45XhHPFFlNIt876Mh9EMRyf0XgwBAAAAGE/tKs=
        area 0.0.0.2 virtual-link 2.2.2.2
            encryption ipsec spi 3456 md5 ciphertext
AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8 null
        exit
        area 0.0.0.3 virtual-link 3.3.3.3
            encryption ipsec spi 5678 sha1 ciphertext
AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8 des
ciphertext AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8
        exit
    exit-address-family
router ospfv3-af 2
    router-id 1.1.1.3
    distance 88
    max-metric router-lsa
    throttle spf start-time 100 hold-time 1000 max-wait-time 10000
    address-family ipv4 unicast
        router-id 1.1.1.3
        distance 88
        area 0.0.0.0
        area 0.0.0.1
        passive-interface default
        redistribute ospfv3-af 1 route-map ospfv3_af4_routes
        max-metric router-lsa
        throttle spf start-time 100 hold-time 1000 max-wait-time 10000
        area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext
AQBapfwg9p/z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwulfvA==
    des ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkEgAAAE0etJaie2d/i2XF8pKkRpQvpw==
    exit-address-family
    address-family ipv6 unicast
        router-id 1.1.1.4
        distance 88
        area 0.0.0.0
        area 0.0.0.1
        passive-interface default
        redistribute ospfv3-af 1 route-map ospfv3_af6_routes

```

```

        max-metric router-lsa
        throttle spf start-time 100 hold-time 1000 max-wait-time 10000
        area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext
AQBapfwg9p/z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwulfvA==
des ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/i2XF8pKkRpQvpw==
        exit-address-family
router ospfv3-af 1 vrf red
    reference-bandwidth 40000
    graceful-restart restart-interval 88
    no graceful-restart helper
    distance intra-area 34 inter-area 66 external 3
    address-family ipv4 unicast
        area 0.0.0.1
        redistribute ospf 2
        redistribute ospfv3-af 2
        reference-bandwidth 40000
        graceful-restart restart-interval 88
        no graceful-restart helper
        distance intra-area 34 inter-area 66 external 3
    exit-address-family
    address-family ipv6 unicast
        area 0.0.0.1
        redistribute ospfv3 2
        redistribute ospfv3-af 2
        reference-bandwidth 40000
        graceful-restart restart-interval 88
max-metric router-lsa include-stub
graceful-restart restart-interval 88
no graceful-restart helper
graceful-restart ignore-lost-interface
distance intra-area 34 inter-area 66
address-family ipv4 unicast
    router-id 1.1.1.8
    area 0.0.0.1
    max-metric router-lsa include-stub
    graceful-restart restart-interval 88
    no graceful-restart helper
    graceful-restart ignore-lost-interface
    distance intra-area 34 inter-area 66
exit-address-family
address-family ipv6 unicast
    router-id 1.1.1.9
    area 0.0.0.1
    graceful-restart restart-interval 88
    no graceful-restart helper strict-lsa-check
    max-metric router-lsa include-stub
    distance intra-area 34 inter-area 66
exit-address-family
router ospfv3-af 1 vrf blue
    graceful-restart restart-interval 99
    no graceful-restart helper
    max-metric router-lsa on-startup 100
    address-family ipv4 unicast
        graceful-restart restart-interval 99
        no graceful-restart helper
        area 0.0.0.0
        max-metric router-lsa on-startup 100
    exit-address-family
    address-family ipv6 unicast
        graceful-restart restart-interval 99
        no graceful-restart helper

```

```

        area 0.0.0.0
        max-metric router-lsa on-startup 100
    exit-address-family
router ospfv3-af 1 vrf green
    max-metric router-lsa include-stub on-startup 100
    address-family ipv4 unicast
        area 0.0.0.0
        max-metric router-lsa include-stub on-startup 100
    exit-address-family
    address-family ipv6 unicast
        area 0.0.0.0
        max-metric router-lsa include-stub on-startup 100
    exit-address-family
switch#

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show running-config all

```
show running-config all
```

### Description

Displays the current running configuration, including default values for all protocols such as OSPFv3 AF.

### Examples

Showing running-config without default values:

```

switch# show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.05.0001BC-26-g94067ba
!export-password: default
led locator on
vrf blue
vrf green
vrf red
!
!
!

```

```

!
ssh server vrf mgmt
vlan 1
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/1
    no shutdown
    ip address 1.1.1.1/24
    ipv6 address 111::1/64
    ipv6 address 99::1/64
    ospfv3-af 1 ipv4 area 0.0.0.0
interface 1/1/2
    no shutdown
interface loopback 2
    ipv6 address 929::1/64
    ospfv3-af 1 ipv4 area 0.0.0.0
interface loopback 3
    ipv6 address 123::1/64
    ospfv3-af 1 ipv4 area 0.0.0.0
    ospfv3-af 1 ipv6 area 0.0.0.0
!
!
!
!
!
router ospfv3-af 1 vrf blue
    max-metric router-lsa on-startup 100
    address-family ipv4 unicast
        distribute-list prefix list2 in
        max-metric router-lsa on-startup 100
        active-backbone stub-default-route
        area 0.0.0.0
    exit-address-family
    address-family ipv6 unicast
        distribute-list prefix list2 in
        max-metric router-lsa on-startup 100
        active-backbone stub-default-route
        area 0.0.0.0
    exit-address-family
router ospfv3-af 1
    router-id 1.1.1.1
    reference-bandwidth 4000
    address-family ipv4 unicast
        router-id 1.1.1.2
        reference-bandwidth 4000
        redistribute host-routes
        redistribute static
        area 0.0.0.0
        area 0.0.0.1
    exit-address-family
    address-family ipv6 unicast
        router-id 1.1.1.3
        reference-bandwidth 4000
        redistribute host-routes
        redistribute static
        area 0.0.0.0
        area 0.0.0.1
        area 0.0.0.1 virtual-link 1.1.1.2
            dead-interval 60
            hello-interval 15
            authentication ipsec spi 256 md5 ciphertext

```



```

        AQBapYH+i/6HV2w+yyJCb45XhHPFFlNIt876Mh9EMRyf0XgwBAAAAGE/tKs=
        exit
    area 0.0.0.2
    area 0.0.0.2 virtual-link 2.2.2.2
        encryption ipsec spi 3456 md5 ciphertext
        AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8 null
    exit
    area 0.0.0.3
    area 0.0.0.3 virtual-link 3.3.3.3
        encryption ipsec spi 5678 sha1 ciphertext
        AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8 des
        ciphertext AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+
        Xod/lZX8
    exit
    exit-address-family
router ospfv3-af 2
    router-id 1.1.1.3
    max-metric router-lsa
    address-family ipv4 unicast
        router-id 1.1.1.4
        max-metric router-lsa
        area 0.0.0.0
        area 0.0.0.1
        area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext

AQBapfwg9p/z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwulfvA==des
ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/
    i2XF8pKkRpQvpw==
    exit-address-family
    address-family ipv6 unicast
        router-id 1.1.1.5
        max-metric router-lsa
        area 0.0.0.0
        area 0.0.0.1
        area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext

AQBapfwg9p/z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwulfvA==des
ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/
    i2XF8pKkRpQvpw==
    exit-address-family
router ospfv3-af 1 vrf green
    max-metric router-lsa include-stub on-startup 100
    address-family ipv4 unicast
        max-metric router-lsa include-stub on-startup 100
        area 0.0.0.0
    exit-address-family
    address-family ipv6 unicast
        max-metric router-lsa include-stub on-startup 100
        area 0.0.0.0
    exit-address-family
router ospfv3-af 1 vrf red
    reference-bandwidth 40000
    address-family ipv4 unicast
        reference-bandwidth 40000
        redistribute ospfv3 2
        area 0.0.0.1
    exit-address-family
    address-family ipv6 unicast
        reference-bandwidth 40000
        redistribute ospfv3 2

```

```

        area 0.0.0.1
        exit-address-family
router ospfv3-af 2 vrf red
    router-id 1.1.1.7
    max-metric router-lsa include-stub
    address-family ipv4 unicast
        router-id 1.1.1.7
        max-metric router-lsa include-stub
    area 0.0.0.1
    exit-address-family
    address-family ipv6 unicast
        router-id 1.1.1.7
        max-metric router-lsa include-stub
    area 0.0.0.1
    exit-address-family
https-server vrf mgmt

```

Showing running-config with default values:

```

switch# show running-config all
Current configuration:
!
!
!
interface 1/1/1
    no shutdown
    mtu 1500
    qos trust none
    apply qos schedule-profile factory-default
    lldp transmit
    lldp receive
    lldp med poe
    no lldp med poe priority-override
    lldp dot3 poe
    lldp dot3 macphy
    lldp med network-policy
    lldp med capability
    lldp med topology-trap
    cdp
    sflow
    routing
    vrf attach default
    ip mtu 1500
    ip address 1.1.1.1/24
    ipv6 address 111::1/64 tag 0
    ipv6 address 99::1/64 tag 0
    no ip urpf-check
    ipv6 nd mtu 1500
    no ip proxy-arp
    ipv6 nd dad attempts 1
    ipv6 nd hop-limit 64
    ipv6 nd ns-interval 1000
    ipv6 nd ra min-interval 200
    ipv6 nd ra max-interval 600
    ipv6 nd ra lifetime 1800
    ipv6 nd ra reachable-time 0
    no ipv6 address autoconfig
    ipv6 nd ra retrans-timer 0
    no ipv6 nd ra managed-config-flag
    no ipv6 nd ra other-config-flag
    ipv6 nd suppress-ra

```

```

ipv6 nd router-preference medium
no ip irdp
ip irdp multicast
ip irdp minadvertinterval 450
ip irdp maxadvertinterval 600
ip irdp holdtime 1800
ip irdp preference 0
arp timeout 1800
ipv6 neighbor timeout 1800
power-over-ethernet
no power-over-ethernet pre-std-detect
power-over-ethernet priority low
power-over-ethernet allocate-by usage
power-over-ethernet assigned-class 4
ospfv3-af 1 ipv4 area 0.0.0.0
ospfv3-af ipv4 dead-interval 40
ospfv3-af ipv4 hello-interval 10
ospfv3-af ipv4 retransmit-interval 5
ospfv3-af ipv4 transit-delay 1
ospfv3-af ipv4 priority 1
ospfv3-af ipv4 cost 4
ospfv3-af ipv4 network broadcast
!
!
!
interface loopback 2
vrf attach default
ipv6 address 929::1/64 tag 0
no ip urpf-check
arp timeout 1800
ipv6 neighbor timeout 1800
ospfv3-af 1 ipv4 area 0.0.0.0
ospfv3-af ipv4 dead-interval 40
ospfv3-af ipv4 hello-interval 10
ospfv3-af ipv4 retransmit-interval 5
ospfv3-af ipv4 transit-delay 1
ospfv3-af ipv4 priority 1
ospfv3-af ipv4 network broadcast
!
!
!
interface loopback 3
vrf attach default
ipv6 address 123::1/64 tag 0
no ip urpf-check
arp timeout 1800
ipv6 neighbor timeout 1800
ospfv3-af 1 ipv4 area 0.0.0.0
ospfv3-af ipv4 dead-interval 40
ospfv3-af ipv4 hello-interval 10
ospfv3-af ipv4 retransmit-interval 5
ospfv3-af ipv4 transit-delay 1
ospfv3-af ipv4 priority 1
ospfv3-af ipv4 network broadcast
ospfv3-af 1 ipv6 area 0.0.0.0
ospfv3-af ipv6 dead-interval 40
ospfv3-af ipv6 hello-interval 10
ospfv3-af ipv6 retransmit-interval 5
ospfv3-af ipv6 transit-delay 1
ospfv3-af ipv6 priority 1
ospfv3-af ipv6 network broadcast
!

```

```

!
!
router ospfv3-af 1 vrf blue
    enable
    reference-bandwidth 100000
    max-metric router-lsa on-startup 100
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    distance 110
    address-family ipv4 unicast
        enable
        reference-bandwidth 100000
        max-metric router-lsa on-startup 100
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        active-backbone stub-default-route
        area 0.0.0.0
        area 0.0.0.0 default-metric 1
    exit-address-family
    address-family ipv6 unicast
        enable
        reference-bandwidth 100000
        max-metric router-lsa on-startup 100
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        active-backbone stub-default-route
        area 0.0.0.0
        area 0.0.0.0 default-metric 1
    exit-address-family
router ospfv3-af 1 vrf default
    enable
    router-id 1.1.1.1
    reference-bandwidth 4000
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    distance 110
    address-family ipv4 unicast
        enable
        router-id 1.1.1.1
        reference-bandwidth 4000
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        redistribute static

```

```

no active-backbone stub-default-route
area 0.0.0.0
area 0.0.0.0 default-metric 1
area 0.0.0.1
area 0.0.0.1 default-metric 1
area 0.0.0.2
area 0.0.0.2 default-metric 1
area 0.0.0.3
area 0.0.0.3 default-metric 1
exit-address-family
address-family ipv6 unicast
enable
router-id 1.1.1.1
reference-bandwidth 4000
maximum-paths 4
timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
timers lsa-arrival 1000
graceful-restart restart-interval 120
default-metric 25
distance 110
redistribute static
no active-backbone stub-default-route
area 0.0.0.0
area 0.0.0.0 default-metric 1
area 0.0.0.1
area 0.0.0.1 default-metric 1
area 0.0.0.1 virtual-link 1.1.1.2
    dead-interval 60
    hello-interval 15
    retransmit-interval 5
    transit-delay 1
    authentication ipsec spi 256 md5 ciphertext
    AQBapYH+i/6HV2w+yyJCb45XhHPFFfLNIt876Mh9EMRyf0XgwBAAAAGE/tKs=
    exit
area 0.0.0.2
area 0.0.0.2 default-metric 1
area 0.0.0.2 virtual-link 2.2.2.2
    dead-interval 40
    hello-interval 10
    retransmit-interval 5
    transit-delay 1
    encryption ipsec spi 3456 md5 ciphertext
    AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8 null
    exit
area 0.0.0.3
area 0.0.0.3 default-metric 1
area 0.0.0.3 virtual-link 3.3.3.3
    dead-interval 40
    hello-interval 10
    retransmit-interval 5
    transit-delay 1
    encryption ipsec spi 5678 sha1 ciphertext
    AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8 des
    ciphertext
AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8
exit
exit-address-family
router ospfv3-af 2 vrf default
enable
router-id 1.1.1.3
reference-bandwidth 100000

```

```

max-metric router-lsa
timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
timers lsa-arrival 1000
graceful-restart restart-interval 120
distance 110
address-family ipv4 unicast
    enable
    router-id 1.1.1.3
    reference-bandwidth 100000
    max-metric router-lsa
    maximum-paths 4
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    default-metric 25
    distance 110
    no active-backbone stub-default-route
    area 0.0.0.0
    area 0.0.0.0 default-metric 1
    area 0.0.0.1
    area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext

AQBapfwg9p/z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwulfvA==des
ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/
i2XF8pKkRpQvpw==
    area 0.0.0.1 default-metric 1
exit-address-family
address-family ipv6 unicast
    enable
    router-id 1.1.1.3
    reference-bandwidth 100000
    max-metric router-lsa
    maximum-paths 4
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    default-metric 25
    distance 110
    no active-backbone stub-default-route
    area 0.0.0.0
    area 0.0.0.0 default-metric 1
    area 0.0.0.1
    area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext

AQBapfwg9p/z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwulfvA==des
ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/
i2XF8pKkRpQvpw==
    area 0.0.0.1 default-metric 1
exit-address-family
router ospfv3 af 1 vrf green
    enable
    reference-bandwidth 100000
    max-metric router-lsa include-stub on-startup 100
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120

```

```

distance 110
address-family ipv4 unicast
    enable
    reference-bandwidth 100000
    max-metric router-lsa include-stub on-startup 100
    maximum-paths 4
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    default-metric 25
    distance 110
    no active-backbone stub-default-route
    area 0.0.0.0
    area 0.0.0.0 default-metric 1
exit-address-family
address-family ipv6 unicast
    enable
    reference-bandwidth 100000
    max-metric router-lsa include-stub on-startup 100
    maximum-paths 4
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    default-metric 25
    distance 110
    no active-backbone stub-default-route
    area 0.0.0.0
    area 0.0.0.0 default-metric 1
exit-address-family
router ospfv3-af 1 vrf red
    enable
    reference-bandwidth 40000
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    distance 110
    address-family ipv4 unicast
        enable
        reference-bandwidth 40000
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        redistribute ospfv3 2
        no active-backbone stub-default-route
        area 0.0.0.1
        area 0.0.0.1 default-metric 1
    exit-address-family
    address-family ipv6 unicast
        enable
        reference-bandwidth 40000
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120

```

```

        default-metric 25
        distance 110
        redistribute ospfv3 2
        no active-backbone stub-default-route
        area 0.0.0.1
        area 0.0.0.1 default-metric 1
    exit-address-family
router ospfv3-af 2 vrf red
    enable
    router-id 1.1.1.7
    reference-bandwidth 100000
    max-metric router-lsa include-stub
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
distance 110
address-family ipv4 unicast
    enable
    router-id 1.1.1.7
    reference-bandwidth 100000
    max-metric router-lsa include-stub
    maximum-paths 4
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    default-metric 25
    distance 110
    no active-backbone stub-default-route
    area 0.0.0.1
    area 0.0.0.1 default-metric 1
exit-address-family
address-family ipv6 unicast
    enable
    router-id 1.1.1.7
    reference-bandwidth 100000
    max-metric router-lsa include-stub
    maximum-paths 4
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    default-metric 25
    distance 110
    no active-backbone stub-default-route
    area 0.0.0.1
    area 0.0.0.1 default-metric 1
exit-address-family
!
!
!
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |



## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show tech ospfv3-af

show tech ospfv3-af [local-file]

### Description

Shows the output of the following OSPFv3 AF commands:

- **show ospfv3-af all vrfs**
- **show ospfv3-af interface all-vrfs**
- **show ospfv3-af neighbors detail all-vrfs**
- **show ospfv3-af neighbors summary all-vrfs**
- **show ospfv3-af route all-vrfs**
- **show ospfv3-af statistics all-vrfs**
- **show ospfv3-af statistics interface all-vrfs**
- **show ospfv3-af border-routers all-vrfs**
- **show ospfv3-af lsdB all-vrfs database-summary**
- **show ospfv3-af virtual-links all-vrfs**

| Parameter  | Description                                              |
|------------|----------------------------------------------------------|
| local-file | Captures the command output into a local file. Optional. |

### Examples

```
switch(config-if)# show tech ospfv3-af
=====
Show Tech executed on Tue Sep 12 09:42:42 2017
=====
[Begin] Feature ospfv3-af
=====

*****
Command : show ospfv3-af ipv6 neighbor summary all-vrfs
*****
Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Neighbor Summary
=====

Interface      Down Attempt Init TwoWay ExStart Exchange Loading Full Total
-----
1/1/1          0    0      0    0      0      0      0      1    1
```

|       |   |   |   |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|---|---|---|
| Total | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
|-------|---|---|---|---|---|---|---|---|---|

# OSPFv3 AF Process ID 1 VRF red, Neighbor Summary

| Interface | Down | Attempt | Init | TwoWay | ExStart | Exchange | Loading | Full | Total |
|-----------|------|---------|------|--------|---------|----------|---------|------|-------|
| 1/1/2     | 0    | 0       | 0    | 0      | 0       | 0        | 0       | 1    | 1     |
| Total     | 0    | 0       | 0    | 0      | 0       | 0        | 0       | 1    | 1     |

\*\*\*\*\*  
Command : show ospfv3-af ipv6 statistics all-vrfs  
\*\*\*\*\*  
Address-family : IPv6 Unicast  
-----

# OSPFv3 AF Process ID 1 VRF default, Statistics (cleared 0h 4m 46s ago)

```

Unknown Interface Drops          : 0
Unknown Virtual Interface Drops  : 0
Bad IPv6 Header Length Drops     : 0
Wrong OSPFv3 Version Drops       : 0
Bad Source IPv6 Drops            : 0
Resource Failure Drops           : 0
Bad Header Length Drops          : 0
Total Drops                     : 0

```

# OSPFv3 AF Process ID 2 VRF red, Statistics (cleared 0h 4m 45s ago)

```

Unknown Interface Drops          : 0
Unknown Virtual Interface Drops  : 0
Bad IPv6 Header Length Drops     : 0
Wrong OSPFv3 Version Drops       : 0
Bad Source IPv6 Drops            : 0
Resource Failure Drops           : 0
Bad Header Length Drops          : 0
Total Drops                     : 0

```

\*\*\*\*\*  
Command : show ospfv3-af ipv6 neighbor detail all-vrfs  
\*\*\*\*\*  
Address-family : IPv6 Unicast  
-----

# Neighbor 3.3.3.3, Interface address fe80::7272:cfff:fec6:6e6c

```

Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
  State is FULL, Neighbor priority is 1
  DR is 3.3.3.3, BDR is 1.1.1.3
  Options is 0x13
  Dead timer due in 00:00:34
  Retransmission queue length 0

```

# Neighbor 4.4.4.4, Interface address fe80::7272:cfff:fec6:6e6c

```

Process ID 2 VRF red, in Area 0.0.0.1 via Interface 1/1/2
  State is FULL, Neighbor priority is 1
  DR is 4.4.4.4, BDR is 1.1.1.7

```

```

Options is 0x13
Dead timer due in 00:00:35
Retransmission queue length 0

*****
Command : show ospfv3-af ipv6 route all-vrfs
*****
Address-family : IPv6 Unicast
-----

Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPFv3 AF Process ID 1 VRF default, Routing Table
-----

Total Number of OSPFv3 Routes : 2

111::/64          (i) area:0.0.0.0
    directly attached to interface 1/1/1, cost 1 distance 110
fd00::/64         (i) area:0.0.0.1
    directly attached to interface vlan10, cost 1 distance 110

OSPFv3 AF Process ID 2 VRF red, Routing Table
-----

Total Number of OSPFv3 Routes : 1

222::/64          (i) area:0.0.0.1
    directly attached to interface 1/1/2, cost 1 distance 110

*****
Command : show ospfv3-af ipv6 all-vrfs
*****
Address-family : IPv6 Unicast
-----

Routing Process 1 with ID: 1.1.1.3 VRF default
-----

Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Maximum Paths to Destination: 4
Number of external LSAs 0, checksum sum 0
Number of areas is: 2, 2 normal, 0 stub
Number of active areas is: 2, 2 normal, 0 stub
Area (0.0.0.0) (Active)
    Interfaces in this Area: 1 Active Interfaces: 1
    Passive Interfaces: 0 Loopback Interfaces: 0
    SPF calculation has run 5 times
    Number of LSAs: 5, checksum sum 138261
Area (0.0.0.1) (Active)
    Interfaces in this Area: 1 Active Interfaces: 1
    Passive Interfaces: 0 Loopback Interfaces: 0
    SPF calculation has run 5 times
Area ranges:
    fd00::1/64, inter-area, no-advertise
    Number of LSAs: 4, checksum sum 85028

```

```

Routing Process 2 with ID: 1.1.1.7 VRF red
-----

Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Maximum Paths to Destination: 4
Number of external LSAs 0, checksum sum 0
Number of areas is: 1, 1 normal, 0 stub
Number of active areas is: 1, 1 normal, 0 stub
Area (0.0.0.1) (Active)
  Interfaces in this Area: 1 Active Interfaces: 1
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 4 times
  Number of LSAs: 4, checksum sum 149947

*****
Command : show ospfv3-af ipv6 interface all-vrfs
*****
Address-family : IPv6 Unicast
-----

Interface 1/1/1 is up, Line Protocol is up
-----

IPv6 address fe80::7272:cfff:fea7:c8fa
Process ID 1 VRF default, Area 0.0.0.0
  State Backup-dr, Status up, Network type Broadcast, cost 1
  Transit delay 1 sec, Router priority 1
  Designated Router ID: 3.3.3.3
  Backup Designated Router ID: 1.1.1.3
  Timer Intervals: Hello 10, Dead 40, Retransmit 5
  Number of Link LSAs: 2, checksum sum 60947

Interface 1/1/2 is up, Line Protocol is up
-----

IPv6 address fe80::7272:cfff:fea7:c8fa
Process ID 2 VRF red, Area 0.0.0.1
  State Backup-dr, Status up, Network type Broadcast, cost 1
  Transit delay 1 sec, Router priority 1
  Designated Router ID: 4.4.4.4
  Backup Designated Router ID: 1.1.1.7
  Timer Intervals: Hello 10, Dead 40, Retransmit 5
  Number of Link LSAs: 2, checksum sum 75181

*****
Command : show ospfv3-af ipv6 statistics interface all-vrfs
*****
Address-family : IPv6 Unicast
-----

OSPFv3 AF Process ID 1 VRF default, Interface 1/1/1 Statistics (cleared 0h 4m 45s
ago)

=====
==

Tx Hello packets      : 28                Rx Hello packets      : 28
Tx Hello bytes        : 1116               Rx Hello bytes        : 1116

```

|                       |       |                       |       |
|-----------------------|-------|-----------------------|-------|
| Tx DD packets         | : 3   | Rx DD packets         | : 2   |
| Tx DD bytes           | : 164 | Rx DD bytes           | : 136 |
| Tx LS request packets | : 1   | Rx LS request packets | : 1   |
| Tx LS request bytes   | : 64  | Rx LS request bytes   | : 64  |
| Tx LS update packets  | : 4   | Rx LS update packets  | : 4   |
| Tx LS update bytes    | : 368 | Rx LS update bytes    | : 452 |
| Tx LS ack packets     | : 3   | Rx LS ack packets     | : 2   |
| Tx LS ack bytes       | : 208 | Rx LS ack bytes       | : 152 |

Total IPsec packets processed : 10  
 Total IPsec bytes processed : 6400  
 Total Number of State Changes : 3  
 Number of LSAs : 2  
 LSA Checksum Sum : 60947

Total OSPFv3-AF Packets Discarded: 0

| Reason                      | Packets Dropped |
|-----------------------------|-----------------|
| Invalid Type                | 0               |
| Invalid Length              | 0               |
| Invalid Version             | 0               |
| Bad or Unknown Source       | 0               |
| Area Mismatch               | 0               |
| Self-originated             | 0               |
| Duplicate Router ID         | 0               |
| Interface Standby           | 0               |
| Total Hello Packets Dropped | 0               |
| Hello Interval Mismatch     | 0               |
| Dead Interval Mismatch      | 0               |
| Options Mismatch            | 0               |
| MTU Mismatch                | 0               |
| Neighbor Ignored            | 0               |
| Resource Failures           | 0               |
| Bad LSA Length              | 0               |
| Others                      | 0               |
| IPsec Authentication Errors | 0               |

Total LSAs Ignored : 0

|                  |     |
|------------------|-----|
| Bad Type         | : 0 |
| Bad Length       | : 0 |
| Invalid Data     | : 0 |
| Invalid Checksum | : 0 |

OSPFv3 AF Process ID 2 VRF red, Interface 1/1/2 Statistics (cleared 0h 4m 45s ago)

|                       |        |                       |        |
|-----------------------|--------|-----------------------|--------|
| Tx Hello packets      | : 28   | Rx Hello packets      | : 28   |
| Tx Hello bytes        | : 1116 | Rx Hello bytes        | : 1116 |
| Tx DD packets         | : 3    | Rx DD packets         | : 2    |
| Tx DD bytes           | : 144  | Rx DD bytes           | : 136  |
| Tx LS request packets | : 1    | Rx LS request packets | : 1    |
| Tx LS request bytes   | : 64   | Rx LS request bytes   | : 52   |
| Tx LS update packets  | : 4    | Rx LS update packets  | : 4    |
| Tx LS update bytes    | : 332  | Rx LS update bytes    | : 452  |
| Tx LS ack packets     | : 3    | Rx LS ack packets     | : 2    |
| Tx LS ack bytes       | : 208  | Rx LS ack bytes       | : 132  |

Total IPsec packets processed : 10  
 Total IPsec bytes processed : 6400

```
Total Number of State Changes : 3
Number of LSAs                  : 2
LSA Checksum Sum                : 75181
```

```
Total OSPFv3-AF Packets Discarded: 0
-----
```

| Reason                      | Packets Dropped |
|-----------------------------|-----------------|
| Invalid Type                | 0               |
| Invalid Length              | 0               |
| Invalid Version             | 0               |
| Bad or Unknown Source       | 0               |
| Area Mismatch               | 0               |
| Self-originated             | 0               |
| Duplicate Router ID         | 0               |
| Interface Standby           | 0               |
| Total Hello Packets Dropped | 0               |
| Hello Interval Mismatch     | 0               |
| Dead Interval Mismatch      | 0               |
| Options Mismatch            | 0               |
| MTU Mismatch                | 0               |
| Neighbor Ignored            | 0               |
| Resource Failures           | 0               |
| Bad LSA Length              | 0               |
| Others                      | 0               |
| IPsec Authentication Errors | 0               |

```
Total LSAs Ignored : 0
-----
```

```
Bad Type      : 0
Bad Length    : 0
Invalid Data  : 0
Invalid Checksum : 0
```

```
*****
```

```
Command : show ospfv3-af ipv6 virtual-links all-vrfs
```

```
*****
```

```
Address-family : IPv6 Unicast
-----
```

```
Virtual link to router 40.40.40.40 is up
-----
```

```
Process ID 21 VRF default, Transit area 0.0.0.1
Transit delay 1 sec
Timer Intervals: hello 10, dead 40, retransmit 5
No authentication
Number of Link LSAs: 0, checksum sum 0
4 state changes
```

```
=====
[End] Feature ospfv3-af
=====
```

```
=====
Show Tech commands executed successfully
=====
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## summary-address (OSPFv3 AF)

summary-address <PREFIX/LENGHT> [no-advertise | tag <TAG-VALUE>]  
no summary-address <PREFIX/LENGHT> [no-advertise | tag <TAG-VALUE>]

### Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The **no** form of this command disables route summarization.



This command only works for an ASBR (Autonomous System Boundary Router).

| Parameter       | Description                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <PREFIX/LENGHT> | Specifies the IP prefix/mask for advertising the aggregate route.                                                                                |
| no-advertise    | Do not advertise the aggregate route. Suppress routes that match the specified prefix/mask pair.                                                 |
| tag <TAG-VALUE> | Specify the tag for the aggregate route. The summary prefix will be advertised along with the tag value in External LSAs. Range: 0 to 4294967295 |

### Examples

Setting OSPFv3 AF route summarization for IPv4:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc)# summary-address 10.1.0.0/16
switch(config-ospfv3-ipv4-uc)# summary-address 10.1.0.0/16 no-advertise
switch(config-ospfv3-ipv4-uc)# summary-address 10.1.0.0/16 tag 10
```

Disabling route summarization for IPv4:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-ipv4-uc)# no summary-address 10.1.0.0/16
switch(config-ospfv3-ipv4-uc)# no summary-address 10.1.0.0/16 no-advertise
switch(config-ospfv3-ipv4-uc)# no summary-address 10.1.0.0/16 tag 10
```

Setting OSPFv3 AF route summarization for IPv6:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-ipv6-uc) # summary-address fd00::1/64  
switch(config-ospfv3-ipv6-uc) # summary-address fd00::1/64 no-advertise  
switch(config-ospfv3-ipv6-uc) # summary-address fd00::1/64 tag 10
```

Disabling route summarization for IPv6:

```
switch(config)# router ospfv3 1  
switch(config-ospfv3-ipv6-uc) # no summary-address fd00::1/64  
switch(config-ospfv3-ipv6-uc) # no summary-address fd00::1/64 no-advertise  
switch(config-ospfv3-ipv6-uc) # no summary-address fd00::1/64 tag 10
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                                        | Authority                                                                          |
|--------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-ipv4-uc<PROCESS-ID><br>config-ospfv3-ipv6-uc<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## timers lsa-arrival (OSPFv3 AF)

```
timers lsa-arrival <DELAY>  
no timers lsa-arrival
```

### Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command **timers throttle lsa** on the neighbor.

The **no** form of this command sets the LSA timers to default values.



Configuration is supported both globally and in the address-family for ipv4-unicast and ipv6-unicast. If disable is configured for the address-family, it will override the global configuration.

| Parameter | Description                                                             |
|-----------|-------------------------------------------------------------------------|
| <DELAY>   | Specifies the delay in milliseconds. Range: 0 to 600000. Default: 1000. |



## Examples

Setting the LSA arrival timer:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-af-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospfv3-af 1  
switch(config-ospfv3-af-1)# no timers lsa-arrival
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context               | Authority                                                                          |
|--------------|-------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-af-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## timers throttle lsa (OSPFv3 AF)

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-time <WAIT-TIME>  
no timers throttle lsa
```

## Description

Configures the timers for LSA generation.

The **no** form of this command sets the LSA timers to default values.



Configuration is supported both globally and in the address-family for ipv4-unicast and ipv6-unicast. If timers are configured for the address-family, they will override the global configuration.

| Parameter               | Description                                                                                                                                                                                                                                                |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| start-time <START-TIME> | Specifies the initial wait time in milliseconds after which LSAs are generated. When set to 0, the LSAs are generated without any delay. Range: 0 to 600000. Default: 5000.                                                                                |
| hold-time <HOLD-TIME>   | Specifies the amount of time, in milliseconds, between regeneration of an LSA. The hold time doubles each time the same LSA must be regenerated, until <b>max-wait-time</b> is reached. When set to 0, LSA regeneration time is not increased. Range: 0 to |

| Parameter                                    | Description                                                                                                                                                                |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | 600000. Default: 0.                                                                                                                                                        |
| <code>max-wait-time &lt;WAIT-TIME&gt;</code> | Specifies the maximum wait time, in milliseconds, for regeneration of the same LSA. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0. |

## Examples

Setting the LSA timers:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# timers throttle lsa start-time 100 hold-time 1000 max-wait-time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# no timers throttle lsa
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context                                  | Authority                                                                          |
|--------------|--------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | <code>config-ospfv3-af-&lt;PROCESS-ID&gt;</code> | Administrators or local user group members with execution rights for this command. |

## timers throttle spf

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds>
max-wait-time <milliseconds>
no timers throttle spf
```

### Description

Configures timers for SPF calculation. There are three timers:

- **start-time:** Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- **hold-time:** Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the *max-wait-time*.

- **max-wait-time:** Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The **no** form of this command sets all the configured non-default timers to default value.



The configuration is supported globally as well as per address-families ipv4-unicast or ipv6-unicast. If address-family specific timers is configured, it takes precedence over global configuration.

| Parameter                       | Description                                                                                                                                         |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;START-TIME&gt;</code> | Time in milliseconds to set timer for initial SPF delay.<br>Default: 200.<br>Range: 1-600000<br><i>Required</i>                                     |
| <code>&lt;HOLD-TINME&gt;</code> | Time in milliseconds to set the minimum hold time between two consecutive SPF calculations.<br>Default: 1000.<br>Range: 1-600000<br><i>Required</i> |
| <code>&lt;WAIT-TIME&gt;</code>  | Time in milliseconds to set the maximum wait time between two consecutive SPF calculations.<br>Default: 5000.<br>Range: 1-600000<br><i>Required</i> |

## Examples

Setting non-default timer values for SPF throttling:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# timers throttle spf start-time 100 hold-time 1000 max-
wait-time 10000
```

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# no throttle spf
switch(config-ospfv3-af-1)# no timers throttle spf start-time 100 hold-time 1000
max-wait-time 10000
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

## Command Information

| Platforms    | Command context               | Authority                                                                          |
|--------------|-------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-ospfv3-af-<PROCESS-ID> | Administrators or local user group members with execution rights for this command. |

## transit-delay (OSPFv3 AF)

```
transit-delay <seconds>
no transit-delay
```

### Description

Sets the time delay in Link state transmission for virtual links.

The **no** form of this command sets the delay in Link state transmission to the default of 1 second for virtual links.

| Parameter | Description                                                                               |
|-----------|-------------------------------------------------------------------------------------------|
| <seconds> | Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1. |

### Examples

Setting OSPFv3 AF virtual links transit delay:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# transit-delay 30
```

Setting OSPFv3 AF virtual links transit delay to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no transit-delay
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release | Modification        |
|---------|---------------------|
| 10.15   | Command introduced. |

### Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-router-vlink | Administrators or local user group members with execution rights for this command. |

# BGP commands

## aggregate-address

```
aggregate-address <IP-ADDR>/<MASK> [as-set] [summary-only]
                  [suppress-map <MAP-NAME>] [advertise-map <MAP-NAME>]
                  [attribute-map <MAP-NAME>]
```

```
no aggregate-address <IP-ADDR>/<MASK> [as-set] [summary-only]
                  [suppress-map <MAP-NAME>] [advertise-map <MAP-NAME>]
                  [attribute-map <MAP-NAME>]
```

```
[no] aggregate-address split-horizon
```

### Description

Creates an aggregate address entry in the BGP routing table.

The **no** form of this command removes the specified aggregate address entry.

| Parameter                | Description                                                                                                                                                                                                                                                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ADDRESS>                | Specifies an aggregate address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F.                                                                  |
| <MASK>                   | Specifies the number of bits in the address mask in CIDR format ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 128.                                                                                                                                                                             |
| as-set                   | The AS_PATH attribute advertised for this route will contain an AS_SET consisting of all AS numbers contained in all paths that are being summarized.                                                                                                                                                       |
| summary-only             | Creates the aggregate route but also suppresses advertisements of more-specific routes to all neighbors.                                                                                                                                                                                                    |
| suppress-map <MAP-NAME>  | Specifies an aggregate route for creation, but suppresses the advertisement of the created route. Match clauses of route maps can be used to suppress some more-specific routes of the aggregate selectively, and leave others unsuppressed. IP prefix lists and as_path lists match clauses are supported. |
| advertise-map <MAP-NAME> | Specifies routes that will be used to build attributes of the aggregate route, such as AS_SET or community.                                                                                                                                                                                                 |
| attribute-map <MAP-NAME> | Specifies that the attributes of the aggregate route can be changed.                                                                                                                                                                                                                                        |

### Examples

```
switch(config-bgp-ipv4-uc) # aggregate-address 10.0.0.0/8
switch(config-bgp-ipv4-uc) # no aggregate-address 10.0.0.0/8
```

```
switch(config-bgp-ipv6-uc) # aggregate-address 2001:0db8:85a3::8a2e:0370:7334/24
switch(config-bgp-ipv6-uc) # no aggregate-address 2001:0db8:85a3::8a2e:0370:7334/24
```

```
switch(config-bgp-ipv4-uc) # aggregate-address 10.0.0.0/8 as-set summary-only
switch(config-bgp-ipv4-uc) # aggregate-address 10.0.0.0/8 attribute-map RMap
```

```
switch(config-bgp-ipv6-uc) # aggregate-address 2001:0db8:85a3::8a2e:0370:7334/24
as-set summary-only
switch(config-bgp-ipv6-uc) # aggregate-address 2001:0db8:85a3::8a2e:0370:7334/24
attribute-map RMap
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## bgp always-compare-med

bgp always-compare-med  
no bgp always-compare-med

### Description

Enables comparison of the Multi Exit Discriminator (MED) for paths from neighbors in different autonomous systems. Any changes in BGP configuration are applied by restarting the current BGP sessions on the VRFs.

The **no** form of this command sets comparison of MED to the default setting (disabled).

### Usage

- MED is one of the parameters that is considered when selecting the best path among many alternative paths. The path with a lower MED is preferred over a path with a higher MED.
- During the best-path selection process, MED comparison is done only among paths from the same autonomous system. Use the command `bgp always-compare-med` to change this behavior by enforcing MED comparison between all paths, regardless of the autonomous system from which the paths are received.

### Examples

```
switch(config-bgp) # bgp always-compare-med
All current BGP sessions in VRF default will be restarted.
Do you want to continue (y/n)?
```

```
switch(config-bgp)# no bgp always-compare-med
All current BGP sessions in VRF default will be restarted.
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp asnotation dotted

```
bgp asnotation dotted
no bgp asnotation dotted
```

## Description

Specifies that Autonomous System (AS) numbers greater than 65535 be shown in dotted integer format for all show commands, including running-configuration.

The **no** form of this command restores the default format of non-dotted, simple integer.

## Example

```
switch(config-bgp)# bgp asnotation dotted
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp asnotation dotted-plus

bgp asnotation dotted-plus  
no bgp asnotation dotted-plus

### Description

Specifies that all Autonomous System (AS) numbers be shown in dotted integer format for all show commands, including running-configuration.

The **no** form of this command restores the default format of non-dotted, simple integer.

### Example

```
switch(config-bgp)# bgp asnotation dotted-plus
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp bestpath as-path ignore

bgp bestpath as-path ignore  
no bgp bestpath as-path ignore

### Description

Configures BGP to avoid considering the autonomous system (AS) path during best path route selection. By default, the AS-path is considered during BGP best path selection.

Any changes in BGP configuration are applied by restarting the current BGP sessions on the VRFs.

The **no** form of this command restores default behavior which configures BGP to consider the AS-path during route selection.

### Examples

```
switch(config-bgp)# bgp bestpath as-path ignore  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```



```
switch(config-bgp)# no bgp bestpath as-path ignore  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp bestpath as-path multipath-relax

```
bgp bestpath as-path multipath-relax  
no bgp bestpath as-path multipath-relax
```

### Description

Configures Border Gateway Protocol (BGP) to treat two BGP routes as equal cost even if their AS-paths differ, as long as their AS-path lengths and other relevant attributes are the same. This allows routes with different AS-paths to be programmed into the forwarding table as equal cost multipath routes.

Any changes in BGP configuration are applied by restarting the current BGP sessions on the VRFs.

The **no** form of this command restores the default behavior which configures BGP to treat two BGP routes as different costs when their AS-paths differ.

### Examples

```
switch(config-bgp)# bgp bestpath as-path multipath-relax  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```

```
switch(config-bgp)# no bgp bestpath as-path multipath-relax  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp bestpath compare-routerid

```
bgp bestpath compare-routerid
no bgp bestpath compare-routerid
```

### Description

Configures a BGP routing process to compare identical routes received from different external peers during the best path selection process and selects the route with the lowest router ID as the best path. Defaults to disabled.

Any changes in BGP configuration are applied by restarting the current BGP sessions in the VRFs.

The **no** form of this command returns the BGP routing process to the default operation. By default, BGP selects the route that was received first when two routes with identical attributes are received.

### Examples

```
switch(config-bgp)# bgp bestpath compare-routerid
All current BGP sessions in VRF default will be restarted.
Do you want to continue (y/n)?
```

```
switch(config-bgp)# no bgp bestpath compare-routerid
All current BGP sessions in VRF default will be restarted.
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp bestpath med confed

bgp bestpath med confed  
no bgp bestpath med confed

### Description

Compares the identical routes received from the different confederation peers and selects the route with the lowest Multi Exit Discriminator (MED) value as the best path. This behavior is disabled by default.

The **no** form of this command prevents the routing process from considering the MED value.



---

The selection of other attributes like `as-multi-path relax` and `as-path ignore` will not affect the behavior of this command within a confederation.

---

### Examples

Selecting the route with lowest MED value:

```
switch(config-bgp)# bgp bestpath med confed  
All active BGP sessions in the VRF %s will be restarted.  
Do you want to continue (y/n)?
```

Preventing the routing process from selecting the MED value:

```
switch(config-bgp)# no bgp bestpath med confed  
All active BGP sessions in the VRF %s will be restarted.  
Do you want to continue (y/n)?
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp bestpath med missing-as-worst

bgp bestpath med missing-as-worst  
no bgp bestpath med missing-as-worst

### Description

Configures a BGP routing process to assign a value of infinity (max possible) to routes that are missing the Multi Exit Discriminator (MED) attribute. The path without a MED value is the least desirable path. Any changes in BGP configuration are applied by restarting the current BGP sessions in the VRFs. The **no** form of this command restores default behavior. The default behavior assigns a value of 0 to the missing MED.

## Examples

```
switch(config-bgp)# bgp bestpath med missing-as-worst  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```

```
switch(config-bgp)# no bgp bestpath med missing-as-worst  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp cluster id

```
bgp cluster-id {<IPv4-ADDR> | <ID>}  
no bgp cluster-id {<IPv4-ADDR> | <ID>}
```

## Description

Specifies the cluster ID when the BGP router is used as a route-reflector. The cluster ID default is the router ID. Any changes in BGP configuration are applied by restarting the current BGP sessions on the VRFs.

The **no** form of this command sets the cluster ID to the default value, which is the router ID.

| Parameter   | Description                                                                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IPv4-ADDR> | Specifies an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255. You can remove leading zeros. For example, the address <b>192.169.005.100</b> becomes <b>192.168.5.100</b> . |
| <ID>        | Specifies the cluster ID as 32-bit number. Range: 1 to 4294967295.                                                                                                                                                        |

## Examples

```
switch(config-bgp)# bgp cluster-id 2.2.2.2  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```

```
switch(config-bgp)# no bgp cluster-id 2.2.2.2  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp confederation

```
bgp confederation <AS-NUMBER>  
no bgp confederation <AS-NUMBER>
```

### Description

Configures a BGP confederation with the confederation identifier. The group of Autonomous Systems (ASs) will be presented as a single autonomous system with the confederation identifier as the AS number.

The **no** form of the command deletes the BGP confederation identifier.

| Parameter   | Description                                                    |
|-------------|----------------------------------------------------------------|
| <AS-NUMBER> | Sets the identifier for the confederation. Range:1-4294967295. |

## Examples

Configuring the BGP confederation with the AS number:

```
switch(config-bgp)# bgp confederation 100
```

Deleting BGP confederation identifier:

```
switch(config-bgp)# no bgp confederation 100
This will delete BGP confederation identifier on this device.
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp confederation peers

```
bgp confederation peers <AS NUMBER>
no bgp bgp confederation peers <AS NUMBER>
```

### Description

Configures BGP confederation peers with both same and different sub-autonomous system to establish an eBGP membership. You can configure a list of AS numbers separated by spaces.

The **no** form of this command disables the peer session and deletes the peer information.

| Parameter   | Description                                                                                  |
|-------------|----------------------------------------------------------------------------------------------|
| <AS NUMBER> | Specifies the autonomous system numbers to establish an eBGP membership. Range: 64512-65535. |

## Examples

```
switch(config-bgp)# bgp confederation peers 64512 64513
```

```
switch(config-bgp)# no bgp confederation peers 64512
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp dampening

```
bgp dampening {{half-life <HALF-TIME> reuse <LOW-THRESHOLD> suppress <HI-THRESHOLD> max-
suppress-time <MAX-TIME>}}
| route-map <NAME>}
```

## Description

Enables route flap dampening which reduces the propagation of unstable routes in the network.

| Parameter                    | Description                                                                                                                                                   |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| half-life <HALF-TIME>        | Specifies the half-life time in minutes. When the time expires, the penalty on a route gets reduced exponentially to half its current value. Default: 15.     |
| reuse <LOW-THRESHOLD>        | Specifies the lower threshold of penalty. On a suppressed route, when the penalty on a route falls below this value, the route is unsuppressed. Default: 750. |
| suppress <HI-THRESHOLD>      | Specifies the upper threshold of penalty. When the penalty on a flapping route exceeds this value, the route is suppressed. Default: 2000.                    |
| max-suppress-time <MAX-TIME> | Specifies the maximum time to keep a route suppressed in minutes. Once this timer expires, the route is unsuppressed. Default: 60.                            |
| route-map <NAME>             | Specifies the name of a route map.                                                                                                                            |

These parameters can be configured at the router level for specific address families or the same parameters can be configured under a route map which can be applied to dampening command.

## Usage

The dampening algorithm assigns a penalty of 1000 to a flapping route every time the route gets withdrawn. The penalty values accumulate on the route every time it flaps. However, the penalty decays and is reduced to half its value by the half-life time.



This feature is not applicable on IBGP routes.

## Example

```

switch(config)# router bgp 1
switch(config-bgp)# address-family ipv4 unicast
switch(config-bgp-ipv4-uc)# bgp dampening

switch(config-bgp-ipv4-uc)# bgp dampening route-map abc

switch(config-bgp-ipv4-uc)# bgp dampening route-map xyz

switch(config-bgp-ipv4-uc)# bgp dampening half-life 10 reuse 100 suppress 250 max-suppress-time 45

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context    | Authority                                                                          |
|-----------|--------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc | Administrators or local user group members with execution rights for this command. |

## bgp default local-preference

```

bgp default local-preference <NUMBER>
no bgp default local-preference

```

### Description

Default local preference value for BGP learned routes. Any changes in BGP configuration are applied by restarting the current BGP sessions on the VRFs.

The **no** form of this command sets the local preference to the default value of 100.

| Parameter | Description                                                                 |
|-----------|-----------------------------------------------------------------------------|
| <NUMBER>  | Specifies the local preference value. Range: 0 to 4294967295. Default: 100. |

## Examples

```

switch(config-bgp)# bgp default local-preference 20
All current BGP sessions in VRF default will be restarted.
Do you want to continue (y/n)?

```

```

switch(config-bgp)# no bgp default local-preference
All current BGP sessions in VRF default will be restarted.

```



Do you want to continue (y/n)?



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp deterministic-med

```
bgp deterministic-med
no bgp deterministic-med
```

### Description

Enables comparison of the Multi-Exit Discriminator (MED) attribute when selecting routes advertised by different peers in the same autonomous system. Any changes in BGP configuration are applied by restarting the current BGP sessions on the VRFs.

The **no** form of this command sets MED comparison to the default setting of disabled.

### Examples

```
switch(config-bgp)# bgp deterministic-med
All current BGP sessions in VRF default will be restarted.
Do you want to continue (y/n)?
```

```
switch(config-bgp)# no bgp deterministic-med
All current BGP sessions in VRF default will be restarted.
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp fast-external-fallover

```
bgp fast-external-fallover
no bgp fast-external-fallover
```

### Description

Sets the switch to reset the BGP sessions of any directly adjacent external peers when the connected link goes down. It is enabled by default.

The **no** form of this command restores the default behavior where BGP waits until the hold time expires before closing sessions.

### Examples

```
switch(config-bgp) # bgp fast-external-fallover
switch(config-bgp) # no bgp fast-external-fallover
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification                                                                                                                                                                                           |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.08            | The default behavior has been changed from disabled to enabled state.<br><br><b>NOTE:</b><br>When upgrading, the feature will remain in the state it was (disabled or enabled) in the earlier release. |
| 10.07 or earlier | --                                                                                                                                                                                                     |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp graceful-restart restart-time

```
bgp graceful-restart restart-time <DELAY>
no bgp graceful-restart restart-time
```

### Description

Sets the graceful restart timer which determines how long the switch waits for a graceful-restart capable neighbor to re-establish BGP peering. Any changes in BGP configuration are applied by restarting the current BGP sessions on the VRFs.

The **no** form of this command resets to the default value of 120 seconds.

| Parameter | Description                                                               |
|-----------|---------------------------------------------------------------------------|
| <DELAY>   | Graceful restart timer delay in seconds. Range: 1 to 3600. Default: 1500. |

## Usage

- Graceful restart functionality is enabled by default, and there is no command to disable the functionality at the protocol level.
- However, the graceful-restart functionality can be disabled globally using the command **router graceful-restart**.

## Examples

```
switch(config-bgp)# bgp graceful-restart restart-time 150  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```

```
switch(config-bgp)# no bgp graceful-restart restart-time  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp graceful-restart stalepath-time

```
bgp graceful-restart stalepath-time <TIME>  
no bgp graceful-restart stalepath-time
```

## Description

Sets the stale path timer. This timer determines how long BGP keeps stale routes from the restarting BGP peer. Any changes in BGP configuration are applied by restarting the current BGP sessions on the VRFs.

The **no** form of this command resets to the stale path timer to the default of 300 seconds.

| Parameter                 | Description                                                                   |
|---------------------------|-------------------------------------------------------------------------------|
| <code>&lt;TIME&gt;</code> | Specifies the stale path timer in seconds. Range: 1 to 3600.<br>Default: 300. |

## Examples

```
switch(config-bgp)# bgp graceful-restart stalepath-time 300  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```

```
switch(config-bgp)# no bgp graceful-restart stalepath-time  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp log-neighbor-changes

```
bgp log-neighbor-changes  
no bgp log-neighbor-changes
```

### Description

Enables logging of BGP neighbor session state changes.

The **no** form of this command disables logging of changes in BGP neighbor adjacencies.

### Examples

```
switch(config-bgp)# bgp log-neighbor-changes  
switch(config-bgp)# no bgp log-neighbor-changes
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## bgp maxas-limit

bgp maxas-limit <LENGTH>  
no bgp maxas-limit

### Description

Specifies the maximum size of AS paths in update messages. Routes with AS paths greater than the specified length are discarded.

The **no** form of this command sets the limit to the default of 32.

| Parameter | Description                                                                   |
|-----------|-------------------------------------------------------------------------------|
| <LENGTH>  | Specifies the number of AS segments. Length: 1 to 32 characters. Default: 32. |

## Example

```
switch(config-bgp)# bgp maxas-limit 20  
switch(config-bgp)# no bgp maxas-limit
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

# bgp router-id

```
bgp router-id <ROUTER-ID>  
no bgp router-id <ROUTER-ID>
```

## Description

Configures a fixed router ID for the BGP peer process running on the router. Any changes in BGP configuration are applied by restarting the current BGP sessions on the VRFs.

The **no** form of this command removes the fixed router ID from the running configuration and restores the default router ID selection.

| Parameter   | Description                                                                                                                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ROUTER-ID> | Specifies the router ID in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.<br>If router-id is changed, then all the active BGP peer sessions go down and restart with the newly configured router-id. |

## Usage

BGP determines the router ID as follows:

1. The address configured with the command `bgp router-id`.
2. The highest IP address on all the loopback interfaces.
3. The highest IP address on any interface.

## Examples

```
switch(config-bgp)# bgp router-id 1.1.1.1  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```

```
switch(config-bgp)# no bgp router-id 1.1.1.1  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

# bgp update-group

bgp update-group  
no bgp update-group

## Description

Enables the update-group feature. Update-group is used to optimize BGP processing. With update-group enabled, neighbors that share same outbound policy are grouped under single update-group and update messages are shared to all peers within that group. Disabled by default.

On enabling/disabling update-group, all the current BGP sessions including dynamic peer sessions will be restarted. The dynamic peer sessions will get re-established based on the connect-retry timer configured on the peer device.

BGP determines the AS number size to be used with each individual peer based on the configured AS octet-size on the peers. If BGP is configured to use 2 octet ASN with peer A and 4 octet ASN with peer B, then BGP updates sends to peer A and peer B will be different. Therefore, they will form into two different update-groups.

This feature is supported for IBGP alone. It is not recommended to enable update-group on VPNv4 address families or configure update-group if an VPNv4 address family is configured. Update-group indices are created automatically and cannot be configured. Indices are ephemeral in nature. Update-group indices will not be retained across process restart. When update-group is configured, route reflector clients and non route reflector clients are parted into different update-groups. The use of update-groups imposes the following limitations in BGP processing:

1. While advertising routes to peers of the update-group, the check on route's next-hop is ignored. This means that, BGP routes can get advertised to a peer even if the next-hop IP of the routes is same as peer's IP, and the peer should discard such routes.
2. The use of ORF is restricted. BGP will continue to send outbound route filters to peers, but received filters are ignored.
3. Sender loop detection is delayed until the point where UPDATE messages are sent to the members of the update-group. This means route dropped by the sender due to loop detection will still count as advertised, as they have been advertised to the update-groups.

It is recommended to enable BGP update-group globally before configuring BGP neighbors. This ensures that toggling of BGP neighborship can be avoided and memory of the routing process can be optimized.

The **no** form of this command disables the update-group feature.

## Examples

Enabling BGP update-group:

```
switch(config-bgp)# bgp update-group  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```

Disabling BGP update-group:

```
switch(config-bgp)# no bgp update-group  
All current BGP sessions in VRF default will be restarted.  
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.13   | Command introduced. |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## clear bgp

```
clear bgp [vrf <VRF-NAME>][ip unicast | ipv6 unicast | l2vpn evpn | all]
        { * | <NEIGHBOR-IP-ADDRESS> | <PEER-GROUP-NAME> } [soft {in | out} ]
```

## Description

Clears BGP with a hard or soft reset. If the address-family is specified, then the neighbors that negotiated the respective family will be reset or a route refresh message will be sent.

A route refresh message is sent when **soft in** or **soft out** is specified.

| Parameter                                     | Description                                                                                                                  |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| ip unicast                                    | Specifies the IPv4 address family.                                                                                           |
| ipv6 unicast                                  | Specifies the IPv6 address family.                                                                                           |
| l2vpn evpn                                    | Selects the L2VPN EVPN address family                                                                                        |
| vrf <VRF-NAME>                                | Specifies a VRF name. By default, the default BGP instance information is displayed if the VRF is not specified.             |
| all                                           | Specifies all VRFs and address families.                                                                                     |
| *   <NEIGHBOR-IP-ADDRESS>   <PEER-GROUP-NAME> | Specifies a neighbor IP address or peer group name for which peer sessions are to be reset, or * to reset all sessions.      |
| soft in                                       | Clears the BGP in the inbound direction without a hard reset of the session. This will send a route refresh request message. |
| soft out                                      | Clears the BGP in the outbound direction without a hard reset of the session. This will send a route refresh message.        |

## Examples



```
switch(config)# clear bgp all *
```

```
switch# clear bgp all PEER_GRP_1 soft in  
switch# clear bgp all PEER_GRP_1 soft out
```

```
switch# clear bgp ip unicast 9.0.0.2 soft in  
switch# clear bgp ip unicast 9.0.0.2 soft out  
switch# clear bgp ip unicast PEER_GRP_1 soft in  
switch# clear bgp ip unicast PEER_GRP_1 soft out
```

```
switch# clear bgp vrf v1 ip unicast 9.0.0.2 soft in  
switch# clear bgp vrf v1 ip unicast 9.0.0.2 soft out  
switch# clear bgp vrf v1 ip unicast PEER_GRP_1 soft in  
switch# clear bgp vrf v1 ip unicast PEER_GRP_1 soft out  
switch# clear bgp vrf v1 all PEER_GRP_1 soft in  
switch# clear bgp vrf v1 all PEER_GRP_1 soft out
```

```
switch# clear bgp ipv6 unicast 9000::2 soft in  
switch# clear bgp ipv6 unicast 9000::2 soft out  
switch# clear bgp vrf v1 ipv6 unicast 9000::2 soft in  
switch# clear bgp vrf v1 ipv6 unicast 9000::2 soft out
```

```
switch# clear bgp l2vpn evpn * soft in  
switch# clear bgp l2vpn evpn * soft out  
switch# clear bgp l2vpn evpn 9.0.0.2 soft in  
switch# clear bgp l2vpn evpn 9.0.0.2 soft out  
switch# clear bgp l2vpn evpn PEER_GRP_1 soft in  
switch# clear bgp l2vpn evpn PEER_GRP_1 soft out
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                                                     |
|------------------|----------------------------------------------------------------------------------|
| 10.15            | <b>soft out</b> and <b>&lt;PEER-GROUP-NAME&gt; soft in</b> parameters are added. |
| 10.14            | Replaced IPv4 with IP in commands. IPv4 is deprecated.                           |
| 10.07 or earlier | --                                                                               |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# disable | enable

disable | enable

## Description

This command disables or enables the BGP instance while retaining the configuration. Disable and enable of the BGP instance may result in a change of the router ID.

By default the BGP instance is enabled.

## Examples

```
switch(config)# router bgp 100
switch(config-bgp)# disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

# distance bgp

```
distance bgp <EXTERNAL> <INTERNAL>
no distance bgp <EXTERNAL> <INTERNAL>
```

## Description

Configures the administrative distance for BGP.

The **no** form of this command restores the default settings, 20 for eBGP and 200 for iBGP,

| Parameter  | Description                                                                           |
|------------|---------------------------------------------------------------------------------------|
| <EXTERNAL> | Specifies the administrative distance for eBGP routes. Range: 1 to 255. Default: 20.  |
| <INTERNAL> | Specifies the administrative distance for iBGP routes. Range: 1 to 255. Default: 200. |

## Example

```
switch(config-bgp-ipv4-uc) # distance bgp 100 150
switch(config-bgp-ipv4-uc) # no distance bgp 100 150
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## maximum-paths (router bgp)

```
maximum-paths <MAXPATHS>
no maximum-paths <MAXPATHS>
```

### Description

Configures the maximum number of paths that BGP adds to the route table for equal-cost multipath (ECMP) load balancing for routes learned from both internal and external BGP. Any changes in BGP configuration are applied by restarting the current BGP sessions on the VRFs.

On HPE Aruba Networking 8325, 8325H, 8325P and 10000 switch series, a maximum-paths configuration is supported globally as well as for **ipv4-unicast**, **ipv6-unicast** and **l2vpn-evpn** address families. For HPE Aruba Networking 8325, 8325H, and 10000 switch series, this configuration is supported under the **l2vpn-evpn** for L2/L3 Overlay ECMP feature. On HPE Aruba Networking 6300, 6400, 8100 and 8360 switch series, this command is supported under the **l2vpn-evpn** for L2 Overlay ECMP feature.

If address-family specific maximum-paths are configured, they take precedence over global configuration. Greater than eight maximum-paths at can be configured within the global context, but for **l2vpn-evpn** routes the supported maximum limit is still eight.

The **no** form of this command restores the default setting of 4.

| Parameter  | Description                                                                                                                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <MAXPATHS> | Specifies the maximum number of paths. 1 to 32 paths can be set globally. 1 to 8 paths can be set for <b>l2vpn-evpn</b> address family. 1 to 32 paths can be set for <b>ipv4-unicast</b> and <b>ipv6-unicast</b> . |

### Usage

When both global and address-family maximum paths are configured then the address family value takes precedence.

| Global max-path | l2evpn, ipv4-unicast, or ipv6-unicast address-family Max-path | Operational Max-path | Comments                                              |
|-----------------|---------------------------------------------------------------|----------------------|-------------------------------------------------------|
| Default( 4)     | Default (4)                                                   | 4                    | --                                                    |
| Configured (4)  | configured (4)                                                | 4                    | Four is the configured value for address family paths |
| Default (4)     | 8 (max)                                                       | 8                    | The address family value takes precedence             |
| 32 (max)        | 8 (max)                                                       | 8                    | CLI configuration is limited to eight paths.          |
| Default (8)     | configured (4)                                                | 4                    | Four is the configured value for address family paths |

## EVPN Multi-Homing Interoperability

### Examples

Configuring the maximum number of global paths:

```
switch(config)# router bgp 1
switch(config-bgp)# maximum-paths 32
All current BGP sessions in VRF default will be restarted.
Do you want to continue (y/n)? y
```

Configuring the maximum-paths for the l2vpn-evpn address family:

```
switch(config-bgp)# address-family l2vpn evpn
switch(config-bgp-l2vpn-evpn)# maximum-paths 6
```

Removing the global maximum paths setting:

```
switch(config-bgp)# no maximum-paths
All current BGP sessions in VRF default will be restarted.
Do you want to continue (y/n)? y
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release    | Modification                                                                                                                                                                                                                                                      |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.16.1000 | Adds support to HPE Aruba Networking 8325, 8325H, 10000 switch series for L2 and L3 Overlay ECMP features.<br>On HPE Aruba Networking 6300, 6400, 8100 and 8360 switch series, this command is supported under the <b>l2vpn-evpn</b> for L2 Overlay ECMP feature. |

| Release          | Modification                                                                                                                                                                                                                                                                                           |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.13.1000       | On 8325, 8325H, 8325P and 10000 switch series, configuration is supported on <b>ipv4-unicast</b> , <b>ipv6-unicast</b> and <b>l2vpn-evpn</b> address families. For HPE Aruba Networking 6300, 6400, 8100, 8360, 8400 and 9300 Switch series, this configuration is supported at the global level only. |
| 10.10            | Increased upper limit of range of <i>&lt;MAXPATHS&gt;</i> parameter to 32.                                                                                                                                                                                                                             |
| 10.07 or earlier | --                                                                                                                                                                                                                                                                                                     |

## Command Information

| Platforms | Command context                     | Authority                                                                          |
|-----------|-------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp<br>config-bgp-l2vpn-evpn | Administrators or local user group members with execution rights for this command. |

## neighbor activate

```
neighbor {<IP-ADDR>|<IFNAME>} activate
no neighbor {<IP-ADDR>|<IFNAME>} activate
```

## Description

This command enables the address-family capability and exchange of information specific to an address family with a BGP neighbor.

The **no** form of this command removes the address-family capability and disables the exchange of routes for the specified address-family with the BGP neighbor.

| Parameter              | Description                                                                                                                                                                                                                                                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>&lt;IP-ADDR&gt;</i> | Specifies an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F. Specifies the unnumbered interface name of the neighbor. |
| <i>&lt;IFNAME&gt;</i>  | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                                                                                     |



VPNv4 address-family, peer group, and BFD are not supported for BGP unnumbered.

## Examples

Enabling and removing the address-family capability and exchange of information on 1/1/1.

```
switch(config-bgp)# neighbor 1/1/1 remote-as 1
switch(config-bgp-ipv4-uc)# neighbor 1/1/1 activate
switch(config-bgp-ipv4-uc)# no neighbor 1/1/1 activate
```

Enabling and removing the address-family capability and exchange of information on 1/1/1.1.

```
switch(config-bgp)# neighbor 1/1/1.1
switch(config-bgp-ipv6-uc)# neighbor 1/1/1.1 activate
switch(config-bgp-ipv6-uc)# no neighbor 1/1/1.1 activate
```

Enabling and removing the address-family capability and exchange of information on 1/1/1:1.

```
switch(config-bgp)# neighbor 1/1/1:1
switch(config-bgp-ipv4-uc)# neighbor 1/1/1:1 activate
switch(config-bgp-ipv4-uc)# no neighbor 1/1/1:1 activate
```

Enabling and removing the address-family capability and exchange of information on 1.1.1.1.

```
switch(config-bgp-l2vpn-evpn)# neighbor 1.1.1.1 activate
switch(config-bgp-l2vpn-evpn)# no neighbor 1.1.1.1 activate
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                                                   | Authority                                                                          |
|-----------|-------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc<br>config-bgp-l2vpn-evpn | Administrators or local user group members with execution rights for this command. |

## neighbor advertisement-interval

```
neighbor {<IP-ADDRESS>|<IFNAME>|<PEER-GROUP-NAME>} advertisement-interval <INTERVAL>
no neighbor {<IP-ADDRESS>|<IFNAME>|<PEER-GROUP-NAME>} advertisement-interval
```

### Description

Sets the advertisement interval, which defines the length of time between transmission of BGP routing updates.

The **no** form of this command restores the default value. Default values are 30 seconds for external BGP peer and 5 seconds for internal BGP peer.

| Parameter | Description                                                                                                                                                                                         |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDR> | Specifies an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a |

| Parameter         | Description                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------|
|                   | hexadecimal number from 0 to F.                                                                                                  |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.                                                                         |
| <PEER-GROUP-NAME> | Specifies a Peer-Group. This is not supported on BGP unnumbered.                                                                 |
| <INTERVAL>        | Specifies the advertisement interval in seconds. Range: 0 to 600. Default: 30 for external BGP peer and 5 for internal BGP peer. |

## Examples

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 advertisement-interval 20
switch(config-bgp-ipv4-uc) # no neighbor 1.1.1.1 advertisement-interval
```

```
switch(config-bgp-ipv4-uc) # neighbor pg advertisement-interval 50
switch(config-bgp-ipv4-uc) # no neighbor pg advertisement-interval
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## neighbor add-paths

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} add-paths {send | rcv | both}
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} add-paths {send | rcv | both}
```

## Description

Enables additional path capability for BGP as described in RFC 7911. This allows BGP peer to send, receive, or send and receive multiple paths for the same address prefix without the subsequent advertisements implicitly replacing any previous paths. The additional path includes the first (N-1) best paths, which means that the total paths for an address prefix received by a BGP speaker will include the best path and the additional paths determined by its BGP peer.

With additional path feature, each path is identified by a path identifier in addition to the address prefix. To use this command, the backup path of BGP next-hop must be different than the primary path.

The **no** form of this command disables the additional path feature.

| Parameter                      | Description                                                                                                                                                                                                                                                   |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDR>                      | Specifies an IP address.                                                                                                                                                                                                                                      |
| <IFNAME>                       | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                                                      |
| <PEER-GROUP-NAME>              | Specifies a peer group. This is not supported on BGP unnumbered.                                                                                                                                                                                              |
| add-paths {send   recv   both} | Configures the additional paths in one of the following ways:<br>send—Enables the neighbor to send the additional paths.<br>recv—Enables the neighbor to receive the additional paths.<br>both—Enables the neighbor to send and receive the additional paths. |

## Examples

Enabling BGP neighbor to send the additional paths:

```
switch(config)# router bgp 100
switch(config-bgp)# address-family ipv4 unicast
switch(config-bgp-ipv4-uc)# neighbor 1.1.1.1 add-paths send
```

Disabling BGP neighbor to send the additional paths:

```
switch(config)# router bgp 100
switch(config-bgp)# address-family ipv4 unicast
switch(config-bgp-ipv4-uc)# no neighbor 1.1.1.1 add-paths send
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## neighbor add-paths advertise-best

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} add-paths advertise-best <2-4>
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} add-paths advertise-best <2-4>
```



## Description

Controls the number of best BGP Paths to be advertised by a BGP speaker to a BGP peer. When enabled, it allows BGP speaker to advertise more than one best paths for the same address prefix. The total paths for an address prefix will include the best path and the additional paths.

The **no** form of this command removes the advertise best path configuration.

| Parameter            | Description                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------|
| <IP-ADDR>            | Specifies an IP address.                                                                          |
| <IFNAME>             | Specifies the unnumbered interface name of the neighbor.                                          |
| <PEER-GROUP-NAME>    | Specifies a peer group. This is not supported on BGP unnumbered.                                  |
| advertise-best <2-4> | Specifies the number of best BGP paths to be advertised to a BGP Peer. Range: 2 to 4. Default: 2. |

## Examples

Setting the number of best paths to send to the neighbor:

```
switch(config)# router bgp 100
switch(config-bgp)# address-family ipv4 unicast
switch(config-bgp-ipv4-uc)# neighbor 1.1.1.1 add-paths advertise-best 3
```

Removing the advertise best path configuration:

```
switch(config)# router bgp 100
switch(config-bgp)# address-family ipv4 unicast
switch(config-bgp-ipv4-uc)# no neighbor 1.1.1.1 add-paths advertise-best 3
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## neighbor allowas-in

neighbor {<IP-ADDRESS>|<IFNAME>|<LIMIT>

no neighbor {<IP-ADDRESS>|<IFNAME>|<LIMIT>

## Description

Specifies the number of times that the AS path of a received route can contain the AS number of the recipient BGP speaker and still be accepted. When this configuration is applied to a peer-group, all the neighbors that are part of the peer-group inherit this setting.

The **no** form of this command restores the default setting, which is to reject as a loop any route where the path contains the speaker AS number.

| Parameter         | Description                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDRESS>      | Specifies the neighbor IP address in the IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or in the IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F.<br><br><b>NOTE:</b> IPv6 MP-BGP peering must not be used for L2VPN EVPN address family, because VXLAN tunnel interface does not support IPv6 addresses. |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                                                                                                                                                                                                         |
| <PEER-GROUP-NAME> | Specifies a peer group.                                                                                                                                                                                                                                                                                                                                                                                          |
| <LIMIT>           | Specifies the number of times that the AS path of a received route can contain the AS number of the recipient BGP. Range: 1 to 10.                                                                                                                                                                                                                                                                               |

## Examples

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 allowas-in 5
switch(config-bgp-ipv4-uc) # no neighbor 1.1.1.1 allowas-in
```

```
switch(config-bgp-ipv6-uc) # neighbor 2001:0db8:85a3::8a2e:0370:7334 allowas-in 5
switch(config-bgp-ipv6-uc) # no neighbor 2001:0db8:85a3::8a2e:0370:7334 allowas-in
```

```
switch(config-bgp-ipv4-uc) # neighbor PG allowas-in 5
switch(config-bgp-ipv4-uc) # no neighbor PG allowas-in
```

```
switch(config-bgp-l2vpn-evpn) # neighbor 1.1.1.1 allowas-in 5
switch(config-bgp-l2vpn-evpn) # no neighbor 1.1.1.1 allowas-in
```

```
switch(config-bgp-l2vpn-evpn) # neighbor PG allowas-in 5
switch(config-bgp-l2vpn-evpn) # no neighbor PG allowas-in
```

```
switch(config-bgp-l2vpn-evpn) # neighbor 2001:0db8:85a3::8a2e:0370:7334 allowas-in
5
switch(config-bgp-l2vpn-evpn) # no neighbor 2001:0db8:85a3::8a2e:0370:7334 allowas-
in
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                                                   | Authority                                                                          |
|-----------|-------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc<br>config-bgp-l2vpn-evpn | Administrators or local user group members with execution rights for this command. |

## neighbor ao

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} ao <keychain-name>
  accept-ao-mismatch
  include-tcp-options
  no ...
```

## Description

Enables TCP Authentication Option (TCP-AO) authentication on a TCP connection between two BGP neighbors. To disable this function, use the **no** form of this command.

| Parameter          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDRESS>       | Specifies an IP address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <IFNAME>           | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <PEER-GROUP-NAME>  | Specifies a peer group. This is not supported on BGP unnumbered.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <keychain-name>    | Name of the keychain for the neighbor.<br>The keychain allows keys to be configured with different valid lifetimes.<br>This mechanism provides a way for a set of keys to be rotated and hence protect against long-lived-key attacks. At any given time only one key is selected as active-key and keys are valid for a duration of the defined send-lifetime. If the send-lifetime and the accept-lifetimes are not configured for the key, the key is considered to be valid for infinite lifetime.<br>When multiple keys are configured, its recommended that keys overlap in their send-lifetimes so that the key rollover occurs at the start of the next key's send-lifetime. This allows for a continuous key usage by TCP-AO. |
| accept-ao-mismatch | Accept incoming TCP segments without TCP-AO option.<br>If enabled, the device will accept a connection from the peer <i>even</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Parameter                        | Description                                                                                                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <i>if the received TCP packets do not contain the TCP-AO option.</i>                                                                                                  |
| <code>include-tcp-options</code> | Include the TCP header options for MAC calculation. Note that enabling this setting will immediately reset the neighbor session. This setting is disabled by default. |
| <code>no ...</code>              | Negates any configured parameter.                                                                                                                                     |

## Usage

TCP-AO authentication can not be used with the [neighbor password](#) feature. When TCP-AO is applied to a peer-group, all the neighbors in peer-group will inherit the peer-group configuration unless there is a configuration specific to an individual neighbor. If a peer-group is configured with the neighbor password feature but the neighbors that belong to that peer-group are configured with TCP-AO, the TCP-AO configuration will be rejected. Similarly, If a peer-group is configured to use TCP-AO authentication, the neighbors that belong to that peer-group will reject the neighbor password.

The TCP-AO feature takes a keychain as a parameter. The key will not be valid until a [Recv-D](#), [Send-ID](#), and [send lifetime](#) is configured. The supported cryptographic algorithms for TCP-AO are:

- HMAC-SHA-1-96 based on [RFC2104] and [FIPS-180-3]
- AES-128-CMAC-96 based on [NIST-SP800-38B][FIPS197]

## Examples

```
switch(config)# keychain bgpkeys
switch(config-keychain)# key 1
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2022 duration
infinite
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2022
duration infinite
switch(config-keychain-key)# send-id 10
switch(config-keychain-key)# recv-id 10
switch(config-keychain-key)# cryptographic-algorithm aes-cmac-128
switch(config-keychain-key)# key-string plaintext qwer
switch(config)# router bgp 1
switch(config-bgp)# neighbor 1.1.1.1 ao bgpkeys
switch(config-bgp)# no neighbor 1.1.1.1 ao accept-ao-mismatch
switch(config-bgp)# no neighbor 1.1.1.1 ao include-tcp-option
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.11 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor capability orf prefix-list

```
neighbor {<IP-ADDRESS>|<IFNAME>} capability orf prefix-list {send | receive | both}
no neighbor {<IP-ADDRESS>|<IFNAME>} capability orf prefix-list {send | receive | both}
```

### Description

Enables the Outbound Route filtering (ORF) capability with the neighbor in one of the three available modes. The available modes are send, receive, and both. The ORF capability is executed based on prefix list only.

The Outbound Route Filtering (ORF) capability provides a mechanism for a BGP speaker to send a set of Outbound Route Filters (ORFs) that can be used by its BGP peer to filter its outbound routing updates to the speaker. This is a filtering method used to reduce the computation on the router receiving the route.

The **no** form of this command disables the ORF capability.

| Parameter                                             | Description                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDRESS>                                          | Specifies an IP address.                                                                                                                                                                                                                                                                                                                                                                      |
| <IFNAME>                                              | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                                                                                                                                                                                      |
| capability orf prefix-list<br>{send   receive   both} | Enables ORF prefix list capability with the neighbor in one of the following modes: <ul style="list-style-type: none"> <li>▪ <b>send</b> - Enables the ORF prefix list capability in send mode.</li> <li>▪ <b>receive</b> - Enables the ORF prefix list capability in receive mode.</li> <li>▪ <b>both</b> - Enables the ORF prefix list capability in both send and receive mode.</li> </ul> |

### Examples

Enabling the ORF prefix list capability in both send and receive mode:

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 capability orf prefix-list both
```

Enabling the ORF prefix list capability in send mode:

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 capability orf prefix-list send
```

Disabling the ORF prefix list capability in both send and receive mode: :

```
switch(config-bgp-ipv4-uc) # no neighbor 1.1.1.1 capability orf prefix-list both
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## neighbor default-originate

neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} default-originate [route-map <MAP-NAME>]  
no neighbor {<IP-ADDRESS>|<IFNAME>|<PEER-GROUP-NAME>} default-originate [route-map <MAP-NAME>]

### Description

Enables the local router to send the default route 0.0.0.0 to a neighbor. The neighbor can then use this route to reach the router when all other routes are unavailable. Use the `route-map` option to configure the route map to modify the default route attributes.

The **no** form of this command disables this feature.

| Parameter         | Description                                                                                                                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDR>         | Specifies an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F. |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                            |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered.                                                                                                                                                                    |
| <MAP-NAME>        | Sets the route map to modify the default route attributes.                                                                                                                                                                          |

### Examples

```
switch(config-bgp-ipv4-uc)# neighbor 1.1.1.1 default-originate  
switch(config-bgp-ipv4-uc)# no neighbor 1.1.1.1 default-originate
```

```
switch(config-bgp-ipv4-uc)# neighbor PG default-originate  
switch(config-bgp-ipv4-uc)# no neighbor PG default-originate
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## neighbor ebgp-multihop

neighbor {<IP-ADDR> | <PEER-GROUP-NAME>} ebgp-multihop <ttl>  
no neighbor {<IP-ADDR> | <PEER-GROUP-NAME>} ebgp-multihop <ttl>

### Description

Enables BGP to establish a session with external peers residing on networks that are not directly connected. By default, BGP can only establish sessions with external BGP peers that are directly connected. If external BGP peer are not connected directly, then ebgp-multihop is enabled to initiate the connection with that external peer. If ebgp-multihop is disabled and external BGP peers are indirectly connected, then BGP peer session is not established.

The **no** form of this command disables the peer ebgp-multihop feature.

| Parameter         | Description              |
|-------------------|--------------------------|
| <IP-ADDR>         | Specifies an IP address. |
| <PEER-GROUP-NAME> | Specifies a peer group.  |
| <ttl>             | Specifies the ttl value. |

### Examples

Enabling BGP to establish connection with external peers residing on networks that are not directly connected:

```
switch(config-bgp)# neighbor 1.1.1.1 ebgp-multihop 5
```

Disabling BGP to establish connection with external peers residing on networks that are not directly connected:

```
switch(config-bgp)# no neighbor 1.1.1.1 ebgp-multihop
```

Enabling BGP to establish connection with external peers residing on networks that are not directly connected:

```
switch(config-bgp)# neighbor pg ebgp-multihop 5
```

Disabling BGP to establish connection with external peers residing on networks that are not directly connected:

```
switch(config-bgp)# no neighbor pg ebgp-multihop 5
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                        |
|------------------|-----------------------------------------------------|
| 10.16            | Syntax updated to use <ttl> instead of <HOP-COUNT>. |
| 10.07 or earlier | --                                                  |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor fall-over

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} fall-over  
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} fall-over
```

## Description

Enables BGP fast peering session deactivation. When neighbor fall-over is configured, the BGP process monitors the RIB and if the route to peer is not present in the routing table, it immediately deactivates the peer session without waiting for the hold down timer. It is disabled by default.

The **no** form of this command disables this feature.

| Parameter         | Description                                                      |
|-------------------|------------------------------------------------------------------|
| <IP-ADDR>         | Specifies an IP address.                                         |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.         |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered. |

## Usage

Neighbor fall-over does not track connected or static routes to peers. However, this is not an issue when IBGP peering is using a loopback interface. To force a fall-over for connected and static routes, use the command `neighbor fall-over bfd`.



## Examples

```
switch(config-bgp)# neighbor 1.1.1.1 fall-over
switch(config-bgp)# no neighbor 1.1.1.1 fall-over
```

```
switch(config-bgp)# neighbor PG fall-over
switch(config-bgp)# no neighbor PG fall-over
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor fall-over bfd (context: config-bgp)

```
neighbor {<IPv4-ADDR>|<IPv6-ADDR>|<PEER-GROUP-NAME>} fall-over bfd
no neighbor {<IPv4-ADDR>|<IPv6-ADDR>|<PEER-GROUP-NAME>} fall-over bfd
```

## Description

Enables BGP to register with BFD to receive fast peering session deactivation messages from BFD. You can either configure BFD support for BGP per neighbor or peer-group.

The **no** form of this command disables BGP for BFD.



Multihop BFD is not supported for BGP.

| Parameter         | Description                                                                                                                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IPv4-ADDR>       | Specifies the neighbor address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.                                                                                                                    |
| <IPv6-ADDR>       | Specifies the neighbor address in IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F. This parameter applies only to HPE Aruba Networking 6300 and 6400 Switch series. |
| <PEER-GROUP-NAME> | Specifies a peer group.                                                                                                                                                                                                                |

## Examples

Enabling BFD for BGP neighbor with IPv4 address:

```
switch(config-bgp)# neighbor 1.1.1.1 fall-over bfd
```

Enabling BFD for BGP neighbor with IPv6 address (applies only to HPE Aruba Networking 6300 and 6400 Switch series):

```
switch(config-bgp)# neighbor 1000::1 fall-over bfd
```

Enabling BFD for peer group:

```
switch(config-bgp)# neighbor PG fall-over bfd
```

Disabling BFD for BGP per neighbor IPv4 address:

```
switch(config-bgp)# no neighbor 1.1.1.1 fall-over bfd
```

Disabling BFD for BGP per neighbor with IPv6 address (applies only to HPE Aruba Networking 6300 and 6400 Switch series):

```
switch(config-bgp)# no neighbor 1000::1 fall-over bfd
```

Disabling BFD for peer group:

```
switch(config-bgp)# no neighbor PG fall-over bfd
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor graceful-shutdown

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} graceful-shutdown  
[local-preference <LOCAL-PREF>
```

```

| <CONFIG-DELAY>
| <LOCAL-PREF> ]
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} graceful-shutdown
[local-preference <LOCAL-PREF>
| <CONFIG-DELAY>
| <LOCAL-PREF> ]

```

## Description

Configures the wait time before shutting down the BGP neighbor session, and can also configure the local preference value to be advertised before graceful shutdown.

The **no** form of this command sets the wait time to the default value of 180 seconds and the local-preference value to the default of 0.

| Parameter                    | Description                                                                                                                                                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDR>                    | Specifies an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F.                   |
| <IFNAME>                     | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                                              |
| <PEER-GROUP-NAME>            | Specifies a peer group. This is not supported on BGP unnumbered.                                                                                                                                                                                      |
| local-preference<LOCAL-PREF> | Specifies the local preference value for exporting the iBGP routes during a graceful shutdown. The lower the value, the lower the local preference. A value of 0 indicates that the route is the least preferred. Range: 0 to 4294967295. Default: 0. |
| <CONFIG-DELAY>               | Specifies the time to wait before shutting down the neighbor in seconds. Range: 10 to 1200. Default: 180.                                                                                                                                             |

## Usage

If the graceful shutdown timer has already started and the administrator configures a command that triggers a session restart, traffic loss can occur if the graceful shutdown delay is not sufficient for the BGP peers to converge to a new route.



On each Autonomous System Boundary Router (ASBR) supporting the graceful shutdown receiver procedure, an inbound BGP route policy must be applied on all EBGP sessions of the ASBR.

The policy must match the GSHUT community and lower the precedence of the route by changing the route attributes.

The Graceful-Shutdown feature does not work for reflected routes because the route reflector (RR) does not modify local-preference attribute. The routes, originated by the RR, carry the GSHUT local-preference value. As per the RFC 4456, when an RR reflects a route, it should not modify the following path attributes:

- NEXT-HOP
- AS-PATH
- LOCAL-PREF
- MED

Their modification could potentially result in routing loops. In this situation, apply on the RR an inbound BGP route policy, meeting the following conditions:

- Match the graceful-shutdown community.
- Set the local preference attributes of the paths tagged with the graceful-shutdown community to a lower value than other routes to the same destination.

## Examples

Setting the wait time delay:

```
switch(config-bgp) # neighbor 1.1.1.1 graceful-shutdown 10
```

Setting the local-preference value:

```
switch(config-bgp) # neighbor 1.1.1.1 graceful-shutdown local-preference 100
```

Setting the wait time delay and local-preference value:

```
switch(config-bgp) # neighbor 1.1.1.1 graceful-shutdown 10 local-preference 100
```

Setting the wait time delay to the default of 180 seconds:

```
switch(config-bgp) # no neighbor 1.1.1.1 graceful-shutdown 10
```

Setting the local-preference value to default of 0:

```
switch(config-bgp) # no neighbor 1.1.1.1 graceful-shutdown local-preference 100
```

Setting the wait time delay and local-preference value to defaults:

```
switch(config-bgp) # no neighbor 1.1.1.1 graceful-shutdown 10 local-preference 100
```

Complete deletion:

```
switch(config-bgp) # no neighbor 1.1.1.1 graceful-shutdown
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor interface

```
neighbor {<IP-ADDR>|<IFNAME>} interface <ifname>  
[no] neighbor {<IP-ADDR>|<IFNAME>} interface <ifname>
```

### Description

Configure a BGP neighbor by specifying the interface name on which the neighbor is connected instead of the neighbor's IP address. When an unnumbered interface is specified to the **neighbor** command, a static route is auto-provisioned for the neighbor with the unnumbered interface as a destination. A static route is also auto-provisioned if the neighbor IP address is learned via LLDP.



If OSPF is enabled on an unnumbered interface then the auto-provisioning is not supported as the route to the BGP peer is expected to be learned via OSPF.

Peer-group is not supported for BGP peering on the IP unnumbered interface.

The **no** form of this command disables this feature.

| Parameter | Description              |
|-----------|--------------------------|
| <IP-ADDR> | Specifies an IP address. |

### Examples

Configuring unnumbered BGP peering on interface 1/1/1:

```
switch(config)# router bgp 100  
switch(config-bgp)# neighbor 1.1.1.1 remote-as 100  
switch(config-bgp)# neighbor 1.1.1.1 interface 1/1/1
```

Disabling this feature:

```
switch(config)# router bgp 100  
switch(config-bgp)# no neighbor 1.1.1.1 interface 1/1/1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release    | Modification       |
|------------|--------------------|
| 10.14.1000 | Command introduced |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor listen ip-range

```
neighbor <PEER-GROUP-NAME> listen ip-range <IP-ADDR>|<MASK> {as-range <AS-RANGE>|limit <LIMIT>}
no neighbor <PEER-GROUP-NAME> listen ip-range <IP-ADDR>|<MASK> {as-range <AS-RANGE>|limit <LIMIT>}
```

### Description

Configures BGP dynamic neighbors as ranges of remote addresses with associated peer groups. The **no** form of this command restores default behavior.



- Each range of remote addresses is configured as a remote address prefix.
- Any BGP peer with a remote address that matches the remote address prefix becomes a member of the associated peer group.

| Parameter         | Description                                                    |
|-------------------|----------------------------------------------------------------|
| <PEER-GROUP-NAME> | Specifies peer group. This is not supported on BGP unnumbered. |
| <IP-ADDR> <MASK>  | Specifies subnet range.                                        |
| <AS-RANGE>        | Specifies AS number as a range in integer or dotted format.    |
| <LIMIT>           | Specifies maximum number of peers.                             |

### Restrictions

- Dynamic peers are always passive. Outbound connections to dynamic peers are not supported.
- Dynamic BGP peering is only compatible with peer-groups
- Disabling partial AS range is not supported. The exact value that is configured must be used.
  - When disabling AS range, CLI must use the same AS range that was used when first configured. For example, if AS range "1-4" is configured, when disabling, "1-4" must be used ("1,2,3,4" is not supported).



Configuring overlapping peer ranges with different remote address prefix lengths is not recommended. Peer range configuration is recommended when peer ranges do not overlap.

### Usage

- All supported address-families are activated on a dynamic peer for negotiation by default.
- If an incoming connection matches multiple peer range entries, the entry with the longest remote address prefix is selected.

- AS ranges are used to match remote AS presented by connecting peers. Remote AS matching with ASes or AS ranges in this list will be accepted.
  - AS range only applies to dynamic peers.
- The limit option is used to set the maximum number of dynamic BGP peers within the peer range. The default is 512 if no limit is set.
  - If the limit is reached, BGP rejects incoming connections from new dynamic BGP peers until BGP session termination causes the number of dynamic BGP peers to fall below the limit.
  - If the limit is reduced below the current number of dynamic BGP peers, BGP will reject incoming connections from new dynamic BGP peers until the number of dynamic BGP peers falls below the new limit. BGP will not terminate existing BGP sessions with dynamic BGP peers in this case. If an existing BGP session gets terminated, that session will not re-establish until the number of BGP sessions falls below the limit.
- After dynamic peer is configured, additional configuration is required on the peer-group as a whole. Individual member groups are incompatible. For example, the neighbor shutdown command can be executed on a peer-group, but not on individual members of the peer-group.
- When a peer is configured as dynamic and is in an established state, a shutdown is required before reconfiguring as static.
- Connect-retry interval is recommended to be configured with a smaller value than the default value on the active peer.
- When a set of valid and invalid AS values are issued (separated by commas), only the valid values are accepted.
- When the AS range parameter is not explicitly configured in dynamic bgp peering, iBGP session comes up, eBGP session does not. If there are no configured remote AS or AS list entries, DC-BGP assumes that any peer is an iBGP peer.

## Examples

```
switch(config-bgp)# neighbor pg listen ip-range 192.168.0.0/16
```

```
switch(config-bgp)# no neighbor pg listen ip-range 192.168.0.0/16
```

## Command History

| Release | Modification       |
|---------|--------------------|
| 10.11   | Command introduced |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor local-as

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} local-as <AS-NUMBER> [no-prepend[replace-as[dual-as]]]
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} local-as [no-prepend[replace-as[dual-as]]]
```

## Description

Configures an alternate local AS number that can be used to establish a session with a peer, allowing a router to appear to be a member of a second autonomous system (AS), and its real AS.

Local AS allows two autonomous systems to merge without modifying peering arrangements. This command is valid only for external peers.

The **no** form of this command restores the default, which is for a peering session to be established using the primary AS (primary AS is the AS number specified at the time of neighbor creation using the command `neighbor remote-as`).

| Parameter                               | Description                                                                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;IP-ADDR&gt;</code>            | Specifies an IP address.                                                                                                                              |
| <code>&lt;IFNAME&gt;</code>             | Specifies the unnumbered interface name of the neighbor.                                                                                              |
| <code>&lt;PEER-GROUP-NAME&gt;</code>    | Specifies a peer group. This is not supported on BGP unnumbered.                                                                                      |
| <code>local-as &lt;AS-NUMBER&gt;</code> | Specifies a 4-byte AS number in asplain format (z), or asdot format (x.y), where z is a number from 1 to 4294967295 and x and y are 16-bit numbers.   |
| <code>no-prepend</code>                 | Optional. When you include this parameter, the switch will not prepend local ASN value to the AS path list for all the routes received from the peer. |
| <code>replace-as</code>                 | Optional. Replaces the configured router AS with the local AS in the EBGp updates.                                                                    |
| <code>dual-as</code>                    | Optional. Use this option to form the neighborhood with any of the two AS numbers; the configured router ASN or the local ASN.                        |

## Examples

Configuring an alternate local AS number that can be used to establish a session with a peer:

```
switch(config-bgp)# neighbor 1.1.1.1 local-as 200
```

Removing the alternate local AS number that can be used to establish a session with a peer:

```
switch(config-bgp)# no neighbor 1.1.1.1 local-as 200
```

Configuring an alternate local AS number that can be used to establish a session with a peer:

```
switch(config-bgp)# neighbor pg local-as 200 no-prepend
```

Removing the alternate local AS number that can be used to establish a session with a peer:

```
switch(config-bgp)# no neighbor pg local-as 200 no-prepend
```

Configuring an alternate local AS number that can be used to establish a session with a peer:



```
switch(config-bgp)# neighbor pg local-as 200 no-prepend replace-as
```

Removing the alternate local AS number that can be used to establish a session with a peer:

```
switch(config-bgp)# no neighbor pg local-as 200 no-prepend replace-as
```

Configuring an alternate local AS number that can be used to establish a session with a peer:

```
switch(config-bgp)# neighbor pg local-as 200 no-prepend replace-as dual-as
```

Removing the alternate local AS number that can be used to establish a session with a peer:

```
switch(config-bgp)# no neighbor pg local-as 200 no-prepend replace-as dual-as
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor maximum-prefix

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>}  
    maximum-prefix <MAXIMUM> [threshold <THRESHOLD>]  
    [restart <INTERVAL>] [warning-only]  
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} maximum-prefix<MAXIMUM> [threshold  
<THRESHOLD>]  
    [restart <INTERVAL>] [warning-only]
```

## Description

Sets the maximum number of prefixes that can be received from a neighbor.

By default, the device accepts 128,000 prefixes from a BGP neighbor with a threshold value of 75%. A warning message is generated when the number of prefixes per neighbor reaches 75% of default prefix limit. Another warning message is generated when the default prefix limit is reached.

The session is re-established only if the number of routes received from the BGP peer does not exceed the configured prefix limit. When the restart timer is configured, sessions are automatically re-established when the timer expires.

The **no** form of this command disables the maximum number of prefixes limit.

| Parameter                                   | Description                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;IP-ADDR&gt;</code>                | Specifies the IP address of the neighbor in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F.                                                                       |
| <code>&lt;IFNAME&gt;</code>                 | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                                                                                                                   |
| <code>&lt;PEER-GROUP-NAME&gt;</code>        | Specifies a Peer-Group. This is not supported on BGP unnumbered.                                                                                                                                                                                                                                                           |
| <code>maximum-prefix &lt;MAXIMUM&gt;</code> | Specifies the maximum number of prefixes allowed from the specified neighbor. Range: 1 to 128000. Default: 128000.                                                                                                                                                                                                         |
| <code>threshold &lt;THRESHOLD&gt;</code>    | Specifies at what percentage of <b>MAXIMUM</b> a warning message is generated. Range: 1 to 100. Default: 75.<br>For example, if <b>MAXIMUM</b> is set to 1000 and threshold is 70, the router generates a warning message when the number of BGP learned routes from the neighbor exceeds 70 percent of 1000 (700) routes. |
| <code>restart &lt;INTERVAL&gt;</code>       | Specifies interval in seconds for restarting the BGP connection when the prefix limit is exceeded. Range: 30 to 65535.                                                                                                                                                                                                     |
| <code>warning-only</code>                   | Specifies generating and logging a warning message without disconnecting the BGP session when the prefix limit is exceeded.                                                                                                                                                                                                |

## Examples

Setting the prefix limit to 1000 prefixes:

```
switch(config-bgp-ipv4-uc) # neighbor 10.0.0.1 maximum-prefix 1000
```

Enabling logging of a warning message when more than 1000 prefixes are received:

```
switch(config-bgp-ipv4-uc) # neighbor 10.0.0.1 maximum-prefix 1000 warning-only
```

Setting the prefix limit to 1000 prefixes and enabling logging of a warning message when 500 prefixes are received:

```
switch(config-bgp-ipv4-uc) # neighbor 10.0.0.1 maximum-prefix 1000 threshold 50
```

Setting the prefix limit to 1000 prefixes and enabling logging of a warning message when 500 prefixes are received and a second warning when the prefix limit is exceeded without disconnecting the session:

```
switch(config-bgp-ipv4-uc) # neighbor 10.0.0.1 maximum-prefix 1000 threshold 50  
warning-only
```

Removing the threshold value:

```
switch(config-bgp-ipv4-uc) # no neighbor 10.0.0.1 maximum-prefix 1000 threshold 50
```

Disabling the maximum-prefix feature:

```
switch(config-bgp-ipv4-uc) # no neighbor 10.0.0.1 maximum-prefix
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## neighbor next-hop-self

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>}next-hop-self  
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} next-hop-self
```

## Description

Configures the router as the next hop for a BGP-speaking neighbor or peer group, and enables BGP to send itself as the next hop for advertised routes.

The **no** form of this command resets the peer next-hop-self status to default. The next hop is generated based on the IP.

| Parameter         | Description                                                                                                                                                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDR>         | Specifies the neighbor's IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F. |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                                        |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered.                                                                                                                                                                                |
| all               | Applies the configuration to all route-reflector clients.                                                                                                                                                                                       |

## Usage

- An administrator uses this command to make a BGP speaker fill its address when advertising routes to a BGP peer.
- This command is useful in non-meshed networks where BGP neighbors may not have direct access to all other neighbors on the same IP subnet.
- While advertising routes to eBGP peers, the next-hop is set to self IP by default. The default behavior can be changed by configuring next-hop-unchanged.
- While advertising routes to iBGP peers, the next-hop is kept unchanged by default. The default behavior can be changed by configuring next-hop-self.

## Examples

Setting and resetting the router as the next hop self for neighbor 1.1.1.1:

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 next-hop-self
switch(config-bgp-ipv4-uc) # no neighbor 1.1.1.1 next-hop-self
```

Setting and resetting the router as the next hop self for its peer group:

```
switch(config-bgp-ipv4-uc) # neighbor pg next-hop-self
switch(config-bgp-ipv4-uc) # no neighbor pg next-hop-self
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## neighbor next-hop-unchanged

```
neighbor {<IP-ADDRESS>|<IFNAME>} next-hop-unchanged
no neighbor {<IP-ADDRESS>|<IFNAME>} next-hop-unchanged
```

## Description

Enables the neighbor to preserve next-hop while advertising routes to eBGP peers, in the L2VPN EVPN address-family.

The **no** form of this command resets the peer next-hop-unchanged status to default.

| Parameter                       | Description                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;IP-ADDRESS&gt;</code> | Specifies the neighbor IP address in the IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or in the IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F.<br><br><b>NOTE:</b> IPv6 MP-BGP peering must not be used for L2VPN EVPN address family, because VXLAN tunnel interface does not support IPv6 addresses. |
| <code>&lt;IFNAME&gt;</code>     | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                                                                                                                                                                                                         |

## Examples

```
switch(config-bgp-l2vpn-evpn) # neighbor 1.1.1.1 next-hop-unchanged
```

```
switch(config-bgp-l2vpn-evpn) # neighbor 2001:0db8:85a3::8a2e:0370:7334 next-hop-unchanged
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context       | Authority                                                                          |
|-----------|-----------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-l2vpn-evpn | Administrators or local user group members with execution rights for this command. |

## neighbor orf prefix-list in

```
neighbor {<IP-ADDRESS>|<IFNAME>} orf-prefix-list <PREFIX-LIST-NAME> in
no neighbor {<IP-ADDRESS>|<IFNAME>} orf-prefix-list <PREFIX-LIST-NAME> in
```

## Description

Applies an inbound prefix list filter to filter the distribution of BGP neighbor information. The **no** form of this command restores the default behavior of not applying the prefix list filter.



This command must be used only along with the ORF capability to take effect.

| Parameter                                 | Description                                              |
|-------------------------------------------|----------------------------------------------------------|
| <IP-ADDRESS>                              | Specifies an IP address.                                 |
| <IFNAME>                                  | Specifies the unnumbered interface name of the neighbor. |
| orf-prefix-list <i>PREFIX-LIST-NAME</i> > | Sends the prefix list name to be filtered.               |

## Usage

To use this command, the following conditions must be met:

- If route-map inbound is also applied on multiple neighbors along with ORF, then the route-map name must be common on all the neighbors.
- If route-map inbound is also applied on an IPv6 AF BGP neighbor, then the route-map sequence number with value 1 cannot be used.

## Examples

Applying the inbound prefix list filter:

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 orf-prefix-list ABC in
```

Removing the inbound prefix list filter:

```
switch(config-bgp-ipv4-uc) # no neighbor 1.1.1.1 orf-prefix-list ABC in
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## neighbor passive

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} passive
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} passive
```

## Description

Configures a specific neighbor, or all neighbors in a peer-group, as passive, which means that they will not initiate the TCP session.

The **no** form of this command enables the neighbor to initiate the TCP session.

| Parameter                            | Description                                              |
|--------------------------------------|----------------------------------------------------------|
| <code>&lt;IP-ADDRESS&gt;</code>      | Specifies an IP address.                                 |
| <code>&lt;IFNAME&gt;</code>          | Specifies the unnumbered interface name of the neighbor. |
| <code>&lt;PEER-GROUP-NAME&gt;</code> | Specifies a peer group.                                  |

## Examples

Configure a specific neighbor as passive:

```
switch(config-bgp) # neighbor 1.1.1.1 passive
```

Enabling the local router to initiate the TCP session for a specified neighbor:

```
switch(config-bgp) # no neighbor 1.1.1.1 passive
```

Configure a specific peer group as passive:

```
switch(config-bgp) # neighbor pg passive
```

Enabling the local router to initiate the TCP session for a specified peer group:

```
switch(config-bgp) # no neighbor pg passive
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor password

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} password [{ciphertext | plaintext}
<PASSWORD>]
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>}
password [ciphertext <PASSWORD>]
```

## Description

Enables message digest5 (MD5) authentication on a TCP connection between two BGP neighbors. When the password is applied to a peer-group, all the neighbors that are part of peer-group inherit the configured setting.

The **no** form of this command removes the neighbor password.

| Parameter                | Description                                                      |
|--------------------------|------------------------------------------------------------------|
| <IP-ADDR>                | Specifies an IP address.                                         |
| <IFNAME>                 | Specifies the unnumbered interface name of the neighbor.         |
| <PEER-GROUP-NAME>        | Specifies a Peer-Group. This is not supported on BGP unnumbered. |
| {ciphertext   plaintext} | Selects the password format.                                     |
| <PASSWORD>               | Specifies the password.                                          |



When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

## Examples

Enabling message digest5 (MD5) authentication for a neighbor with a provided plaintext password:

```
switch(config-bgp)# neighbor 1.1.1.1 password plaintext doubt_Plane#93
```

Enabling message digest5 (MD5) authentication for a neighbor with a prompted plaintext password:

```
switch(config-bgp)# neighbor 1.1.1.5 password
Enter the neighbor password: *****
Re-Enter the neighbor password: *****
```

Enabling message digest5 (MD5) authentication for a peer group with a provided plaintext password:

```
switch(config-bgp)# neighbor pg_3 password plaintext doubt_Plane#93
```

Disabling message digest5 (MD5) authentication for a neighbor:

```
switch(config-bgp)# no neighbor 1.1.1.5 password
```

Disabling message digest5 (MD5) authentication for a peer group:



```
switch(config-bgp) # no neighbor pg_3 password
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor port

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} port <NUMBER>  
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} port <NUMBER>
```

## Description

Sets a custom TCP port on which to communicate with the BGP neighbor.

When this configuration is applied for peer-group, all the neighbors that are part of peer-group will inherit this setting. Though the neighbor inherits the configuration from the peer-group, the neighbor-specific command, if configured, takes precedence.

This setting only takes effect after a hard reset of the session.

The **no** form of this command allows a random TCP port to be selected for the communication with the BGP neighbor.

| Parameter         | Description                                                      |
|-------------------|------------------------------------------------------------------|
| <IP-ADDRESS>      | Specifies an IP address.                                         |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.         |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered. |
| port <NUMBER>     | Specifies a TCP port number. Range: 0 to 65535.                  |

## Examples

```
switch(config-bgp) # neighbor 1.1.1.1 port 1500  
switch(config-bgp) # no neighbor 1.1.1.1 port
```

```
switch(config-bgp) # neighbor PG port 1500
switch(config-bgp) # no neighbor PG port
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor remote-as

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} remote-as <AS-NUMBER>
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} remote-as <AS-NUMBER>
```

## Description

Creates a peer, initiates the connection to the peer, and adds an entry to the BGP neighbor table. Specifies a neighbor with an autonomous system (AS) number that identifies the neighbor as internal to the local autonomous system. Otherwise, the neighbor is considered as external. By default, neighbors that are defined using this command, exchange only unicast address prefixes.

The **no** form of this command disables the peer session and deletes the peer information.

| Parameter             | Description                                                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDR>             | Specifies an IP address.                                                                                                                                                    |
| <IFNAME>              | Specifies the unnumbered interface name of the neighbor.                                                                                                                    |
| <PEER-GROUP-NAME>     | Specifies a peer group. This is not supported on BGP unnumbered.                                                                                                            |
| remote-as <AS-NUMBER> | Specifies a 4-byte AS number in asplain format (z), or asdot format (x.y), where z is a number from 1 to 4294967295 and x and y are 16-bit numbers in the range 0 to 65535. |

## Usage

The configured peer AS number is compared with the AS number received in the open message and a peer session is initiated only if both the AS numbers match.

## Examples

```
switch(config-bgp) # neighbor 1.1.1.1 remote-as 1
switch(config-bgp) # no neighbor 1.1.1.1 remote-as 1
```

```
switch(config-bgp) # neighbor pg remote-as 1
switch(config-bgp) # no neighbor pg remote-as 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor remove-private-AS

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} remove-private-AS
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} remove-private-AS
```

## Description

This command forces the BGP neighbor to drop the private AS numbers. When the outbound update contains a sequence of private AS numbers, this sequence is dropped. If the command is configured for peer-group, then all the neighbors that are part of peer-group will remove the private-AS before sending the BGP update message.

The **no** form of this command allows the private-AS number to be carried in BGP update message.

| Parameter         | Description                                                      |
|-------------------|------------------------------------------------------------------|
| <IP-ADDRESS>      | Specifies an IP address.                                         |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.         |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered. |

## Examples

Forcing the BGP neighbor to drop the private AS numbers:

```
switch(config-bgp) # neighbor 1.1.1.1 remove-private-AS
```

Allowing the private AS number to be carried in the BGP update message:

```
switch(config-bgp)# no neighbor 1.1.1.1 remove-private-AS
```

Forcing the BGP neighbor to drop the private AS numbers:

```
switch(config-bgp)# neighbor PG remove-private-AS
```

Allowing the private AS number to be carried in the BGP update message:

```
switch(config-bgp)# no neighbor PG remove-private-AS
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor remote-as

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} remote-as <AS-NUMBER>  
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} remote-as <AS-NUMBER>
```

## Description

Creates a peer, initiates the connection to the peer, and adds an entry to the BGP neighbor table. Specifies a neighbor with an autonomous system (AS) number that identifies the neighbor as internal to the local autonomous system. Otherwise, the neighbor is considered as external. By default, neighbors that are defined using this command, exchange only unicast address prefixes.

The **no** form of this command disables the peer session and deletes the peer information.

| Parameter         | Description                                              |
|-------------------|----------------------------------------------------------|
| <IP-ADDR>         | Specifies an IP address.                                 |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor. |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on         |

| Parameter                                | Description                                                                                                                                                                 |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | BGP unnumbered.                                                                                                                                                             |
| <code>remote-as &lt;AS-NUMBER&gt;</code> | Specifies a 4-byte AS number in asplain format (z), or asdot format (x.y), where z is a number from 1 to 4294967295 and x and y are 16-bit numbers in the range 0 to 65535. |

## Usage

The configured peer AS number is compared with the AS number received in the open message and a peer session is initiated only if both the AS numbers match.

## Examples

```
switch(config-bgp)# neighbor 1.1.1.1 remote-as 1
switch(config-bgp)# no neighbor 1.1.1.1 remote-as 1
```

```
switch(config-bgp)# neighbor pg remote-as 1
switch(config-bgp)# no neighbor pg remote-as 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor route-map

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} route-map <MAP-NAME> {in|out}
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} route-map <MAP-NAME> {in|out}
```

## Description

This command applies a route map to incoming or outgoing routes. It configures the route map for modifying the default attributes of the route.



When both peer group and neighbor configuration have route maps associated, then the following configuration applies:

- For outbound route maps, peer group configuration will override the configuration of the neighbor.
- For inbound route maps, neighbor configuration will override the peer group configuration.

The **no** form of this command removes a route map.

| Parameter         | Description                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDRESS>      | Specifies an IP address.                                                                                                                |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.                                                                                |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered.                                                                        |
| <MAP-NAME>        | Specifies the name of the route map.                                                                                                    |
| in out            | Sets the route map policy to apply to either the received routes from the neighbor (in) or the advertised routes to the neighbor (out). |

## Examples

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 route-map HPE in
switch(config-bgp-ipv4-uc) # no neighbor 1.1.1.1 route-map HPE in
```

```
switch(config-bgp-ipv4-uc) # neighbor PG route-map HPE in
switch(config-bgp-ipv4-uc) # no neighbor PG route-map HPE in
```

```
switch(config) # route-map Rmap permit seq 10
switch(config-route-map-Rmap-10) # match metric 100
switch(config-route-map-bgp-10) # router bgp 100
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 remote-as 100
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 route-map Rmap out
```

Configuring inbound route maps in L2VPN EVPN address family.

```
switch(config) # router bgp 100
switch(config-bgp) # neighbor 2.1.1.1 remote-as 100
switch(config-bgp) # neighbor 2.1.1.1 update-source loopback 1
switch(config-bgp) # address-family l2vpn evpn
switch(config-bgp-l2vpn-evpn) # neighbor 2.1.1.1 activate
switch(config-bgp-l2vpn-evpn) # neighbor 2.1.1.1 route-map Rmap in
switch(config-bgp-l2vpn-evpn) # neighbor 2.1.1.1 send-community extended
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                                                   | Authority                                                                          |
|-----------|-------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc<br>config-bgp-l2vpn-evpn | Administrators or local user group members with execution rights for this command. |

## neighbor route-reflector-client

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} route-reflector-client
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} route-reflector-client
```

## Description

This command configures the router as a BGP route reflector and the specified peer as its client. The **no** form of this command disables this function.

| Parameter                      | Description                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>&lt;IP-ADDRESS&gt;</i>      | Specifies the neighbor IP address in the IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or in the IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F.<br><br><b>NOTE:</b> Pv6 MP-BGP peering must not be used for L2VPN EVPN address family, because VXLAN tunnel interface does not support IPv6 addresses. |
| <i>&lt;IFNAME&gt;</i>          | Specifies the unnumbered interface name of the neighbor.                                                                                                                                                                                                                                                                                                                                                        |
| <i>&lt;PEER-GROUP-NAME&gt;</i> | Specifies a peer group. This is not supported on BGP unnumbered.                                                                                                                                                                                                                                                                                                                                                |

## Examples

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 route-reflector-client
switch(config-bgp-ipv4-uc) # no neighbor 1.1.1.1 route-reflector-client
```

```
switch(config-bgp-ipv4-uc) # neighbor PG route-reflector-client
switch(config-bgp-ipv4-uc) # no neighbor PG route-reflector-client
```

```
switch(config-bgp-l2vpn-evpn) # neighbor 1.1.1.1 route-reflector-client
switch(config-bgp-l2vpn-evpn) # no neighbor 1.1.1.1 route-reflector-client
```

```
switch(config-bgp-l2vpn-evpn) # neighbor PG route-reflector-client
switch(config-bgp-l2vpn-evpn) # no neighbor PG route-reflector-client
```

```
switch(config-bgp-l2vpn-evpn) # neighbor 2001:0db8:85a3::8a2e:0370:7334 route-
reflector-client
switch(config-bgp-l2vpn-evpn) # no neighbor 2001:0db8:85a3::8a2e:0370:7334 route-
reflector-client
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                                                   | Authority                                                                          |
|-----------|-------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc<br>config-bgp-l2vpn-evpn | Administrators or local user group members with execution rights for this command. |

## neighbor send-community

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} send-community [standard | extended]
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} send-community [standard | extended]
```

## Description

This command allows community values to be sent to a specific neighbor. When this command is configured for the peer-group, then all the neighbors that are part of peer-group will send the community values to the peers. The parameters `standard` and `extended` send only the respective community numbers. When the command is issued without either of these parameters, both standard and extended communities will be sent to the neighbor.

The **no** form of this command will not allow the neighbor to send community values to the specific neighbors that are part of peer-group.

| Parameter | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDR> | Specifies the neighbor IP address in the IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255, or in the IPv6 format ( <b>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</b> ), where <b>x</b> is a hexadecimal number from 0 to F.<br><br><b>NOTE:</b> IPv6 MP-BGP peering must not be used for L2VPN EVPN address family, because VXLAN tunnel interface does not support |



| Parameter         | Description                                                      |
|-------------------|------------------------------------------------------------------|
|                   | IPv6 addresses.                                                  |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.         |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered. |

## Examples

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 send-community standard
switch(config-bgp-ipv4-uc) # no neighbor 1.1.1.1 send-community standard
```

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 send-community extended
switch(config-bgp-ipv4-uc) # no neighbor 1.1.1.1 send-community
```

```
switch(config-bgp-ipv4-uc) # neighbor PG send-community standard
switch(config-bgp-ipv4-uc) # no neighbor PG send-community standard
```

```
switch(config-bgp-ipv4-uc) # neighbor PG send-community extended
switch(config-bgp-ipv4-uc) # no neighbor PG send-community
```

```
switch(config-bgp-l2vpn-evpn) # neighbor 1.1.1.1 send-community standard
switch(config-bgp-l2vpn-evpn) # no neighbor 1.1.1.1 send-community standard
```

```
switch(config-bgp-l2vpn-evpn) # neighbor 1.1.1.1 send-community extended
switch(config-bgp-l2vpn-evpn) # no neighbor 1.1.1.1 send-community
```

```
switch(config-bgp-l2vpn-evpn) # neighbor PG send-community standard
switch(config-bgp-l2vpn-evpn) # no neighbor PG send-community standard
```

```
switch(config-bgp-l2vpn-evpn) # neighbor PG send-community extended
switch(config-bgp-l2vpn-evpn) # no neighbor PG send-community
```

```
switch(config-bgp-l2vpn-evpn) # neighbor 2001:0db8:85a3::8a2e:0370:7334 send-
community extended
switch(config-bgp-l2vpn-evpn) # no neighbor 2001:0db8:85a3::8a2e:0370:7334 send-
community
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                                                   | Authority                                                                          |
|-----------|-------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc<br>config-bgp-l2vpn-evpn | Administrators or local user group members with execution rights for this command. |

## neighbor shutdown

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} shutdown
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} shutdown
```

### Description

This command disables the peer session, terminates any active session for the specified neighbor or peer group, and removes all associated routing information. This action can cause the sudden termination of many peering sessions.

The **no** form of this command enables the peer session for the specified neighbor.

| Parameter         | Description                                                      |
|-------------------|------------------------------------------------------------------|
| <IP-ADDRESS>      | Specifies an IP address.                                         |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.         |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered. |

### Usage

Sessions are gracefully shut down when graceful-shutdown is enabled. Enter the `neighbor graceful-shutdown` command to enable graceful-shutdown. If graceful-shutdown is configured without delay or local-preference, the default values are used.

### Examples

Disable the peer session, terminate any active session for the specified neighbor or peer group, and removing all associated routing information. Then enable the peer session for the specified neighbor:

```
switch(config-bgp)# neighbor 1.1.1.1 shutdown
switch(config-bgp)# no neighbor 1.1.1.1 shutdown
```

Disable the peer session, terminate any active session for the specified neighbor or peer group, and removing all associated routing information. Then enable the peer session for the specified neighbor:

```
switch(config-bgp)# neighbor pg shutdown
switch(config-bgp)# no neighbor pg shutdown
```

Establish a peer graceful shutdown:

```
switch(config-bgp)# neighbor 1.1.1.1 graceful-shutdown 10
switch(config-bgp)# neighbor 1.1.1.1 shutdown
Peer 1.1.1.1 will be shutdown in 10 seconds. Do not modify BGP configurations
until the shutdown process completes.
```

Enter an unsuccessful peer session because the gshut timer is running:

```
switch(config-bgp)# no neighbor 1.1.1.1 shutdown
Peer 1.1.1.1 is being gracefully shutdown. Peer cannot be brought up until the
shutdown process completes.
```

Enter an successful peer session because the gshut timer is running:

```
switch(config-bgp)# no neighbor pg shutdown
```

Reset the timer to the default value:

```
switch(config-bgp)# no neighbor pg graceful-shutdown 10
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                        |
|------------------|-----------------------------------------------------|
| 10.16            | Added support for BGP unnumbered.                   |
| 10.16            | Added graceful shutdown examples and configuration. |
| 10.07 or earlier | --                                                  |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor slow-peer

```
neighbor {<IP-ADDR>|<IPv6-ADDR>|<IFNAME>} slow-peer
no neighbor {<IPv4-ADDR>|<IPv6-ADDR>|<IFNAME>} slow-peer
```

## Description

Slow peer configuration moves the peer from its normal update group to a slow update group. Removing slow peers allows the normal update group to function without being slowed down and enables faster convergence. Slow peer configuration is applicable only to explicitly configured BGP neighbors. All configured slow peers will be grouped into a slow update group. This group will converge based on the slowest group peer. Dynamic detection and splitting isn't supported. Peers can be statically configured as slow.

The **no** form of this command removes the slow peer configuration.

| Parameter                      | Description                        |
|--------------------------------|------------------------------------|
| <code>&lt;IP-ADDR&gt;</code>   | Specifies a neighbor IP address.   |
| <code>&lt;IPv6-ADDR&gt;</code> | Specifies a neighbor IPv6 address. |

## Examples

Configuring a neighbor with an IP address of **1.1.1.1** as a slow peer:

```
switch(config-bgp) # neighbor 1.1.1.1 slow-peer
```

Removing the slow peer configuration for a neighbor with an IP address of **1.1.1.1**:

```
switch(config-bgp) # no neighbor 1.1.1.1 slow-peer
```

Configuring a neighbor with an IPv6 address of **2001::2** as a slow peer:

```
switch(config-bgp) # neighbor 2001::2 slow-peer
```

Removing the slow peer configuration for a neighbor with an IPv6 address of **2001::2**:

```
switch(config-bgp) # no neighbor 2001::2 slow-peer
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.13   | Command introduced. |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor soft-reconfiguration inbound

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} soft-reconfiguration inbound
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} soft-reconfiguration inbound
```

## Description

This command enables you to generate inbound updates from a neighbor and change and activate BGP policies without clearing the BGP session. Changes in BGP policies require the BGP session to be cleared which can have a large negative impact on network operations.

The **no** form of this command disables this setting.

| Parameter         | Description                                                      |
|-------------------|------------------------------------------------------------------|
| <IP-ADDRESS>      | Specifies an IP address.                                         |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.         |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered. |

## Usage

- To perform inbound soft reconfiguration, the BGP speaker must store all received route updates, regardless of the current inbound policy.
- When inbound soft reconfiguration is enabled, the stored updates are processed by the new policy configuration to create new inbound updates.

## Examples

```
switch(config-bgp-ipv4-uc) # neighbor 1.1.1.1 soft-reconfiguration inbound
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## neighbor timers

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} timers <KEEPALIVE> <HOLDTIME>
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} timers <KEEPALIVE> <HOLDTIME>
```

## Description

This command sets the timers for a specific BGP neighbor or peer group. When the timer is applied to peer-group then all the neighbors that are part of peer-group will inherit the value configured.

The **no** form of this command clears the timers for a specific BGP neighbor or peer group.

| Parameter                            | Description                                                                                 |
|--------------------------------------|---------------------------------------------------------------------------------------------|
| <code>&lt;IP-ADDRESS&gt;</code>      | Specifies an IP address.                                                                    |
| <code>&lt;IFNAME&gt;</code>          | Specifies the unnumbered interface name of the neighbor.                                    |
| <code>&lt;PEER-GROUP-NAME&gt;</code> | Specifies a peer group. This is not supported on BGP unnumbered.                            |
| <code>&lt;KEEPALIVE&gt;</code>       | Specifies the Keep-Alive timer value for the neighbor. Default: 60 seconds. Range: 0-65535. |
| <code>&lt;HOLDTIME&gt;</code>        | Specifies the Hold-timer value. Default: 180 seconds. Range: 0-65535.                       |

## Examples

```
switch(config-bgp) # neighbor 1.1.1.1 timers 120 360
switch(config-bgp) # no neighbor 1.1.1.1 timers 120 360
```

```
switch(config-bgp) # neighbor pg timers 120 360
switch(config-bgp) # no neighbor pg timers 120 360
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor timers connect-retry

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} timers connect-retry <RETRYTIME>
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} timers connect-retry <RETRYTIME>
```

## Description

This command sets the connect-retry timer for a specific BGP neighbor or peer group.  
The **no** form of this command clears the connect-retry timers for a specific BGP neighbor or peer group.

| Parameter         | Description                                                      |
|-------------------|------------------------------------------------------------------|
| <IP-ADDR>         | Specifies an IP address.                                         |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.         |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered. |
| <RETRYTIME>       | Specifies the retry timer value.                                 |

## Usage

When the timer is applied to peer-group, then all the neighbors that are part of peer-group inherit the value configured. A connect-retry timer configured for a specific peer takes precedence over that configured for the peer group of this peer. In addition, a connect-retry timer configured for a specific peer or peer group takes precedence over that configured on the global level.

## Examples

Setting the connect-retry timer to 50:

```
switch(config-bgp)# neighbor 1.1.1.1 timers connect-retry 50
```

Removing the connect-retry timer:

```
switch(config-bgp)# no neighbor 1.1.1.1 timers connect-retry 50
```

Setting the connect-retry timer to 80:

```
switch(config-bgp)# neighbor pg timers connect-retry 80
```

Removing the connect-retry timer:

```
switch(config-bgp)# no neighbor pg timers connect-retry 80
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.16   | Command introduced. |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor ttl-security-hops

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} ttl-security-hops <HOP-COUNT>
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} ttl-security-hops <HOP-COUNT>
```

### Description

This command enables BGP to establish connection with external peers residing on networks that are not directly connected. By enabling this feature, the received TTL from a BGP peer is compared with the difference "255 - hop-count". BGP messages coming with a TTL less than this value are not accepted. BGP peering will not be established if the TTL in the session establishment is received with a lower value. Also, by enabling this feature the router will send BGP packets with TTL value of 255 to the neighbor. For a neighbor, either TTL security or `ebgp-multihop` can be configured, not both together. If there are multiple paths to reach the node, then the hop count should be configured considering the longest route.

The **no** form of this command disables the peer ttl-security-hop feature.

| Parameter         | Description                                                                       |
|-------------------|-----------------------------------------------------------------------------------|
| <IP-ADDR>         | Specifies an IP address.                                                          |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered.                  |
| <HOP-COUNT>       | Specifies the hop count to reach the neighbor for the eBGP session. Range: 1-255. |

### Examples

```
switch(config-bgp)# neighbor 1.1.1.1 ttl-security-hops 10
switch(config-bgp)# no neighbor 1.1.1.1 ttl-security-hops
```

```
switch(config-bgp)# neighbor pg ttl-security-hops 5
switch(config-bgp)# no neighbor pg ttl-security-hops
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information



| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor update-source

```
neighbor {<IP-ADDRESS>|<PEER-GROUP-NAME>}
    update-source {<IPv4>|<IPv6> | loopback <NUMBER>}
no neighbor {<IP-ADDRESS>|<PEER-GROUP-NAME>}
    update-source [<IPv4>|<IPv6> | loopback <NUMBER>]
```

### Description

This command specifies the source address to reach the neighbor.

An iBGP connection can occur as long as there is a TCP/IP path between the routers. If multiple paths exist between the iBGP routers, using a loopback interface as the neighbor address can add stability to the network. With this command, stability can be achieved by providing the loopback interface address as the source address of the TCP/IP session.

The **no** form of this command negates the route updates of the neighbor.

| Parameter         | Description                             |
|-------------------|-----------------------------------------|
| <IP-ADDRESS>      | Specifies an IP address.                |
| <PEER-GROUP-NAME> | Specifies a peer group.                 |
| <IPv4>            | Specifies an interface by IPv4 address. |
| <IPv6>            | Specifies an interface by IPv6 address. |
| loopback <NUMBER> | Specifies a loopback interface number.  |

### Examples

```
switch(config-bgp)# neighbor 1.1.1.1 update-source loopback 1
switch(config-bgp)# no neighbor 1.1.1.1 update-source
```

```
switch(config-bgp)# neighbor PG update-source loopback 1
switch(config-bgp)# no neighbor PG update-source
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## neighbor weight

```
neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} weight <WEIGHT-VALUE>
no neighbor {<IP-ADDR>|<IFNAME>|<PEER-GROUP-NAME>} weight <WEIGHT-VALUE>
```

### Description

This command assigns a weight to a neighbor connection. When the weight is applied to a peer-group then all the neighbors that are part of the peer-group will inherit the value configured.

The **no** form of this command removes a weight assignment.

| Parameter         | Description                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------|
| <IP-ADDR>         | Specifies an IP address.                                                                         |
| <IFNAME>          | Specifies the unnumbered interface name of the neighbor.                                         |
| <PEER-GROUP-NAME> | Specifies a peer group. This is not supported on BGP unnumbered.                                 |
| <WEIGHT-VALUE>    | Specifies the weigh to be associated with the routes received from the neighbor. Range: 0-65535. |

### Examples

```
switch(config-bgp)# neighbor 1.1.1.1 weight 500
switch(config-bgp)# no neighbor 1.1.1.1 weight
```

```
switch(config-bgp)# neighbor pg weight 600
switch(config-bgp)# no neighbor pg weight
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification                      |
|------------------|-----------------------------------|
| 10.16            | Added support for BGP unnumbered. |
| 10.07 or earlier | --                                |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

# network

```
network { [<IPV4/MASK> | <IPV6/MASK> ] } [route-map <ROUTE-MAP-NAME>]  
no network { [<IPV4/MASK> | <IPV6/MASK> ] } [route-map <ROUTE-MAP-NAME>]
```

## Description

This command specifies the networks to be advertised by the Border Gateway Protocol (BGP) routing processes.

The **no** form of this command removes an entry from the routing table.

| Parameter                  | Description                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| <IPV4/MASK>                | Specifies the IPv4 network with mask. For example: 1.1.1.1/24                                                     |
| <IPV6/MASK>                | Specifies the IPv6 network with mask. For example: 2001:0db8:85a3::8a2e:0370:7334/24                              |
| route-map <ROUTE-MAP-NAME> | Optional parameter. Specifies a route map to apply to the prefixes advertised by this specific network statement. |

## Usage

- This command is used to advertise prefixes currently installed in the routing table into the BGP table.
- Use the `route-map` keyword to apply the specified route map to network advertisements. The mask length as configured in the network statement must match the mask length of prefixes in the routing table.

## Examples

```
switch(config-bgp-ipv4-uc) # network 11.11.11.0/24  
switch(config-bgp-ipv4-uc) # no network 11.11.11.0/24
```

```
switch(config-bgp-ipv6-uc) # network 2001:0db8:85a3::8a2e:0370:7334/24  
switch(config-bgp-ipv6-uc) # no network 2001:0db8:85a3::8a2e:0370:7334/24
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context                          | Authority                                                                          |
|-----------|------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp-ipv4-uc<br>config-bgp-ipv6-uc | Administrators or local user group members with execution rights for this command. |

## redistribute (bgp)

```
redistribute {connected|ospf <process ID> |static|local loopback} [route-map <ROUTE-MAP-NAME>]  
no redistribute {connected|ospf <process ID> |static|local loopback} [route-map <ROUTE-MAP-NAME>]
```

### Description

This command specifies routes to import into BGP. This command causes routes from the specified protocol to be considered for redistribution into BGP.

The **no** form of this command specifies no redistribution into BGP.

| Parameter                  | Description                                                                                                                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| connected                  | Redistributes directly attached networks (directly attached to the subnet or host).                                                                                                                                                                        |
| ospf                       | Redistributes Open Shortest Path First (OSPFv2) routes. It is optional to mention the process ID. Range: <1-65535>                                                                                                                                         |
| static                     | Redistributes statically configured routes .                                                                                                                                                                                                               |
| local loopback             | Performs the following functions: <ul style="list-style-type: none"><li>▪ Redistributes local routes on loopback interfaces.</li><li>▪ For EVPN enabled VRFs, it advertises the IP address of loopback interfaces as a EVPN Type-5 prefix route.</li></ul> |
| route-map <ROUTE-MAP-NAME> | Optional. Specifies a route map to match for redistribution.                                                                                                                                                                                               |

### Usage

- If a route map is specified, then routes that pass the match clause specified in the route map will be imported into the BGP peer Routing Information Base (RIB).
- Route-maps must be configured prior to being referenced in redistribution statements.
- Redistribute connected is required to redistribute connected subnet even if redistribute local loopback is already configured.

### Examples

Redistribute directly attached networks:

```
switch(config-bgp-ipv4-uc) # redistribute connected  
switch(config-bgp-ipv4-uc) # no redistribute connected
```

Redistributing local routes on loopback interfaces:

```
switch(config-bgp-ipv4-uc) # redistribute local loopback  
switch(config-bgp-ipv4-uc) # no redistribute local loopback
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.12            | Added support for the <code>host-routes</code> parameter. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms | Command context                                                    | Authority                                                                          |
|-----------|--------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420      | <code>config-bgp-ipv4-uc</code><br><code>config-bgp-ipv6-uc</code> | Administrators or local user group members with execution rights for this command. |

## router bgp

```
router bgp <AS-NUMBER>
no router bgp <AS-NUMBER>
```

### Description

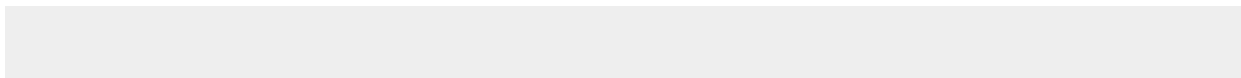
This command configures the BGP instance on the router, configures the AS (Autonomous System) the router belongs to, and enters into the BGP router configuration mode. Only a single BGP AS number can be assigned for the entire system.

The **no** form of the command deletes the BGP instance from the router.

| Parameter        | Description                                                                                                        |
|------------------|--------------------------------------------------------------------------------------------------------------------|
| <i>AS-NUMBER</i> | Specifies a 4-byte AS number in the range 1-4294967295 in integer format or from 0.1-65535.65535 in dotted format. |

### Examples

Configuring the BGP instance with the AS number:



Deleting BGP configurations:

```
switch(config)# no router bgp 100
This will delete all BGP configurations on this device.
Continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config          | Administrators or local user group members with execution rights for this command. |

## show bgp

```
show bgp [{vrf <VRF-NAME>|all-vrf}] [{ipv4 unicast|ipv6 unicast  
|all}][update-group [<INDEX>]]  
show bgp l2vpn evpn
```

## Description

This command shows entries in the BGP routing table.

| Parameter              | Description                                                                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ipv4                   | Selects the IPv4 address family.                                                                                                                                                   |
| ipv6                   | Selects the IPv6 address family.                                                                                                                                                   |
| unicast                | The subaddress family identifier.                                                                                                                                                  |
| vrf <VRF-NAME>         | Select to display information by VRFs by specifying the VRF name.                                                                                                                  |
| all-vrf                | Select to display the BGP summary information for all VRFs and address-families.                                                                                                   |
| l2vpn evpn             | Shows the information for L2VPN EVPN address family.                                                                                                                               |
| update-group [<INDEX>] | Select to display information about update-group and peers which are member of that update-group for given afi/safi. Specifying an integer defines the desired update-group index. |

## Examples

Showing BGP routing table information for VRF 1 IPv4 unicast:

```
switch# show bgp vrf v1 ipv4 unicast  
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,  
               i internal, e external S Stale, R Removed, a additional-paths  
Origin codes: i - IGP, e - EGP, ? - incomplete  
  
VRF : v1  
Local Router-ID 9.0.0.1  
  
      Network          Nexthop      Metric      LocPrf      Weight Path  
-----
```

```

*>e 9.0.0.0/24          9.0.0.2          0          100          0          65534.65535 3.4
18.54934 3574.8570 5.6 ?
*>e 100.0.0.0/24         9.0.0.2          0          100          0          200 ?
*>e 100.0.1.0/24         9.0.0.2          0          100          0          200 ?
*>e 100.0.2.0/24         9.0.0.2          0          100          0          200 ?
*>e 100.0.3.0/24         9.0.0.2          0          100          0          200 ?
*ae 100.0.3.0/24         9.0.0.3          0          100          0          200 ?
Total number of entries 6

```

## Showing BGP routing table information for L2VPN EVPN:

```

switch# show bgp l2vpn evpn

Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN Route-Type 2 prefix: [2]:[ESI]:[EthTag]:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]

VRF : default
Router-ID not configured

      Network
Metric LocPrf Weight Path
-----
Route Distinguisher: 10.1.1.54:32967 (L2VNI 30000)
*> [2]:[0]:[0]:[00:06:f6:3f:e3:c1]:[] 1.1.1.20 0
   100 32768 i
*> [2]:[0]:[0]:[8c:60:4f:f2:f5:41]:[] 1.1.1.10 0
   100 0 i
*> [3]:[0]:[1.1.1.1] 0.0.0.0 0
   100 0 ?
Total number of entries 3

```

```

switch# show bgp l2vpn evpn neighbors 1920:1680:1:1::8
Codes: ^ Inherited from peer-group, * Dynamic Neighbor
VRF : default
BGP Neighbor 1920:1680:1:1::8 (Internal)
Description          : RR peer-group^
Peer-group           : RRV6
Remote Router Id      : 192.168.1.8      Local Router Id    : 192.168.1.1
Remote AS             : 65001             Local AS           : 65001
Remote Port           : 42423             Local Port         : 179
State                 : Established        Admin Status       : Up
Conn. Established     : 5                  Conn. Dropped      : 4
Passive               : No                 Update-Source       : loopback0^
Cfg. Hold Time        : 180                 Cfg. Keep Alive    : 60
Neg. Hold Time        : 180                 Neg. Keep Alive    : 60
Up/Down Time          : 06h:46m:13s        Connect-Retry Time : 120
Local-AS Prepend      : No                 Alt. Local-AS      : 0
BFD                   : Disabled            Slow Peer          : Yes
Password              :
Last Err Sent         : No Error
Last SubErr Sent      : No Error
Last Err Rcvd         : No Error
Last SubErr Rcvd      : No Error
Graceful-Restart       : Enabled             Gr. Restart Time   : 120
Gr. Stalepath Time    : 300                 Remove Private-AS  : No

```

```

TTL                : 255                      Local Cluster-ID   :
Weight             : 0                        Fall-over       : No
Confederation-Peers : No

Message statistics      Sent      Rcvd
-----
Open                   8          7
Notification           3          1
Updates                20730      91332
Keepalives             1153        952
Route Refresh          0           0
Total                  21894      92292

Capability              Advertised      Received
-----
Route Refresh           Yes             Yes
Graceful Restart        Yes             Yes
Add-Path                No             No
Four Octet ASN          Yes             Yes
Address family IPv4 Unicast No             No
Address family IPv6 Unicast No             No
Address family VPNv4 Unicast No             No
Address family L2VPN EVPN Yes             Yes
Address Family : L2VPN EVPN

-----
Rt. Reflect. Client : No                Send Community    : extended^
Allow-AS in         : 0                  Advt. Interval     : 30
Max. Prefix         : 64000              Soft Reconfig In   :
Nexthop-Self        :                    Default-Originate :
Cfg. Add-Path        :
Neg. Add-Path        :
Routemap In         :
Routemap Out        :
ORF type             : Prefix-list
ORF capability        :

```

Showing all BGP update groups:

```

switch# show bgp all update-group

VRF : default

BGP Update-Group 1
  Address Family : ipv4-unicast      Peering Type : internal
  Peer Count     : 5                 Slow Group   : No

  Members
  -----
  110.162.100.221, 120.221.221.221, 130.121.21.111, 140.131.131.131,
  150.100.101.2

BGP Update-Group 2
  Address Family : ipv4-unicast      Peering Type : external
  Peer Count     : 2                 Slow Group   : Yes

  Members
  -----
  20.1.1.1, 30.1.1.1

BGP Update-Group 3
  Address Family : ipv6-unicast      Peering Type : external

```



```
Peer Count      : 1                      Slow Group    : No

Members
-----
20ab::cd:08
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                           |
|------------------|--------------------------------------------------------|
| 10.14            | Replaced IPv4 with IP in commands. IPv4 is deprecated. |
| 10.07 or earlier | --                                                     |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp

```
show bgp
[vrf <VRF-NAME>] {ipv4 unicast <A.B.C.D/M>|{ipv6 unicast <X::Y/M>}
l2vpn evpn
  EVPN Route-Type 1 prefix: [1]:[ESI]:[EthTag]
  RD-[ROUTE_TYPE]:[ESI]:[EthTag]:[MAC]:[OrigIP]
  RD-[ROUTE_TYPE]:[EthTag]:[OrigIP]
  RD-[ROUTE_TYPE]:[ESI]:[EthTag]:[IPAddrLen]:[IPAddr]
```

## Description

This command displays entries in the BGP routing table that are part of the specified network. For EVPN Route-type 2 with MAC only prefix as an input, displays all the prefixes containing the specific MAC address (MAC route, MAC/IP route, Host route).

| Parameter                | Description                                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| vrf <VRF-NAME>           | Shows the information for a specified VRF.                                                                                                  |
| ipv4 unicast <A.B.C.D/M> | Shows the information for an IPv4 unicast family with an IP prefix (network/length such as 35.0.0.0/8) in the BGP routing table to display. |
| ipv6 unicast <X::Y/M>    | Shows the information for an IPv6 unicast family an IPv6 prefix in the                                                                      |

| Parameter                                           | Description                                          |
|-----------------------------------------------------|------------------------------------------------------|
|                                                     | BGP routing table to display.                        |
| l2vpn evpn                                          | Shows the information for L2VPN EVPN address family. |
| RD-[ROUTE_TYPE]:[ESI]:[EthTag]:[MAC]:[OrigIP]       | EVPN Route-Type 2 prefix.                            |
| RD-[ROUTE_TYPE]:[EthTag]:[OrigIP]                   | EVPN Route-Type 3 prefix.                            |
| RD-[ROUTE_TYPE]:[ESI]:[EthTag]:[IPAddrLen]:[IPAddr] | EVPN Route-Type 5 prefix.                            |

## Examples

Showing the entries in the BGP routing table that are part of an IPv4 unicast network

```
switch# show bgp ipv4 unicast 10.0.0.0/16

VRF : default
BGP Local AS 2          BGP Router-ID 1.1.1.2

  Network      : 10.0.0.0/16          Nexthop      : 1.1.1.1
  Peer         : 1.1.1.1              Origin       : IGP
  Metric       : 0                    Local Pref   : 100
  Weight       : 0                    Calc. Local Pref : 100
  Best         : Yes                  Valid        : Yes
  Type         : external              Stale       : No
  Originator ID : 0.0.0.0             Path ID      : 0
  Aggregator ID :
  Aggregator AS :
  Atomic Aggregate :
  RFD Flaps    : 0                    RFD Penalty  : 0

  AS-Path      : 1
  Cluster List :
  Communities  :
50:100,50:101,50:102,50:103,50:104,50:105,50:106,50:107,50:108,50:109,50:110,50:1
  Extd. Communities :
```

Showing the entries in the BGP routing table that are part of L2VPN EVPN

```
switch# show bgp l2vpn evpn vni 30000
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN Route-Type 2 prefix: [2]:[ESI]:[EthTag]:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]

VRF : default
Router-ID not configured

      Network                                          Nexthop
Metric LocPrf Weight Path
-----
Route Distinguisher: 10.1.1.54:32967      (L2VNI 30000)
*>   [2]:[0]:[0]:[00:06:f6:3f:e3:c1]:[]          1.1.1.20      0
    100      32768 i
```

```
*> [2]:[0]:[0]:[8c:60:4f:f2:f5:41]:[] 1.1.1.10 0
100 0 i
*> [3]:[0]:[1.1.1.1] 0.0.0.0 0
100 0 ?
Total number of entries 3
```

Showing the entries in the BGP routing table for EVPN route-type 2

```
switch# show bgp l2vpn evpn 2:2-[2]:[0]:[0]:[00:50:56:96:6d:6f]:[20.20.1.10]
VRF : default
BGP Local AS 1 BGP Router-id 3.3.3.3
Network : 2:2-[2]:[0]:[0]:[00:50:56:96:6d:6f]:[20.20.1.10]
Nexthop : 1.1.1.1
vni : 2 vni_type : L2VNI
Peer : 2.2.2.2 Origin : incomplete
Metric : 0 Local Pref : 100
Weight : 0 Calc. Local Pref : 100
Best : Yes Valid : Yes
Type : internal Stale : No
Originator ID : 1.1.1.1 Aggregator ID :
Aggregator AS :
Atomic Aggregate :
AS-Path :
Cluster List :
Communities :
Ext-Communities : RT: 2:2 RT: 10:10 Router MAC: 00:00:00:00:00:11
Network : 2:2-[2]:[0]:[0]:[00:50:56:96:6d:6f]:[20.20.1.10]
Nexthop : 1.1.1.1
vni : 10000 vni_type : L3VNI
Peer : 2.2.2.2 Origin : incomplete
Metric : 0 Local Pref : 100
Weight : 0 Calc. Local Pref : 100
Best : Yes Valid : Yes
Type : internal Stale : No
Originator ID : 1.1.1.1 Aggregator ID :
Aggregator AS :
Atomic Aggregate :
AS-Path :
Cluster List :
Communities :
Ext-Communities : RT: 2:2 RT: 10:10 Router MAC: 00:00:00:00:00:11
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                           |
|------------------|--------------------------------------------------------|
| 10.14            | Replaced IPv4 with IP in commands. IPv4 is deprecated. |
| 10.08            | Added l2vpn evpn route types                           |
| 10.07 or earlier | --                                                     |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp community

```
show bgp [{ipv4 | ipv6 | ipv4 {vrf <VRF-NAME>}} unicast] community [<VALUE> | <TYPE>]
```

### Description

This command shows routes that belong to BGP communities. Optionally you can specify displaying information by a specific community or by VRF.

| Parameter           | Description                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ipv4                | Shows the information for an IPv4 address family.                                                                                                                                                                                                                                                                                                                  |
| ipv6                | Shows the information for an IPv6 address family.                                                                                                                                                                                                                                                                                                                  |
| unicast             | Shows the information for a subaddress family identifier.                                                                                                                                                                                                                                                                                                          |
| ipv4 vrf <VRF-NAME> | Shows the information for a specified VRF.                                                                                                                                                                                                                                                                                                                         |
| <VALUE>             | Shows the information for a community number. Specify the information in aa:nn format.                                                                                                                                                                                                                                                                             |
| <TYPE>              | Shows a specified community type. Select the following well-known communities, as well as others:<br>internet<br>Advertise the prefix to all BGP neighbors.<br>local-as<br>Do not advertise the prefix outside the sub-AS.<br>no-advertise<br>Do not advertise the prefix to any BGP neighbors.<br>no-export<br>Do not advertise the prefix to any eBGP neighbors. |

### Examples

Showing information for routes that belong to all BGP communities:

```
switch# show bgp ipv4 unicast community
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath, i
internal, e external S Stale, R Removed, a additional-paths VRF : default Local
Router-ID 9.0.0.1
Network                Nexthop      Metric    LocPrf    Weight Path
-----
*>e 9.0.0.0/24          9.0.0.2      0         100       0       200 ?
*>e 100.0.0.0/24         9.0.0.2      0         100       0       200 ?
*>e 100.0.1.0/24         9.0.0.2      0         100       0       200 ?
*>e 100.0.2.0/24         9.0.0.2      0         100       0       200 ?
*>e 100.0.3.0/24         9.0.0.2      0         100       0       200 ?
*ae 100.0.3.0/24        9.0.0.3      0         100       0       200 ?
Total number of entries 6
```

Showing information for routes that belong to the 200:20 BGP community number:

```
switch# show bgp ipv4 unicast community 200:20
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
i internal, e external S Stale, R RemovedOrigin codes: i - IGP, e - EGP, ? -
incomplete
VRF : defaultLocal Router-ID 9.0.0.1
```

| Network          | Nexthop | Metric | LocPrf | Weight | Path  |
|------------------|---------|--------|--------|--------|-------|
| *>e 9.0.0.0/24   | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.0.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.1.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.2.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.3.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *ae 100.0.3.0/24 | 9.0.0.3 | 0      | 100    | 0      | 200 ? |

Total number of entries 6

Showing information for routes that belong to the Internet BGP community type:

```
switch# show bgp ipv4 unicast community internet
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete
VRF : default
Local Router-ID 9.0.0.1
```

| Network          | Nexthop | Metric | LocPrf | Weight | Path  |
|------------------|---------|--------|--------|--------|-------|
| *>e 9.0.0.0/24   | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.0.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.1.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.2.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.3.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *ae 100.0.3.0/24 | 9.0.0.3 | 0      | 100    | 0      | 200 ? |

Total number of entries 6

Showing information for routes that belong to the local-as BGP community type:

```
switch# show bgp ipv4 unicast community local-as
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete
VRF : default

Local Router-ID 9.0.0.1
```

| Network          | Nexthop | Metric | LocPrf | Weight | Path  |
|------------------|---------|--------|--------|--------|-------|
| *>e 9.0.0.0/24   | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.0.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.1.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.2.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *>e 100.0.3.0/24 | 9.0.0.2 | 0      | 100    | 0      | 200 ? |
| *ae 100.0.3.0/24 | 9.0.0.3 | 0      | 100    | 0      | 200 ? |

Total number of entries 6

Showing information for routes that belong to the no-advertise BGP community type:

```

switch# show bgp ipv4 unicast community no-advertise
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
i internal, e external S Stale, R Removed, a additional-paths Origin codes: i -
IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 9.0.0.1

```

| Network          | Nextthop | Metric | LocPrf | Weight | Path  |
|------------------|----------|--------|--------|--------|-------|
| *>e 9.0.0.0/24   | 9.0.0.2  | 0      | 100    | 0      | 200 ? |
| *>e 100.0.0.0/24 | 9.0.0.2  | 0      | 100    | 0      | 200 ? |
| *>e 100.0.1.0/24 | 9.0.0.2  | 0      | 100    | 0      | 200 ? |
| *>e 100.0.2.0/24 | 9.0.0.2  | 0      | 100    | 0      | 200 ? |
| *>e 100.0.3.0/24 | 9.0.0.2  | 0      | 100    | 0      | 200 ? |
| *ae 100.0.3.0/24 | 9.0.0.2  | 0      | 100    | 0      | 200 ? |

Total number of entries 6

Showing information for routes that belong to the no-export BGP community type:

```

switch# show bgp ipv4 unicast community no-export
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,i
internal, e external S Stale, R Removed, a additional-paths Origin codes: i - IGP,
e - EGP, ? - incomplete

VRF : default
Local Router-ID 9.0.0.1

```

| Network          | Nextthop | Metric | LocPrf | Weight | Path  |
|------------------|----------|--------|--------|--------|-------|
| *>e 9.0.0.0/24   | 9.0.0.2  | 0      | 100    | 0      | 200 ? |
| *>e 100.0.0.0/24 | 9.0.0.2  | 0      | 100    | 0      | 200 ? |
| *>e 100.0.1.0/24 | 9.0.0.2  | 0      | 100    | 0      | 200 ? |
| *>e 100.0.2.0/24 | 9.0.0.2  | 0      | 100    | 0      | 200 ? |
| *>e 100.0.3.0/24 | 9.0.0.2  | 0      | 100    | 0      | 200 ? |
| *ae 100.0.3.0/24 | 9.0.0.3  | 0      | 100    | 0      | 200 ? |

Total number of entries 6

Showing information for routes that belong to the gshut BGP community type:

```

switch# show bgp ipv4 unicast community gshut
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
i internal, e external S Stale, R Removed Origin codes: i - IGP, e - EGP, ? -
incomplete

VRF : default
Local Router-ID 1.1.1.2

```

| Network        | Nextthop | Metric | LocPrf | Weight | Path |
|----------------|----------|--------|--------|--------|------|
| *>e 1.1.1.0/24 | 10.1.1.2 | 0      | 0      | 0      | 2 i  |

Total number of entries 1

```

switch#
switch# show bgp ipv6 unicast community gshut

Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,i
internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete

```

```
VRF : default
Local Router-ID 1.1.1.2
```

| Network                   | Nexthop | Metric | LocPrf | Weight | Path |
|---------------------------|---------|--------|--------|--------|------|
| -----                     |         |        |        |        |      |
| *>e 1::/64                | 10::2   |        |        |        |      |
| fe80::98f2:b300:1368:e882 |         | 0      | 0      | 0      | 2 i  |
| Total number of entries 1 |         |        |        |        |      |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp flap-statistics

```
show bgp {vrf <VRF_NAME> | all-vrf} [{ipv4 unicast | ipv6 unicast | all}] flap-statistics
```

### Description

Displays all the flapped and suppressed routes.

### Usage

Status of the route with dampening enabled:

- If the route is available, the history flag is unset.
- If route has been flapping, is not suppressed and is withdrawn; the state of the route is h
- If route is currently available but is suppressed due to dampening, the state of the route is d
- If the route is unsuppressed and currently withdrawn, the state of the route is h

### Examples

Showing all the flapped and suppressed routes:

```
switch# show bgp all
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
```

```
Local Router-ID 1.1.1.1
```

```
Address-family : IPv4 Unicast
```

```
-----
      Network           Nexthop           Metric      LocPrf      Weight Path
*>i 2.2.2.0/24          2.2.2.2             0           100         0        ?
*>i 11.1.1.0/24         2.2.2.2             0           100         0        ?
*ai 11.1.1.0/24        2.2.2.3             0           100         0        ?
Total number of entries 3
```

```
Address-family : IPv6 Unicast
```

```
-----
      Network           Nexthop           Metric      LocPrf      Weight Path
Total number of entries 0
```

```
switch# show bgp ipv4 unicast flap-statistics
```

```
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete
```

```
VRF : default
```

```
Local Router-ID 20.0.0.1
```

```
      Network           Nexthop           Flaps           Reuse           Path
*>e 2.2.2.0/24          20.0.0.2           1              00h:00m:00s     300 ?
de 3.3.3.0/24          20.0.0.2           2              00h:29m:31s     300 ?
Total number of entries 2
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                          |
|-----------|-----------------------------|------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## show bgp neighbor advertised-routes

```
show bgp [vrf <VRF-NAME>] {ipv4 unicast | ipv6 unicast
    | l2vpn evpn | all} neighbors <IP-ADDRESS>
    advertised-routes
show bgp l2vpn evpn neighbors <IP-ADDRESS> advertised-routes
```

## Description

Shows all routes that have been advertised to the specified neighbor.



| Parameter      | Description                                                             |
|----------------|-------------------------------------------------------------------------|
| vrf <VRF-NAME> | Shows the information for a specified VRF.                              |
| ipv4 unicast   | Shows the information for an IPv4 unicast address family.               |
| ipv6 unicast   | Shows the information for an IPv6 unicast address family.               |
| all            | Shows the information for all address families and subaddress families. |
| <IP-ADDRESS>   | Shows the information for a neighbor IP address.                        |

## Examples

Showing routes that have been advertised to the specified IPv4 unicast neighbor:

```
switch# show bgp ipv4 unicast neighbors 9.0.0.1 advertised-routes
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 9.0.0.2
```

| Network                      | Nexthop | Metric | LocPrf | Weight | Path            |
|------------------------------|---------|--------|--------|--------|-----------------|
| *>e 9.0.0.0/24               | 9.0.0.2 | 0      | 0      | 0      | 200 65534.65535 |
| 3.4 18.54934 3574.8570 5.6 ? |         |        |        |        |                 |
| *>e 100.0.0.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *>e 100.0.1.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *>e 100.0.2.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *>e 100.0.3.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |

```
Total number of entries 5
```

Showing routes that have been advertised to the specified L2VPN EVPN neighbor:

```
switch# show bgp l2vpn evpn neighbor 9.0.0.1 advertised-routes
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN Route-Type 2 prefix: [2]:[ESI]:[EthTag]:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]

VRF : default
Local Router-ID 9.0.0.2
```

| Network                                            | Nexthop  | Metric |
|----------------------------------------------------|----------|--------|
| LocPrf Weight Path                                 |          |        |
| -----                                              |          |        |
| Route Distinguisher: 10.1.1.54:32967 (L2VNI 30000) |          |        |
| *> [2]:[0]:[0]:[00:06:f6:3f:e3:c1]:[]              | 1.1.1.20 | 0 100  |
| 32768 i                                            |          |        |
| *> [2]:[0]:[0]:[8c:60:4f:f2:f5:41]:[]              | 1.1.1.10 | 0 100  |
| 0 i                                                |          |        |

```
Total number of entries 2
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp neighbor paths

```
show bgp [vrf <VRF-NAME>] {ipv4 unicast | ipv6 unicast  
| all} neighbors <IP-ADDRESS> paths  
show bgp l2vpn evpn neighbors <IP-ADDRESS> paths
```

## Description

Shows autonomous system paths learned from the specified neighbor.

| Parameter      | Description                                                             |
|----------------|-------------------------------------------------------------------------|
| vrf <VRF-NAME> | Shows the information for a specified VRF.                              |
| ipv4 unicast   | Shows the information for an IPv4 unicast address family.               |
| ipv6 unicast   | Shows the information for an IPv6 unicast address family.               |
| all            | Shows the information for all address families and subaddress families. |
| <IP-ADDRESS>   | Shows the information for a neighbor IP address.                        |
| l2vpn evpn     | Shows the information for L2VPN EVPN address family.                    |

## Examples

Showing autonomous system paths learned from the specified IPv4 unicast neighbor:

```
switch# show bgp ipv4 unicast neighbors 192.168.12.2 paths  
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,  
i internal, e external S Stale, R Removed  
VRF : default  
Local Router-ID 9.0.0.1  
  
      Network          Nexthop          Path  
-----  
*>e 9.0.0.0/24        9.0.0.2          200 65534.65535 3.4 18.54934 3574.8570 5.6
```

```
*>e 100.0.0.0/24      9.0.0.2      200
*>e 100.0.1.0/24      9.0.0.2      200
*>e 100.0.2.0/24      9.0.0.2      200
*>e 100.0.3.0/24      9.0.0.2      200
Total number of entries 5
```

Showing autonomous system paths learned from the specified L2VPN EVPN neighbor:

```
switch# show bgp l2vpn evpn neighbors 192.168.12.1 paths

Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN Route-Type 2 prefix: [2]:[ESI]:[EthTag]:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]

VRF : default
Local Router-ID 9.0.0.2
```

| Network                                            | Nexthop  | Path |
|----------------------------------------------------|----------|------|
| -----                                              |          |      |
| -                                                  |          |      |
| Route Distinguisher: 10.1.1.54:32967 (L2VNI 30000) |          |      |
| *> [2]:[0]:[0]:[00:06:f6:3f:e3:c1]:[]              | 1.1.1.20 | 100  |
| *> [2]:[0]:[0]:[8c:60:4f:f2:f5:41]:[]              | 1.1.1.10 | 100  |
| Total number of entries 2                          |          |      |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp neighbor received orf-prefix-list

```
show bgp [vrf <VRF-NAME>]{ipv4 unicast | ipv6 unicast | all} neighbors <IP-ADDRESS>
received orf-prefix-list
```

## Description

Shows all the prefix lists received from the specified neighbor.

| Parameter      | Description                                                             |
|----------------|-------------------------------------------------------------------------|
| vrf <VRF-NAME> | Shows the information for a specified VRF.                              |
| ipv4 unicast   | Shows the information for an IPv4 unicast address family.               |
| ipv6 unicast   | Shows the information for an IPv6 unicast address family.               |
| all            | Shows the information for all address families and subaddress families. |
| <IP-ADDRESS>   | Shows the information for a neighbor IP address.                        |

## Examples

Showing received prefix list from the specified neighbor:

```
switch# show bgp ipv4 unicast neighbors A.B.C.D received orf-prefix-list
Address family: IPv4 Unicast

ip prefix-list 10.0.0.200: 4 entries

    seq 10 permit 28.119.16.0/24

    seq 15 deny 28.119.19.0/24

    seq 20 permit 28.119.17.0/24

Address family: IPv6 Unicast

ip prefix-list 10.0.0.200: 4 entries

    seq 30 permit 2000::/64

    seq 35 deny 3000::/64

    seq 40 permit 4000:0/64
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp neighbor received-routes

```
show bgp [vrf <VRF-NAME>] {ipv4 unicast | ipv6 unicast
| all} neighbors <IP-ADDRESS> received-routes
show bgp l2vpn evpn neighbors <IP-ADDRESS> received-routes
```

## Description

Shows received routes from the specified neighbor.

| Parameter      | Description                                                             |
|----------------|-------------------------------------------------------------------------|
| vrf <VRF-NAME> | Shows the information for a specified VRF.                              |
| ipv4 unicast   | Shows the information for an IPv4 unicast address family.               |
| ipv6 unicast   | Shows the information for an IPv6 unicast address family.               |
| all            | Shows the information for all address families and subaddress families. |
| <IP-ADDRESS>   | Shows the information for a neighbor IP address.                        |
| l2vpn evpn     | Shows the information for L2VPN EVPN address family.                    |

## Examples

Showing received routes from the specified IPv4 unicast neighbor:

```
switch# show bgp ipv4 unicast neighbors 192.168.12.1 received-routes
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 9.0.0.2
```

| Network                      | Nexthop | Metric | LocPrf | Weight | Path            |
|------------------------------|---------|--------|--------|--------|-----------------|
| *>e 9.0.0.0/24               | 9.0.0.2 | 0      | 0      | 0      | 200 65534.65535 |
| 3.4 18.54934 3574.8570 5.6 ? |         |        |        |        |                 |
| *>e 100.0.0.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *>e 100.0.1.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *>e 100.0.2.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *>e 100.0.3.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *ae 100.0.3.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |

```
Total number of entries 6
```

Showing received routes from the specified L2VPN EVPN neighbor:

```
switch# show bgp l2vpn evpn neighbors 192.168.12.1 received-routes

Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN Route-Type 2 prefix: [2]:[ESI]:[EthTag]:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]

VRF : default
Local Router-ID 9.0.0.2
```

| Network                                            | Nexthop  | Metric |
|----------------------------------------------------|----------|--------|
| LocPrf Weight Path                                 |          |        |
| -----                                              |          |        |
| Route Distinguisher: 10.1.1.54:32967 (L2VNI 30000) |          |        |
| *> [2]:[0]:[0]:[00:06:f6:3f:e3:c1]:[]<br>32768 i   | 1.1.1.20 | 0 100  |
| *> [2]:[0]:[0]:[8c:60:4f:f2:f5:41]:[]<br>0 i       | 1.1.1.10 | 0 100  |
| Total number of entries 2                          |          |        |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp neighbor routes

```
show bgp [vrf <VRF-NAME>] {ipv4 unicast | ipv6 unicast
| all} neighbors <IP-ADDRESS> routes
show bgp l2vpn evpn neighbors <IP-ADDRESS>
```

## Description

This command shows routes that are received and accepted from the specified neighbor.

| Parameter      | Description                                                             |
|----------------|-------------------------------------------------------------------------|
| vrf <VRF-NAME> | Shows the information for a specified VRF.                              |
| ipv4 unicast   | Shows the information for an IPv4 unicast address family.               |
| ipv6 unicast   | Shows the information for an IPv6 unicast address family.               |
| all            | Shows the information for all address families and subaddress families. |
| <IP-ADDRESS>   | Shows the information for a neighbor IP address.                        |
| l2vpn evpn     | Shows the information for L2VPN EVPN address family.                    |

## Examples

Showing all routes that are received and accepted from the specified neighbor:

```
switch# show bgp ipv4 unicast neighbors 9.0.0.2 routes
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 9.0.0.1
```

| Network                      | Nexthop | Metric | LocPrf | Weight | Path            |
|------------------------------|---------|--------|--------|--------|-----------------|
| *>e 9.0.0.0/24               | 9.0.0.2 | 0      | 100    | 0      | 200 65534.65535 |
| 3.4 18.54934 3574.8570 5.6 ? |         |        |        |        |                 |
| *>e 100.0.0.0/24             | 9.0.0.2 | 0      | 100    | 0      | 200 ?           |
| *>e 100.0.1.0/24             | 9.0.0.2 | 0      | 100    | 0      | 200 ?           |
| *>e 100.0.2.0/24             | 9.0.0.2 | 0      | 100    | 0      | 200 ?           |
| *>e 100.0.3.0/24             | 9.0.0.2 | 0      | 100    | 0      | 200 ?           |
| *ae 100.0.3.0/24             | 9.0.0.3 | 0      | 100    | 0      | 200 ?           |
| Total number of entries 6    |         |        |        |        |                 |

Showing 12 VPN EVPN routes that are received and accepted from the specified neighbor:

```
switch# show bgp l2vpn evpn neighbor 9.0.0.2 routes

Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN Route-Type 2 prefix: [2]:[ESI]:[EthTag]:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]

VRF : default
Local Router-id: 9.0.0.1
```

| Network                      | Nexthop | Metric | LocPrf | Weight | Path            |
|------------------------------|---------|--------|--------|--------|-----------------|
| *>e 9.0.0.0/24               | 9.0.0.2 | 0      | 0      | 0      | 200 65534.65535 |
| 3.4 18.54934 3574.8570 5.6 ? |         |        |        |        |                 |
| *>e 100.0.0.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *>e 100.0.1.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *>e 100.0.2.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *>e 100.0.3.0/24             | 9.0.0.2 | 0      | 0      | 0      | 200 ?           |
| *ae 100.0.3.0/24             | 9.0.0.3 | 0      | 100    | 0      | 200 ?           |
| Total number of entries 6    |         |        |        |        |                 |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp neighbors

```
show bgp [vrf <VRF-NAME>] {ipv4 unicast | ipv6 unicast | all}
neighbors
```

### Description

This command shows information about BGP and TCP connections to neighbors. If neighbors are member of a peer-group, the command shows the configured values inherited from the peer-group. The configured values are postfixed with a caret (^) for inherited values.

| Parameter      | Description                                                             |
|----------------|-------------------------------------------------------------------------|
| vrf <VRF-NAME> | Shows the information for a specified VRF.                              |
| ipv4 unicast   | Shows the information for an IPv4 unicast address family.               |
| ipv6 unicast   | Shows the information for an IPv6 unicast address family.               |
| all            | Shows the information for all address families and subaddress families. |

### Examples

Showing all information about BGP and TCP connections to neighbors:

```
switch# show bgp all neighbors
Codes: ^ Inherited from peer-group
VRF : default

BGP Neighbor 10.1.1.2 (Internal)
  Description      :
  Peer-group       :

  Remote Router Id : 10.1.1.2      Local Router Id : 1.0.0.1
  Remote AS        : 1             Local AS         : 1
  Remote Port      : 0             Local Port       : 0
  State            : Established   Admin Status     : Up
  Conn. Established : 1            Conn. Dropped    : 0
  Passive          : No            Update-Source    :
  Cfg. Hold Time   : 180           Cfg. Keep Alive  : 60
  Neg. Hold Time   : 0             Neg. Keep Alive  : 0
  Up/Down Time     : 00h:00m:00s   Connect-Retry Time : 120
  Local-AS Prepend : No            Alt. Local-AS    : 0
  BFD              : Disabled      Slow Peer        : No
  Password         :
  Last Err Sent    : No Error
  Last SubErr Sent : No Error
  Last Err Rcvd    : No Error
  Last SubErr Rcvd : No Error

  Graceful-Restart : Enabled        Gr. Restart Time : 120
```



|                    |       |                   |      |
|--------------------|-------|-------------------|------|
| Gr. Stalepath Time | : 150 | Remove Private-AS | : No |
| TTL                | : 255 | Local Cluster-ID  | :    |
| Weight             | : 0   | Fall-over         | : No |

| Message statistics | Sent  | Rcvd  |
|--------------------|-------|-------|
| -----              | ----- | ----- |
| Open               | 7     | 6     |
| Notification       | 5     | 2     |
| Updates            | 3     | 2     |
| Keepalives         | 12    | 10    |
| Route Refresh      | 0     | 0     |
| Total              | 28    | 20    |

| Capability                  | Advertised | Received |
|-----------------------------|------------|----------|
| -----                       | -----      | -----    |
| Route Refresh               | Yes        | No       |
| Graceful Restart            | Yes        | No       |
| Four Octet ASN              | Yes        | No       |
| Add-Path                    | Yes        | Yes      |
| Address family IPv4 Unicast | Yes        | No       |
| Address family IPv6 Unicast | No         | No       |

#### Address Family : IPv4 Unicast

|                     |         |                   |      |
|---------------------|---------|-------------------|------|
| Rt. Reflect. Client | : No    | Send Community    | :    |
| Allow-AS in         | : 0     | Advt. Interval    | : 30 |
| Max. Prefix         | : 64000 | Soft Reconfig In  | :    |
| Nexthop-Self        | :       | Default-Originate | :    |
| Update-Group        | : 2     |                   |      |

|                |                    |
|----------------|--------------------|
| Cfg. Add-Path  | : Send and Receive |
| Neg. Add-Path  | : Send and Receive |
| Routemap In    | :                  |
| Routemap Out   | :                  |
| ORF type       | : Prefix-list      |
| ORF capability | : Receive          |

#### Address Family : IPv6 Unicast

|                     |         |                   |      |
|---------------------|---------|-------------------|------|
| Rt. Reflect. Client | : No    | Send Community    | :    |
| Allow-AS in         | : 0     | Advt. Interval    | : 30 |
| Max. Prefix         | : 64000 | Soft Reconfig In  | :    |
| Nexthop-Self        | :       | Default-Originate | :    |
| Update-Group        | : 3     |                   |      |

|                |                    |
|----------------|--------------------|
| Cfg. Add-Path  | : Send and Receive |
| Neg. Add-Path  | : Send and Receive |
| Routemap In    | :                  |
| Routemap Out   | :                  |
| ORF type       | : Prefix-list      |
| ORF capability | : Receive          |

Showing information for BGP IPv4 unicast neighbors:

```
switch# show bgp ipv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : default
```

# BGP Neighbor 10.1.1.2 (Internal)

```

Description      :
Peer-group       :

Remote Router Id  : 10.1.1.2      Local Router Id   : 1.0.0.1
Remote AS         : 1             Local AS           : 1
Remote Port      : 0             Local Port         : 0
State            : Idle           Admin Status       : Up
Conn. Established : 0             Conn. Dropped      : 0
Passive          : No             Update-Source      :
Cfg. Hold Time   : 180            Cfg. Keep Alive    : 60
Neg. Hold Time   : 0             Neg. Keep Alive    : 0
Up/Down Time     : 00h:00m:00s   Connect-Retry Time : 120
Local-AS Prepend : No            Alt. Local-AS      : 0
BFD              : Disabled       Slow Peer          : Yes
Password         :
Last Err Sent    : No Error
Last SubErr Sent : No Error
Last Err Rcvd    : No Error
Last SubErr Rcvd : No Error

Graceful-Restart  : Enabled       Gr. Restart Time   : 120
Gr. Stalepath Time : 150         Remove Private-AS : No
TTL              : 255          Local Cluster-ID   :
Weight           : 0            Fall-over          : No

```

| Message statistics | Sent | Rcvd |
|--------------------|------|------|
| Open               | 0    | 0    |
| Notification       | 0    | 0    |
| Updates            | 0    | 0    |
| Keepalives         | 0    | 0    |
| Route Refresh      | 0    | 0    |
| Total              | 0    | 0    |

| Capability                  | Advertised | Received |
|-----------------------------|------------|----------|
| Route Refresh               | Yes        | No       |
| Graceful Restart            | Yes        | No       |
| Four Octet ASN              | Yes        | No       |
| Add-Path                    | Yes        | Yes      |
| Address family IPv4 Unicast | Yes        | No       |
| Address family IPv6 Unicast | No         | No       |

## Address Family : IPv4 Unicast

```

Rt. Reflect. Client : No          Send Community    :
Allow-AS in         : 0           Advt. Interval    : 30
Max. Prefix         : 64000       Soft Reconfig In  :
Nexthop-Self        :            Default-Originate   :
Update-Group        : 12

Routemap In         :
Routemap Out        :

ORF Type            : Prefix-list
ORF Capability       : Send

Cfg. Add-Path       : Send

```

Neg. Add-Path : Send



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp paths

```
show bgp [vrf <VRF-NAME>] {ipv4 unicast | ipv6 unicast | all}
    paths
show bgp l2vpn evpn paths
```

## Description

Shows received BGP path information in the database.

| Parameter      | Description                                                             |
|----------------|-------------------------------------------------------------------------|
| vrf <VRF-NAME> | Shows the information for a specified VRF.                              |
| ipv4 unicast   | Shows the information for an IPv4 unicast address family.               |
| ipv6 unicast   | Shows the information for an IPv6 unicast address family.               |
| all            | Shows the information for all address families and subaddress families. |

## Examples

Showing received BGP path information from the specified IPv4 unicast neighbor:

```
switch# show bgp ipv4 unicast paths
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
VRF : default
Local Router-ID 9.0.0.1

  Network          Nexthop          PathID          Path
  -----
*>e 9.0.0.0/24      9.0.0.2           0              200 65534.65535 3.4 18.54934
3574.8570 5.6
```

```
*>e 100.0.0.0/24      9.0.0.2      0      200
*>e 100.0.1.0/24      9.0.0.2      0      200
*>e 100.0.2.0/24      9.0.0.2      0      200
*>e 100.0.3.0/24      9.0.0.2      10     200
*ae 100.0.3.0/24      9.0.0.3      5      200
Total number of entries 6
```

Showing received BGP path information from the specified L2VPN EVPN neighbor:

```
switch# show bgp l2vpn evpn paths

Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN Route-Type 2 prefix: [2]:[ESI]:[EthTag]:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]

VRF : default
Local Router-ID 9.0.0.2

      Network                                     Nexthop      Path
-----
Route Distinguisher: 10.1.1.54:32967 (L2VNI 30000)
*> [2]:[0]:[0]:[00:06:f6:3f:e3:c1]:[]          1.1.1.20      100
*> [2]:[0]:[0]:[8c:60:4f:f2:f5:41]:[]          1.1.1.10      100
Total number of entries 2
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp peer-group summary

```
show bgp [vrf <VRF-NAME>] {ipv4 unicast |
    ipv6 unicast | all} peer-group <PEER-GROUP-NAME>
    summary
show bgp l2vpn evpn peer-group <PEER-GROUP-NAME> summary
```

## Description

This command shows the peer-group information in the database.

| Parameter         | Description                                                             |
|-------------------|-------------------------------------------------------------------------|
| vrf <VRF-NAME>    | Shows the information for a specified VRF.                              |
| ipv4 unicast      | Shows the information for an IPv4 unicast address family.               |
| ipv6 unicast      | Shows the information for an IPv6 unicast address family.               |
| all               | Shows the information for all address families and subaddress families. |
| <PEER-GROUP-NAME> | Shows the information for the BGP peer-group for the BGP instance.      |

## Examples

Showing the information from IPv4 unicast address families in `pg_name1` peer-group:

```
switch# show bgp ipv4 unicast peer-group pg_name1 summary
Codes: * Dynamic Neighbor
VRF : default
BGP Peer-Group Summary
=====
Local AS           : 1           BGP Router Identifier : 2.2.2.2
Peers              : 1           Dynamic Peer Count   : 3
Cfg. Hold Time     : 180        Cfg. Keep Alive      : 60

Neighbor    Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
10.0.0.1     1             8        10       00h:00m:58s   Established Up
*10.1.1.5    11            15       14       00h:10m:24s   Established Up
```

Showing the information from all address families in `pg_name1` peer-group:

```
switch# show bgp all unicast peer-group pg_name1 summary
Codes: * Dynamic Neighbor
VRF : default
BGP Peer-Group Summary
=====
Local AS           : 1           BGP Router Identifier : 2.2.2.2
Peers              : 1           Dynamic Peer Count   : 3
Cfg. Hold Time     : 180        Cfg. Keep Alive      : 60
Confederation Id   : 0

For address family: IPv4 Unicast
Neighbor    Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
10.0.0.1     1             8        10       00h:00m:58s   Established Up
*10.1.1.5    11            15       14       00h:10m:24s   Established Up

For address family: IPv6 Unicast
Neighbor    Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
1001::1002  11          12        12       00h:00m:07s   Established Up
2001::2002  11          12        12       00h:00m:07s   Established Up

For address family: L2VPN EVPN
Neighbor    Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
10.0.0.1     1             8        10       00h:00m:58s   Established Up
10.1.1.6     11            15       14       00h:10m:24s   Established Up
```

Showing the information from L2VPN EVPN address families in `pg_name1` peer-group:

```

switch# show bgp l2vpn evpn peer-group pg_name1 summary
VRF : default
BGP Peer-Group Summary
=====
Local AS           : 1           BGP Router Identifier : 2.2.2.2
Peers              : 1           Dynamic Peer Count    : 3
Cfg. Hold Time     : 180        Cfg. Keep Alive       : 60
Confederation Id   : 0

Neighbor      Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
10.0.0.1      1          8        10       00h:00m:58s   Established Up
*10.1.1.6     11         15       14       00h:10m:24s   Established Up

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp summary

```

show bgp [vrf <VRF-NAME>] {ipv4 unicast | ipv6 unicast | all}
    summary
show bgp l2vpn evpn summary

```

## Description

This command shows a summary of the status of Border Gateway Protocol (BGP) connections.

| Parameter      | Description                                                                               |
|----------------|-------------------------------------------------------------------------------------------|
| ipv4 unicast   | Selects to display the BGP summary information for the IPv4 subaddress family identifier. |
| ipv6 unicast   | Selects to display the BGP summary information for the IPv6 subaddress family identifier. |
| all            | Selects to display the BGP summary information for all VRFs and address-families.         |
| vrf <VRF-NAME> | Selects to display information by VRFs by specifying the VRF name.                        |

## Examples

Showing BGP summary information for all address-families:

```

switch(config-bgp)# show bgp all summary
Codes: * Dynamic Neighbor
VRF : default
BGP Summary
  Local AS           : 100           BGP Router Identifier : 9.0.0.1
  Peers              : 1             Log Neighbor Changes  : No
  Cfg. Hold Time     : 180          Cfg. Keep Alive       : 60
  Confederation Id    : 0

Address-family : IPv4 Unicast
-----
Neighbor      Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
9.0.0.2        200         25       23       00h:17m:50s  Established Up
*10.1.1.5      11          26       24       00h:20m:26s  Established Up

Address-family : IPv6 Unicast
-----
Neighbor      Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
*2001::2002    11         3         3        00h:00m:14s  Established Up
9000::2        200        25        23       00h:17m:50s  Established Up

Address-family : VPNv4 Unicast
-----
Neighbor      Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
1.1.1.1        100        207       208      02h:54m:18s  Established Up
*3.3.3.4       11         26        24       00h:20m:26s  Established Up

Address-family : L2VPN EVPN
-----
Neighbor      Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
10.0.0.2       200        25        23       00h:17m:50s  Established Up
*10.1.1.6      11         26        24       00h:20m:26s  Established Up
VRF : v1
BGP Summary
  Local AS           : 100           BGP Router Identifier : 9.0.0.1
  Peers              : 1             Log Neighbor Changes  : No
  Cfg. Hold Time     : 180          Cfg. Keep Alive       : 60

Address-family : IPv4 Unicast
-----
Neighbor      Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
*4.4.4.4       11         26        24       00h:20m:26s  Established Up
9.0.0.2        200        25        23       00h:17m:50s  Established Up

Address-family : IPv6 Unicast
-----
Neighbor      Remote-AS  MsgRcvd  MsgSent  Up/Down Time  State      AdminStatus
*3001::3002    11         3         3        00h:00m:14s  Established Up
9000::2        200        25        23       00h:17m:50s  Established Up

```

Showing BGP summary information for a specific VRF for IPv4 unicast network:

```

switch(config-bgp)# show bgp ipv4 unicast vrf v1 summary
Codes: * Dynamic Neighbor
VRF : v1
BGP Summary
  Local AS           : 100           BGP Router Identifier : 9.0.0.1
  Peers              : 1             Log Neighbor Changes  : No
  Cfg. Hold Time     : 180          Cfg. Keep Alive       : 60

```

| Neighbor  | Remote-AS | MsgRcvd | MsgSent | Up/Down     | Time | State       | AdminStatus |
|-----------|-----------|---------|---------|-------------|------|-------------|-------------|
| 9.0.0.2   | 200       | 25      | 23      | 00h:17m:50s |      | Established | Up          |
| *10.1.1.5 | 11        | 26      | 24      | 00h:20m:26s |      | Established | Up          |

Showing BGP summary information for L2VPN EVPN:

```
switch(config-bgp)# do show bgp l2vpn evpn summary
Codes: * Dynamic Neighbor
VRF : default
BGP Summary
  Local AS           : 100           BGP Router Identifier : 9.0.0.1
  Peers              : 1             Log Neighbor Changes  : No
  Cfg. Hold Time     : 180          Cfg. Keep Alive       : 60

Neighbor    Remote-AS  MsgRcvd  MsgSent  Up/Down  Time  State      AdminStatus
10.0.0.2    200        25       23       00h:17m:50s  Established  Up
10.1.1.6    11         26       24       00h:20m:26s  Established  Up
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show bgp l2vpn evpn

show bgp l2vpn evpn <ROUTE-TYPE>

### Description

Shows the BGP L2VPN information..

| Parameter    | Description                                       |
|--------------|---------------------------------------------------|
| <ROUTE-TYPE> | Specifies the routes filtered by NLRI route type. |

## Examples

Showing BGP L2VPN information for extcommunity:



```

switch# show bgp l2vpn evpn extcommunity
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete

EVPN Route-Type 1 prefix: [1]:[ESI]:[EthTag]
EVPN Route-Type 2 prefix: [2]:{ESI}:{EthTag}:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]
EVPN Route-Type 4 prefic: [4]:[ESI]:[OrigIP]
EVPN Route-Type 5 prefix: [5]:[ESI]:[EthTag]:[IPAddrLen]:[IPAddr]
EVPN Route-Type 6 prefix: [6]:[EthTag]:McastSrcIP]:[McastGrpIP]:[OrigIP]

VRF : default
Local Router-ID 2.2.2.2

```

| Network                                           | Ext-Community                 | Nextthop |
|---------------------------------------------------|-------------------------------|----------|
| -----                                             |                               |          |
| Route Distinguisher: 2:2 (L2VNI 1)                |                               |          |
| *>i [2]:[0]:[0]:[00:00:00:00:00:aa]:[100.100.1.2] | 1:1,100:100,00:00:00:00:00:11 | 1.1.1.1  |
| *>i [3]:[0]:[1.1.1.1]                             | 1:1                           | 1.1.1.1  |
| Route Distinguisher: 2:2 (L2VNI 2)                |                               |          |
| *>i [2]:[0]:[0]:[00:00:00:00:00:aa]:[200.200.1.2] | 2:2,100:100,00:00:00:00:00:22 | 3.3.3.3  |
| *>i [3]:[0]:[3.3.3.3]                             | 2:2                           | 3.3.3.3  |
| Route Distinguisher: 1:1 (L3VNI 10000)            |                               |          |
| *>i [2]:[0]:[0]:[00:00:00:00:00:aa]:[100.100.1.2] | 1:1,100:100,00:00:00:00:00:11 | 1.1.1.1  |
| Route Distinguisher: 2:2 (L3VNI 10000)            |                               |          |
| *>i [2]:[0]:[0]:[00:00:00:00:00:aa]:[200.200.1.2] | 2:2,100:100,00:00:00:00:00:22 | 3.3.3.3  |
| Route Distinguisher: 4.4.4.4:40 (L2VNI 400)       |                               |          |
| *>i [3]:[0]:[4.4.4.4]                             | 4:4,IGMP-MLD Proxy            | 4.4.4.4  |
| Total number of entries 7                         |                               |          |

### Showing BGP L2VPN information:

```

switch# show bgp l2vpn evpn 5.5.5.5:41-[6]:[0]:[30.1.36.30]:[232.0.0.8]:
VRF : default
BGP Local AS 65000      BGP Router-id 3.3.3.3

Network      : 5.5.5.5:41-[6]:[0]:[30.1.36.30]:[232.0.0.8]:[5.5.5.5]
Nextthop     : 5.5.5.5
vni          : 0                      vni_type      : L2VNI
Peer         : 2.2.2.2                Origin        : incomplete
Metric       : 0                      Local Pref    : 100
Weight       : 0                      Calc. Local Pref : 100
Best         : Yes                    Valid         : Yes
Type         : internal                Stale         : No

```

```

Originator ID      : 5.5.5.5
Aggregator ID      :
Aggregator AS      :
Atomic Aggregate   :

AS-Path            :
Cluster List       : 2.2.2.2
Communities        :
Ext-Communities    : RT: 139:139

```

Showing BGP L2VPN information for route type 6:

```

switch# show bgp l2vpn evpn route-type 6

Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN Route-Type 6 prefix: [6]:[EthTag]:[McastSrcIP]:[McastGrpIP]:[OrigIP]

VRF: default
Router-ID not configured

Network                               Nexthop   Metric LocPrf  Weight
  Path
-----
--
Route Distinguisher: 10.1.1.54:32967
*> [6]:[0]:[0.0.0.0]:[239.1.1.1]:[1.1.1.1]  1.1.1.1   0      100    32768
    i
*> [6]:[0]:[10.1.1.30]:[225.1.1.1]:[1.1.1.1]  1.1.1.1   0      100     0
    i
*> [6]:[0]:[10.1.1.40]:[225.1.1.2]:[1.1.1.1]  1.1.1.1   0      100     0
    ?
*> [6]:[0]:[0.0.0.0]:[0.0.0.0]:[10.1.1.1]    10.1.1.1  0      100     0
    i

Route Distinguisher: 2:2
*> [6]:[0]:[10.1.1.30]:[232.1.1.1]:[1.1.1.1]  1.1.1.1   0      100     0
    ?
*> [6]:[0]:[10.1.1.40]:[239.1.1.1]:[2.2.2.2]  2.2.2.2   0      100
    0
Total number of entries 5

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                       |
|------------------|----------------------------------------------------|
| 10.16            | Expanded route type range to include route type 6. |
| 10.07 or earlier | --                                                 |

## Command Information

| Platforms | Command context             | Authority                                                                          |
|-----------|-----------------------------|------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## show bgp l2vpn evpn vni route-type

show bgp l2vpn evpn vni <VNI-Value> route-type <ROUTE-TYPE-Value>

### Description

Shows the BGP L2VPN information for the particular EVPN VNI and routes type.

| Parameter          | Description                                       |
|--------------------|---------------------------------------------------|
| <VNI-Value>        | Specifies the VNI.                                |
| <ROUTE-TYPE-Value> | Specifies the routes filtered by NLRI route type. |

### Examples

Showing BGP L2VPN information for the particular EVPN VNI and route type:

```
switch# show bgp l2vpn evpn vni 30000 route-type 5
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN Route-Type 5 prefix: [5]:[ESI]:[EthTag]:[IPAddrLen]:[IPAddr]

VRF : default
Router-ID not configured
Network                               Nexthop           Metric LocPrf
Weight Path
-----
Route Distinguisher: 1:100             (L3VNI 10000)
*>i [5]:[0]:[0]:[24]:[32.32.32.0]      3.3.3.3           0      100    0
?
*> [5]:[0]:[0]:[24]:[52.52.52.0]      1.1.1.1           0      100    0
?
*>i [5]:[0]:[0]:[64]:[aaa::]          3.3.3.3           0      100    0
?

Total number of entries 3
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification                                       |
|------------------|----------------------------------------------------|
| 10.16            | Expanded route type range to include route type 6. |
| 10.07 or earlier | --                                                 |

### Command Information

| Platforms | Command context             | Authority                                                                          |
|-----------|-----------------------------|------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## show bgp l2vpn evpn vtep

show bgp l2vpn evpn vtep <IP-address>

### Description

Shows the BGP L2VPN information for the particular EVPN VTEP IP address.

| Parameter    | Description                    |
|--------------|--------------------------------|
| <IP-address> | Specifies the VTEP IP address. |

### Examples

Showing BGP L2VPN information for the particular EVPN VTEP IP:

```
switch# show bgp l2vpn evpn vtep 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete

EVPN Route-Type 2 prefix: [2]:[ESI]:[EthTag]:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]
EVPN Route-Type 5 prefix: [5]:[ESI]:[EthTag]:[IPAddrLen]:[IPAddr]
VRF : default
Local Router-ID 2.2.2.2
```

| Network                                         | LocPrf | Weight | Path | NextHop | Metric |
|-------------------------------------------------|--------|--------|------|---------|--------|
| -----                                           |        |        |      |         |        |
| Route Distinguisher: 1.1.1.1:2 (L2VNI 2)        |        |        |      |         |        |
| *>i [2]:[0]:[0]:[00:00:00:00:00:33]:[10.1.1.10] |        |        |      | 1.1.1.1 | 0      |
| 100                                             | 0      | ?      |      |         |        |
| *>i [2]:[0]:[0]:[00:00:00:00:00:33]:[1000::10]  |        |        |      | 1.1.1.1 | 0      |
| 100                                             | 0      | ?      |      |         |        |
| *>i [2]:[0]:[0]:[00:50:56:96:15:1c]:[10.1.1.1]  |        |        |      | 1.1.1.1 | 0      |
| 100                                             | 0      | ?      |      |         |        |
| *>i [2]:[0]:[0]:[00:50:56:96:15:1c]:[]          |        |        |      | 1.1.1.1 | 0      |
| 100                                             | 0      | ?      |      |         |        |
| *>i [3]:[0]:[1.1.1.1]                           |        |        |      | 1.1.1.1 | 0      |
| 100                                             | 0      | ?      |      |         |        |
| Route Distinguisher: 1.1.1.1:2 (L3VNI 10000)    |        |        |      |         |        |
| *>i [2]:[0]:[0]:[00:00:00:00:00:33]:[10.1.1.10] |        |        |      | 1.1.1.1 | 0      |
| 100                                             | 0      | ?      |      |         |        |
| *>i [2]:[0]:[0]:[00:00:00:00:00:33]:[1000::10]  |        |        |      | 1.1.1.1 | 0      |
| 100                                             | 0      | ?      |      |         |        |
| *>i [2]:[0]:[0]:[00:50:56:96:15:1c]:[10.1.1.1]  |        |        |      | 1.1.1.1 | 0      |
| 100                                             | 0      | ?      |      |         |        |
| *>i [2]:[0]:[0]:[00:50:56:96:15:1c]:[]          |        |        |      | 1.1.1.1 | 0      |
| 100                                             | 0      | ?      |      |         |        |
| Route Distinguisher: 1:100 (L3VNI 10000)        |        |        |      |         |        |

```
*>i [5]:[0]:[0]:[24]:[10.1.1.0] 1.1.1.1 0
100 0 ?
*>i [5]:[0]:[0]:[64]:[1000::] 1.1.1.1 0
100 0 ?
Total number of entries 11
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                          |
|-----------|-----------------------------|------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## show bgp l2vpn evpn vtep route-type

show bgp l2vpn evpn vtep <IP-address> route-type <ROUTE-TYPE-Value>

### Description

Shows the BGP L2VPN information for the particular EVPN VTEP IP address and routes type.

| Parameter          | Description                                       |
|--------------------|---------------------------------------------------|
| <IP-address>       | Specifies the VTEP IP address.                    |
| <ROUTE-TYPE-Value> | Specifies the routes filtered by NLRI route type. |

## Examples

Showing BGP L2VPN information for the particular EVPN VTEP and route type:

```
switch# show bgp l2vpn evpn vtep 1.1.1.1 route-type 5
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN Route-Type 5 prefix: [5]:[ESI]:[EthTag]:[IPAddrLen]:[IPAddr]

VRF : default
Router-ID not configured

      Network                                          Nexthop
Metric LocPrf Weight Path
-----
Route Distinguisher: 1:100                          (L3VNI 10000)
```

```
*>i [5]:[0]:[0]:[24]:[32.32.32.0] 1.1.1.1 0
100 0 ?
*> [5]:[0]:[0]:[24]:[52.52.52.0] 1.1.1.1 0
100 0 ?
*>i [5]:[0]:[0]:[64]:[aaa::] 1.1.1.1 0
100 0 ?

Total number of entries 3
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                          |
|-----------|-----------------------------|------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## show bgp l2vpn evpn vtep vni

show bgp l2vpn evpn vtep <IP-address> vni <VNI-Value>

### Description

Shows the BGP L2VPN information for the particular EVPN VTEP IP address and VNI.

| Parameter    | Description                    |
|--------------|--------------------------------|
| <IP-address> | Specifies the VTEP IP address. |
| <VNI-Value>  | Specifies the VNI.             |

## Examples

Showing BGP L2VPN information for the particular EVPN VTEP IP and VNI:

```
switch# show bgp l2vpn evpn vtep 1.1.1.1 vni 10000
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete

EVPN Route-Type 2 prefix: [2]:[ESI]:[EthTag]:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]
EVPN Route-Type 5 prefix: [5]:[ESI]:[EthTag]:[IPAddrLen]:[IPAddr]
VRF : default
Local Router-ID 2.2.2.2
```

| Network                                         | LocPrf | Weight | Path | NextHop | Metric |
|-------------------------------------------------|--------|--------|------|---------|--------|
| -----                                           |        |        |      |         |        |
| Route Distinguisher: 1.1.1.1:2 (L3VNI 10000)    |        |        |      |         |        |
| *>i [2]:[0]:[0]:[00:00:00:00:00:33]:[10.1.1.10] |        |        |      | 1.1.1.1 | 0      |
| 100                                             |        | 0      | ?    |         |        |
| *>i [2]:[0]:[0]:[00:00:00:00:00:33]:[1000::10]  |        |        |      | 1.1.1.1 | 0      |
| 100                                             |        | 0      | ?    |         |        |
| *>i [2]:[0]:[0]:[00:50:56:96:15:1c]:[10.1.1.1]  |        |        |      | 1.1.1.1 | 0      |
| 100                                             |        | 0      | ?    |         |        |
| *>i [2]:[0]:[0]:[00:50:56:96:15:1c]:[]          |        |        |      | 1.1.1.1 | 0      |
| 100                                             |        | 0      | ?    |         |        |
| Route Distinguisher: 1:100 (L3VNI 10000)        |        |        |      |         |        |
| *>i [5]:[0]:[0]:[24]:[10.1.1.0]                 |        |        |      | 1.1.1.1 | 0      |
| 100                                             |        | 0      | ?    |         |        |
| *>i [5]:[0]:[0]:[64]:[1000::]                   |        |        |      | 1.1.1.1 | 0      |
| 100                                             |        | 0      | ?    |         |        |
| Total number of entries 6                       |        |        |      |         |        |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                          |
|-----------|-----------------------------|------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## show bgp l2vpn evpn vtep vni route-type

show bgp l2vpn evpn vtep <VTEP-ID> vni <VNI-Value> route-type <ROUTE-TYPE-Value>

## Description

Shows the BGP L2VPN information for the particular EVPN VTEP, VNI, and route type.

| Parameter          | Description               |
|--------------------|---------------------------|
| <VTEP-ID>          | Specifies the VTEP.       |
| <VNI-Value>        | Specifies the VNI.        |
| <ROUTE-TYPE-Value> | Specifies the route type. |

## Examples

Showing BGP L2VPN information for the particular EVPN VTEP, VNI and route type 2:

```

switch# show bgp l2vpn evpn vtep 1.1.1.1 vni 10000 route-type 2
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, e external S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete

EVPN Route-Type 2 prefix: [2]:[ESI]:[EthTag]:[MAC]:[OrigIP]
EVPN Route-Type 3 prefix: [3]:[EthTag]:[OrigIP]
EVPN Route-Type 5 prefix: [5]:[ESI]:[EthTag]:[IPAddrLen]:[IPAddr]
VRF : default
Local Router-ID 2.2.2.2

      Network
      Weight  Path
-----
Route Distinguisher: 1.1.1.1:2 (L3VNI 10000)
*>i [2]:[0]:[0]:[00:50:56:96:7d:03]:[10.1.1.1] 1.1.1.1 0 100 0
      ?
*>i [2]:[0]:[0]:[00:50:56:96:7d:03]:[] 1.1.1.1 0 100 0
      ?

Total number of entries 3

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                          |
|-----------|-----------------------------|------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## show ip ospf routes

```

show ip ospf [<PROCESS-ID>] routes
             [<IPV4-ADDR>/<MASK>] protected [all-vrfs | vrf <VRF-NAME>]

```

## Description

Displays OSPF routing information.

| Parameter   | Description                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| <IPV4-ADDR> | Specify an IP address in IPv4 format ( <b>x.x.x.x</b> ), where <b>x</b> is a decimal number from 0 to 255.                     |
| <MASK>      | Specifies the number of bits in the address mask in CIDR format ( <b>x</b> ), where <b>x</b> is a decimal number from 0 to 32. |



| Parameter      | Description                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| <PROCESS-ID>   | Enter an OSPF process ID to display OSPF neighbor information for the particular OSPF process. Range: 1 to 65535. |
| all-vrfs       | Select to display neighbor information for all VRFs.                                                              |
| vrf <VRF-NAME> | Specify the name of a VRF. Default: default.                                                                      |
| protected      | Selects a protected route. Specifying an IPv4 address will select protected route for that prefix.                |

## Examples

Showing OSPF routing table information:

```
switch# show ip ospf routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1 VRF default, Routing Table
-----

Total Number of Routes : 2

10.1.1.0/24      (i) area: 0.0.0.0
    directly attached to interface 1/1/1, cost 1 distance 110
20.1.1.0/24      (I)
    via 10.1.1.2 interface 1/1/1, cost 2 distance 110
```

Showing OSPF routing table information for a specific subnet:

```
switch# show ip ospf routes 10.1.1.0/2
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1 VRF default, Routing Table for prefixes 10.1.1.0/24
-----

Total Number of Routes : 1

10.1.1.0/24      (i) area: 0.0.0.0
    directly attached to interface 1/1/1, cost 1 distance 110
```

Showing OSPF routing table information for a specific protected subnet on the default VRF:

```
switch# show ip ospf routes 211.0.0.0/24 protected vrf default
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1, VRF default, Protected Routes for prefixes 211.0.0.0/24
-----

211.0.0.0/24      (i) area: 0.0.0.0
    via 110.110.110.1, interface vlan20, cost 122
    bkp 130.130.130.1, interface vlan22, cost 132
```

```
Total Number of Routes : 1
```

Showing OSPF routing table information for a specific protected subnet on all VRFs

```
switch# show ip ospf routes 211.0.0.0/24 protected all-vrfs
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1, VRF default, Protected Routes for prefixes 211.0.0.0/24
-----

211.0.0.0/24      (i) area: 0.0.0.0
                  via 110.110.110.1, interface vlan20, cost 122
                  bkp 130.130.130.1, interface vlan22, cost 132

Total Number of Routes : 1
```

Showing OSPF routing table information for a specific protected subnet:

```
switch# show ip ospf routes 211.0.0.0/24 protected
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1, VRF default, Protected Routes for prefixes 211.0.0.0/24
-----

211.0.0.0/24      (i) area: 0.0.0.0
                  via 110.110.110.1, interface vlan20, cost 122
                  bkp 130.130.130.1, interface vlan22, cost 132

Total Number of Routes : 1
```

Showing OSPF routing table information for all protected routes:

```
switch# show ip ospf routes 211.0.0.0/24 protected
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1, VRF default, Protected Routes
-----

120.120.120.0/24  (i) area: 0.0.0.0
                  via 110.110.110.1, interface vlan20, cost 120
                  bkp 130.130.130.1, interface vlan22, cost 230

140.140.140.0/24  (i) area: 0.0.0.0
                  via 110.110.110.1, interface vlan20, cost 130
                  bkp 130.130.130.1, interface vlan22, cost 220

211.0.0.0/24      (i) area: 0.0.0.0
                  via 110.110.110.1, interface vlan20, cost 122
                  bkp 130.130.130.1, interface vlan22, cost 132

Total Number of Routes : 3
```

```
OSPFs Process ID 1, VRF red, Protected Routes
```

```
-----  
Total Number of Protected Routes: 0
```

Showing OSPF routing table information for protected routes on the default VRF:

```
switch# show ip ospf routes protected vrf default  
Codes: i - Intra-area route, I - Inter-area route  
       E1 - External type-1, E2 - External type-2  
  
OSPF Process ID 1, VRF default, Protected Routes  
-----  
  
140.140.140.0/24      (i) area: 0.0.0.0  
    via 130.130.130.1, interface vlan22, cost 130  
    bkp 110.110.110.1, interface vlan20, cost 220  
  
211.0.0.0/24         (i) area: 0.0.0.0  
    via 110.110.110.1, interface vlan20, cost 122  
    bkp 130.130.130.1, interface vlan22, cost 132  
  
120.120.120.0/24     (i) area: 0.0.0.0  
    via 110.110.110.1, interface vlan20, cost 120  
    bkp 130.130.130.1, interface vlan22, cost 230  
  
Total Number of Routes : 3
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.15            | Introduces the <b>protected</b> parameter.                |
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show running-config bgp

show running-config bgp

## Description

This command shows all configured BGP commands.

## Examples

```
switch# show running-config bgp
router bgp 65534.65535
  bgp asnotation dotted
  network 2.2.2.0/24
  neighbor 1.1.1.2 remote-as 65533.65535
  address-family ipv4 unicast
    neighbor 1.1.1.2 activate
    neighbor 1.1.1.2 route-map A out
  vrf v1
  address-family l2vpn evpn
    neighbor 1.1.1.2 activate
    neighbor 1.1.1.2 send-community extended
  exit-address-family
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420      | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## timers bgp

```
timers bgp <KEEPALIVE> <HOLDTIME>
no timers bgp <KEEPALIVE> <HOLDTIME>
```

## Description

The command adjusts BGP network timers.

The **no** form of this command resets the BGP timers to defaults of 60 seconds for the keepalive timer and 180 seconds for the holdtime timer.

| Parameter   | Description                                                              |
|-------------|--------------------------------------------------------------------------|
| <KEEPALIVE> | Sets the value for keepalive timer. Default: 60 seconds. Range: 0-65535. |
| <HOLDTIME>  | Sets the value for holdtime timer. Default: 180 seconds. Range: 0-65535. |

## Usage

- The keepalive timer is the number of seconds a BGP peer waits for a keep-alive message from a BGP peer before deciding the connection is down.

The holdtime timer is the number of seconds a BGP peer waits after not receiving a keepalive, update, or notification message before declaring that a connection with BGP peer is down.

- When a session is started, BGP negotiates holdtime with the neighbor, and selects the smaller value. The keepalive timer is then set based on the negotiated holdtime and the configured keepalive time.

## Examples

```
switch(config-bgp)# timers bgp 100 150
switch(config-bgp)# no timers bgp
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config-bgp      | Administrators or local user group members with execution rights for this command. |

## set extcommunity

```
set extcommunity "[rt <VALUE> | evpn-rmac <MAC-ADDRESS>]"
no set extcommunity [rt | evpn-mac]
```

## Description

Sets the extended community number attribute for a route matching the route map. Extended communities are supported only on Route Targets. This command is applicable to OSPF, static, and connected routes which will be redistributed to the BGP protocol.

The **no** form of this command restores the default behavior of not modifying the extended community number attribute of the route.

| Parameter     | Description                                                                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <VALUE>       | Sets the extended community number attribute. Specify the information in asn:nn format.                                                                                                  |
| <MAC-ADDRESS> | Specifies MAC address of the Router-MAC extended community. When configuring a route map and using the <b>set ip next-hop</b> command, this value can be set for the OUT direction only. |

## Usage

- Multiple community numbers can be configured within the double quotes.
- 2-byte and 4-byte ASN values are supported in the global administrator component of the extended community attribute.
- 4-byte ASN values must be within the range of 1-4294967295.
- 4-byte ASN values do not support dotted notation.
- Extended communities are only supported on route targets.
- The

## Examples

Configuring a set clause in a route-map to modify the community number attribute of the route:

```
switch(config)# route-map abc permit seq 10
switch(config-route-map-abc-10)# set extcommunity rt "1:1 2:2"
```

Configuring a set clause in a route-map to modify the router mac:

```
switch(config)# route-map abc permit seq 1
switch(config-route-map-abc-1)# set extcommunity evpn-mac 00:01:01:90:90:01
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification       |
|---------|--------------------|
| 10.10   | Command introduced |

## Command Information

| Platforms | Command context         | Authority                                                                          |
|-----------|-------------------------|------------------------------------------------------------------------------------|
| 5420      | config-route-map-abc-10 | Administrators or local user group members with execution rights for this command. |

## Chapter 7

# Equal Cost Multipath (ECMP)

---



---

ECMP is not supported on the HPE Aruba Networking 4100i Switch Series.

---



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ECMP is not supported on the HPE Aruba Networking 6000 and 6100 Switch Series.

---

Equal Cost Multipath (ECMP) allows a router to load balance traffic destined to some network that is reachable through multiple equal cost route nexthops. ECMP functionality is dependent on L3 routing being enabled on the router.

## Overview

The nexthop chosen out of the ECMP group of nexthops for a given packet is typically determined by its header information, usually by hashing source and destination IP addresses as well as potentially source and destination L4 ports. The hashing algorithm used is dependent on the type of switch in use, and user configuration as well as whether the packet is flowing through the slow-path or fast-path.

By default all of the following fields are enabled and utilized for the load balancing decision.

- L3 Destination IP address
- L3 Source IP address
- L4 Destination port
- L4 Source port

ECMP support is NOT dependent on address family. The ECMP feature applies to both IPv4 and IPv6 traffic flowing through the router.



---

If two different routing protocols are (mis)configured to use the same Administrative Distance, then they have the opportunity to create ECMP groups together.

---

## ECMP commands

### show ip ecmp

```
show ip ecmp
```

#### Description

Displays the Equal Cost Multipath (ECMP) configuration.

#### Examples

Showing output for HPE Aruba Networking 5420, 6200, 6300,6400, 8320,8325 and 8360 Switch series:

```
switch# show ip ecmp

ECMP Configuration
-----

ECMP Status          : Enabled

ECMP Load Balancing by
-----
Source IP             : Enabled
Destination IP        : Enabled
Source Port           : Enabled
Destination Port      : Enabled
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |



# Chapter 8

## Virtual Router Redundancy Protocol

---



VRRP is not supported on the HPE Aruba Networking 4100i Switch Series.



VRRP is not supported on the HPE Aruba Networking 6000 and 6100 Switch Series.

This section describes how to configure and verify VRRP configuration and operational states.

- All VRRP configurations work in VRRP context.
- VRRP can be configured on physical ports, VLAN interfaces, and LAG interfaces.
- VRRP mandates the associated interface to be routing interface.

### Overview

In many networks, edge devices are configured to send packets to a statically configured default router. If this router becomes unavailable, devices that use it as a first-hop router become isolated from the network.

VRRP uses dynamic failover to ensure the availability of an end node default router. The IP address used as the default route is assigned to a virtual router (VR). The VR includes:

- An Active router assigned to forward traffic designated for the virtual route
- One or more prioritized Standby routers (If a Standby is forwarding traffic for the VR, it has replaced the Owner as the Active router.)

This redundancy provides a backup for gateway IP addresses (first-hop routers). If a VR Active router becomes unavailable, the traffic it supports will be transferred to a Standby router without major delays or operator intervention. This operation can eliminate single-point-of-failure problems and provide dynamic failover (and failback) support. As long as one physical router in a VR configuration is available, IP addresses assigned to the VR are always available. Edge devices can send packets to these IP addresses without interruption.

Advantages to using VRRP include:

- Minimizing failover time and bandwidth overhead if a primary router becomes unavailable
- Minimizing service disruptions during a failover
- Providing backup for a load-balanced routing solution
- Avoiding the need to make configuration changes in the end nodes if a gateway router fails
- Eliminating the need for router discovery protocols to support failover operation

### Terminology

**Owner router:** The Owner router is automatically configured with the highest VRRP router priority in the VR (255). It operates as the Active router for the VR unless it becomes unavailable to the network. All

Virtual Router members can be configured so virtual IP is not the same as physical (or real) IP. Such virtual address can be called pure virtual IP address.

**Active router:** The Owner or Standby router that is the physical forwarding agent for routed traffic using the VR as a gateway. There can be only one router operating as the Active for a network. If the router configured as the Owner for a VR is available to the network, it will also be the Active. If the Owner fails or loses availability to the network, the highest-priority Standby becomes the Active.

**Standby:** A router configured in a VR as a Standby to the Owner/Active for the same VR. There must be a minimum of one Standby in a VR to support VRRP operation if the Owner/Active fails. Every Standby is created with a configurable priority (default: 100). This priority determines the precedence for becoming the Active of the VR if the Owner or another Standby operating as Active becomes unavailable.

When the current Active router is no longer available, the Standby router with highest priority will become the current Active. When a router with higher priority becomes available, it is switched to Active. The user can override this behavior by disabling preemption mode.

**Virtual Router (VR):** Consists of one Owner/Active router and one or more Standby routers that belong to the same network. The Owner is the router that owns any IP addresses associated with the VR. The VR has one virtual IP address (multiple virtual IP addresses for multinetted interfaces) that correspond to a real IP address on the Owner. A VR includes the following:

- A VR identification (VRID) configured on all VRRP routers in the same network. For a multinetted interface, the VRID is configured on all routers in the same subnet.
- The same VIP configured on each instance of the same VR.
- The status of either Owner or Standby configured on each instance of the same VR. There can be one Owner and one or more Standbys configured on each VR.
- The priority level configured on each instance of the VR. On the Owner router, the highest priority setting of 255 is automatically fixed. On Standbys, the default priority setting is 100 and configurable.
- A VR MAC address that is not configurable.

**Virtual IP address (VIP):** The VIP associated with a VR must be a real IP address already configured in the associated interface on the Owner router.

- If the VIP is an IPv6 address, a link-local address must be configured before adding a global IPv6 address.
- The Owner and all Standby routers belonging to the VR have this IP address configured in their VRID contexts as the VIP.
- A subnetted interface allows multiple VIPs. If there are fewer IP addresses in an interface and a user wants VRRP support on multiple subnets, configure a separate VR instance for each IP address in the interface.

**VRID:** The identifier for a VR configured on an interface. A VRID can be used for only one VR in any interface on a router. It can be used again for a different VR in a different interface.

## VRRP operation

VRRP supports router redundancy through a prioritized election process among routers configured as members of the same virtual router (VR).

A VR includes two or more member routers configured with:

- A virtual IP address that is also configured as a real IP address on one of the routers
- A virtual router MAC address

The router that owns the IP address is configured to operate as the Owner of the VR for traffic-forwarding purposes. By default, this router has the highest VRRP priority in the VR. Any other routers in the VR have a lower priority and are configured to operate as Standbys if the Owner router becomes unavailable.

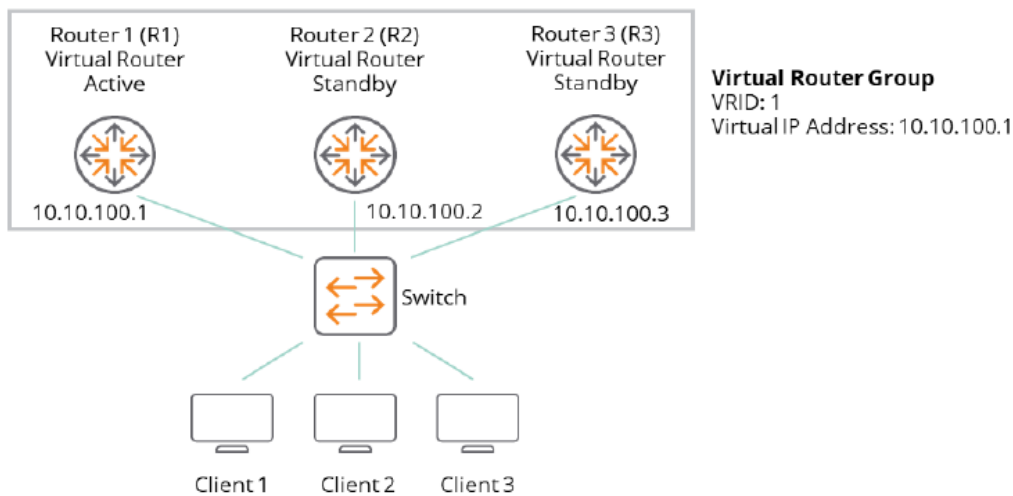
The Owner normally operates as the Active for a VR. If the Owner becomes unavailable, a failover to a Standby router belonging to the same VR occurs. This Standby becomes the Active. If the Owner recovers, a failback occurs, and Active status reverts to the Owner.



- Using more than one Standby provides additional redundancy. If the Owner and the highest-priority Standby fail, another lower-priority Standby can take over as Active.
- In a VRRP owner scenario, duplicate address detection (DAD) should be disabled for VRRP to work.

The image shown here depicts a basic VLAN topology. In this example, Routers 1, 2 and 3 form a VRRP group. The IP address of the group is the same address configured for the SVI interface of R1 (10.10.100.1).

**Figure 1** *Basic VLAN Topology*



Because the virtual IP address uses the IP address of the SVI interface of Router R1, Router R1 is the Active (also known as IP address Owner).

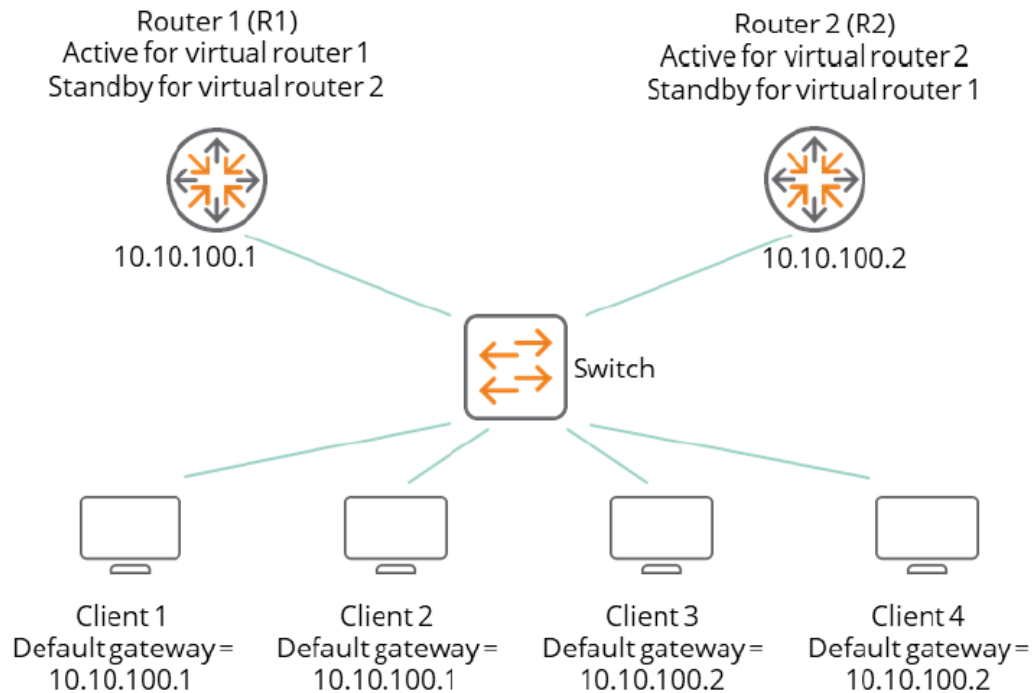
- As the Active, R1 owns the virtual IP address of the VRRP group and forwards packets sent to this IP address.
- Clients 1 through 3 are configured with the default gateway IP address of 10.10.100.1.
- Routers R2 and R3 function as Standbys. If the Active fails, the Standby router with highest priority becomes the Active and takes over the virtual IP address to provide uninterrupted service for the clients connected to the L2 Switch. When Router R1 recovers, it becomes the Active router again.

## Multiple VRRP groups

A user can configure multiple VRRP groups on a router (or L3) interface (Physical, LAG, or SVI interfaces). In a topology in which multiple VRRP groups are configured on a router interface, the interface can act as an Active for one VRRP group and a Standby for one or more other VRRP groups.

This image displays a topology in which VRRP is configured so that Routers R1 and R2 share the traffic to and from clients 1 through 4. Routers R1 and R2 act as Standbys to each other if either router fails.

**Figure 1** *Routers share the traffic to and from clients 1 through 4.*



This topology contains two virtual IP addresses for two VRRP groups that overlap. For VRRP group 1, Router R1 is the owner of IP address 10.10.100.1 and is the Active. Router R2 is the Standby to Router R1. Clients 1 and 2 are configured with the default gateway IP address of 10.10.100.1.

For VRRP group 2, Router R2 is the owner of IP address 10.10.100.2 and is the Active. Router R1 is the Standby to Router R2. Clients 3 and 4 are configured with the default gateway IP address of 10.0.0.2.

## VRRP priority

In a Standby router VR configuration, the virtual router priority defaults to 100. The priority for the configured Owner is automatically set to the highest value of 255.

In a VR in which there are two or more Standby routers, priority settings can be reconfigured to define the order in which Standbys will be reassigned as Active if a failover occurs.

## VRRP preemption

In a Standby router VR configuration, the virtual router priority defaults to 100. The priority for the configured Owner is automatically set to the highest value of 255.

When multiple Standby routers exist in a VR:

- If the current Active fails and the highest-priority Standby is not available, VRRP selects the next-highest priority Standby to operate as Active.
- If the highest-priority Standby later becomes available, it preempts the lower-priority Standby and takes over the Active function.

To remove this preemptive ability on a VR, disable it with the `no preempt` command.



---

Preemption applies only to VRRP routers configured as Standbys.

---

## Virtual Router MAC address

When a Virtual Router (VR) instance is configured, the protocol automatically assigns a MAC address based on the standard MAC prefix for VRRP packets plus the VRID number (as described in RFC 3768). The first five octets form the standard MAC prefix for VRRP and the last octet is the configured VRID. For example: 00-00-5E-00-01-<VRid>

## VRRP and ARP

The current Active for a VR responds to ARP requests for the virtual IP addresses with the VR-assigned MAC address. The virtual MAC address is also used as source MAC address for periodic advertisements sent by the current Active.

The VRRP router responds to ARP requests for non-virtual IP addresses with the system MAC address. Non-virtual IP addresses are not configured as virtual IP addresses for any VR on the interface.

## VRRP tracking

VRRP supports object tracking. This functionality allows monitoring of the state of a configured object. The state determines the priority of the VRRP router in a VRRP group. A track object can be created using the `track <OBJECT-ID>` command and the object can be associated with an interface. A change in interface state will then affect the priority of a VRRP group.

## High availability

VRRP supports high availability through stateful restarts and stateful switchovers.

- A stateful restart occurs when the VRRP process (or daemon) fails and is restarted.
- A stateful switchover occurs when the active management module (AMM) switches to the standby management module (SMM).

## VRRP and Neighbor Discovery for IPv6

Neighbor Discovery (ND) is the IPv6 equivalent of the IPv4 ARP for layer 2 address resolution. It uses IPv6 ICMP messages to do the following:

- Determine the link-layer address of neighbors
- Verify that a neighbor is reachable
- Track neighbor (local) routers

Neighbor Discovery enables functions such as:

- Router and neighbor solicitations and discovery
- Detecting address changes for devices
- Identifying a replacement for a router or router path that has become unavailable
- Duplicate address detection (DAD)
- Router Advertisement processing
- Neighbor reachability

- Resolution of destination addresses
- Changes to link-layer addresses

An instance of Neighbor Discovery is triggered on a device when a new or changed IPv6 address is detected. VRRPv3 provides a faster failover to a Standby router by not using standard ND procedures. A failover to a Standby router can occur in approximately three seconds without any interaction with hosts and a minimum of VRRPv3 traffic.

## Duplicate address detection (DAD)

Duplicate address detection verifies that a configured unicast IPv6 address is unique before it is assigned to an interface. When the Owner router fails, the Standby VRRP router assumes the Active role. When the Owner router becomes operational, DAD will fail because a Standby VRRP router is in the Active role that responds to the DAD request. To avoid this, on virtual routers that are in Owner mode (priority = 255) DAD needs to be disabled for the interface on which the Owner VR is configured.

## Guidelines and limitations

- VRRP must be enabled at the global level using the `router vrrp enable` command. It must also be enabled at the interface level at which you configure VRRP.
- A maximum of 8 IPv4 and 8 IPv6 VRRP groups are supported on an interface.
- A maximum of 256 VRRP groups is supported on a router. The groups can be IPv4 or IPv6.
- Supported virtual router identification (VRID) range is 1-255.
- Proxy-ARP and Active-Gateway are not supported in conjunction with VRRP.

## VRRP commands

### address (vrrp)

```
address <IP-ADDR> [ primary | secondary ]
no address <IP-ADDR> [ primary | secondary ]
```

### Description

Configures a primary or secondary IPv4 or IPv6 address for the VRRP group. To use secondary IP addresses in a VRRP group, you must first configure a primary IP address on the same group. A maximum of 16 IP addresses per IPv4 VRRP group and 8 IPv6 addresses per IPv6 VRRP group are supported.




---

Do not configure an IPv4 VRRP group using addresses from the /30, /31, and /32 subnets of the interface IP address.

---

16 Virtual IP addresses per IPv4 VR and 8 Virtual IP addresses per IPv6 VR are supported.

---




---

The total number of VIPs supported by a switch is:

- 1024 VIPs for IPv4 VRs
  - 512 VIPs for IPv6 VRs
-

The **no** form of this command deletes a primary or secondary IPv4 or IPv6 address from the VRRP group.

| Parameter | Description                          |
|-----------|--------------------------------------|
| <IP-ADDR> | Configures the IPv4 or IPv6 address. |
| primary   | Configures a primary address.        |
| secondary | Configures a secondary address.      |

## Examples

```
switch(config)# system internal-vlan-range 4041-4050
switch(config)# router vrrp enable
```

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# address 10.0.0.1 primary
```

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv6
switch(config-if-vrrp)# address fe80::1 primary
```

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no address 10.0.0.1 primary
```

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv6
switch(config-if-vrrp)# no address fe80::1 primary
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## authentication (vrrp)

```
authentication {text | md5} [{plaintext | ciphertext} <KEY>]
```

no authentication

## Description

This command enables authentication mode and the authentication key for VRRP groups on an interface. VRRP members or routers of the same VRRP group must use the same authentication mode and authentication key.

The no form of this command disables authentication mode and the authentication key for VRRP groups on an interface.

IPv4 VRRPv3 and IPv6 VRRPv3 do not support VRRP packet authentication. Authentication mode and key configuration take effect only in VRRPv2 (IPv4 only - RFC2338).

In VRRPv3, authentication mode and authentication key settings do not take effect because VRRP Authentication was removed from RFC5798.

VRRP provides the following authentication modes as described in RFC2338:

### Simple authentication

The sender fills an authentication key into the VRRP packet and the receiver compares the received authentication key with its local authentication key.

If the two authentication keys match, the received VRRP packet is legitimate. Otherwise, the received packet is illegitimate and is discarded.



---

Authentication key text is sent in the clear and can be seen in a packet trace. This makes MD5 authentication more secure than text.

---

### MD5 authentication

The sender computes a digest for the packet that will be sent using the authentication key and MD5 algorithm, and saves the result in the VRRP packet.

The receiver performs the same operation with the authentication key and MD5 algorithm, and compares the result with the content in the authentication header. If the results match, the received VRRP packet is legitimate. Otherwise, the received packet is illegitimate and is discarded.

| Parameter  | Description                                              |
|------------|----------------------------------------------------------|
| text       | Configures the simple authentication type.               |
| md5        | Configures the MD5 (message-digest) authentication type. |
| plaintext  | Specifies that the key is provided as plaintext.         |
| ciphertext | Specifies that the key is provided as ciphertext.        |
| <KEY>      | Specifies the key in the chosen format.                  |



---

When the key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

---

## Examples

Enabling VRRP authentication using MD5 with a provided plaintext key:



```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# authentication md5 plaintext testvrrpkey
```

Enabling VRRP authentication using MD5 with a prompted plaintext key:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# authentication md5
Enter the authentication key: *****
Re-Enter the authentication key: *****
```

Enabling VRRP authentication using MD5 with a provided ciphertext key:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# authentication md5 ciphertext
AQBapfciFZ/P...biBAAAAOjc0a8=
```

Disabling VRRP authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# no authentication
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if-vrrp  | Administrators or local user group members with execution rights for this command. |

## preempt

preempt  
no preempt

## Description

Enables the preempt option. The default value is enabled. In default mode, a Standby router with a higher priority than another Standby that is operating as Active will take over the Active function.

Applies to VRRP Standby routers only and is used to minimize network disruption caused by unnecessary preemption of the Active operation among Standby routers.

The **no** form of this command disables the preempt option, thus preventing the higher-priority Standby from taking over the Active operation from a lower-priority Standby. This command does not prevent an owner router from resuming the Active function after recovering from being unavailable.

## Examples

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# preempt
```

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no preempt
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## preempt delay minimum

```
preempt delay minimum <DELAY-IN-SECONDS>
no preempt delay minimum <DELAY-IN-SECONDS>
```

## Description

Sets the time in seconds (1-3600) that the router will wait before taking control of the virtual IP and starting to route packets.

The **no** form of this command sets the preempt delay for the VRRP group to the default preempt delay of 0 seconds.

The VRRP Preempt Delay Timer (PDT) allows admin users to configure a period of time before the VR takes control of the virtual IP address. It does not transition to the Active state until the timer period expires.

The timer value configured should be long enough to allow upper layer protocol to converge. The PDT is applied during initialization and down/up events of the router.

| Parameter          | Description                                                 |
|--------------------|-------------------------------------------------------------|
| <DELAY-IN-SECONDS> | Selects the time in seconds (1-3600). Default is 0 seconds. |

## Examples

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# preempt delay minimum 30

switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no preempt delay
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## priority

```
priority <1-254>
no priority
```

## Description

Sets the priority for the VRRP group.

The **no** form of this command sets the priority for the VRRP group as default priority.

- The default value for non-Owner virtual routers is 100.
- The default value for Owner virtual router is 255, which cannot be changed.

## Examples

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# priority 150
```

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no priority
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## router vrrp {enable | disable}

```
router vrrp {enable | disable}
no router vrrp {enable | disable}
```

## Description

Enables or disables VRRP protocol globally. You must globally enable the VRRP feature for VRRP virtual router.

**no router vrrp enable** disables VRRP protocol globally but does not remove all VRRP configurations.

**no router vrrp disable** enables VRRP protocol globally.

## Example

Enabling VRRP protocol globally:

```
switch(config)# router vrrp enable
```

Disable VRRP protocol globally:

```
switch(config)# router vrrp disable
```

Disable VRRP protocol globally:

```
switch(config)# no router vrrp enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## no router vrrp

no router vrrp

### Description

Removes VRRP configuration and VRRP global protocol. If **auto-confirm** is enabled or VRRP is not configured on any interface, this command will not ask for user confirmation.

### Examples

Removing VRRP configuration:

```
switch(config)# no router vrrp
All VRRP configuration will be deleted.
Do you want to continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## show track

show track [brief | <OBJECT-ID>]

### Description

Shows all or specific track object information.

| Parameter   | Description                                                    |
|-------------|----------------------------------------------------------------|
| brief       | Displays brief information about all or specific track objects |
| <OBJECT-ID> | Displays information about a specified track object (1-128)    |

## Examples

```
switch# show track
Track 1
  interface 1/1/1
  Interface is DOWN
```

```
switch# show track brief
Track  Interface      State
1       1/1/1         Down
2       None          Down
3       1/1/2         Up
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                                                                                                              |
|--------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Manager (#)     | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show track brief

show track brief

### Description

Shows brief information for all track objects.

### Examples

Showing brief information for all track objects:

```
switch# show track brief
Track  Interface      State
1       1/1/1         Down
```

|   |       |      |
|---|-------|------|
| 2 | None  | Down |
| 3 | 1/1/2 | Up   |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                                                                                                              |
|--------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Manager (#)     | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show vrrp

```
show vrrp [brief | detail | interface <INTERFACE-NAME> | interface <LAG-NAME> | interface
<VLAN-NAME> | ip | ipv6 | statistics | statistics interface <INTERFACE-NAME> | statistics
interface <LAG-NAME> | statistics interface <VLAN-NAME>]
```

## Description

Shows all VRRP virtual routers information.

| Parameter                  | Description                                                                                                                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| brief                      | Displays brief output of all VRRP virtual routers<br>Keywords used in displayed information:<br>Grp: VRRP virtual router group ID.<br>A-F: Address Family.<br>Pri: Priority.<br>Time: Uptime of VRRP virtual router since it moved out of INIT state.<br>Pre: Preempt mode (Y is enabled, N if not enabled). |
| detail                     | Displays detailed output of all VRRP virtual routers                                                                                                                                                                                                                                                         |
| interface <INTERFACE-NAME> | Displays VRRP information for a specific interface                                                                                                                                                                                                                                                           |
| interface <LAG-NAME>       | Displays VRRP information for a specific LAG interface                                                                                                                                                                                                                                                       |
| interface <VLAN-NAME>      | Displays VRRP information for a specific VLAN interface                                                                                                                                                                                                                                                      |
| ip                         | Displays the IP address family                                                                                                                                                                                                                                                                               |
| ipv6                       | Displays IPv6 address family                                                                                                                                                                                                                                                                                 |
| statistics                 | Displays VRRP statistics information for all interfaces                                                                                                                                                                                                                                                      |

| Parameter                                | Description                                                        |
|------------------------------------------|--------------------------------------------------------------------|
| statistics interface<br><INTERFACE-NAME> | Displays VRRP statistics information for a specific interface      |
| statistics interface<br><LAG-NAME>       | Displays VRRP statistics information for a specific LAG interface  |
| statistics interface<br><VLAN-NAME>      | Displays VRRP statistics information for a specific VLAN interface |

## Examples

```
switch# show vrrp
VRRP is enabled

Interface 1/1/1 - Group 1 - Address-Family IPv
    State is ACTIVE
    State duration 56 mins 57.826 secs
    Virtual IP address is 10.0.0.1
    Virtual MAC address is 00:00:5e:00:01:01
    Advertisement interval is 1000 msec
    Preemption enabled
    Priority is 100
    Active Router is 10.0.0.2 (local), priority is 100
    Active Advertisement interval is 1000 msec
    Active Down interval is unknown
    Tracked object ID is 1, and state Down

Interface 1/1/2 - Group 1 - Address-Family IPv4
    State is INIT (Interface Down)
    State duration 45 mins 28.313 secs
    Virtual IP address is no address
    Virtual MAC address is 00:00:5e:00:01:01
    Advertisement interval is 1000 msec
    Preemption enabled
    Priority is 100
    Active Router is unknown, priority is unknown
    Active Advertisement interval is unknown
    Active Down interval is unknown

Interface 1/1/2 - Group 1 - Address-Family IPv6
    State is INIT (Group Disabled)
    State duration 20 mins 19.794 secs
    Virtual IP address is no address
    Virtual secondary IP addresses:
        2201:13::110:4
    Virtual MAC address is 00:00:5e:00:02:01
    Advertisement interval is 1000 msec
    Preemption enabled
    Priority is 100
    Active Router is unknown, priority is unknown
    Active Advertisement interval is unknown
    Active Down interval is unknown
```

```
switch# show vrrp brief

VRRP is enabled
```



| Interface | Grp | A-F  | Pri | Time | Owner | Pre | State  | Active addr/Group addr   |
|-----------|-----|------|-----|------|-------|-----|--------|--------------------------|
| 1/1/1     | 1   | IPv4 | 100 | 0    | N     | Y   | ACTIVE | 10.0.0.2(local) 10.0.0.1 |
| 1/1/2     | 1   | IPv4 | 100 | 0    | N     | Y   | INIT   | AF-UNDEFINED no address  |
| 1/1/2     | 1   | IPv6 | 100 | 0    | N     | Y   | INIT   | AF-UNDEFINED no address  |

```
switch# show vrrp detail
```

```
VRRP is enabled
```

```
Interface 1/1/1 - VRRPv2 Statistics
```

```
Invalid group ID packets received : 0
```

```
Invalid version packets received : 0
```

```
Invalid checksum packets received : 0
```

```
Interface 1/1/1 - VRRPv3 Statistics
```

```
Invalid group ID packets received : 0
```

```
Invalid version packets received : 0
```

```
Invalid checksum packets received : 0
```

```
Interface 1/1/1 - Group 1 - Address-Family IPv4
```

```
State is ACTIVE
```

```
State duration 1 mins 35.486 secs
```

```
Virtual IP address is 10.0.0.1
```

```
Virtual MAC address is 00:00:5e:00:01:01
```

```
Advertisement interval is 1000 msec
```

```
Version 3
```

```
Preemption enabled
```

```
Priority is 100
```

```
Active Router is 10.0.0.2 (local), priority is 100
```

```
Active Advertisement interval is 1000 msec
```

```
Active Down interval is unknown
```

```
Tracked object ID is 1, and state Down
```

```
VRRPv3 Advertisements: sent 3931 (errors 0) - rcvd 0
```

```
VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
```

```
Group Discarded Packets: 3537
```

```
IP address Owner conflicts: 0
```

```
IP address configuration mismatch : 3537
```

```
Advert Interval errors : 0
```

```
Adverts received in Init state: 0
```

```
Invalid group other reason: 0
```

```
Group State transition:
```

```
Init to active: 0
```

```
Init to standby: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
```

```
Standby to active: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
```

```
Active to standby: 0
```

```
Active to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
```

```
Standby to init: 0
```

```
Interface 1/1/2 - VRRPv2 Statistics
```

```
Invalid group ID packets received : 0
```

```
Invalid version packets received : 0
```

```
Invalid checksum packets received : 0
```

```
Interface 1/1/2 - VRRPv3 Statistics
```

```
Invalid group ID packets received : 0
```

```
Invalid version packets received : 0
```

```
Invalid checksum packets received : 0
```

```
Interface 1/1/2 - Group 1 - Address-Family IPv4
```

```
State is INIT (Interface Down)
```

```
State duration 49 mins 23.507 secs
```

```

Virtual IP address is no address
Virtual MAC address is 00:00:5e:00:01:01
Advertisement interval is 1000 msec
Version 3
Preemption enabled
Priority is 100
Active Router is unknown, priority is unknown
Active Advertisement interval is unknown
Active Down interval is unknown
VRRPv3 Advertisements: sent 0 (errors 0) - rcvd 0
VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
Group Discarded Packets: 0
  IP address Owner conflicts: 0
  IP address configuration mismatch : 0
  Advert Interval errors: 0
  Adverts received in Init state: 0
  Invalid group other reason: 0
Group State transition:
  Init to active: 0
  Init to standby: 0
  Standby to active: 0
  Active to standby: 0
  Active to init: 0
  Standby to init: 0

Interface 1/1/2 - Group 1 - Address-Family IPv6
State is INIT (Interface Down)
State duration 24 mins 14.988 secs
Virtual IP address is no address
Virtual secondary IP addresses:
  2201:13::110:4
Virtual MAC address is 00:00:5e:00:02:01
Advertisement interval is 1000 msec
Preemption enabled
Priority is 100
Active Router is unknown, priority is unknown
Active Advertisement interval is unknown
Active Down interval is unknown
VRRPv3 Advertisements: sent 0 (errors 0) - rcvd 0
VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
Group Discarded Packets: 0
  VRRPv2 incompatibility: 0
  IP address Owner conflicts: 0
  IP address configuration mismatch : 0
  Advert Interval errors : 0
  Adverts received in Init state: 0
  Invalid group other reason: 0
Group State transition:
  Init to active: 0
  Init to standby: 0
  Standby to active: 0
  Active to standby: 0
  Active to init: 0
  Standby to init: 0

```

```

switch# show vrrp interface 1/1/1
VRRP is enabled

Interface 1/1/1 - Group 1 - Address-Family IPv4
State is ACTIVE
State duration 11 mins 21.617 secs

```

```
Virtual IP address is 10.0.0.1
Virtual MAC address is 00:00:5e:00:01:01
Advertisement interval is 1000 msec
Version 3
Preemption enabled
Priority is 100
Active Router is 10.0.0.2 (local), priority is 100
Active Advertisement interval is 1000 msec
Active Down interval is unknown
```

```
switch# show vrrp interface lag10
VRRP is enabled
```

```
Interface lag10 - Group 1 - Address-Family IPv4
State is ACTIVE
State duration 11 mins 21.617 secs
Virtual IP address is 10.0.0.1
Virtual MAC address is 00:00:5e:00:01:01
Advertisement interval is 1000 msec
Version 3
Preemption enabled
Priority is 100
Active Router is 10.0.0.2 (local), priority is 100
Active Advertisement interval is 1000 msec
Active Down interval is unknown
```

```
switch# show vrrp interface vlan100
VRRP is enabled
```

```
Interface vlan100 - Group 1 - Address-Family IPv4
State is ACTIVE
State duration 11 mins 21.617 secs
Virtual IP address is 10.0.0.1
Virtual MAC address is 00:00:5e:00:01:01
Advertisement interval is 1000 msec
Version 3
Preemption enabled
Priority is 100
Active Router is 10.0.0.2 (local), priority is 100
Active Advertisement interval is 1000 msec
Active Down interval is unknown
```

```
switch# show vrrp statistics
```

```
VRRP is enabled
```

```
Interface 1/1/1 - VRRPv2 Statistics
Invalid group ID packets received : 0
Invalid version packets received : 0
Invalid checksum packets received : 0
```

```
Interface 1/1/1 - VRRPv3 Statistics
Invalid group ID packets received : 0
Invalid version packets received : 0
Invalid checksum packets received : 0
```

```
VRRP Statistics for interface 1/1/1 - Group 1 - Address-Family IPv4
State is ACTIVE
```

```

State duration 6 mins 55.006 secs
VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
Group Discarded Packets: 3856
  IP address Owner conflicts: 0
  IP address configuration mismatch : 0
  Advert Interval errors : 0
  Adverts received in Init state: 0
  Invalid group other reason: 0
Group State transition:
  Init to active: 0
  Init to standby: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
  Standby to active: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
  Active to standby: 0
  Active to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
  Standby to init: 0

Interface 1/1/2 - VRRPv2 Statistics
  Invalid group ID packets received : 0
  Invalid version packets received : 0
  Invalid checksum packets received : 0

Interface 1/1/2 - VRRPv3 Statistics
  Invalid group ID packets received : 0
  Invalid version packets received : 0
  Invalid checksum packets received : 0

VRRP Statistics for Interface 1/1/2 - Group 1 - Address-Family IPv4
State is INIT (No Primary Group Address)
State duration 54 mins 43.027 secs
VRRPv3 Advertisements: sent 0 (errors 0) - rcvd 0
VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
Group Discarded Packets: 0
  IP address Owner conflicts: 0
  Invalid address count: 0
  IP address configuration mismatch : 0
  Advert Interval errors : 0
  Adverts received in Init state: 0
  Invalid group other reason: 0
Group State transition:
  Init to active: 0
  Init to standby: 0
  Standby to active: 0
  Active to standby: 0
  Active to init: 0
  Standby to init: 0

VRRP Statistics for Interface 1/1/2 - Group 1 - Address-Family IPv6
State is INIT (Interface Down)
State duration 29 mins 34.508 secs
VRRPv3 Advertisements: sent 0 (errors 0) - rcvd 0
VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
Group Discarded Packets: 0
  IP address Owner conflicts: 0
  IP address configuration mismatch : 0
  Advert Interval errors: 0
  Adverts received in Init state: 0
  Invalid group other reason: 0
Group State transition:
  Init to active: 0
  Init to standby: 0
  Standby to active: 0

```

```
Active to standby: 0
Active to init: 0
Standby to init: 0
```

```
switch# show vrrp statistics interface 1/1/1
```

```
VRRP is enabled
```

```
Interface 1/1/1 - VRRPv2 Statistics
```

```
Invalid group ID packets received : 0
Invalid version packets received : 0
Invalid checksum packets received : 0
```

```
Interface 1/1/1 - VRRPv3 Statistics
```

```
Invalid group ID packets received : 0
Invalid version packets received : 0
Invalid checksum packets received : 0
```

```
VRRP Statistics for interface 1/1/1 - Group 1 - Address-Family IPv4
```

```
State is ACTIVE
```

```
State duration 6 mins 55.006 secs
```

```
VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
```

```
VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
```

```
Group Discarded Packets: 3856
```

```
IP address Owner conflicts: 0
```

```
IP address configuration mismatch : 0
```

```
Advert Interval errors : 0
```

```
Adverts received in Init state: 0
```

```
Invalid group other reason: 0
```

```
Group State transition:
```

```
Init to active: 0
```

```
Init to standby: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
```

```
Standby to active: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
```

```
Active to standby: 0
```

```
Active to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
```

```
Standby to init: 0
```

```
switch# show vrrp statistics interface lag10
```

```
VRRP is enabled
```

```
Interface lag10 - VRRPv2 Statistics
```

```
Invalid group ID packets received : 0
Invalid version packets received : 0
Invalid checksum packets received : 0
```

```
Interface lag10 - VRRPv3 Statistics
```

```
Invalid group ID packets received : 0
Invalid version packets received : 0
Invalid checksum packets received : 0
```

```
VRRP Statistics for interface lag10 - Group 1 - Address-Family IPv4
```

```
State is ACTIVE
```

```
State duration 6 mins 55.006 secs
```

```
VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
```

```
VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
```

```
Group Discarded Packets: 3856
```

```
IP address Owner conflicts: 0
```

```
IP address configuration mismatch : 0
```

```
Advert Interval errors : 0
```

```
Adverts received in Init state: 0
Invalid group other reason: 0
Group State transition:
Init to active: 0
Init to standby: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
Standby to active: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
Active to standby: 0
Active to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
Standby to init: 0
```

```
switch# show vrrp statistics interface vlan100
```

VRRP is enabled

Interface vlan100 - VRRPv2 Statistics

```
Invalid group ID packets received : 0
Invalid version packets received : 0
Invalid checksum packets received : 0
```

Interface vlan100 - VRRPv3 Statistics

```
Invalid group ID packets received : 0
Invalid version packets received : 0
Invalid checksum packets received : 0
```

VRRP Statistics for interface vlan100 - Group 1 - Address-Family IPv4

State is ACTIVE

State duration 6 mins 55.006 secs

VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0

VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0

Group Discarded Packets: 3856

IP address Owner conflicts: 0

IP address configuration mismatch : 0

Advert Interval errors : 0

Adverts received in Init state: 0

Invalid group other reason: 0

Group State transition:

Init to active: 0

Init to standby: 2 (Last change Mon Jun 16 11:19:36.316 UTC)

Standby to active: 2 (Last change Mon Jun 16 11:19:39.926 UTC)

Active to standby: 0

Active to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)

Standby to init: 0



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

## Command History

| Release          | Modification                                                                                              |
|------------------|-----------------------------------------------------------------------------------------------------------|
| 10.14            | Replaced the <b>ipv4</b> parameter with the <b>ip</b> parameter. The <b>ipv4</b> parameter is deprecated. |
| 10.08            | Updated command output for inclusive language                                                             |
| 10.07 or earlier | --                                                                                                        |

## Command Information

| Platforms    | Command context | Authority                                                                                                                                                              |
|--------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Manager (#)     | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## shutdown (vrrp)

shutdown  
no shutdown

### Description

Enables standby VRRP groups on the interface to route the traffic sent to virtual router's MAC address. It is recommended to enable this mode on VLANs where mc-lags are configured.

Disabled by default.

The **no** form of this command disables standby VRRP groups on the interface to route the traffic sent to virtual router's MAC address.



Only supported on SVI interfaces.

### Examples

Enabling standby VRRP groups on interface VLAN 10 to route traffic sent to the router's MAC address:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# vrrp dual-active-forwarding
```

Disabling the ability of standby VRRP groups on interface VLAN 10 to route traffic sent to the router's MAC address:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no vrrp dual-active-forwarding
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release    | Modification        |
|------------|---------------------|
| 10.12.1000 | Command introduced. |

### Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## timers advertise

```
timers advertise <ADVERTISE-IN-MILLISECONDS>
no timers advertise
```

## Description

Sets the advertisement interval in ms (100-40950). The default value is 1000. Advertisement interval can be configured in multiples of 1,000 ms.

The **no** form of this command sets the advertisement interval in ms to the default value of 1000.



---

This release does not support sub-second timer for VRRPv3.

---

## Examples

Setting the advertisement interval in ms to 2000:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# timers advertise 2000
```

Setting the advertisement interval in ms to the default value of 1000:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no timers advertise
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## track (VRRP group)

```
track <OBJECT-ID>
no track <OBJECT-ID>
```

## Description

Sets the track object ID (1-128) for the group. The track object is first configured globally for the interface and then attached to the VRRP virtual router.



---

The track object must not track the same interface for which a VRRP group is configured.

---



The **no** form of this command removes the track object ID from the group.

## Examples

Setting the track object ID for the group:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# track 1
```

Removing the track object ID from the group:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no track 1
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## track (VRRP virtual router)

```
track <OBJECT-ID>
no track <OBJECT-ID>
```

## Description

Configures a track object that can be associated with an interface. A change in interface state will then affect the priority of a VRRP group. By default, no interface is associated to a track object, so state is down.

The **no** form of this command deletes a tracked object for an interface. If it is not associated with a VRRP virtual router, a track object cannot be deleted.



---

Track cannot be configured by using port with no routing.

---

---

When all tracked interfaces go down on a virtual router, priority is automatically set to zero instead of its configured value. Owner virtual routers always use a default priority of 255.

---

| Parameter                      | Description                                         |
|--------------------------------|-----------------------------------------------------|
| <code>&lt;OBJECT-ID&gt;</code> | Specify the track object ID value. Range: 1 to 128. |

## Examples

Configuring a tracked object:

```
switch(config)# track 1
```

Deleting a tracked object:

```
switch(config)# no track 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## track by

```
track by <OBJECT-ID>
no track by <OBJECT-ID>
```

## Description

Specifies an interface to be tracked when changes in the state of the interface affect the priority of a VRRP group. Once track is associated with an interface, the track state reflects the interface forwarding state.

The **no** form of this command removes an interface from tracking, affecting VRRP states of any interfaces associated with VRRP groups.



The VLAN interface 1 is always tracked.

| Parameter                      | Description                                           |
|--------------------------------|-------------------------------------------------------|
| <code>&lt;OBJECT-ID&gt;</code> | Specifies the track object ID value. Range: 1 to 128. |

## Example

Specifying an interface to be tracked:

```
switch (config)# interface 1/1/1
switch (config-if)# track by 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## version

version <VERSION-NUMBER>

## Description

Sets the protocol version for the VRRP group. Version change is allowed only for the IPv4 address-family. The default value is 2, which supports IPv4 with minimum 1 second advertisement interval. Value 3 supports IPv4 and IPv6 with minimum 1 second advertisement interval.

| Parameter        | Description                                                                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <VERSION-NUMBER> | Specifies the VRRP protocol version. Possible values: 2 or 3. The default value is 2, which supports IPv4 with a minimum 1 second advertisement interval. |

## Example

Setting the protocol version for the VRRP group to 3:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# version 3
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## vrrp

```
vrrp <VRID> address-family {ip | ipv6}
no vrrp <VRID> address-family {ip | ipv6}
```

## Description

Creates a VRRP group and establishes VRRP group configuration context.

- A maximum of 16 VRRP groups, including both IPv4 and IPv6, are supported on an interface.
- A maximum of 256 VRRP groups is supported on a router. The groups can be IPv4 or IPv6 on a first come first serve basis.

The **no** form of this command deletes a VRRP group.

| Parameter                 | Description                                        |
|---------------------------|----------------------------------------------------|
| <VRID>                    | Selects the VRRP router ID value. Range: 1 to 255. |
| address-family [IP  IPv6] | Specifies which address family to use, IP or IPv6. |

## Examples

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if-vrrp)# vrrp 1 address-family ip
switch(config-if-vrrp)#

switch(config-if)# no vrrp 1 address-family ip
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                                                                              |
|------------------|-----------------------------------------------------------------------------------------------------------|
| 10.14            | Replaced the <b>ipv4</b> parameter with the <b>ip</b> parameter. The <b>ipv4</b> parameter is deprecated. |
| 10.07 or earlier | --                                                                                                        |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## router bgp

```
router bgp <AS-NUMBER>
no router bgp <AS-NUMBER>
```

### Description

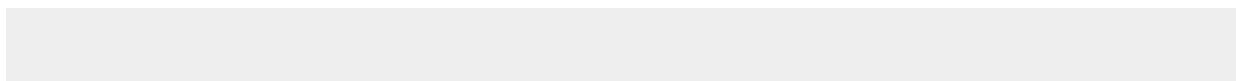
This command configures the BGP instance on the router, configures the AS (Autonomous System) the router belongs to, and enters into the BGP router configuration mode. Only a single BGP AS number can be assigned for the entire system.

The **no** form of the command deletes the BGP instance from the router.

| Parameter        | Description                                                                                                        |
|------------------|--------------------------------------------------------------------------------------------------------------------|
| <i>AS-NUMBER</i> | Specifies a 4-byte AS number in the range 1-4294967295 in integer format or from 0.1-65535.65535 in dotted format. |

### Examples

Configuring the BGP instance with the AS number:



Deleting BGP configurations:

```
switch(config)# no router bgp 100
This will delete all BGP configurations on this device.
Continue (y/n)?
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 5420      | config          | Administrators or local user group members with execution rights for this command. |

## Chapter 9

# Policy Based Routing (PBR)

---



---

PBR is not supported on the 4100i Switch Series.

---



---

PBR is not supported on the 6000 and 6100 Switch Series.

---

Policy Based routing (PBR) lets you manipulate the path of a packet based on the various attributes of the packet. Packets that can be manipulated by PBR are packets that are already routing through the system at Layer 3, with a destination IP address that is on a network other than the packet ingress Layer 3 interface. PBR is an extension of the existing classifier policy system where traffic to be manipulated is matched by a classifier **class**, and policy **actions** to be executed on the matching traffic. Matching traffic with the same destination can be routed over different paths so that different types of traffic, such as VoIP or traffic with special security requirements, can be better managed.



---

PBR's ability to influence a packet path is limited to the current router in that the next router to which the traffic is redirected makes an independent decision on where to forward traffic to next, if anywhere. Packets not matched by a PBR policy entry are not affected and take routes specified in the system route table.

---

## PBR actions

- **interface null:** equivalent to the policy **drop** policing action. Any packets matching the class criteria for that policy entry will be dropped and not routed any further.
- **nexthop:** allows for overriding the routing table's longest prefix match next-hop router for matching packets. If no such routing table entry exists for matching packets, (default or not) this action still affects matching packets.
- **default-nexthop:** allows for specifying a next-hop router for matching packets when there is no longest prefix match for those packets in the routing table. Such a default-nexthop overrides a system default route if already configured and also applies if there is no system default route.



- 
- Next-hop and default-nexthop facilitate routing matching packets where they otherwise might not be, due to the absence of routing table entries.
  - Unlike next-hop, default-nexthop only applies if there is no destination lookup match in the main routing table for matching packets.
- 

## PBR policy action and action list

Entries in a policy use a class to specify the criteria on what packets to match and a set of one or more policy actions to take on the matching packets such as dscp or mirror.

PBR is configured for use as another such policy action in a policy entry by using the `pbr` keyword followed by the name of a user configured PBR action list. This list may contain up to eight PBR action entries which are listed according to priority by sequence number. The [PBR actions](#) section lists the different action entry choices.

In a given action list applied to an interface, zero entries may be available (that is, reachable). Of the entries that are available, only one can be active for that action list. Due to real-time changes in the network operating environment, list entry availability can change at any moment. At any given time, the highest priority available entry in the list is selected as the active PBR action for that list. If the active entry becomes unavailable, the next highest available entry is automatically selected and replaces that now-unavailable entry as the active entry. If a higher priority entry (other than the current active entry) becomes available, this entry is selected and replaces the current active entry, as the new active entry (even though that current entry is still available). If no entries are available or the only available entry becomes unavailable, routing occurs based on the routing table and is not influenced by PBR.



---

The **interface null** action is always considered available. If there are no higher priority PBR actions available, it will be selected as the active entry when configured in a PBR action list, and applied to an interface.

---

## PBR action list maximum entries

There is a limit of eight entries per action list and a maximum of 16 action lists in the system. Only one entry from an action list can be 'active' at any given moment, so the maximum number of unique active entries in the system is also 16.

For example, a given next-hop specified in an action list applied to an interface counts against the limit once. If that same next-hop address is entered in a different action list and applied to another interface in the same VRF, this next-hop is not unique in the VRF and does not affect the limit further. If one of those action lists (that specify the address) is applied to an interface in a different VRF, the entry will affect the limit since addresses are unique to each VRF.

Another example is if two PBR action lists with eight unique next-hop entries (for a total of 16 unique next-hops) are configured as PBR actions in a policy. If that policy is applied to 32 route-only ports (each of which is in a different VRF), the entry supply will be exhausted.



- 
- PBR action list maximum capacity is 16.
- 

## IP versions in an action list

Configuring PBR actions of different IP versions in the same PBR action list is not supported. This limitation applies to the next-hop and default-nexthop actions specifically. If an action of one IP version is already configured, configuring an action with a different IP version in the same action list will be blocked. This limitation applies regardless of whether the action list is part of a policy, applied to an interface or whether any of the entries are selected or not.

Furthermore, use cases similar to the following are not supported:

- Applying a PBR action list with entries of one IP version (for example IPv4 next-hop) to an interface of the other IP version (for example a route-only port with an IPv6 address only).
- Creating a policy entry with a class of one IP version and a PBR action list with entries of the other IP version.

Configuration mismatches while not prevented are not supported and behavior is not defined. At best, traffic will not reach its intended destination.

## Specifying valid next-hop and default-nexthop addresses

PBR supports all types of directly connected hosts as nexthop or default-nexthop actions.

This means that while you may specify any IPv4 or IPv6 address as a nexthop or default-nexthop, after application only those reachable through a directly connected network will be installed as an active PBR nexthop. Even if a specified address is reachable from the router, if it is not on a directly connected subnet it will not be installed.

## Hardware path PBR versus software path PBR

Traffic to be routed that cannot be handled by the switching ASIC (for example IPv4 packets with IP options), is handled by the switch operating system kernel routing software.

The characteristics of network usage determine the proportion of traffic handled by software path PBR relative to the hardware path. Under normal conditions, software path PBR is likely to range from a fraction of a percent of all traffic to almost none.

Packets with IP options are sent to the hardware ASIC as well as the CPU. Since the software path PBR actions are applied to the packets in the CPU, a default entry is programmed in the hardware to take no action on those packets. This hardware entry count can be seen in diagnostics with the command `diag-dump policy basic`.

## Hardware versus software path for default-nexthop action

When a policy containing an entry with PBR is applied to an interface and the active action for that entry action list is 'default-nexthop', packets that match the class criteria and have no explicit route table hit (that is, no destination address longest prefix match, a.k.a 'route-miss'), they are forwarded to the specified active PBR default-nexthop. This is true for packets that follow the hardware and software paths through the switch.

There is a difference between PBR hardware and software path behavior when a route table hit occurs for class-matched packets (when the policy entry is a PBR default-nexthop).

Hardware path match criteria for a PBR policy entry with default-nexthop is extended to include the route table miss along with the class qualifiers, resulting in a **policy entry hit** for that policy entry. Conversely when there is a route table hit, the result is a policy entry miss in hardware path. When a policy entry miss occurs, policy processing moves on to the next entry in the policy and takes whatever action is specified, if any exist. This can include a different PBR routing action including interface null or no PBR action at all, for example.

In software path, the class match criteria is the only criteria required to achieve a policy entry hit. When that occurs, policy processing will stop. When there is a class match and a route table hit (with PBR action default-nexthop), the packet is forwarded according to that route table entry, not the PBR default-nexthop entry, nor is it influenced by any subsequent policy entries.

This difference in behavior is due to a limitation in the software path matching and routing engine, relative to hardware.



**Table 1:** *default-nexthop behaviors*

| Routing packet | System route miss               | System route hit                                                                                                                                     |
|----------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware       | Take PBR default-nexthop route. | Policy entry miss, continue policy processing with next entry, if present. No further matches result in system default route being used, if present. |
| Software       | Take PBR default-nexthop route. | Policy entry hit is overridden by system route hit. System route entry used, policy processing stops.                                                |

## Software path and system default route

AOS-CX default behavior (for packets with destination networks that lack entries in the system route table with a reachable next-hop address), sends packets to the system CPU for special handling, if any are specified (for example, sending an ICMP unreachable reply message).

A side effect of this CPU routing is that it has higher priority than an applied PBR next-hop action. Even though a policy is applied with an entry that matches the traffic and specifies a PBR next-hop action which is reachable, the traffic will still be routed to the system CPU (due to the absence of a reachable next-hop in the system route table), and not through the desired PBR hardware path. With traffic routing to the system CPU, it will be properly routed by PBR software path, but it will also be rate limited by the control plane policing feature. Loss can occur at higher traffic rates.



The workaround for this issue is to create a default next-hop route in the system with a reachable next-hop router/host. This will result in a route hit, or reachable next-hop detected for the traffic with no further need to route traffic to the CPU. The PBR hardware path next-hop action will then occur, as desired.

## PBR, ECMP, and routing protocols

When a policy with PBR actions is applied to an interface, the highest priority action from the action list is applied to matching traffic on that interface and overrides any static, ECMP or dynamically installed routes in the system.

## PBR, VSX, and VLAN ACLs

Multichassis Link Aggregation Group (MLAG) with VSX is a high-availability feature where a switch containing LAG members is connected to multiple switches to allow for node-level redundancy on that link. If one of the other switches goes down the LAG remains up and can continue to carry all the LAG traffic, bandwidth permitting.

A LAG (and an MLAG) can be a member of a VLAN and a Layer 3 VSI (virtual switch interface), which can be created for that VLAN for the purposes of routing a policy with Layer 3 specific actions (that is, PBR), which can therefore be applied to that interface to influence routing decisions for matching Layer 3 packets on the MLAG; like any other route-only port or VSI. Under such a configuration, it is likely that it is desirable to use VLAN ACLs applied to the VLAN the MLAG is a member of.



---

There is a limit with this particular combination when the VLAN ACL specifies both IPv4 and IPv6 entries and the PBR policy has entries with IPv4 and IPv6 classes. This configuration will exhaust the switching ASIC resources and will fail to apply. To achieve a similar configuration, you may use port ACLs for IPv4 traffic (applying the ACL to all ports individually in the VLAN) while preserving the rest of the configuration.

---

## PBR software path, VSX, and VRRP

Due to the dynamic nature of the VSX and VRRP protocols, applying a policy with PBR to VSX or VRRP Layer 3 kernel interfaces and then making further changes to those protocols may not result in expected behavior. Apply policies with PBR to VSX and VRRP Layer 3 interfaces after all changes have been made. If further changes are necessary the policy should be removed first and reapplied when configuration updates are complete.

## PBR and next-hop router reachability

When a policy with PBR actions is applied to an interface, all nexthop and default-nexthop entries in the associated action list are continuously monitored for reachability. A probe occurs every 5 seconds for each entry. These probes guarantee that the reaction to the loss of reachability or detection of the new reachability of any such entry will be 5 seconds or less.

For example, if there are two reachable nexthops in an applied action list and the active nexthop (highest priority by lowest sequence number) entry becomes unreachable, the switchover to the lower priority nexthop will occur within 5 seconds. If there is only one nexthop or default-nexthop entry and no entries of any other type in the action list, routing decisions will return to the system routing table within 5 seconds.

Conversely if an applied action list has a single unreachable nexthop and that nexthop at some point becomes reachable, the switch back to routing to that nexthop will occur within 5 seconds of the new reachability of the nexthop on the network. Or in another scenario an action list has multiple nexthops and the active nexthop is not the highest priority entry. When the nexthop with the higher priority becomes available, the promotion of that now reachable nexthop to active will occur within 5 seconds of its new reachability on the network.

## PBR nexthop over VXLAN L3VNI

In general PBR does not support remote (recursive) nexthops. However, PBR supports configuring remote non-connected hosts as PBR nexthops or default-nexthop actions that are learned over EVPN-distributed subnets across a routed VXLAN L3VNI tunnel. These nexthops are not guaranteed to exist, but are supported as a PBR nexthop or default-nexthop action as long as there is a resolved EVPN-BGP route that matches the PBR nexthop address. As a result, hosts learned over EVPN-distributed subnets across a routed VXLAN L3VNI tunnel will not be probed for reachability.



---

For more information on VXLAN or EVPN, refer to the VXLAN Guide.

---

A use-case of PBR over VxLAN L3VNI is to route packets to a dedicated machine over a tunnel such as a firewall or load balancer. The incoming packet will hit a VTEP and PBR policy will send the packet into the tunnel (ingress). On the other end, the packet will exit the tunnel and PBR policy will route the packet to the correct nexthop (egress). It is important to note that



---

In order to configure the end-to-end solution, a policy needs to be configured on both VTEPs, ingress and egress.

---

## Ingress policy

In order to route traffic to a nexthop learned over VXLAN L3VNI, a policy needs to be applied to the ingress VTEP's L3 interface associated with the arriving traffic that needs to be redirected.

In the following example the policy is applied to **vlan 20**, (where **nexthop 100.1.1.4** is learned over VxLAN L3VNI):

```
class ip class1
  10 match any any any
pbr-action-list pbr1
  10 nexthop 100.1.1.4
policy p1
  10 class ip class1 action pbr pbr1
interface vlan 20
  vrf attach red
  ip address 200.1.1.2/16
  active-gateway ip mac aa:ee:ff:00:00:20
  active-gateway ip 200.1.1.1
  apply policy p1 routed-in
```

## Egress policy

A policy must be applied to the egress VTEP's L3VNI that is receiving the traffic from the VXLAN tunnel.

In the following example the policy is applied to VNI 10000:

```
class ip class1
  10 match any any any
pbr-action-list pbr1
  10 nexthop 100.1.1.4
policy p1
  10 class ip class1 action pbr pbr1
interface vxlan 1
  source ip 1.1.1.1
  no shutdown
  vni 10
    vlan 10
  vni 10000
    vrf red
    routing
    apply policy p1 routed-in
```

## L2VNI vs L3VNI

There is an important distinction between PBR nexthops learned over L2VNI and L3VNI. Any L2VNI is treated the same as a locally connected nexthop. It will be probed periodically for reachability. If this L2VNI nexthop is determined to be unreachable, then PBR will take action on the next item in the action list. If the nexthop becomes reachable again and it has a higher precedence than the currently configured action, then it will take over. See [PBR policy action and action list](#) for more details.

For recursive nexthops (learned over L3VNI), the connected nexthop to reach the recursive nexthop is constantly probed for reachability. However, the recursive PBR nexthop itself will not be periodically

probed. They are not guaranteed to be reachable. The PBR action list will use the nexthop as long as there is a EVPN-BGP route that matches.

## CLI errors

**Table 1:** *Error messages and triggering events*

| Message                                                                               | Event                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Failed to create action list                                                          | A PBR action list create operation was unable to create the OVSDB row.                                                                                                                                            |
| PBR action list '%s' doesn't exist                                                    | A PBR action list update or delete operation was unable to locate the list OVSDB row.<br>OR<br>A PBR action list entry create, update, or delete operation was unable to locate its parent action list OVSDB row. |
| Unable to automatically set sequence number                                           | The automatically generated sequence number for a new PBR action list entry is higher than the maximum permitted sequence number.                                                                                 |
| Unable to add PBR action list entry                                                   | A PBR action list entry create failed to create an entry row in OVSDB.                                                                                                                                            |
| Configuration of IPv4 and IPv6 addresses in the same PBR action list is not supported | An attempt to create both IPv4 and IPv6 addresses in the same action list was made.                                                                                                                               |
| Invalid sequence number                                                               | A PBR action list entry delete did not specify the entry sequence number.                                                                                                                                         |
| Cannot add entry because its automatic sequence number would exceed the maximum       | The automatic sequence number exceeded the maximum.                                                                                                                                                               |
| Action list entry does not exist                                                      | A PBR action list entry delete could not locate the OVSDB entry row.                                                                                                                                              |
| Action list name required                                                             | A PBR action list delete was passed an invalid list name.                                                                                                                                                         |
| Invalid action list name length                                                       | A PBR action list name was longer than the permitted length.                                                                                                                                                      |

## PBR commands

### apply policy (interface)

```
apply policy <POLICY-NAME> routed-in  
no apply policy <POLICY-NAME> routed-in
```

### Description

Applies a classifier policy containing a PBR action to an interface. A policy with PBR actions is only applicable to L3/routing interfaces.

The **no** form of this command removes a classifier policy containing a PBR action from an interface.  
config-if

| Parameter                        | Description                   |
|----------------------------------|-------------------------------|
| <code>&lt;POLICY-NAME&gt;</code> | Specifies name of the policy. |

## Restrictions

- Layer 3 LAGs are not supported on 5420 or 6200 series switches. PBR policies cannot be applied to Layer 3 LAGs.

## Usage

To use route-only ports (ROPs) as Layer 3 interfaces, an internal VLAN range must be configured first. A policy with PBR actions can be applied to ROPs.

## Example

Applying a policy to an interface:

```
switch(config)# interface 1/1/10
switch(config-if)# routing
switch(config-if)# apply policy pbr_policy routed-in
switch(config-if)# exit
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                              |
|------------------|-------------------------------------------|
| 10.08            | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | --                                        |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## pbr-action-list

pbr-action-list `<ACTION-LIST-NAME>`

```
[<SEQUENCE-NUMBER>]
{nexthop | default-nexthop} <NEXT-HOP-IP-ADDR>
interface null

no [<SEQUENCE-NUMBER>]
```

```
{nexthop | default-nexthop} <IP-ADDR>
interface null
```

```
no pbr-action-list <ACTION-LIST-NAME>
```

## Description

Creates a PBR action list or modifies its entries.

The **no** form of this command can be used to delete an action list or an individual action list entry.

| Parameter          | Description                                                                                                                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ACTION-LIST-NAME> | Specifies the action list name. An action list name can be 1 to 64 alphanumeric characters.                                                                                                                                                                                                                             |
| <SEQUENCE-NUMBER>  | Specifies list entry sequence number. Range: 1-4294967295<br>{nexthop   default-nexthop}<br>Selects a regular next-hop ( <b>nexthop</b> ) or a default next-hop ( <b>default-nexthop</b> ). These parameters specify the address of a next-hop router to forward traffic matched by a class under different conditions. |
| nexthop            | Sets the next hop for routing the packet.                                                                                                                                                                                                                                                                               |
| default-nexthop    | Sets the next hop for routing the packet when there is no explicit route for its destination.                                                                                                                                                                                                                           |
| <NEXTHOP-IP-ADDR>  | Specifies IPv4 or IPv6 address of the next-hop router.                                                                                                                                                                                                                                                                  |
| interface null     | Selects keyword interface: <b>null</b> .                                                                                                                                                                                                                                                                                |
| null               | Specifies to drop matching traffic.                                                                                                                                                                                                                                                                                     |

## Restrictions

Reachability of next-hop routers in the list is not guaranteed. Such reachability can change at any time due to the dynamic nature of the network environment.

## Usage

Each action list may contain up to four entries of three different entry types:

- interface null
- nexthop
- default-nexthop

List entries have a unique sequence number which, if not user specified, are automatically assigned beginning at 10 and continuing at intervals of 10 for each subsequent new list entry, for example 20, 30, and 40. Sequence numbers of any value can be specified manually, a different interval may be set, and new entries can be added to (or removed from) any location in the list at any time.

Specifying an existing sequence number causes the existing list entry to be replaced by the new details. The list entry with the lowest sequence number has the highest priority entry in the list. The sequence numbers may be renumbered with the [pbr-action-list resequence](#) command.

Only one next-hop router or interface from the list is used per packet matched. This router or interface is defined as the highest priority list entry that is reachable or available at the time of the traffic match. If the highest priority list entry next-hop router is reachable - that list entry is chosen, the search is stopped, and the traffic is forwarded to the next-hop router or interface for the entry. If the highest

priority list entry next-hop router is not reachable, the next highest priority list entry reachability is determined and used if reachable, otherwise the process continues down the list. If none of the routers in the list are reachable, the packet may be dropped (through the null interface entry if configured) or forwarded according to a system route table entry.



---

An action list that contains a next-hop of one IP version cannot also contain an entry of another IP version. For example, an action list must contain only IPv4 or IPv6 next-hop addresses.

---

## Examples

The list name is included in the context prompt for easy current-list identification. Any list name over 10 characters will be truncated at 10 characters and terminated with the tilde character (~) to indicate a reduced list name display. This reduction affects the prompt display of the list name only:

```
switch(config)# pbr-action-list eighteenchars
switch(config-pbr-action-list-eightench~)#
```

The following example creates an action list with two IPv4 next-hops, a default IPv4 next-hop, and a null interface. The example uses default sequence numbering for its list entries.

```
switch(config)# pbr-action-list test1
switch(config-pbr-action-list-test1)# nexthop 1.1.1.1
switch(config-pbr-action-list-test1)# nexthop 2.2.2.2
switch(config-pbr-action-list-test1)# default-nexthop 9.9.9.9
switch(config-pbr-action-list-test1)# interface null
switch(config-pbr-action-list-test1)# end

switch(config)# show pbr-action-list test1
```

| Sequence | Name<br>Type    | Address/Interface |
|----------|-----------------|-------------------|
| -----    |                 |                   |
|          | test1           |                   |
| 10       | nexthop         | 1.1.1.1           |
| 20       | nexthop         | 2.2.2.2           |
| 30       | default-nexthop | 9.9.9.9           |
| 40       | interface       | null              |

The following example creates an action list with an IPv4 next-hop and a null interface with manual sequence numbers for its entries.

```
switch(config)# pbr-action-list test2
switch(config-pbr-action-list-test2)# 6 default-nexthop 4.4.4.4
switch(config-pbr-action-list-test2)# 1 interface null
switch(config-pbr-action-list-test2)# exit

switch(config)# show pbr-action-list test2
```

| Sequence | Name<br>Type    | Address/Interface |
|----------|-----------------|-------------------|
| -----    |                 |                   |
|          | test2           |                   |
| 1        | interface       | null              |
| 6        | default-nexthop | 4.4.4.4           |

The following example creates an action list with two IPv6 next-hops and the null interface, with manual sequence numbers.

```
switch(config)# pbr-action-list test4
switch(config-pbr-action-list-test4)# 5 nexthop 2000:abcd::cccc:dddd
switch(config-pbr-action-list-test4)# 6 nexthop 1000:abcd::1234:5678
switch(config-pbr-action-list-test4)# 7 interface null
switch(config-pbr-action-list-test4)# end
```

```
switch(config)# show pbr-action-list test4
```

| Sequence | Name<br>Type | Address/Interface    |
|----------|--------------|----------------------|
|          | test4        |                      |
| 5        | nexthop      | 2000:abcd::cccc:dddd |
| 6        | nexthop      | 1000:abcd::1234:5678 |
| 7        | interface    | null                 |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                              |
|------------------|-------------------------------------------|
| 10.08            | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | --                                        |

## Command Information

| Platforms    | Command context                                                                                                                                                                                                                                | Authority                                                                          |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config<br>The <code>pbr-action-list</code><br><ACTION-LIST-NAME><br>command takes you into<br>the <code>config-pbr-</code><br><code>action-list-</code><br><ACTION-LIST-NAME><br>context where you<br>modify entries for a PBR<br>action list. | Administrators or local user group members with execution rights for this command. |

## pbr-action-list copy

```
pbr-action-list <ACTION-LIST-NAME> copy <DESTINATION-ACTION-LIST-NAME>
```

## Description

Copies an existing PBR action list.



| Parameter                                         | Description                                                                                                          |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <code>&lt;ACTION-LIST-NAME&gt;</code>             | Specifies the action list name to be copied.                                                                         |
| <code>&lt;DESTINATION-ACTION-LIST-NAME&gt;</code> | Specifies the name of the copied action list. A destination action list name can be 1 to 64 alphanumeric characters. |

## Examples

The following example copies test4 action list to test 5.

```
switch(config)# show pbr-action-list test4
```

| Sequence | Name<br>Type | Address/Interface    |
|----------|--------------|----------------------|
|          | test4        |                      |
| 5        | nexthop      | 2000:abcd::cccc:dddd |
| 6        | nexthop      | 1000:abcd::1234:5678 |
| 7        | interface    | null                 |

```
switch(config)# pbr-action-list test4 copy test5
switch(config-pbr-action-list-test4)# show pbr-action-list test5
```

| Sequence | Name<br>Type | Address/Interface    |
|----------|--------------|----------------------|
|          | test4        |                      |
| 1        | nexthop      | 2000.abcd::cccc.dddd |
| 11       | nexthop      | 1000.abcd::1234.5678 |
| 21       | interface    | null                 |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                              |
|------------------|-------------------------------------------|
| 10.08            | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | --                                        |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## pbr-action-list resequence

`pbr-action-list <ACTION-LIST-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>`

## Description

Renums the entries in an action list. The list entry with the lowest sequence number has the highest priority entry in the list.

| Parameter                                     | Description                                                      |
|-----------------------------------------------|------------------------------------------------------------------|
| <code>&lt;ACTION-LIST-NAME&gt;</code>         | Specifies the action list name to have its entries resequenced.  |
| <code>&lt;STARTING-SEQUENCE-NUMBER&gt;</code> | Specifies the starting sequence number. Range: 1-4294967295      |
| <code>&lt;INCREMENT&gt;</code>                | Specifies the increment of the resequencing. Range: 1-4294967295 |

## Examples

The following command shows how a PBR action list is resequenced. In the following example, an action list named **test4** is resequenced so that instead of its entries starting at 5 and being numbered sequentially, its entries start now at 1 and they are numbered in increments of 10:

```
switch(config)# show pbr-action-list test4
```

| Sequence | Name<br>Type | Address/Interface    |
|----------|--------------|----------------------|
| -----    |              |                      |
|          | test4        |                      |
| 5        | nexthop      | 2000.abcd::cccc.ddd  |
| 6        | nexthop      | 1000.abcd::1234.5678 |
| 7        | interface    | null                 |

```
switch(config)# pbr-action-list test4 resequence 1 10
```

| Sequence | Name<br>Type | Address/Interface    |
|----------|--------------|----------------------|
| -----    |              |                      |
|          | test4        |                      |
| 1        | nexthop      | 2000.abcd::cccc.ddd  |
| 11       | nexthop      | 1000.abcd::1234.5678 |
| 21       | interface    | null                 |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                              |
|------------------|-------------------------------------------|
| 10.08            | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | --                                        |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## pbr-action-list reset

pbr-action-list <ACTION-LIST-NAME> reset

### Description

Resets a specified PBR action list to its last successful configuration.

| Parameter          | Description                                 |
|--------------------|---------------------------------------------|
| <ACTION-LIST-NAME> | Specifies the action list name to be reset. |

### Examples

```
switch(config)# pbr-action-list test reset
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification                              |
|------------------|-------------------------------------------|
| 10.08            | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | --                                        |

### Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## policy (pbr)

policy <POLICY-NAME>

```
[<SEQUENCE-NUMBER>]
  class {ip|ipv6|mac} <CLASS-NAME>
  action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}
  [{<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}]

[<SEQUENCE-NUMBER>]
  comment ...
```

no policy <POLICY-NAME>

### Description

Creates, modifies, or deletes a classifier policy. A policy contains one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6/MAC class and one or more policy actions associated with it. An applied policy processes a packet sequentially against policy entries in the list until the last entry in the list has been evaluated or the packet matches an entry. If a match occurs the related entry, actions are taken.

The **no** form of this command is used to delete a policy or an individual policy entry.

| Parameter                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;POLICY-NAME&gt;</code>                    | Specifies the name of the policy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>&lt;SEQUENCE-NUMBER&gt;</code>                | Specifies a sequence number for the policy entry. Optional.<br>Range: 1 to 4294967295.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>comment</code>                                | Stores the remaining entered text as a policy entry comment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>class {ip ipv6 mac} &lt;CLASS-NAME&gt;</code> | Specifies a type of class, <b>ip</b> for IPv4, <b>ipv6</b> for IPv6 and <b>mac</b> for a MAC policy. And specifies a class name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>&lt;REMARK-ACTIONS&gt;</code>                 | <p>Remark actions can be any of the following options: <b>{pbr &lt;ACTION-LIST&gt;   pcp &lt;PRIORITY&gt;   ip-precedence &lt;IP-PRECEDENCE-VALUE&gt;   dscp &lt;DSCP-VALUE&gt;   local-priority &lt;LOCAL-PRIORITY-VALUE&gt;}</b> where:</p> <p><b>pbr &lt;ACTION-LIST&gt;</b><br/>Specifies the PBR action list to be used.</p> <p><b>pcp &lt;PCP-VALUE&gt;</b><br/>Specifies Priority Code Point (PCP) value. Range: 0 to 7.</p> <p><b>ip-precedence &lt;IP-PRECEDENCE-VALUE&gt;</b><br/>Specifies the numeric IP precedence value. Range: 0 to 7.</p> <p><b>dscp &lt;DSCP-VALUE&gt;</b><br/>Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows:</p> <ul style="list-style-type: none"><li><b>AF11</b> - DSCP 10 (Assured Forwarding Class 1, low drop probability)</li><li><b>AF12</b> - DSCP 12 (Assured Forwarding Class 1, medium drop probability)</li><li><b>AF13</b> - DSCP 14 (Assured Forwarding Class 1, high drop probability)</li><li><b>AF21</b> - DSCP 18 (Assured Forwarding Class 2, low drop probability)</li><li><b>AF22</b> - DSCP 20 (Assured Forwarding Class 2, medium drop probability)</li><li><b>AF23</b> - DSCP 22 (Assured Forwarding Class 2, high drop probability)</li><li><b>AF31</b> - DSCP 26 (Assured Forwarding Class 3, low drop probability)</li><li><b>AF32</b> - DSCP 28 (Assured Forwarding Class 3, medium drop probability)</li><li><b>AF33</b> - DSCP 30 (Assured Forwarding Class 3, high drop probability)</li><li><b>AF41</b> - DSCP 34 (Assured Forwarding Class 4, low drop probability)</li><li><b>AF42</b> - DSCP 36 (Assured Forwarding Class 4, medium drop probability)</li><li><b>AF43</b> - DSCP 38 (Assured Forwarding Class 4, high drop probability)</li><li><b>CS0</b> - DSCP 0 (Class Selector 0: Default)</li><li><b>CS1</b> - DSCP 8 (Class Selector 1: Scavenger)</li></ul> |

| Parameter                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <b>CS2</b> - DSCP 16 (Class Selector 2: OAM)<br><b>CS3</b> - DSCP 24 (Class Selector 3: Signaling)<br><b>CS4</b> - DSCP 32 (Class Selector 4: Real time)<br><b>CS5</b> - DSCP 40 (Class Selector 5: Broadcast video)<br><b>CS6</b> - DSCP 48 (Class Selector 6: Network control)<br><b>CS7</b> - DSCP 56 (Class Selector 7)<br><b>EF</b> - DSCP 46 (Expedited Forwarding)<br><code>local-priority &lt;LOCAL-PRIORITY-VALUE&gt;</code><br>Specifies a local priority value. Range: 0 to 7. |
| <code>&lt;POLICE-ACTIONS&gt;</code> | Police actions can be the following <b>{cir &lt;RATE-BPS&gt; cbs &lt;BYTES&gt; exceed}</b> where:<br><code>cir &lt;RATE-BPS&gt;</code><br>Specifies a Committed Information Rate value in Kilobits per second. Range: 1 to 4294967295.<br><code>cbs &lt;BYTES&gt;</code><br>Specifies a Committed Burst Size value in bytes. Range: 1 to 4294967295.<br><code>exceed</code><br>Specifies action to take on packets that exceed the rate limit.                                            |
| <code>&lt;OTHER-ACTIONS&gt;</code>  | Other actions can be the following:<br><code>drop</code><br>Specifies drop traffic.                                                                                                                                                                                                                                                                                                                                                                                                       |

## Restrictions

MAC classes are not applicable to policies containing PBR actions. Applying such policies to an interface are blocked.

## Usage

- For Policy Based Routing, the policy action keyword is **pbr** which itself takes the name of a PBR action list as a parameter.
- A policy entry that contains a PBR action can contain other action types as well.
- An applied policy processes a packet sequentially against policy entries in the list until the last policy entry in the list has been evaluated or the packet matches an entry.
- Entering an existing **<POLICY-NAME>** value will cause the existing policy to be modified, with any new **<SEQUENCE-NUMBER>** value creating an additional policy entry, and any existing **<SEQUENCE-NUMBER>** value replacing the existing policy entry with the same sequence number.
- If no sequence number is specified, a new policy entry is appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10.

## Examples

Create a policy with two PBR actions:

```
switch(config)# policy pbr_policy
switch (config-policy)# 10 class ip v4_class action pbr action_list1
switch (config-policy)# 20 class ipv6 v6_class action pbr action_list2
switch (config-policy)# exit
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                              |
|------------------|-------------------------------------------|
| 10.08            | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | --                                        |

## Command Information

| Platforms    | Command context                                                                                                                     | Authority                                                                          |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config<br>The <code>policy</code> command takes you into the <code>config-policy</code> context where you enter the policy entries. | Administrators or local user group members with execution rights for this command. |

## show pbr

```
show pbr {interface <INTERFACE-NAME>|vrf <VRF-NAME>|summary}
```

## Description

Shows a detailed view of Policy Based Routing (PBR) in the system.

| Parameter        | Description                                               |
|------------------|-----------------------------------------------------------|
| <VRF-NAME>       | Specifies name of a VRF.                                  |
| <INTERFACE-NAME> | Specifies an interface. Format: <b>member/slot/port</b> . |

## Usage

Show commands can only reference the default VRF.

## Examples

Showing PBR summary information when there is no active next-hop in the system:

```
switch# show pbr summary
VRF      Port    Policy    PBR      Seq  Type    Nexthop
-----
No active PBR nexthop found
-----
```

Showing PBR summary information when there are active next-hops in the system:

```
switch# show pbr summary
```

| VRF     | Port  | Policy   | PBR   | Seq | Type    | Nexthop          |
|---------|-------|----------|-------|-----|---------|------------------|
| default | 1/1/1 | policy_1 | pbr_1 | 10  | nexthop | 1.1.1.1 (active) |
|         | 1/1/2 | policy_2 | pbr_2 | 20  | nexthop | 5.5.5.5 (active) |

Showing PBR summary information when displaying a policy with a **pbr-action-list** applied on a VxLAN L3VNI:

```
switch# configure terminal
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 10000
switch(config-vni-10000)# apply policy p1 routed-in
switch(config-vni-10000)# show pbr summary
VRF
    Port
        Policy
            Class
                PBR
                    Sequence  Type
                        Nexthop
-----
red
    vni10000
        p1
            c1
                pbr1
                    10  nexthop
                        11.2.1.4
(active)
-----
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification                              |
|------------------|-------------------------------------------|
| 10.08            | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | --                                        |

### Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

### show pbr-action-list

show pbr-action-list [<ACTION-LIST-NAME>] [commands] [configuration] [vsx-peer]

### Description

Shows the current PBR action list configuration. Action list entries are displayed in ascending order of their sequence number.

| Parameter          | Description                            |
|--------------------|----------------------------------------|
| <ACTION-LIST-NAME> | Specifies the PBR action list name.    |
| commands           | Formats output as CLI commands.        |
| configuration      | Displays user-specified configuration. |
| vsx-peer           | Displays VSX peer switch information.  |

## Restrictions

If an action list entry is modified to an invalid value (for example through the REST interface), this command will indicate a mismatch for that action entry when run. In this event, use the **pbr-action-list <NAME> reset** command to restore it to the previous valid value.

## Usage

- This command does not indicate whether the action list is configured in a policy or applied to an interface. Use the **show pbr** command for PBR status involving action lists.
- A single action list is shown by specifying its name or you can show all action lists by omitting a name argument.
- Using the additional commands keyword, you can change the tabulated output to a configuration style output for single or all list display.

## Examples

Create two PBR action lists then run **show pbr-action-list** to display all configured action lists in the default configuration mode:

```
switch(config)# pbr-action-list v4_pbr
switch(config-pbr-action-list-v4_pbr)# 1 nexthop 1.1.1.1
switch(config-pbr-action-list-v4_pbr)# 5 default-nexthop 2.2.2.2
switch(config-pbr-action-list-v4_pbr)# 10 interface null
switch(config-pbr-action-list-v4_pbr)# exit
switch(config)#
switch(config)# pbr-action-list v6_pbr
switch(config-pbr-action-list-v6_pbr)# 20 nexthop 2000:abcd::cccc:dddd
switch(config-pbr-action-list-v6_pbr)# 40 default-nexthop 1000:abcd::1234:5678
switch(config-pbr-action-list-v6_pbr)# 60 interface null
switch(config-pbr-action-list-v6_pbr)# exit
switch#

switch# show pbr-action-list
      Name
Additional PBR-Action-List Parameters
Sequence   Type                Nexthop
-----
-----
1          v4_pbr
          nexthop            1.1.1.1
5          default-nexthop    2.2.2.2
10         interface          null
```



```

20      v6_pbr
40      nexthop          2000:abcd::cccc:dddd
60      default-nexthop  1000:abcd::1234:5678
        interface        null

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                              |
|------------------|-------------------------------------------|
| 10.08            | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | --                                        |

## Command Information

| Platforms    | Command context | Authority                                                                                                                                                              |
|--------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show running-config current-context

show running-config current-context

### Description

Displays the configuration of the PBR action list in the current configuration context, in commands mode.

| Parameter       | Description                                                      |
|-----------------|------------------------------------------------------------------|
| running-config  | Shows configuration currently running on switch.                 |
| current-context | Limits display to current config context only, in commands mode. |

### Usage

Useful for reexamining entries previously entered into the action list after its entries have scrolled off the terminal due to other output or upon reentering the context of an existing action list.

### Examples

Creating two PBR action lists and running **show running-configuration curent-context** to display the action list configuration in commands mode:

```

switch(config)# pbr-action-list v4_pbr
switch(config-pbr-action-list-v4_pbr)# 1 nexthop 1.1.1.1
switch(config-pbr-action-list-v4_pbr)# 5 default-nexthop 2.2.2.2

```

```

switch(config-pbr-action-list-v4_pbr)# 10 interface null
switch(config-pbr-action-list-v4_pbr)# exit
switch(config)#
switch(config)# pbr-action-list v6_pbr
switch(config-pbr-action-list-v6_pbr)# 20 nexthop 2000:abcd::cccc:dddd
switch(config-pbr-action-list-v6_pbr)# 40 default-nexthop 1000:abcd::1234:5678
switch(config-pbr-action-list-v6_pbr)# 60 interface null
switch(config-pbr-action-list-v6_pbr)# show running-config current-context
pbr-action-list v6_pbr
  20 nexthop 2000:abcd::cccc:dddd
  40 default-nexthop 1000:abcd::1234:5678
  60 interface null

```

Switching context back to the first action list and running the same command:

```

switch(config-pbr-action-list-v6_pbr)# pbr-action-list v4_pbr
switch(config-pbr-action-list-v4_pbr)#
switch(config-pbr-action-list-v4_pbr)# show running-config current-context
pbr-action-list v4_pbr
  1 nexthop 1.1.1.1
  5 default-nexthop 2.2.2.2
  10 interface null

```

Removing action list entry number 5 and running the command again:

```

switch(config-pbr-action-list-v4_pbr)# no 5
switch(config-pbr-action-list-v4_pbr)# show running-config current-context
pbr-action-list v4_pbr
  1 nexthop 1.1.1.1
  10 interface null

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                              |
|------------------|-------------------------------------------|
| 10.08            | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | --                                        |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |



---

IP Directed Broadcast is only supported on the 5420 and 6200 Switch Series. It is not supported on the 4100i, 6000, or 6100 Switch Series.

---

IP Directed Broadcast is a feature by which remote administration tasks such as backups and wake-on-LAN (WOL) application can be achieved by sending directed broadcast packets for hosts/servers residing on a different subnet. The datagram is routed by normal mechanisms until it reaches a gateway attached to the destination hardware network, at which point it is broadcast. This class of broadcasting is also known as 'directed broadcasting'. IP Directed Broadcast packets are Layer 3 subnet broadcasts.

For example, for the broadcast IP address 192.168.1.255, the corresponding egress interface Subnet IP would be 192.168.1.x/24. Classless subnetting (CIDR) is also supported. For example, for the broadcast IP address 10.10.15.255, the broadcast address for the corresponding interface IP would be 10.10.15.x/21.

IP Directed Broadcast is supported on Route Only Port (ROP), Switched Virtual Interface (SVI), Layer 3 Link Aggregation Group (L3LAG) and Virtual Extensible Lan (VxLAN) interfaces, but is disabled by default. The feature is supported for the associated interface addresses for both the primary and secondary addresses. IP directed broadcast can be enabled on all the configured L3 interfaces. It is only supported for IPv4 addresses.



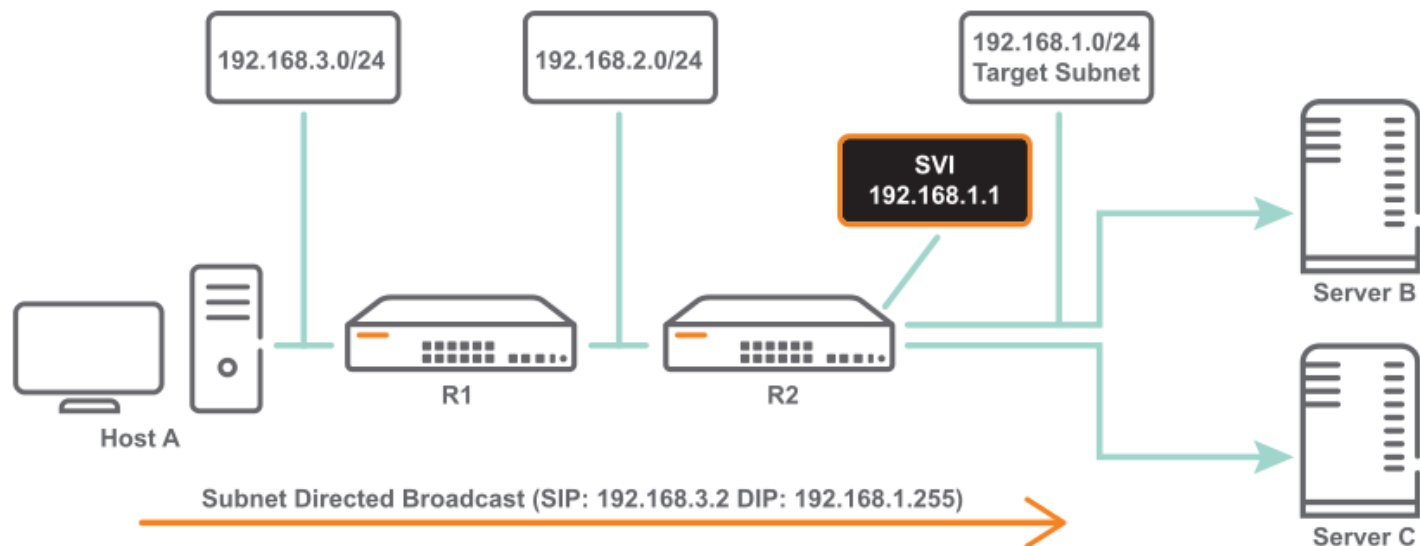
- VxLAN is only supported on the 6300, 6400, 8100 and 8360 Switch Series.
  - VxLAN interfaces require IP Directed Broadcast to be enabled on a Switched Virtual Interface associated with the VxLAN interface.
  - IP Directed Broadcast is not supported with Dynamic VRF Route Leak.
- 

For more information on this feature, see the related video on the [HPE Aruba Networking AirHeads Broadcasting Channel](#).

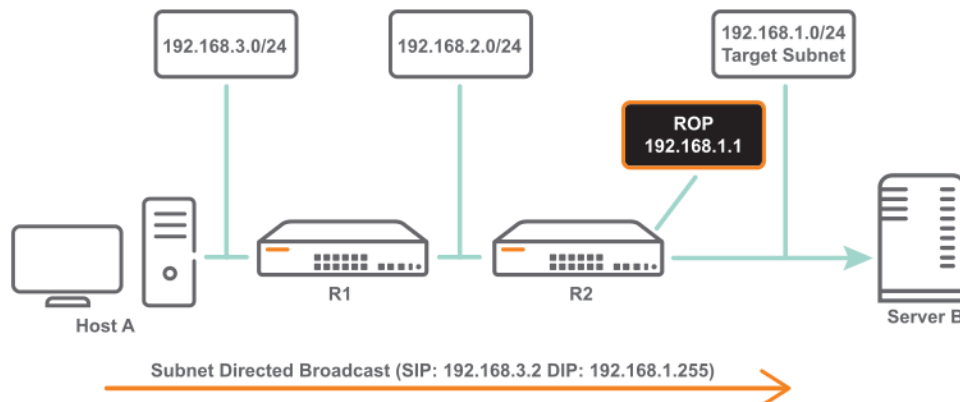
## IP Directed Broadcast configuration example

The following are sample topology diagrams for an IP Directed Broadcast configuration. Figure 1 shows when the egress interface is an SVI, while figure 2 shows the diagram when the egress interface is an ROP.

**Figure 1** SVI: IP Directed Broadcast



**Figure 2** ROP: IP Directed Broadcast



Intermediate routers forward IP Directed Broadcast packets as Unicast. The IP directed broadcast packet is broadcast or flood in the target subnet (DA MAC: All 0xFFs) only after the last hop router. Host A in subnet 192.168.3.0/24 wants to inject a IP Directed Broadcast (192.168.1.255) packet into Target Subnet 192.168.1.0/24. Router R1 forwards the IP Datagram with DIP 192.168.1.255 as a regular Unicast Datagram. Router R2 then floods the IP Datagram over egress ROP, SVI or VxLAN interface with Destination MAC as all 0xFFs.

At Ingress, Port Based ACLs (PACL) and VLAN Based ACLs (VACL) can be used to restrict/allow IP Directed Broadcast traffic. Existing Port based ACLs (PACL) can be used to allow or disallow certain IP Directed Broadcast Traffic.

An ACL can be configured using the `access-list ip <ACL-NAME>` command and then applied using the `apply access-list ip <ACL-NAME>` command as shown in the following output.

```
switch(config)# access-list ip ipdbacl

switch(config)# interface 1/1/1
switch(config-if)# apply access-list ipdbacl
in Inbound (ingress) traffic
```

```
out Outbound (egress) traffic

switch(config-if)# int lag 10
switch(config-lag-if)# apply access-list ipdbacl
in Inbound (ingress) traffic
out Outbound (egress) traffic
```

The following is an example of the `show running-config` command on an ROP interface.

```
switch(config)# interface 1/1/1
no shutdown
ip address 192.168.1.1/24
ip directed-broadcast
```

The following is an example of the `show running-config` command on an SVI interface.

```
switch(config)# vlan 10
interface vlan10
no shutdown
ip address 192.168.1.1/24
ip directed-broadcast
```

The following is an example of the `show running-config` command on an L3LAG interface.

```
switch(config)# interface lag 3
no shutdown
ip address 192.168.1.1/24
ip directed-broadcast
```



---

Note: Currently egress ACL is supported only on ROP and LAG interfaces, and not on an SVI interface.

---

## IP Directed Broadcast commands

### copy support-file feature

`copy support-file feature 13`

#### Description

Captures support logs to debug any IP Directed Broadcast issues.

#### Examples

Capturing the support logs into a local file:

```
switch# copy support-file feature 13 sftp
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

### Command History

| Release | Modification                                   |
|---------|------------------------------------------------|
| 10.09   | Command introduced for the 6200 Switch Series. |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | Manager (#)     | Administrators or local user group members with execution rights for this command. |

## ip directed-broadcast

```
ip directed-broadcast
no ip directed-broadcast
```

### Description

Turns on IP Directed Broadcast for the specified interface. The **no** form of this command turns it off. This command is disabled by default.

### Examples

Enabling and disabling IP Directed Broadcast on an physical interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip directed-broadcast
switch(config-if)# no ip directed-broadcast
```

Enabling and disabling IP Directed Broadcast on a VLAN interface:

```
switch(config)# interface vlan 100
switch(config-if-vlan)# ip directed-broadcast
switch(config-if-vlan)# no ip directed-broadcast
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification                                   |
|---------|------------------------------------------------|
| 10.09   | Command introduced for the 6200 Switch Series. |

## Command Information

| Platforms    | Command context | Authority                                                                                                                                                              |
|--------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | config-if       | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show arp (IPDB)

show arp

## Description

Shows IP directed broadcast verification.



IP Directed Broadcast is not supported on subinterfaces on 8100, 6300, 6400, 8325, 8360 and 10000 Switch series.

## Examples

Showing IP directed broadcast verification:

```
switch# show arp
IPv4 Address      MAC                Port              Physical Port      State
-----
1.1.1.255         FF:FF:FF:FF:FF:FF  1/1/1            1/1/1              permanent
3.1.1.255         FF:FF:FF:FF:FF:FF  vlan10           vlan10              permanent

Total Number Of ARP Entries Listed: 2.
-----
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification                                   |
|---------|------------------------------------------------|
| 10.09   | Command introduced for the 6200 Switch Series. |

## Command Information

| Platforms    | Command context | Authority                                                                                                                                                              |
|--------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip interface (Directed Broadcast)

show ip interface <INTERFACE-NAME>

## Description

Displays the status of IP Directed Broadcast on the specified interface along with other interface related attributes.

| Parameter        | Description                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------|
| <INTERFACE-NAME> | Specifies the interface to use as a source for displaying the status of the IP Directed Broadcast. |

## Examples

Displaying the IP Directed Broadcast status on the specified interface:

```
switch# show ip interface vlan30

Interface vlan30 is up
Admin state is up
Hardware: Ethernet, MAC Address: 94:f1:28:21:63:00
IP MTU 1500
IP Directed Broadcast is Enabled
IPv4 address 192.168.3.1/24
L3 Counters: Rx Disabled, Tx Disabled

Statistics                RX                TX                Total
-----
L3 Packets                0                0                0
L3 Bytes                  0                0                0
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification                                   |
|---------|------------------------------------------------|
| 10.09   | Command introduced for the 6200 Switch Series. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip directed-broadcast

show ip directed-broadcast

### Description

Displays the summary of the interfaces on which IP Directed Broadcast is enabled.

### Examples

*On the HPE Aruba Networking 6400 Switch Series, interface identification differs.*

Displaying the summary of the interfaces on which IP Directed Broadcast is enabled:

```
switch# show ip directed-broadcast

IPv4 Directed Broadcast Configuration

Interface      Status
-----
1/1/1          Enabled
vlan10         Enabled
```



vlan30                      Enabled

Displaying IP Directed Broadcast Host entries installed in Neighbor cache:

```
switch# show arp state permanent
IPv4 Address      MAC                Port      Physical Port  State
-----
52.1.1.255        FF:FF:FF:FF:FF:FF  1/1/1     1/1/1          permanent
40.0.0.255        FF:FF:FF:FF:FF:FF  vlan20    vlan20          permanent
Total Number Of ARP Entries Listed- 2.
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification                                   |
|---------|------------------------------------------------|
| 10.09   | Command introduced for the 6200 Switch Series. |

## Command Information

| Platforms    | Command context             | Authority                                                                                                                                                              |
|--------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |



Key chain is not supported on the HPE Aruba Networking 6000 and 6100 Switch Series.

This chapter provides details for configuring and verifying the functions of a key chain. Key chain configurations are spread across `config` context, `keychain` context, and `keychain-key` context.

A key chain provides seamless authentication-key rollover. When authentication method is enabled, using key chain allows OSPF to overcome static key configuration to change the key periodically. Using key chain also reduces risks of keys being guessed by configuring rotating keys.



A key chain must be used by an application like OSPF that communicates by using the keys with its peers.

## Key chain commands

### accept-lifetime

```
accept-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds> | infinite}
| end-time <time> <month>/<day>/<year>}
```

#### Description

Configures the duration for which the key is valid for receiving packets.

The **no** form of this command configures the key packet receiving duration to the default value of an infinite time.

| Parameter  | Description                                                              |
|------------|--------------------------------------------------------------------------|
| start-time | Time at which the key chain lifetime starts. Required. Format: HH:MM:SS  |
| end-time   | Time at which the key chain lifetime expires. Required. Format: HH:MM:SS |
| day        | Day of the month. Required. Range: 1-31.                                 |
| month      | Month of the year. Required.                                             |
| year       | Year. Required. Range: 2020-2050                                         |
| duration   | Time in seconds. Optional. Range: 1-2147483646.                          |
| infinite   | Specifies infinite time for the key. Optional.                           |

#### Examples

Configuring the duration for which the key is valid for receiving packets:

```

switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 end-
time 10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020
duration 1000
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020
duration infinite
switch(config-keychain-key)# accept-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime duration 1000
switch(config-keychain-key)# accept-lifetime duration infinite

```

Configuring the key packet receiving duration to the default value of an infinite time:

```

switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no accept-lifetime

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context     | Authority                                                                                                                                                              |
|---------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config-keychain-key | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## cryptographic-algorithm

```

recv-id cryptographic-algorithm {aes-cmac-128 | hmac-sha-1 | hmac-sha-256 | hmac-sha-384 |
hmac-sha-512 | md5}
no cryptographic-algorithm

```

## Description

Configures the recv-id cryptographic algorithm for the key. The key will not be valid until the [receive ID](#), the [send ID](#), and [send lifetime](#) is configured for [TCP-AO](#). Choose one of the authentication algorithms from the following parameters. The **no** form of this command configures the default cryptographic algorithm for a key, md5.



TCP Authentication Option (TCP-AO) authentication supports only the *aes-cmac-128* and *hmac-sha-1* algorithms. If you are configuring TCP-AO, you must select one of these options.

| Parameter    | Description                                                                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| aes-cmac-128 | Sets the authentication algorithm for the key to AES-CMAC-128. This parameter is only supported for TCP-AO.                                                                               |
| hmac-sha-1   | Sets the authentication algorithm for the key to SHA-1. This parameter is also supported for TCP-AO.                                                                                      |
| hmac-sha-256 | Sets the authentication algorithm for the key to SHA-256.                                                                                                                                 |
| hmac-sha-384 | Sets the authentication algorithm for the key to SHA-384.                                                                                                                                 |
| hmac-sha-512 | Sets the authentication algorithm for the key to SHA-512.                                                                                                                                 |
| md5          | Sets the authentication algorithm for the key to md5. Maximum length of the key string supported: 16 bytes ( <b>config-if context</b> ), 64 bytes ( <b>config-keychain-key context</b> ). |

## Examples

Set the authentication algorithm for the key to **SHA-384**:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# recv-id cryptographic-algorithm hmac-sha-384
```

Set the authentication algorithm to the default, **md5**:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no recv-id cryptographic-algorithm
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                     |
|------------------|--------------------------------------------------|
| 10.11            | The <b>aes-cmac-128</b> parameter is introduced. |
| 10.07 or earlier | --                                               |

## Command Information

| Platforms    | Command context     | Authority                                                                          |
|--------------|---------------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config-keychain-key | Administrators or local user group members with execution rights for this command. |

## key

key <KEY-ID>

## Description

Creates the key for a key chain and enters the key chain key context. A maximum of 64 keys can be configured per key chain.

The **no** form of this command deletes the key from the key chain.

| Parameter | Description                            |
|-----------|----------------------------------------|
| <KEY-ID>  | ID of the key. Required. Range: 1-255. |

## Examples

Creating a key for a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
```

Deleting a key from a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# no key 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                                                                                                              |
|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config-keychain | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## keychain (global)

keychain <KEYCHAIN-NAME>

## Description

Creates the key chain and enters the key chain context. A maximum of 64 key chains can be configured in the system.

The **no** form of this command removes the key chain if it is not used by any subscribers.

| Parameter       | Description                      |
|-----------------|----------------------------------|
| <KEYCHAIN-NAME> | Name of the key chain. Required. |

## Examples

Creating a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

Removing a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                                                                                                              |
|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config          | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## key-string

key-string [{ciphertext | plaintext} <PASSWORD>]

### Description

Sets the key password. The password is internally stored in encrypted form. The key is not valid until its password has been set.

The **no** form of this command deletes the password used for the key.

| Parameter  | Description                                                |
|------------|------------------------------------------------------------|
| ciphertext | Specifies that the key password is provided as ciphertext. |
| plaintext  | Specifies that the key password is provided as plaintext.  |
| <PASSWORD> | Specifies the key password.                                |



When the key password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

### Examples

Setting the key password with plaintext:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string plaintext F82#450bHP
```

Setting the key password with plaintext prompting:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string
Enter the key password: *****
Re-Enter the key password: *****
```

Setting the key password with ciphertext:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string ciphertext AQBpfcifZ/P...biAAAOjc0a8=
```

Deleting the password for the key:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no key-string
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms     | Command context     | Authority                                                                                                                                                              |
|---------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config-keychain-key | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

### name

name <KEY-NAME>

no name <KEY-NAME>

## Description

Configures a name for a numbered key in a key chain.

The **no** form of this command removes the name of the key.

| Parameter  | Description                                                            |
|------------|------------------------------------------------------------------------|
| <KEY-NAME> | Specifies the name of the key in alphanumeric characters. Range: 1-64. |

## Examples

Creating a name for a key in a key chain called **abcdef123456**:

```
switch# configure terminal
switch(config)# keychain macsec_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# name abcdef123456
```

Removing the name of the key named **abcdef123456**:

```
switch# configure terminal
switch(config)# keychain macsec_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no name abcdef123456
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification   |
|---------|----------------|
| 10.11   | Command added. |

## Command Information

| Platforms     | Command context     | Authority                                                                                                                                                              |
|---------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config-keychain-key | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## recv-id

recv-id <0-255>

## Description

Configures the receive ID for a keychain key. The receive ID has to be unique across keys in the keychain.



The **no** form of this command configures removes the recv-id value. The receive ID can not be changed for an active key of a keychain which is associated with BGP neighbor.

| Parameter | Description                                                                       |
|-----------|-----------------------------------------------------------------------------------|
| <0-255>   | Set the receive ID corresponding to the keychain key. Supported values are 0-255. |

## Examples

Configuring the receive ID for the keychain key.

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# recv-id 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.11   | Command introduced. |

## Command Information

| Platforms     | Command context     | Authority                                                                                                                                                              |
|---------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config-keychain-key | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## send-id

send-id <0-255>

## Description

Configures the send ID for a keychain key. The send ID has to be unique across keys in the keychain. The **no** form of this command configures removes the send-id value. The send id can not be changed for an active key of a keychain which is associated with BGP neighbor.

| Parameter | Description                                                                    |
|-----------|--------------------------------------------------------------------------------|
| <0-255>   | Set the send ID corresponding to the keychain key. Supported values are 0-255. |

## Examples

Configuring the send ID for the keychain key.

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# send-id 218
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.11   | Command introduced. |

## Command Information

| Platforms     | Command context     | Authority                                                                                                                                                              |
|---------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config-keychain-key | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## send-lifetime

```
send-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds> | infinite} |
end-time <time> <month>/<day>/<year>}
```

## Description

Configures the duration for which the key is valid for sending packets.

The **no** form of this command configures the key packet sending duration to the default value of an infinite time.

| Parameter  | Description                                                              |
|------------|--------------------------------------------------------------------------|
| start-time | Time at which the key chain lifetime starts. Required. Format: HH:MM:SS  |
| end-time   | Time at which the key chain lifetime expires. Required. Format: HH:MM:SS |
| day        | Day of the month. Required. Range: 1-31.                                 |
| month      | Month of the year. Required.                                             |
| year       | Year. Required. Range: 2020-2050                                         |
| duration   | Time in seconds. Optional. Range: 1-2147483646.                          |
| infinite   | Specifies infinite time for the key. Optional.                           |

## Examples

Configuring the duration for which the key is valid for sending packets:

```

switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 end-time
10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 duration
1000
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 duration
infinite
switch(config-keychain-key)# send-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime duration 1000
switch(config-keychain-key)# send-lifetime duration infinite

```

Configuring the key packet sending duration to the default value of an infinite time:

```

switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no send-lifetime

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context     | Authority                                                                                                                                                              |
|---------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | config-keychain-key | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show capacities keychain

show capacities keychain

### Description

Shows the maximum number of key chains and keys configurable in a key chain.

### Example

```

switch# show capacities keychain

System Capacities: Filter Keychain
Capacities Name
Value
-----
-----

```

```
Maximum number of keychains supported in the system
64
Maximum number of Keys supported in a single Keychain
64
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | Operator (>)    | Administrators or local user group members with execution rights for this command. |

## show keychain

```
show keychain [<KEYCHAIN-NAME>]
```

## Description

Shows information about configured and active keys of a named key chain or (if **keychain-name** is not specified) all configured key chains.

| Parameter       | Description                      |
|-----------------|----------------------------------|
| <KEYCHAIN-NAME> | Name of the key chain. Optional. |

## Example

```
switch# show keychain
Keychain Name : macsec_keys
  Number of Keys      : 1
  Active Send Key ID  :
  Active Recv Key IDs :

  Key ID : 1
    Key name      : abcdef123456
    Key string     :
  AQBapYa+0qQDzcakbB1TopeX0AMYDDWDW015orkH5mY3qJDaBAAAADASiBQ=
    Send Key Validity : 00:00:00 01/01/2020 to Infinite
    Recv Key Validity : 00:00:00 01/01/2020 to Infinite

Keychain Name : ospf_keys
  Number of Keys      : 2
  Active Send Key ID  : 7
  Active Recv Key IDs : 7, 200
```

```

Key ID          : 7
Key name        : -
Key string      :
AQBapZlOHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Crypto-Algorithm : sha256
Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/1/2020 to infinite
Key ID          : 200
Key name        : -
Key string      :
AQBapZlOHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Crypto-Algorithm : sha512
Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021

Keychain Name : bgp_keys
Number of Keys : 2
Active Send Key ID : 7
Active Recv Key IDs : 7

Key ID          : 7
Key name        : -
Key string      :
AQBapZlOHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Crypto-Algorithm : md5
Send Key Validity : 00:00:01 10/26/2020 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/22/2020 to infinite
Key ID          : 8
Key name        : -
Key string      :
AQBapZlOHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Crypto-Algorithm : sha384
Send Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021
...
...

Keychain Name : ospf_keys
Number of Keys : 2
Active Send Key ID : 7
Active Recv Key IDs : 7, 200

Key ID          : 7
Key name        : -
Key string      :
AQBapZlOHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Crypto-Algorithm : sha256
Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/1/2020 to infinite
Key ID          : 200
Key name        : -
Key string      :
AQBapZlOHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Crypto-Algorithm : sha512
Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | Operator (>)    | Administrators or local user group members with execution rights for this command. |

## show running-config keychain

show running-config keychain

### Description

Shows the configurations for key chain protocol.

### Example

```
switch# show running-config keychain
keychain ospf_keys
  key 1
    key-string ciphertext
    AQBapZl0HiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    cryptographic-algorithm md5
    accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
    send-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
  key 45
    key-string ciphertext
    AQBapZl0HiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
  key 33
keychain macsec_keys
  key 1
    name abcdef123456
    key-string ciphertext
    AQBapYa+0qQDzcakbB1TopeX0AMYDDWDW015orkH5mY3qJDaBAAAADASiBQ=
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | Operator (>)    | Administrators or local user group members with execution rights for this command. |

The client IP address tracking feature will learn and update the IP addresses of the access devices and clients connected to the switch. It can track addresses of directly connected clients, as well as clients connected to a downstream device such as a wireless access point.

IP Client tracker can be enabled for user-based tunneling (UBT) clients to learn the IP addresses on the required VLANs and interfaces.

Enabling IP Client Tracker based on UBT modes:

- If the UBT mode is Local VLAN, enable IP Client tracker for the UBT client VLAN defined using the `ubt-client-vlan` command.
- If the UBT mode is VLAN extend, enable IP Client tracker for the UBT client VLAN defined under role.

## IP Client Tracker commands

### client track ip

```
client track ip
```

#### Description

Enables client IP address tracking on the switch. The default is disabled on global and VLAN levels.

Admin users can enable client IP address tracking at the VLAN level.



---

Tracking enabling will take effect only if the client IP address tracking is enabled at system and VLAN level.

---

The **no** form of the command disables client IP address tracking. If tracking is disabled at switch level, it will be stopped even if it is enabled at VLAN or port level.

#### Example

Enable client IP address tracking at switch level:

```
switch(config)# client track ip
```

Enable client IP address tracking on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# client track ip
```

Enable client IP address tracking on VLANs 10 to 100:

```
switch(config)# vlan 10-100
switch(config-vlan-<10-100>)# client track ip
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms                             | Command context | Authority                                                                                                                                                              |
|---------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i<br>5420<br>6000<br>6100<br>6200 | config          | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## client track ip { enable | disable | auto }

client track ip { enable | disable | auto }

### Description

Enables client IP address tracking on the specified set of interfaces. Tracking will take effect only if client IP address tracking is enabled at both the system level and for the VLAN to which the port belongs. Default: auto.

The **no** form of the command disables client IP address tracking on the specified set of interfaces.

| Parameter | Description                                                                                                                                                                                                                                     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| enable    | Specifies that all client IP addresses will be tracked in the port.                                                                                                                                                                             |
| disable   | Specifies that client IP addresses will not be tracked in the port.                                                                                                                                                                             |
| auto      | Specifies the following:<br>For LLDP devices: Only the specified client IP address will be tracked in the port and other client IP addresses will not be tracked.<br>For non-LLDP devices: All client IP addresses will be tracked in the port. |

### Example

Enable client IP address tracking on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip enable
```

Enable client IP address tracking on interfaces 1/1/1 to 1/1/5:



```
switch(config)# interface 1/1/1-1/1/5
switch(config-if-<1/1/1-1/1/5>)# client track ip enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms                             | Command context | Authority                                                                                                                                                              |
|---------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i<br>5420<br>6000<br>6100<br>6200 | config-if       | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## client track ip client-limit

```
client track ip client-limit <CLIENT-LIMIT>
```

### Description

Configures the maximum number of clients to be tracked on the specified set of interfaces.

For the HPE Aruba Networking 6000 and 6100 Switch Series the **no** form of the command resets the client limit to the default value of 128.

For the HPE Aruba Networking 5420 and 6200 Switch Series the **no** form of the command resets the client limit to the default value of 2048.

| Parameter           | Description                                                                                                                                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>CLIENT-LIMIT</i> | Specifies the maximum number of clients tracked on a port. Required. For the HPE Aruba Networking 6000 and 6100 Switch Series, Range: 1-128. Default: 128. For the HPE Aruba Networking 5420 and 6200 Switch Series, Range: 1-2048. Default: 2048. |

### Example

Configure the maximum number of clients to be tracked on interface 1/1/5:

```
switch(config)# interface 1/1/5
switch(config-if)# client track ip client-limit 32
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms                             | Command context | Authority                                                                                                                                                              |
|---------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i<br>5420<br>6000<br>6100<br>6200 | config-if       | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## client track ip update-interval

client track ip update-interval <INTERVAL>

### Description

Configures how often client IP addresses are updated.

The **no** form of the command resets the update interval to the default of 1800 seconds.

| Parameter       | Description                                                                         |
|-----------------|-------------------------------------------------------------------------------------|
| <i>INTERVAL</i> | Specifies the update interval in seconds. Required. Range: 60-28000. Default: 1800. |

### Example

Configure the update interval for an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip update-interval 600
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms                             | Command context | Authority                                                                                                                                                              |
|---------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i<br>5420<br>6000<br>6100<br>6200 | config-if       | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# client track ip update-method probe

client track ip update-method probe

## Description

Enables probing the client to update the IP address.

The probe is sent to all clients on the tracking list that have an IP address in the following scenarios:

1. IP packets are not received from the clients during the IP address update cycle.
2. There is no IP packet from a learned IP address. In this case, a probe will be sent for the IP address to confirm if it is still owned by that client.

The **no** form of the command disables probing.

## Example

Disable probing to update the client IP address:

```
switch(config)# no client track ip update-method probe
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms                             | Command context | Authority                                                                                                                                                              |
|---------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i<br>5420<br>6000<br>6100<br>6200 | config          | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# show capacities

show capacities

## Description

Shows the capacities configured on the switch.

## Example

```
switch# show capacities

System Capacities:
Capacities Name                                     Value
```

```

-----
Maximum number of Access Control Entries configurable in a system      14336
Maximum number of Access Control Lists configurable in a system      1024
Maximum number of class entries configurable in a system             1024
Maximum number of classes configurable in a system                   512
Maximum number of entries in an Access Control List                  1024
Maximum number of entries in a class                                 1024
Maximum number of entries in a policy                                1024
Maximum number of classifier policies configurable in a system        512
Maximum number of policy entries configurable in a system            1024
Maximum number of clients supported for tracking the IP address in the system 128

switch# show capacities client-track-ip-client-limit

System Capacities: Filter Client Track IP Client Limit
Capacities Name
Value
-----
-
Maximum number of clients supported for tracking the IP address in the system

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms                             | Command context | Authority                                                                          |
|---------------------------------------|-----------------|------------------------------------------------------------------------------------|
| 4100i<br>5420<br>6000<br>6100<br>6200 | Operator (>)    | Administrators or local user group members with execution rights for this command. |

## show client ip { count | port | vlan }

```
show client ip { count | port | vlan }
```

## Description

Shows number of client IP addresses or information about client IP addresses tracked on ports and VLANs.

| Parameter    | Description                                        |
|--------------|----------------------------------------------------|
| <i>count</i> | Displays number of clients tracked.                |
| <i>port</i>  | Displays client IP addresses tracked on the ports. |

| Parameter         | Description                                        |
|-------------------|----------------------------------------------------|
| <code>vlan</code> | Displays client IP addresses tracked on the VLANs. |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms                             | Command context | Authority                                                                          |
|---------------------------------------|-----------------|------------------------------------------------------------------------------------|
| 4100i<br>5420<br>6000<br>6100<br>6200 | Operator (>)    | Administrators or local user group members with execution rights for this command. |

## Routing Information Protocol (RIP)



Routing Information Protocol (RIP, RIPv2 and RIPv6) is not supported on the HPE Aruba Networking 6000 and 6100 Switch Series.

Routing Information protocol (RIP, RIPv2, RIPv6) is a distance-vector routing protocol that uses hop count as a routing metric. It is useful in small networks. RIP can be configured on L3 ports, LAG, VLAN and Loopback interfaces. All configurations work in the interface context and require the associated interface to be routing interface.

- **RIP/RIPv2:** IPv4 implementation of Routing Information protocol defined in RFC 2453.
- **RIPv6:** IPv6 implementation of Routing Information protocol defined in RFC 2080.

### Overview

RIP characteristics:

- Prevents routing loops by adding a limit to the number of hops allowed in a path from source to destination.
- Allows a maximum of 15 hops which limits the size of the network that RIP can support.
- Uses broadcast UDP data packets to exchange routing information.
- RIPv2 is a classless protocol that supports variable-length subnet mask (VLSM), classless inter-domain routing (CIDR) and route summarization.
- RIPv6 uses ipsec for message authentication.

RIP limitations:

- RIPv1 is not supported.
- Only poison reverse is supported for loop prevention.
- RIP/RIPv6 metric configuration support is not available.
- Redistribute RIP/RIPv6 is not supported in BGP/BGP+.

### RIPv2 (IPv4) commands

#### Configuration commands

##### router rip

```
router rip <PROCESS-ID> [vrf <VRF-NAME>]  
no router rip <PROCESS-ID> [vrf <VRF-NAME>]
```

##### Description

Creates RIP process if not already created and enters the **router rip <PROCESS-ID>** context for the VRF mentioned. If no VRF is mentioned, a default is used. Only one RIP process is allowed per VRF.

The **no** form of this command deletes the RIP instance for the VRF. If no VRF is mentioned the default is deleted.

| Parameter                         | Description                                        |
|-----------------------------------|----------------------------------------------------|
| <code>&lt;PROCESS-ID&gt;</code>   | Specifies name of the RIP process ID. Range: 1-63. |
| <code>vrf &lt;VRF-NAME&gt;</code> | Specifies VRF name.                                |

## Examples

Creating RIP process and naming the VRF:

```
switch(config)# router rip 2 vrf red
```

Deleting RIP process:

```
switch(config)# no router rip 2 vrf red
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## Interface commands

### ip rip

```
ip rip <PROCESS-ID> {all-ip | ip-address}  
no ip rip <PROCESS-ID> {all-ip | ip-address}
```

### Description

Enables RIP process on an interface.

The **no** form of this command deletes the RIP process from an interface.

| Parameter                              | Description                            |
|----------------------------------------|----------------------------------------|
| <code>ip rip &lt;PROCESS-ID&gt;</code> | Specifies RIP process ID. Range: 1-63. |

| Parameter  | Description                                                     |
|------------|-----------------------------------------------------------------|
| all-ip     | Specifies RIP for all IP addresses configured on the interface. |
| ip-address | Specifies IP address for RIP on the interface.                  |

## Usage

- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.
- If **ip rip 1 all-ip** is configured and a new IP address is added to the interface, RIP configurations will not be applicable for the newly added IP address.

## Examples

Configuring RIP for all IP addresses configured on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 all-ip
```

Deleting RIP for all IP addresses configured on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip 1 all-ip
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

### ip rip all-ip enable

```
ip rip all-ip enable
no ip rip all-ip enable
```

## Description

Enables RIP process for all RIP enabled IP addresses configured on interface.

The **no** form of this command disables RIP process on the interface.

## Usage



- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1  
switch(config-if)# ip rip all-ip enable
```

Disabling RIP process on interface:

```
switch(config)# interface 1/1/1  
switch(config-if)# no ip rip all-ip enable
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ip rip all-ip disable

```
ip rip all-ip disable  
no ip rip all-ip disable
```

## Description

Disables RIP process for all RIP enabled IP addresses configured on the interface.

The **no** form of this command enables RIP process for all RIP enabled IP addresses configured on the interface.

## Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip enable
```

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ip rip all-ip send disable

```
ip rip all-ip send disable
no ip rip all-ip send disable
```

## Description

Disables interface from sending RIP packets for all RIP enabled IP addresses.

The **no** form of this command enables interface to send RIP packets for all RIP enabled IP addresses.

## Usage

- Default settings allow an interface to send RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling interface from sending RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip send disable
```

Enabling interface to send RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip send disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## ip rip all-ip receive disable

```
ip rip all-ip receive disable
no ip rip all-ip receive disable
```

## Description

Disables interface from receiving RIP packets for all enabled IP addresses.

The **no** form of this command enables interface to receive RIP packets for all RIP enabled IP addresses.

## Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling interface from receiving RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip receive disable
```

Enabling interface to receive RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip receive disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## Routing commands

### enable (router rip)

enable  
no enable

### Description

Enables RIP process if disabled. By default RIP process is enabled.  
The **no** form of this command disables the RIP process.

### Examples

Enabling RIP process when disabled:

```
switch(config)# router rip 1
switch(config-rip-1)# enable
```

Disabling RIP process when enabled:

```
switch(config)# router rip 1
switch(config-rip-1)# no enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## disable (router rip)

disable  
no disable

### Description

Disables RIP process.

The **no** form of this command enables the RIP process.

### Examples

Disabling RIP process:

```
switch(config)# router rip 1  
switch(config-rip-1)# disable
```

Enabling RIP process:

```
switch(config)# router rip 1  
switch(config-rip-1)# no disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## distance (router rip)

distance <DISTANCE>  
no distance

### Description

Configures administrative distance for RIP. Administrative distance is used as criteria to select the best route when multiple protocols have the same route.

The **no** form of this command sets the RIP administrative distance to the default. Default: 120.

| Parameter  | Description                                             |
|------------|---------------------------------------------------------|
| <DISTANCE> | Specifies RIP administrative distance. Range: 1 to 255. |

## Examples

Configuring administrative distance for RIP:

```
switch(config)# router rip 1  
switch(config-rip-1)# distance 100
```

Setting administrative distance for RIP to default values:

```
switch(config)# router rip 1  
switch(config-rip-1)# no distance
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## maximum-paths (router rip)

```
maximum-paths <MAX-VALUE>  
no maximum-paths
```

## Description

Sets the maximum number of ECMP routes that RIP can support.

The **no** form of this command sets the maximum number of ECMP routes to the default value of 4.

| Parameter   | Description                                     |
|-------------|-------------------------------------------------|
| <MAX-VALUE> | Sets the number of RIP ECMP routes. Range: 1-8. |

## Examples

Setting maximum number of RIP ECMP routes:

```
switch(config)# router rip 1  
switch (config-rip-1)# maximum-paths 8
```

Setting maximum number of RIP ECMP routes to default:

```
switch(config)# router rip 1  
switch (config-rip-1)# no maximum-paths
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## redistribute (rip)

```
redistribute {bgp | connected | ospf <PROCESS-ID> | static}  
no redistribute {bgp | connected | ospf <PROCESS-ID> | static}
```

## Description

Redistributes routes originating from other protocols into RIP.

The **no** form of this command disables redistribution of routes originating from other protocols into RIP.

| Parameter         | Description                                                                             |
|-------------------|-----------------------------------------------------------------------------------------|
| bgp               | Specifies BGP routes to redistribute into RIP.                                          |
| connected         | Specifies connected routes (directly attached subnet or host) to redistribute into RIP. |
| ospf <PROCESS-ID> | Specifies the OSPF route to redistribute into RIP. Range: <1-65535>                     |
| static            | Specifies static route to redistribute into RIP.                                        |



Parameters **ospf** and **bgp** are not supported on 6100 switches.

## Examples

Redistributing BGP routes into RIP:

```
switch(config)# router rip 1  
switch(config-rip-1)# redistribute bgp
```

Disabling BGP routes that originate from other protocols and redistribute into RIP:

```
switch(config)# router rip 1  
switch(config-rip-1)# no redistribute bgp
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## timers update (RIP)

```
timers update <INTERVAL> timeout <DURATION> garbage-collection <PERIOD>  
no timers
```

## Description

Configures RIP timers with specific values.

The **no** form of this command sets all RIP timers to default values.

| Parameter                   | Description                                                                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timers update <INTERVAL>    | Specifies frequency at which RIP sends updates to all of its peers. Range: 1 to 2147484. Default: 30.                                                                |
| timeout <DURATION>          | Specifies timeout duration from the point of the last refresh after a route is received from a peer timeout and is marked as expired. Range: 1 to 255. Default: 180. |
| garbage-collection <PERIOD> | Specifies amount of time route remains in routing table after route expiration. Range: 1 to 255. Default: 120.                                                       |

## Examples

Configuring RIP timers with specific values:

```
switch(config)# router rip 1  
switch(config-rip-1)# timers update 40 timeout 200 garbage-collection 150
```

Configuring RIP timers with default values:



```
switch(config)# router rip 1  
switch(config-rip-1)# no timers
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## RIPv2 clear commands

### clear ip rip statistics

```
clear ip rip [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

### Description

Clears RIP event statistics.

| Parameter    | Description                           |
|--------------|---------------------------------------|
| <PROCESS-ID> | Specifies RIP process ID. Range: 1-63 |
| all-vrfs     | Clears statistics for all VRFs.       |
| vrf          | Selects VRF to clear statistics for.  |
| <VRF-NAME>   | Specifies VRF name.                   |

## Examples

Clearing RIP event statistics:

```
switch# clear ip rip statistics
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                          |
|----------------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## RIPv2 interface commands

### enable (interface ip rip)

enable  
no enable

### Description

Enables RIP process for RIP enabled IP address configured on interface.

The **no** form of this command disables RIP process on interface.

### Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

### Examples

Enabling RIP process for RIP enabled IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# enable
```

Disabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

### disable (interface ip rip)

disable  
no disable

### Description

Disables RIP process for RIP enabled IP addresses configured on interface.

The **no** form of this command enables RIP process on interface.

### Examples

Disabling RIP process for RIP enabled IP addresses configured on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# disable
```

Enabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

### send disable (rip)

send disable  
no send disable

### Description

Disables an interface from sending RIP packets for a specific IP address.

The **no** form of this command enables interface for sending RIP packets for a specific IP address.

## Usage

- Default settings allow an interface to send and receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling interface from sending RIP packets for a specific IP address :

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# send disable
```

Enabling interface to send RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no send disable
```



---

For more information on features that use this command, refer to the IP Routing Guide for your switch model.

---

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

### receive disable (rip)

receive disable  
no receive disable

## Description

Disables interface from receiving RIP packets for a specific IP address.

The **no** form of this command enables interface for receiving RIP packets for a specific IP address.

## Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling interface from receiving RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# receive disable
```

Enabling interface for receiving RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no receive disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## RIPv2 show commands

### show capacities rip

```
show capacities rip
```

### Description

Displays maximum number of RIP interfaces, routes and process.

### Examples

Displaying maximum number of RIP interfaces, routes and process:

```
switch# show capacities rip

System Capacities: Filter RIP
Capacities Name                                         Value
```

```

-----
Maximum number of RIP interfaces configurable in the system      32
Maximum number of RIP processes supported across each VRF      1
Maximum number of routes in RIP supported across all VRFs      2540

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show capacities-status rip

show capacities-status rip

## Description

Displays number of RIP interfaces, routes and process configured in the system.

## Examples

Displaying number of RIP interfaces, routes and process:

```

switch# show capacities-status rip

System Capacities Status: Filter RIP
Capacities Name                                     Value Maximum
-----
Number of RIP interfaces configured in the system    0           32

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ip rip

show ip rip [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]

## Description

Displays general RIP configuration.

| Parameter    | Description                                         |
|--------------|-----------------------------------------------------|
| <PROCESS-ID> | Specifies RIP process ID. Range: 1-63.              |
| all vrfs     | Displays general RIP information for all VRFs.      |
| vrf          | Selects VRF to display general RIP information for. |
| <VRF-NAME>   | Specifies VRF name.                                 |

## Usage

- Parameters display general RIP information for a specific RIP process.
- Parameters display general RIP information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying general RIP configuration for all VRFs:

```
switch# show ip rip 34 all-vrfs
VRF : Default                               Process-ID : 34
-----
RIP Version           : RIPv2               Protocol Status : Enabled
Update Time           : 60 sec               Timeout Time    : 240 sec
Garbage Collection Time : 250 sec            ECMP           : 6
Distance              : 100                 Redistribution : static,
                                           ospf 1

VRF : vrf_1                               Process-ID : 34
-----
RIP Version           : RIPv2               Protocol Status : Enabled
Update Time           : 30 sec               Timeout Time    : 180 sec
Garbage Collection Time : 120 sec            ECMP           : 4
Distance              : 120                 Redistribution : None
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ip rip interface

show ip rip [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief] [all-vrfs | vrf <VRF-NAME>]

## Description

Displays information about RIP enabled interfaces.

| Parameter        | Description                                             |
|------------------|---------------------------------------------------------|
| <PROCESS-ID>     | Specifies RIP process ID. Range: 1-63.                  |
| <INTERFACE-NAME> | Specifies interface.                                    |
| brief            | Shows brief overview information for the RIP interface. |
| all-vrfs         | Displays interface information for all VRFs.            |
| vrf              | Selects specific VRF.                                   |
| <VRF-NAME>       | Specifies VRF.                                          |

## Usage

- Parameters display general RIP information for a specific RIP process.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

```
switch# show ip rip interface
Interface 1/1/1 is up, IP Address is 10.10.10.1/24
-----
VRF           : Default           Process-ID    : 1
Status        : Oper Up           Mode          : Send and Receive
MTU           : 500               Version       : RIPv2
Poision Reverse : Enabled

Interface 1/1/2 is up, IP Address is 20.10.10.1/24
-----
VRF           : Default           Process-ID    : 1
Status        : Admin Down        Mode          : Receive
MTU           : 500               Version       : RIPv2
Poision Reverse : Enabled
```



```
Interface 1/1/3 is up, IP Address is 30.10.10.1/24
```

```
-----  
VRF          : Default          Process-ID    : 1  
Status       : Admin Down      Mode         : Send  
MTU          : 500              Version      : RIPv2  
Poision Reverse : Enabled
```

```
switch# show ip rip interface brief  
VRF : default   Process-ID : 1  
-----
```

```
Total Number of Interfaces: 2
```

```
-----  
Interface    IP-Address/Mask    Status    MTU  
-----  
1/1/1        10.10.10.1/24       up        500  
1/1/2        20.10.10.1/24       up        500  
1/1/3        30.10.10.1/24       up        500
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ip rip neighbors

```
show ip rip [<PROCESS-ID>] neighbors [<IP-ADDRESS>] [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays information about RIP neighbors.

| Parameter    | Description                                                            |
|--------------|------------------------------------------------------------------------|
| <PROCESS-ID> | Specifies RIP process ID. Range: 1-63.                                 |
| <IP-ADDRESS> | Specifies IP address of a specific neighbor to display information on. |
| all-vrfs     | Displays neighbor information for all VRFs.                            |
| vrf          | Selects VRF to display neighbor information.                           |
| <VRF-NAME>   | Specifies VRF name.                                                    |

## Usage

- Parameters display RIP neighbor information for a specific RIP process.
- Parameters display RIP neighbor information for a specific neighbor.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIP neighbor information for all VRFs:

```
switch# show ip rip neighbors all-vrfs
VRF : default          Process-ID : 1
-----

Total Number of Neighbors: 1

Peer-Address    Type    Last-Update    Rcvd-Bad-Pkts    Rcvd-Bad-Routes
-----
1.1.1.2         RIPv2   0:0:7          4                 5
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ip rip routes

```
show ip rip [<PROCESS-ID>] routes [<PREFIX/LENGTH>] [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays RIP routing table for a specific RIP process.

| Parameter       | Description                                                                              |
|-----------------|------------------------------------------------------------------------------------------|
| <PROCESS-ID>    | Specifies RIP process ID to display information for a specific RIP process. Range: 1-63. |
| <PREFIX/LENGTH> | Specifies the network prefix.                                                            |
| all-vrfs        | Displays RIP routing information for all VRFs.                                           |

| Parameter  | Description                                     |
|------------|-------------------------------------------------|
| vrf        | Selects VRF to display RIP routing information. |
| <VRF-NAME> | Specifies VRF name.                             |

## Usage

- <PREFIX/LENGTH> is an optional parameter that displays RIP routing table information for a specific subnet.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIP routing table for all VRFs:

```
switch# show ip rip routes all-vrfs
VRF : default          Process-ID : 1
-----
Total Number of Routes : 6

Prefix          Metric    Interface    Nexthop
-----
10.1.0.0/16      2         1/1/1        30.1.1.2
20.1.2.0/24      3         1/1/1        30.1.1.2
30.1.1.0/24      1         1/1/1

VRF : vrf_1          Process-ID : 34
-----

Prefix          Metric    Interface    Nexthop
-----
20.1.0.0/16      10        1/1/2        50.1.1.2
40.1.2.0/24      14        1/1/2        50.1.1.2
50.1.1.0/24      1         1/1/2
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ip rip statistics

show ip rip [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]

## Description

Displays RIP statistics.

| Parameter                       | Description                                            |
|---------------------------------|--------------------------------------------------------|
| <code>&lt;PROCESS-ID&gt;</code> | Specifies RIP process ID. Range: 1-63.                 |
| <code>all-vrfs</code>           | Displays statistics information for all VRFs.          |
| <code>vrf</code>                | Selects VRF to display RIP statistics information for. |
| <code>&lt;VRF-NAME&gt;</code>   | Specifies VRF name.                                    |

## Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIP statistics for all VRFs:

```
switch# show ip rip statistics all-vrfs
VRF : default   Process-ID : 1
-----

Global Route Changes : 50
Global Queries       : 2
Last Cleared          : 0h 30m 28s ago

VRF : vrf_1     Process-ID : 34
-----

Global Route Changes : 20
Global Queries       : 0
Last Cleared          : 0h 30m 28s ago
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ip rip statistics interface

```
show ip rip [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>] [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays RIP statistics for RIP enabled interfaces.

| Parameter        | Description                                      |
|------------------|--------------------------------------------------|
| <PROCESS-ID>     | Specifies RIP process ID. Range: 1-63.           |
| <INTERFACE-NAME> | Specifies name of interface.                     |
| all-vrfs         | Displays RIP interface statistics for all VRFs.  |
| vrf              | Selects VRF to display RIP interface statistics. |
| <VRF-NAME>       | Specifies VRF name.                              |

## Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIP statistics for a RIP enabled interface:

```
switch# show ip rip statistics interface 1/1/1
VRF : default Process-ID : 1 interface 1/1/1
-----
IP-Address      Trigger-Updates  Rcvd-Bad-Packets  Rcvd-Bad-Routes
-----
10.1.1.1        15                3                  4
Last Cleared : 0h 30m 28s ago
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show running-config

```
show running config
```

## Description

Displays all running configurations for all protocols including RIP.

## Examples

Displaying all running configurations for all protocols including RIP:

```
switch# show running-config
Current configuration:
!
!Version Halon 0.1.0 (Build: genericx86-64-Halon-0.1.0-master-20170309054955-dev)
!Schema version 0.1.8
lldp enable
timezone set utc
vrf blue
vrf green
vrf red
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf red
router rip 1
    maximum-paths 5
    distance 1
router rip 1 vrf red
    default-information originate always
    maximum-paths 7
    distance 5
    redistribute ospf 1
    timers update 40 timeout 200 garbage-collection 120
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ip address 33.1.1.1/24
    ip rip 1 33.1.1.1
        send disable
interface 1/1/1
    no shutdown
    ip address 33.44.1.1/24
    ip address 44.44.1.1/24 secondary
    ip rip 1 33.44.1.1
        send disable
    ip rip 1 44.44.1.1
        send disable
interface 1/1/2
interface loopback 2
```

```
ip address 55.55.55.55/32
ip rip 1 55.55.55.55
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

# RIPng (IPv6) commands

## Configuration commands

### router ripng

```
router ripng <PROCESS-ID> [vrf <VRF-NAME>]
no router ripng <PROCESS-ID> [vrf <VRF-NAME>]
```

### Description

Creates RIPng process if not already created and enters the **router ripng <PROCESS-ID>** context for the VRF mentioned. If no VRF is mentioned, a default is used. Only one RIPng process is allowed per VRF.

The **no** form of this command deletes the RIPng instance for the VRF. If no VRF is mentioned the default is deleted.

| Parameter    | Description                                           |
|--------------|-------------------------------------------------------|
| <PROCESS-ID> | Specifies name of the RIPng process ID. Range: <1-63> |
| vrf          | Sets VRF name for RIPng process.                      |
| <VRF-NAME>   | VRF name for VRF.                                     |

## Examples

Creating RIPng process and naming the VRF:

```
switch(config)# router ripng 2 vrf red
```

Deleting RIPng process:

```
switch(config)# no router ripng 2 vrf red
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## Interface commands

### ipv6 ripng

```
ipv6 ripng <PROCESS-ID>  
no ipv6 ripng <PROCESS-ID>
```

### Description

Enables RIPng process on interface and creates a new context.  
The **no** form of this command deletes RIPng process on interface.

| Parameter    | Description                              |
|--------------|------------------------------------------|
| <PROCESS-ID> | Specifies RIPng process ID. Range: 1-63. |

### Examples

Enabling RIPng process on an interface:

```
switch(config)# interface 1/1/1  
switch (config-if)# ipv6 ripng 1  
switch (config-if-ripng)#
```

Deleting RIPng process on an interface:

```
switch(config)# interface 1/1/1  
switch (config-if)# no ipv6 ripng 1
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.



## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## Routing commands

### enable (router ripng)

enable  
no enable

### Description

Enables RIPng process if disabled. By default RIPng process is enabled.  
The **no** form of this command disables the RIPng process.

### Examples

Enabling RIPng process when disabled:

```
switch(config)# router ripng 1  
switch(config-ripng-1)# enable
```

Disabling RIPng process when enabled:

```
switch(config)# router ripng 1  
switch(config-ripng-1)# no enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                 |
|-----------|-----------------|-----------------------------------------------------------|
| 5420      | config          | Administrators or local user group members with execution |

| Platforms    | Command context | Authority                |
|--------------|-----------------|--------------------------|
| 6100<br>6200 |                 | rights for this command. |

## disable (router ripng)

disable  
no disable

## Description

Disables RIPng process.

The **no** form of this command enables the RIPng process.

## Examples

Disabling RIPng process:

```
switch(config)# router ripng 1
switch(config-ripng-1)# disable
```

Enabling RIPng process:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## distance (router ripng)

distance <DISTANCE>  
no distance

## Description

Configures administrative distance for RIPng. Administrative distance is used as criteria to select the best route when multiple protocols have the same route.

The **no** form of this command sets the RIPng administrative distance to the default. Default: 120.

| Parameter  | Description                                               |
|------------|-----------------------------------------------------------|
| <DISTANCE> | Specifies RIPng administrative distance. Range: 1 to 255. |

## Examples

Configuring administrative distance for RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# distance 100
```

Setting administrative distance for RIPng to default values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no distance
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## maximum-paths (router ripng)

```
maximum-paths <MAX-VALUE>
no maximum-paths
```

## Description

Sets the maximum number of ECMP routes that RIPng can support.

The **no** form of this command sets the maximum number of ECMP routes to the default value of 4.

| Parameter   | Description                                       |
|-------------|---------------------------------------------------|
| <MAX-VALUE> | Sets the number of RIPng ECMP routes. Range: 1-8. |

## Examples

Setting maximum number of RIPng ECMP routes:

```
switch(config)# router ripng 1  
switch (config-ripng-1)# maximum-paths 8
```

Setting maximum number of RIPng ECMP routes to default:

```
switch(config)# router ripng 1  
switch (config-ripng-1)# no maximum-paths
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## redistribute (ripng)

```
redistribute {bgp | connected | ospfv3 <PROCESS-ID> | static}  
no redistribute {bgp | connected | ospfv3 <PROCESS-ID> | static}
```

## Description

Redistributes routes originating from other protocols into RIPng.

The **no** form of this command disables redistribution of routes originating from other protocols into RIPng.

| Parameter           | Description                                                                               |
|---------------------|-------------------------------------------------------------------------------------------|
| bgp                 | Specifies BGP routes to redistribute into RIPng.                                          |
| connected           | Specifies connected routes (directly attached subnet or host) to redistribute into RIPng. |
| ospfv3 <PROCESS-ID> | Specifies the OSPFv3 route to redistribute into RIPng. Range: <1-65535>                   |
| static              | Specifies static route to redistribute into RIPng.                                        |



Parameters **ospfv3** and **bgp** are not supported on 6100 switches.

## Examples

Redistributing BGP routes into RIPng:

```
switch(config)# router ripng 1  
switch(config-ripng-1)# redistribute bgp
```

Disabling BGP routes that originate from other protocols and redistribute into RIPng:

```
switch(config)# router ripng 1  
switch(config-ripng-1)# no redistribute bgp
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification                                              |
|------------------|-----------------------------------------------------------|
| 10.14            | Supported process ID range expanded from 1-63 to 1-65535. |
| 10.07 or earlier | --                                                        |

## Command Information

| Platforms    | Command context | Authority                                                                          |
|--------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## timers update (RIPng)

```
timers update <INTERVAL> timeout <DURATION> garbage-collection <PERIOD>  
no timers
```

## Description

Configures RIPng timers with specific values.

The **no** form of this command sets all RIPng timers to default values.

| Parameter                   | Description                                                                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timers update <INTERVAL>    | Specifies frequency at which RIPng sends updates to all of its peers. Range: 1 to 2147484. Default: 30.                                                              |
| timeout <DURATION>          | Specifies timeout duration from the point of the last refresh after a route is received from a peer timeout and is marked as expired. Range: 1 to 255. Default: 180. |
| garbage-collection <PERIOD> | Specifies amount of time route remains in routing table after route expiration. Range: 1 to 255. Default: 120.                                                       |

## Examples

Configuring RIPng timers with specific values:

```
switch(config)# router ripng 1  
switch(config-ripng-1)# timers update 40 timeout 200 garbage-collection 150
```

Configuring RIPng timers with default values:

```
switch(config)# router ripng 1  
switch(config-ripng-1)# no timers
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config          | Administrators or local user group members with execution rights for this command. |

## RIPng clear commands

### clear ipv6 ripng statistics

```
clear ipv6 ripng [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

### Description

Clears RIPng event statistics.

| Parameter    | Description                             |
|--------------|-----------------------------------------|
| <PROCESS-ID> | Specifies RIPng process ID. Range: 1-63 |
| all-vrfs     | Clears statistics for all VRFs.         |
| vrf          | Selects VRF to clear statistics for.    |
| <VRF-NAME>   | Specifies VRF name.                     |

## Examples

Clearing RIPng event statistics:

```
switch# clear ipv6 ripng statistics
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                          |
|----------------------|-----------------------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## RIPng interface commands

### enable

enable  
no enable

### Description

Enables RIPng process on interface.

The **no** form of this command disables RIPng process on interface.

### Examples

Enabling RIPng process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# enable
```

Disabling RIPng process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# no enable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## disable (interface ipv6 ripng)

disable  
no disable

### Description

Disables RIPng process on interface.

The **no** form of this command enables RIPng process on interface.

### Examples

Disabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# disable
```

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# no disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## send disable (ripng)

send disable  
no send disable

### Description

Disables interface from sending RIPng packets. An interface can send RIPng packets by default.



The **no** form of this command enables interface to send RIPng packets, if disabled.

## Examples

Disabling interface from sending RIPng packets:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)# send disable
```

Enabling interface to send RIPng packets:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)# no send disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## receive disable (ripng)

```
receive disable
no receive disable
```

## Description

Disables interface from receiving RIPng packets for all enabled IP addresses. An interface can receive RIPng packets by default.

The **no** form of this command enables interface to receive RIPng packets, if disabled.

## Examples

Disabling interface from receiving RIPng packets:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch (config-if-ripng)# receive disable
```

Enabling interface to receive RIPng packets when disabled:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch (config-if-ripng)# no receive disable
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | config-if       | Administrators or local user group members with execution rights for this command. |

## RIPng show commands

### show capacities ripng

show capacities ripng

### Description

Displays the maximum number of RIPng interfaces, routes and process.

### Examples

Displaying maximum number of RIPng interfaces, routes and process:

```
switch# show capacities ripng

System Capacities: Filter RIPng
Capacities Name                                     Value
-----
Maximum number of RIPng interfaces configurable in the system 32
Maximum number of RIPng processes supported across each VRF    1
Maximum number of routes in RIPng supported across all VRFs    2540
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show capacities-status ripng

show capacities-status ripng

### Description

Displays number of RIPng interfaces, routes and process configured in the system.

### Examples

Displaying number of RIPng interfaces, routes and process:

```
switch# show capacities-status ripng

System Capacities Status: Filter RIPng
Capacities Name
      Value  Maximum
-----
Number of RIPng interfaces configured in the system
              0       32
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng

show ipv6 ripng [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]

### Description

Displays general RIPng configuration.

| Parameter    | Description                                           |
|--------------|-------------------------------------------------------|
| <PROCESS-ID> | Specifies RIPng process ID. Range: 1-63.              |
| all vrfs     | Displays general RIPng information for all VRFs.      |
| vrf          | Selects VRF to display general RIPng information for. |
| <VRF-NAME>   | Specifies VRF name.                                   |

## Usage

- Parameters display general RIPng information for a specific RIPng process.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying general RIPng configuration for all VRFs:

```
switch# show ipv6 ripng 34 all-vrfs
VRF : Default                               Process-ID : 34
-----
Protocol Status      : Enabled   ECMP           : 6
Update Time         : 60 sec   Timeout Time    : 240 sec
Garbage Collection Time : 250 sec Distance        : 100
Redistribution       : static,
                    ospfv3 1

VRF : vrf_1                               Process-ID : 34
-----
Protocol Status      : Enabled   ECMP           : 4
Update Time         : 30 sec   Timeout Time    : 180 sec
Garbage Collection Time : 120 sec Distance        : 120
Redistribution       : None
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context                | Authority                                                                                                                                                                  |
|----------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager<br>(#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng interface

```
show ipv6 ripng [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief] [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays information about RIPng enabled interfaces.

| Parameter        | Description                                           |
|------------------|-------------------------------------------------------|
| <PROCESS-ID>     | Specifies RIPng process ID. Range: 1-63.              |
| <INTERFACE-NAME> | Specifies interface.                                  |
| brief            | Shows brief overview information for RIPng interface. |
| all-vrfs         | Displays interface information for all VRFs.          |
| vrf              | Selects specific VRF.                                 |
| <VRF-NAME>       | Specifies VRF.                                        |

## Usage

- Parameters display general RIPng information for a specific RIPng process.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

```
switch# show ipv6 ripng interface
Interface 1/1/1 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----
VRF           : Default           Process-ID      : 1
Status        : Oper Up           Mode            : Send and Receive
MTU           : 500               Poision Reverse : Enabled

Interface 1/1/2 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----
VRF           : Default           Process-ID      : 1
Status        : Admin Down        Mode            : Receive
MTU           : 500               Poision Reverse : Enabled

Interface 1/1/3 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----
VRF           : Default           Process-ID      : 1
Status        : Admin Down        Mode            : Send
MTU           : 500               Poision Reverse : Enabled

switch# show ipv6 ripng interface brief
VRF : default   Process-ID : 1
-----

Total Number of Interfaces: 2

Interface      IPv6-Address           Status    MTU
-----
1/1/1          fe80::7272:cfff:fe70:67a  up        500
1/1/2          fe80::7272:cfff:fe71:67a  up        500
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng neighbors

```
show ipv6 ripng [<PROCESS-ID>] neighbors [<LINK-LOCAL-ADDRESS>] [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays information about RIPng neighbors.

| Parameter            | Description                                  |
|----------------------|----------------------------------------------|
| <PROCESS-ID>         | Specifies RIPng process ID. Range: 1-63.     |
| neighbors            | Specifies neighbor IP address.               |
| <LINK-LOCAL-ADDRESS> | Specifies link-local address.                |
| all-vrfs             | Displays neighbor information for all VRFs.  |
| vrf                  | Selects VRF to display neighbor information. |
| <VRF-NAME>           | Specifies VRF name.                          |

## Usage

- Parameters display RIPng neighbor information for a specific RIPng process.
- Parameters display RIPng neighbor information for a specific neighbor.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIPng neighbor information for all VRFs:

```
switch# show ipv6 ripng neighbors all-vrfs
VRF : default          Process-ID : 1
-----
```

```
Total Number of Neighbors: 1
```

| Peer-Address              | Type  | Last-Update | Rcvd-Bad-Pkts | Rcvd-Bad-Routes |
|---------------------------|-------|-------------|---------------|-----------------|
| fe80::7272:cfff:fe70:86ae | RIPng | 0:0:7       | 4             | 5               |



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng routes

```
show ipv6 ripng [<PROCESS-ID>] routes [<PREFIX/LENGTH>] [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays RIPng routing table for a specific RIPng process.

| Parameter       | Description                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------|
| <PROCESS-ID>    | Specifies RIPng process ID to display information for a specific RIPng process. Range: 1-63. |
| <PREFIX/LENGTH> | Specifies the network prefix.                                                                |
| all-vrfs        | Displays RIPng routing information for all VRFs.                                             |
| vrf             | Selects VRF to display RIPng routing information.                                            |
| <VRF-NAME>      | Specifies VRF name.                                                                          |

## Usage

- <PREFIX/LENGTH> is an optional parameter that displays RIPng routing table information for a specific subnet.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIPng routing table for all VRFs:

```

switch# show ipv6 ripng routes all-vrfs
VRF : default                      Process-ID : 1
-----
Prefix                Metric      Interface    Nexthop
-----
2001:DB8:10::/64      2        1/1/1        FE80::2E0:E6FF:FE1B:8242
2002:DB8:10::/64      3        1/1/1        FE80::2E0:E6FF:FE1B:8242
2003:DB8:10::/64      1        1/1/1
VRF : vrf_1                      Process-ID : 34
-----
Prefix                Metric      Interface    Nexthop
-----
3001:DB8:10::/64      10       1/1/2        FE80::2E0:E6FF:FE1B:8232
3002:DB8:10::/64      14       1/1/2        FE80::2E0:E6FF:FE1B:8232
3003:DB8:10::/64      1        1/1/2

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng statistics

show ipv6 ripng [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]

## Description

Displays RIPng statistics.

| Parameter    | Description                                       |
|--------------|---------------------------------------------------|
| <PROCESS-ID> | Specifies RIPng process ID. Range: 1-63.          |
| all-vrfs     | Displays statistics information for all VRFs.     |
| vrf          | Selects VRF and displays RIPng statistics for it. |
| <VRF-NAME>   | Specifies VRF name.                               |

## Usage



- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIPng statistics for all VRFs:

```
switch# show ipv6 ripng statistics all-vrfs
VRF : default   Process-ID : 1
-----

Global Route Changes : 50
Global Queries       : 2
Last Cleared          : 0h 30m 28s ago

VRF : vrf_1     Process-ID : 34
-----

Global Route Changes : 20
Global Queries       : 0
Last Cleared          : 0h 30m 28s ago
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context                | Authority                                                                                                                                                                  |
|----------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager<br>(#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng statistics interface

show ipv6 ripng [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>] [all-vrfs | vrf <VRF-NAME>]

## Description

Displays RIPng statistics for RIPng enabled interfaces.

| Parameter        | Description                                       |
|------------------|---------------------------------------------------|
| <PROCESS-ID>     | Specifies RIPng process ID. Range: 1-63.          |
| <INTERFACE-NAME> | Specifies name of interface.                      |
| all-vrfs         | Displays RIPng interface statistics for all VRFs. |

| Parameter  | Description                                        |
|------------|----------------------------------------------------|
| vrf        | Selects VRF to display RIPng interface statistics. |
| <VRF-NAME> | Specifies VRF name.                                |

## Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIPng statistics for a RIPng enabled interface:

```
switch# show ipv6 ripng statistics interface 1/1/1
VRF : default Process-ID : 1 interface 1/1/1
-----
IPv6-Address      Trigger-Updates    Rcvd-Bad-Packets    Rcvd-Bad-Routes
-----
fe80::7272:cfff:fe70:86ae
                  15                  3                    4

Last Cleared: 0h 30m 28s ago
```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show running-config

```
show running config
```

## Description

Displays all running configurations for all protocols including RIPng.

## Examples

Displaying all running configurations for all protocols including RIPng:

```

switch# show running-config
Current configuration:
!
!Version Halon 0.1.0 (Build: genericx86-64-Halon-0.1.0-master-20170309054955-dev)
!Schema version 0.1.8
lldp enable
timezone set utc
vrf green
vrf red
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospfv3 1
router ripng 1
    maximum-paths 5
    distance 1
    redistribute ospfv3 1
    timers update 40 timeout 200 garbage-collection 150
router ripng 1 vrf red
    default-information originate always
    maximum-paths 7
    distance 5
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ipv6 address link-local
    ipv6 ripng 1
        send disable
interface 1
    no shutdown
    ipv6 address link-local
    ipv6 ripng 1
        receive disable

```



For more information on features that use this command, refer to the IP Routing Guide for your switch model.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                                                                                                                  |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5420<br>6100<br>6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## Accessing HPE Aruba Networking Support

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HPE Aruba Networking Support Services       | <a href="https://www.hpe.com/us/en/networking/hpe-aruba-networking-support-services.html">https://www.hpe.com/us/en/networking/hpe-aruba-networking-support-services.html</a> |
| AOS-CX Switch Software Documentation Portal | <a href="https://arubanetworking.hpe.com/techdocs/AOS-CX/help_portal/Content/home.htm">https://arubanetworking.hpe.com/techdocs/AOS-CX/help_portal/Content/home.htm</a>       |
| HPE Aruba Networking Support Portal         | <a href="https://networkingsupport.hpe.com/home">https://networkingsupport.hpe.com/home</a>                                                                                   |
| North America telephone                     | 1-800-943-4526 (US & Canada Toll-Free Number)<br>+1-650-750-0350 (Backup—Toll Number)                                                                                         |
| International telephone                     | <a href="https://www.hpe.com/psnow/doc/a50011948enw">https://www.hpe.com/psnow/doc/a50011948enw</a>                                                                           |

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

### Other useful sites

Other websites that can be used to find information:

|                                                      |                                                                                                                                                                                                                       |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HPE Aruba Networking Developer Hub                   | <a href="https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about">https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about</a>                                                     |
| Airheads social forums and Knowledge Base            | <a href="https://community.arubanetworks.com/">https://community.arubanetworks.com/</a>                                                                                                                               |
| AOS-CX Software Technical Update channel on YouTube. | Videos on new features introduced in this release:<br><a href="https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS">https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS</a> |

|                                                                     |                                                                                                                                                                                             |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HPE Aruba Networking Hardware Documentation and Translations Portal | <a href="https://arubanetworking.hpe.com/techdocs/hardware/DocumentationPortal/Content/home.htm">https://arubanetworking.hpe.com/techdocs/hardware/DocumentationPortal/Content/home.htm</a> |
| HPE Aruba Networking software                                       | <a href="https://networkingsupport.hpe.com/downloads">https://networkingsupport.hpe.com/downloads</a>                                                                                       |
| Software licensing and Feature Packs                                | <a href="https://licensemanagement.hpe.com/">https://licensemanagement.hpe.com/</a>                                                                                                         |
| End-of-Life information                                             | <a href="https://networkingsupport.hpe.com/end-of-life">https://networkingsupport.hpe.com/end-of-life</a>                                                                                   |

## Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

## Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

## Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

### Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

## Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback

([docsfeedback-switching@hpe.com](mailto:docsfeedback-switching@hpe.com)). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.